

CCPS 209 Computer Science II Labs

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♪ If you're having code problems I feel bad for you son,
I got ninety-nine problems but unit tests ain't one. ♪

— Ilk-K, "99 Java problems"

As far back as I can remember, I always wanted to be a computer scientist. To me, being a computer scientist was better than being the President of Finland. Even before I first wandered into the department for a TA job I knew I wanted to be a part of them. It was a glorious time. Coders were all over the place. It was before the first Internet crash. There was Big Antti, Little Antti, Konsta, Jaco, Manu, Henri, Terhi, and me, Ilkka One-Time, who got that nickname because I always said everything once. You learned the two greatest things in life, never rat on your friend functions, and always keep your functions quiet. You know, when you think of a university, you get pictures in your mind of all those college comedies with rows and rows of students behind desks, doing real drinking and sexting all mixed up like Animal House. All those goody two-shoes writing their essays. It wasn't like that for code guys. We studied alone. We owned the joint. For us, to live any other way was nuts.

— Ilcolas Koleggi, "Codeguy"

*For I am the random and the deterministic
I am the passing one and the failing one.
I am the non compiling one and many are my results.
I am the acceptance test and she who does not demo.
I am the slave of him who forked me.
I am the ruler of my stack and heap.
I am the control flow that is incomprehensible
And the design whose refactoring is frequent.*

— "Asunder, Imperfect Code"

*And the cat purred, aye, she sees the procession of your functions and your modules.
You shall be executing to all eternity, for as it is on compilation so it is in runtime.*

— Ilar Khoyyam, "The Process Runner"

♪ And you may find yourself living in a local stack
And you may find yourself in another part of the heap
And you may find yourself behind the root of a large composite tree
And you may find yourself in a beautiful scope with a beautiful type
And you may ask yourself, well, how did I get declared? ♪

— Talking Headers, "Once in a variable's lifetime"

When the statements come, they are merely empty shells without the runtime. They live as they are executed, for the statements are the body and the runtime the spirit.

— Ildegard of Bongen

Our code is what our thoughts make it.

— Marcus Coccarius, “Indentations”

Outer loops have inner loops, on their bodies to bite them.

Inner loops have inner loops, and so on ad infinitum.

And the outer loops themselves, in turn, have outer loops to go on,

While those again have greater still, and greater still, and so on.

— Markustus De Korgan, “Bugonapthera”

♪ *It's not the side effects of the pattern,*

I'm thinking that it must be null,

It's too late to be stateful,

It's too late to pass time limit again,

It's too late to be resourceful,

Dependency inversion is here! ♪

— The Thin White Koke, “Kether to Malkuth”

♪ *i, j, k are indices,*

n and m fixed integers,

x, y, z for decimals,

u, v, w are nodes of graphs! ♪

— GySgt I. Kortman, “I don’t code but source is gold”

Our lines no longer feel the editor under them.

At ten pages you can’t read our comment.

And where there is enough to start linking

Kremlin test suite is remembered at once.

His fingers on keyboard are bulky and fat like live baits

And his accurate statements are heavy as weights.

The huge laughing declarations on top of his files

The glitter of his boot sequence.

But around his screen a crowd of thin applications

And he plays with the services of these half-utilities.

One compiles, other optimizes,

He is alone typing, poking and peeking.

He forges his statements like horseshoes

Slapping some in the header, in the class, in the program body.

He rolls the executions on his tongue like raspberries.

He wishes he could container them like big friend back home.

— Ilip Kokkelstam, “The Kremniya Epigram”

This document contains the lab specifications for the course [CCPS 209 Computer Science II](#), as compiled and taught over the years by [Ilkka Kokkarinen](#) for the [Chang School of Continuing Education, Toronto Metropolitan University, Canada](#). It offers a veritable buffet of **99 problem specifications** followed by fourteen bonus problems for students to freely choose their hills to rush and capture the flag on. These problems vary greatly in their themes and techniques so that, in spirit of *The Love Boat*, they promise something for everyone. Students who have previously taken this author's first programming course [CCPS 109 Computer Science I](#) taught in Python should also note that unlike in [the problem set used in that course](#), **these problems are not listed in their order of difficulty**, but in the order in which they were originally created. Have no fear of taking a look around all over this document to see which topics interest you enough to raise the urge to hit the keyboard!

These problems drill in various aspects of the Java language and its standard library, and introduce (although occasionally between the lines) many classic ideas, concepts, techniques and algorithms that will surely come handy in your future coding tasks in both school and working life. The first half of these labs mostly contains problems about the straightforward application of basic data types such as integers, arrays and lists, whereas the problems in the second half are inspired by discrete math, combinatorics and computational geometry. Occasionally these problems even reach out into more remote areas such as **fractal art**, **cellular automata** and **political science**, this last topic covered in problems about implementing and analyzing various **voting systems**.

To work on these labs using [IntelliJ IDEA](#), create yourself new project where all your source code files for these labs will go. This will be a different project than the one where you keep [the example programs of this course](#). In your labs project, you will then separately copy the automated **JUnit tests** from the instructor's GitHub repository [CCPS209Labs](#) for each lab that you attempt to solve. First download this entire folder somewhere safe on your computer, and whenever you embark on a new lab problem, copy its corresponding JUnit test class in your lab project folder. (There is no point having tests in your project for labs that you haven't even started.) Once you have implemented all the methods of the lab you are currently working on so that they compile, IntelliJ IDEA will allow you to run either one test or all the tests for that lab, for you to determine whether your work passes the muster. (Incomplete methods can throw [UnsupportedOperationException](#) for the time being, until you stuff their bodies with the meat of actual logic and statements.)

When IntelliJ IDEA initially complains about the `import` statements in the JUnit test class, you should correct the problem once and for all by clicking the red light bulb symbol to open a drop-down menu to add JUnit 4 in your classpath. That is enough for IDEA to know to use JUnit 4 for all the tests in the same project. **Note that these tests are written in JUnit 4, instead of JUnit 5**, so be sure to add the correct framework in your classpath when prompted to do so by IDEA!

The **fuzz acceptance tests** in the JUnit test classes will try out your methods with a large number of **pseudorandomly** generated **test cases**. These test cases are guaranteed to be exactly the same for everyone working on the Java language and its standard libraries, regardless of the particular computer environment they currently happen to be working on. Each pseudorandom fuzz test computes a checksum from the results that your method returns for the test cases given to it. If this checksum equals the expected checksum computed and hardcoded into the tester from the instructor's private model solution, [The Man from Del Monte says yes](#) and grants you the green checkmark to celebrate your beating this level with a nice glass of pineapple juice. Your method is now correct and accepted

as **black box equivalent** to the instructor's private solution, all this testing taking place without ever making the private model solutions available for students.

Any discrepancy in the checksum reveals that your method returned a wrong answer for at least one test case. Unfortunately, this pseudorandom fuzzing scheme makes it impossible to pinpoint the actual test case for which your method failed, or even to determine for how many test cases your method disagrees with the instructor's private model solution of correct answers. To alleviate this "something is wrong but I won't tell you what, you would know if you really loved me" mechanism that some psychologist might deem passive aggressive were it done by a human, the JUnit test classes also feature conventional tests with explicit test cases aimed to flush out the most common bugs and edge case situations that seem to repeat in student solutions.

All these JUnit fuzz tests are designed to run to completion within at most a couple of seconds per each green checkmark, when executed on a modern off-the-shelf desktop computer less than five years old. **None of these automated tests should ever require minutes, let alone hours, to complete.** If any JUnit test gets stuck that way when coupled with your code, that means **your methods are doing something algorithmically very inefficient**, either in time or in space. You should mercilessly root out all such inefficient designs from your method, and replace them with better algorithms that get to the same end result at least an order of magnitude faster!

In all these labs, [silence is golden](#). When it comes to your methods of talking out of school all over the text console, the rules for *omertà* are as strict as they are in the thorny badlands of Sicily. Since these JUnit tests will pummel your code with a large number of pseudo-randomly generated test cases harder than an old time detective giving a suspect the third degree under the hot lights, **all required methods must be absolutely silent** and print absolutely nothing on the console during their execution. **Any lab solution that prints anything at all on the console during its JUnit test will be rejected and receive a zero mark.** Make sure to comment out all your console output statements you used for debugging before submitting your project.

Once you have completed all the labs that you think you are going to complete before the deadline, you will and must **submit all your completed labs in one swoop** as the entire "CCPS 209 Labs" project folder that contains all the source files along with the provided JUnit tests and any necessary text or image data files, compressed into a zip file that you then upload into the assignment tab on D2L. As there is no partial credit given for these labs, **please do not include any solutions that do not pass the tests with flying colours in your submission.** To make it easier for the instructor to tally up your working labs, you should make it clear in the project README file how many lab problems you are claiming to have successfully solved, so that the instructor doesn't have to count with his fingers and toes how many points to award you for your labs.

Students should work on solving these problems independently. Discussion of problems between students is acceptable, as long as this discussion takes place solely at the level of ideas and no actual solution code is passed between the students. Two students who have both independently completed some problem may then compare and contrast their solutions with each other to the heart's content, and are recommended to do so if their solutions differ greatly in their execution efficiency. This problem collection also has an official subreddit [r/ccps209](#) to facilitate such discussion. Note that this subreddit is intended only for the discussion of these problems, and any course management issues should be handled elsewhere through the appropriate channels.

Lab 0(A): Plain Old Integers

JUnit: [P2J1Test.java](#)

The transition labs numbered 0(X) translate your existing Python knowledge into equivalent Java knowledge all the way between your brain and your fingertips. You may already be familiar with some of these problems from this author's earlier course [CCPS 109 Computer Science I](#). Even if you are coming to this second programming course via some other route than this author's, these problems are self-contained and require no specific background knowledge.

But first: if you have trouble with IntelliJ IDEA in that it refuses to run your examples or labs code at all, the issue is most likely with the **Project** or **Module settings**. If the issue is that a Java Virtual Machine (JVM) has not been selected, or that the Java language level is unsupported, you can change these settings to ones that work with the instructions on the page "[Module structure settings](#)" in the IntelliJ IDEA documentation. All the examples and labs of this course can be run on Java version 15. (In fact, even the practically stone age version 8 of Java would be sufficient for these labs, but Java 15 should be a reasonable universal baseline in Java programming in this day and age. We don't need any more modern or any preview features of the later versions of this language.)

Inside your fresh new lab project, create the first class that **must be named exactly P2J1**. If you name your classes and methods anything other than how they are specified in this document, the JUnit tests that we use to automatically check and grade these labs will not even be able to find your methods, let alone test them! Inside the new class, the first method to write is

```
public static long fallingPower(int n, int k)
```

Python has the integer exponentiation operator `**` conveniently built in the language, whereas Java unfortunately does not offer that. Exponentiation would be basically useless anyway in a language that uses fixed size integers that silently hide the overflows that the integer exponentiation will be producing in spades. (You should also note that in both Python and Java, the **caret** character `^` denotes the **bitwise exclusive or** operation that has nothing to do with integer exponentiation.)

Instead of integer exponentiation, your task is to implement the related operation of **falling power** that is useful in many **combinatorial** formulas of [discrete calculus](#). Distinguished syntactically from ordinary powers by underlining the exponent, each term multiplied into the product is always one less than the previous term. For example, the falling power $8^{\underline{3}}$ is computed as $8 \times 7 \times 6 = 336$. Similarly, the falling power $10^{\underline{5}}$ equals $10 \times 9 \times 8 \times 7 \times 6 = 30240$. Nothing essential changes in this calculation when the base is negative; the falling power $(-4)^{\underline{5}}$ is computed with the same formula with $-4 \times -5 \times -6 \times -7 \times -8 = -6720$. Similar to normal exponentiation, $n^{\underline{0}} = 1$ for any nonzero integer n .

This method should return the falling power $n^{\underline{k}}$ where the base n can be any integer, positive or negative or zero, and the exponent k can be any **nonnegative** integer, so also zero. The JUnit fuzz tests are designed so that your method does not have to worry about potential integer overflow... provided that you perform your arithmetic calculations with the `long` kind of 64-bit integers! If you use the bare 32-bit `int` type anywhere in your method, a silent integer **overflow** will make your

method occasionally mess up its interim values and return incorrect results that fail the JUnit fuzz test. (Yes, even if that method worked when you tried it with your own small values of n and k .)

```
public static int[] everyOther(int[] arr)
```

Given an integer array `arr`, create and return a new array that contains precisely the elements in the even-numbered positions in the array `arr`. For example, given the array `{5, 2, 3, 10, 2}`, this method would create and return a new array object `{5, 3, 2}` that contains the elements in the even-numbered positions 0, 2 and 4 of the original array. Make sure that your method works correctly for arrays of both odd and even lengths, and for **empty** arrays and **singleton** arrays with just one element. The length of the result array that you return must also be exactly right so that there are no extra zeros hanging around at the end of that result.

```
public static int[][] createZigZag(int rows, int cols, int start)
```

This method creates and returns a new two-dimensional integer array, which in Java is really just a one-dimensional array whose elements are one-dimensional arrays of type `int[]`. The returned array must have the correct number of rows that each have exactly `cols` columns. This array must contain the numbers `start`, `start+1`, ..., `start+(rows*cols-1)` in its rows in sorted order, except that the elements in each odd-numbered row must be listed in descending order. Note that `start` can be any integer, even a negative one.

For example, when called with `rows=4`, `cols=5` and `start=4`, this method should create and return the two-dimensional array whose contents show up as

4	5	6	7	8
13	12	11	10	9
14	15	16	17	18
23	22	21	20	19

when displayed in the standard **matrix form** that is more readable than the more truthful form `{{4,5,6,7,8},{13,12,11,10,9},{14,15,16,17,18},{23,22,21,20,19}}`, in reality a one-dimensional array whose four elements are themselves one-dimensional arrays of integers that serve as rows of this two-dimensional array. In addition to rectangular grids of values, this scheme can represent arbitrary **ragged arrays** whose rows don't need to all be of the same length.

```
public static int countInversions(int[] arr)
```

Inside an array `arr`, an **inversion** is a pair of two positions $i < j$ so that `arr[i] > arr[j]`. In combinatorics, the inversion count inside an array is a rough measure of how much “out of order” that array is. If an array is sorted in ascending order, it has zero inversions, whereas an n -element array sorted in reverse order has $n(n-1)/2$ inversions, the largest number possible. (You will encounter inversions again in this course, if you work through the labs 45 to 47 that deal with **permutations** and their operations.) This method should count the inversions inside the given array `arr`, and return that count. As always when writing methods that operate on arrays, make sure that your

method works correctly for arrays of any length, including the important special cases of zero and one.

Once you have written all four methods, add the above **JUnit test** class [P2J1Test.java](#) inside the same project. Unlike in Python with its more dynamic nature, **the JUnit test class cannot be compiled until your class contains all four methods exactly as they are specified**. If you want to test one method without also having to first write the other three, you must implement the other three methods as do-nothing **method stubs** that only contain some placeholder `return` statement that returns zero or some other convenient do-nothing value. An even better way to handle this situation would be to have the method waiting to be implemented have a one-liner body

```
throw new UnsupportedOperationException();
```

that tells the caller in a controlled fashion that this method does not even pretend to achieve anything at all, but gives up right away. Of course all such method stubs will immediately fail their respective tests, as they properly should. However, their existence allows the JUnit test class to compile and run cleanly so that you can start testing each one method the moment you have properly replaced the above statement in its body with the actual Java code to solve the specified problem. IntelliJ IDEA conveniently allows you to run all tests for that problem, or just one individual test, just by clicking the green Play button next to each individual test in its source code.

Lab 0(B): Duplications

JUnit: [P2J2Test.java](#)

Create a new class named P2J2 inside the very same project as you placed your P2J1 class in the previous lab. Inside this new class P2J2, write the following four methods.

```
public static String removeDuplicates(String text)
```

Given a `text` string, create and return a brand new string that is otherwise the same as `text` except that every run of equal consecutive characters has been turned into a single character. For example, given the arguments "Kokkarinen" and "aaaabbxxxxaaxa", this method would return "Kokkarinen" and "abxaxa", respectively. Only consecutive duplicate occurrences of the same character are eliminated, but the later occurrences of the same character remain in the result as long as there was some other character between these occurrences. For the purposes of this problem, the upper- and lowercase versions of the same character are considered different, so that removing duplicates from "AaabbBBBCc" should return the result "AabBCc".

```
public static String uniqueCharacters(String text)
```

Create and return a new string that contains each character of `text` only once regardless of its appearance count there, these characters listed in order that they appear in the original string. For example, given the argument strings "Kokkarinen" and "aaaabbxxAxxaaxa", this method should return "Kokkarine" and "abxA", respectively. Same as in the previous problem, the upper- and lowercase versions of the same character are treated as different characters.

You could solve this problem with two nested loops, the outer loop iterating through the positions of the `text`, and the inner loop iterating through all previous positions to look for a previous occurrence of that character. But you can be more efficient and use a [HashSet<Character>](#) to remember which characters you have already seen, and let that data structure help you decide in constant time whether the next character is appended into the result.

```
public static int countSafeSquaresRooks(int n, boolean[][] rooks)
```

A number of chess rooks have been placed on some squares of a generalized n-by-n chessboard. The two-dimensional array `rooks` of `boolean` truth values tells you which squares contain a rook. (This array is guaranteed to be exactly n-by-n in its dimensions.) This method should return the count of how many remaining squares are safe from the rolling wrath of these rampaging rooks, that is, do not contain any rooks in the same row or column.

Probably the best way to do this is to create and fill in two one-dimensional `boolean` arrays named `safeRows` and `safeCols` with `n` elements each. These arrays keep track of which rows and columns of the chessboard are unsafe. Initially all rows and columns are safe as far as we know, since both arrays are initially all `false`. However, to get count of the actual situation in the given chess-

board, loop through all the squares of the chessboard. Whenever you find a rook in some position `(row,col)`, mark that entire `row` and `col` as being unsafe in those respective arrays. After you have done this for all the squares throughout the entire chessboard, loop through the `safeRows` and `safeCols` arrays separately to count how many rows and columns are still safe. Return the product of those two numbers as your final answer.

```
public static int recaman(int n)
```

Compute and return n th term in the [Recamán's sequence](#), starting the count from term $a_1 = 1$. See the definition of this curious sequence on the linked Wolfram Mathworld page, or read more from the page "[Asymptotics of Recaman's sequence](#)". For example, when called with $n = 7$, this method would return 20, and when called with $n = 19$, return 62. (More values for debugging purposes are listed at [OEIS sequence A005132](#).)

To make your function fast and efficient even when computing the sequence element for large values of n , you should use a sufficiently large `boolean[]` to keep track of which integer values are already part of the generated sequence, so that you can generate each element in constant time instead of having to loop through the entire previously generated sequence, the way some "[Shlemiel](#)" would do tiring himself by running back and forth across the same positions. The size `10*n` should be enough for this array. This way this scheme needs only ten measly bits of heap memory for each number generated, plus some spare change.

Storing the numbers encountered so far in the generated sequence in some `HashSet<Integer>` instance instead would be a textbook case of overpaying for overkill, and burn up precious memory at least an order of magnitude harder than the technique of using a simple **bit vector** to remember which values you have already encountered before.

Lab 0(C): Reverse Verse

JUnit: [P2J3Test.java](#)

Here is one more batch of transitional problems adapted from the instructor's old collection of Python graded labs. Only three methods to write this time, though. The JUnit test for these labs uses the [warandpeace.txt](#) text file as its data source, so you should ensure that this text file has been properly copied inside your course project root directory. Unless you have changed this setting in the IntelliJ IDEA project settings, the working directory is the root directory of your project, not the `/src` directory, so this text file should be in this root directory.

Create the class named `P2J3` in your project folder, in which you will be writing three `static` methods. The first one of these three methods to write is

```
public static void reverseAscendingSubarrays(int[] items)
```

Rearrange the elements of the given array of integers **in place** (that is, **do not create and return a new array**) so that the elements of every **maximal strictly ascending subarray** are reversed. For example, given the array `{5, 7, 10, 4, 2, 7, 8, 1, 3}` (the pretty colours indicate the ascending subarrays and are not actually part of the argument), after executing this method, the elements of the array would be `{10, 7, 5, 4, 8, 7, 2, 3, 1}`. Given another argument array `{5, 4, 3, 2, 1}`, the contents of that array would stay as `{5, 4, 3, 2, 1}` seeing that its each element is a maximal ascending subarray of length one.

The second method to write in the same `P2J3` class is

```
public static String pancakeScramble(String text)
```

This nifty little problem is [taken from the excellent Wolfram Challenges problem site](#) where you can also see examples of what the result should be for various arguments. Given a `text` string, construct a new string by reversing its first two characters, then reversing the first three characters of that, and so on, until the last round where you reverse your entire current string.

This problem is an exercise in Java string manipulation. For some mysterious reason, even in this day and age the Java `String` type still does not come with the `reverse` method baked in. The canonical way to reverse a `String` object is to first convert it to mutable `StringBuilder`, reverse the contents of that in place, and convert the result back to an immutable string. That is,

```
text = new StringBuilder(text).reverse().toString();
```

A bit convoluted, but does what is needed without fuss or muss. Maybe one day the Java strings will come with the `reverse` method, just like the string data types of all sensible programming languages these days. Or at least have it available as a utility somewhere in the standard library.

To reach the finish line of this lab, the third method to write inside this same class is

```
public static String reverseVowels(String text)
```

Given a `text` string, create and return a new string of same length where all vowels have been reversed, and all other characters are kept as they were. For simplicity, in this problem only the characters `aeiouAEIOU` are considered vowels, and `y` is never a vowel. For example, given the text string "Uncle Sente lives in Russia", this method should create and return the string "Ancli SuntI levis en Resseu".

Furthermore, to make this problem more interesting and the result look more palatable, this method **must maintain the capitalization of vowels** based on the vowel character that was originally in the position that each new vowel character is moved into. For example, "Ilkka Markus" should become "Ulkka Markis" instead of "ulkka MarkIs". Use the handy character classification methods in the [Character](#) utility class to determine whether some particular character is in upper or lower case, and convert some character to its upper or lowercase version as needed.

As a leisurely mental exercise after completing this method, the reader can try to think up the longest meaningful sentence that does not change in vowel reversal, or try to find the longest meaningful sentence whose vowel reversal is another meaningful sentence. An equally obvious follow up is to write a function to reverse the consonants of the sentence, and perhaps use the resulting mess as an artificial language in your swords and cloaks fantasy universe.

Lab 0(D): Lists Of Integers

JUnit: [P2J4Test.java](#)

This fourth transition lab from Python to Java contains four more interesting problems taken from the [graded labs of the instructor's Python course](#). The four static methods to write this time all deal with `List<Integer>` data type of Java standard library (remember to put the necessary imports to the top of the source code of your class) whose behaviour and operations are essentially identical to those of ordinary Python lists of integers, except with a way more annoying syntax.

Create a class `P2J4` in your lab project directory, and there the first method

```
public static List<Integer> runningMedianOfThree(List<Integer> items)
```

Create and return a new `List<Integer>` instance (use any concrete subtype of `List` of your choice) whose first two elements are the same as that of original `items` (so the result for any list with fewer than three elements will be same as that list), after which each element equals the **median** of the three elements in the original list ending in that position. For example, when called with a list that prints out as `[5, 2, 9, 1, 7, 4, 6, 3, 8]`, this method would return an object of type `List<Integer>` that prints out as `[5, 2, 5, 2, 7, 4, 6, 4, 6]`. The second method to write in this class is

```
public static int firstMissingPositive(List<Integer> items)
```

Given a list whose each element is a positive integer, return the first positive integer missing from this list. For example, given a list `[7, 5, 2, 3, 10, 2, 9999999, 4, 6, 3, 1, 9, 2]`, this method should return 8. Given one of the lists `[ ]`, `[6, 2, 12345678]` or `[42]`, this method should return 1.

If your method compares the `items` to each other for equality, you must use the `equals` method to do so instead of the `==` operator, since these `items` are all `Integer` objects, not primitive integers. However, since the **pigeonhole principle** dictates that the first missing positive integer of an n -element list must necessarily be less than or equal to $n + 1$, a boolean array of that many elements is a superior technique to keep track of which numbers you have seen so far. With its aid, you can solve this problem efficiently in both time and space with two **consecutive** for-loops, instead of being a Shlemiel who does the same job using two **nested** for-loops. For the purpose of minimizing the method running time, these two ways to have two loops are not at all equal!

The third method out of the four methods to write is

```
public static void sortByElementFrequency(List<Integer> items)
```

Sort the elements of the given list in **decreasing order of frequency**, that is, how many times each element appears in this list. If two elements have the same frequency in the parameter list, those

elements should end up in **ascending order of values**, the same way as we do in ordinary sorting of lists of integers. Note that this method does not return anything, but modifies `items` in place.

For example, given a list object that prints out as `[4, 99999, 2, 2, 99999, 4, 4, 4]`, after calling this method that list object would print out as `[4, 4, 4, 4, 2, 2, 99999, 99999]`. The return type of this method is `void`, because this method rearranges the elements of `items` in place instead of creating and returning a separate result list.

As in all computer programming, you should allow the language and its standard library to do your work for you instead of rolling your own logic. The method `Collections.sort` can be given a [Comparator<Integer>](#) strategy object that compares two integers for their ordering. Start by building a local **counter map** of type `Map<Integer, Integer>` used to keep track of how many times each value that you see appears inside the list. Next, define a local class that implements `Comparator<Integer>` to create strategy objects whose `compare` method performs integer order comparisons by consulting this map for the counts of those elements to compute the answer, reverting to ordinary integer order comparison only in the case where these counts are equal. Once you have defined this local class, the actual sorting of `items` takes place with just a single call to the utility method `Collections.sort`.

The fourth method that takes you cross the finish line of this lab is

```
public static List<Integer> factorFactorial(int n)
```

Compute and return the list of prime factors of the **factorial** of `n` (that is, the product of all positive integers up to `n`), with those prime factors sorted in ascending order and with each factor appearing in this list exactly as many times as it would appear in the prime factorization of that factorial. For example, when called with `n=10`, this method would create and return a list of prime factors that prints out as `[2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 5, 5, 7]`. Multiplying all those itty bitty prime numbers together would produce the result of 3,628,800, which equals the product of the first ten positive integers. Remember that when `n` equals either zero or one, its factorial equals one (since there exists exactly one way to arrange the elements of the empty list and the singleton list in a row), and therefore produces an empty list of prime factors.

The factorial `n!` grows exponentially, so when `n` goes up to [not just mere eleven](#) but even past that mythical barrier, these factorials would no longer be representable inside the 31 bits used to house the magnitude of the primitive `int` type of Java. Since this method must be able to work for values of `n` that are in the thousands and their factorials consist of tens of thousands of digits, you should not first compute that entire factorial of `n` and only then start breaking the resulting behemoth down to its prime factors. Instead, you need to build up the list of prime factors as you go, by appending the prime factors of the integer that you are currently multiplying into the factorial. When you have done this for all integers up to `n`, sort the result list in ascending order before returning it.

Lab 0(E): All Integers Great And Small

JUnit: [P2J5Test.java](#)

The Python integer type is **unbounded** so that its magnitude is limited only by your available heap memory. The Python virtual machine silently switches behind the scenes between different efficient representations for small and large integers, so that up here we can concentrate on the actual problem instead of the details of nitty gritty integer arithmetic. When dealing with the primitive integer types of Java, we have to constantly worry about and protect against overflow errors. (This is typically achieved by closing your eyes and ingesting some hopium that nobody will ever pass your methods arguments big enough to make some intermediate result overflow.)

The utility class [java.math.BigInteger](#) allows you to do computations in Java with similarly unlimited integers. This again being Java as originally conceived in the early nineties, the syntax is not quite as nice as the ordinary and natural syntax for primitive `int` type. For example, to add two such integers `a` and `b`, we have to say `a.add(b)` instead of `a+b`, almost as if our nation had lost a war or something. (Explanations for why Java even this blessed day and age still doesn't have **operator overloading** make for a good reading for all those times that you could use a chuckle.) Consult the rest of the methods of `BigInteger` as needed to solve the following problems.

There are only methods to complete in this lab, in class named `P2J5`. The first one is

```
public static List<BigInteger> fibonacciSum(BigInteger n)
```

This method computes some [Fibonacci numbers](#), that dusty old example of a silly recursion still used in many introductory courses on computer science, turn out to have interesting combinatorial properties that also make for more meaningful problems for us to play with. For this lab, we look at [Zeckendorf's theorem](#) that proves that any given positive integer n can be exactly one way into a sum of distinct Fibonacci numbers, given the constraint that no two consecutive Fibonacci numbers appear in this sum. After all, if the sum contains two consecutive Fibonacci numbers F_i and F_{i+1} , these two can always be replaced by F_{i+2} without affecting the total. (To ensure that you have at most one of each F_i at any moment, this conversion should be performed from higher indices down.)

The unique breakdown of n into Fibonacci numbers can be constructed with a **greedy algorithm** that simply finds the largest Fibonacci number f that is less than or equal to n . Add this f to the result list, and break down the rest of the number $n-f$ the same way. This method should create and return an instance of some subtype of `List<BigInteger>` that contains the selected Fibonacci numbers in **descending order**. For example, when called with $n=1000000$, this method would return the list `[832040, 121393, 46368, 144, 55]`.

Your method must remain efficient even if n contains thousands of digits. To achieve this, maintain a static list of Fibonacci numbers that you have generated so far, initialized with

```
private static List<BigInteger> fibs = new ArrayList<>();
static { fibs.add(BigInteger.ONE); fibs.add(BigInteger.ONE); }
```


Whenever the last Fibonacci number stored in this list is not big enough for your current needs, extend the list with the next Fibonacci number, computed by adding up the last two known Fibonacci numbers. Even though this is not needed in this problem, keeping the Fibonacci numbers discovered so far in one sorted list would also allow you to quickly determine whether the given integer is some Fibonacci number with `Collections.binarySearch`.

The second method to implement in this lab is

```
public static BigInteger sevenZero(int n)
```

Since seven is a lucky number in the Western culture, whereas [zero is what nobody wants to be](#), let us briefly bring these two opposites together and look at positive integers that consist of some sequence of sevens, followed by some (possibly empty) sequence of zeros. For example, 0, 7, 77777, 7700000, 77777700, or 70000000000000000000000000000000. All the sevens must come first in one consecutive bunch, followed by all the zeros in another consecutive bunch, the second bunch being possibly empty. This rule keeps the space of such numbers small enough for us to examine through a brute-force search, since there exist only n such numbers with n digits.

In the wonderful MIT OpenCourseWare online textbook “*Mathematics for Computer Science*” ([PDF link](#) to the 2018 version, for anybody interested in that sort of stuff), one of the examples of **non-constructive proofs** used the [pigeonhole principle](#) to prove that for any positive integer n , there exists some seven-zero integer that is divisible by n . This integer made of sevens and zeroes can quickly get gargantuan even for relatively small values of n . For example, for $n=12345$, the smallest seven-zero number divisible by n consists of a whopping 822 copies of the digit seven, followed by a single zero digit tacked to the end.

This method should find and return the smallest seven-zero integer that is divisible by the given n . The easiest way to do this would be to use two nested loops. The outer `while`-loop iterates through all possible digit lengths (that is, how many digits the number contains) of the number. For each digit length, the inner `for`-loop iterates through all legal sequences of consecutive sevens followed by a sufficient number of zeros to make that number to have the desired digit length. Keep going up until you find the first seven-zero number exactly divisible by n . Since your loops iterate through the seven-zero numbers in ascending order, the first number found this way must be the smallest such number that satisfies the requirement.

Furthermore, you can cheerily exploit a handy additional theorem proven in the same book to speed up your search immensely. Unless n is divisible by either 2 or 5, the smallest seven-zero number that is divisible by n is guaranteed to contain only sevens, and no zeros. This realization should speed up your search by at least an order of magnitude for such easy values of n , since the quadratic number of possibilities to examine is pruned down into a single linear branch that goes through the numbers 7, 77, 777, 7777, 77777... in search of the answer. On the other hand, numbers that contain the powers of two and five as their factors require seven-zero numbers whose tails of zero digits in the end grow along the exponents of these two little prime powers. The result for $n = m 2^a 5^b$ where m contains neither prime factor 2 or 5 will have exactly $\max(a, b)$ zeros tacked to the end.

Lab 0(F): Two Branching Recursions

JUnit: [P2J6Test.java](#)

In mainstream computer science education given at university level, teaching of **recursion** tends to nerf down this mighty blade to utter pointlessness. Recursion can be used to cut through an exponential thicket of possibilities, yet only ever seems to be used to solve outright trivial **linear recursions** such as computing factorials. Linear recursions gain nothing over their everyday alternative of ordinary while- and for-loops, the way that any competent coder would have solved those problems anyway whenever their pay check was on the line. This completely needless limitation of vision tends to leave students dazed and confused about the general usefulness of recursion, since from their by that time glazed over point of view, recursion only ever seems to be used to solve toy problems in a needlessly highfalutin manner. (Mainstream teaching of introductory recursion makes even the formal verification community, the only science that has to cheat at toy problems to get any results at all, look like Strong Doge in comparison.)

However, as ought to become amply evident somewhere in the third or fourth year, the power of recursion lies in its **ability to branch to multiple directions** one at a time. This remarkable ability to branch allows a recursive method to explore a potentially **exponentially large branching tree of possibilities** and **backtrack on failure** to the previous choice point. It is as if some program built out of the usual forward-going loops had magically gained the ability to turn back time and take back the assignments that hurt you, as in the song. Even better, such recursions can be trivially modified to either find the first branch that has a working piece of fruit hanging from its end, or keep going to collect all such available fruits from all branches. In this spirit, this last transition lab uses branching recursions to solve two interesting combinatorial problems. So, you handsome maniac you, without any further ado, create a new class P2J6 to write this lab's two methods into.

```
public static List<Integer> sumOfDistinctCubes(int n)
```

Determine whether the given **positive** integer n can be expressed as a sum of cubes of positive integers greater than zero so that all these integers are distinct. For example, $n=1456$ can be expressed as a sum of two cubes $11^3 + 5^3$. This method should return the list of these distinct integers as an object of some subtype of `List<Integer>` that prints out as `[ 11, 5 ]`, the elements listed in descending order. If there is no way to break the given integer n into a sum of distinct cubes, this method should return the empty list. Note that for the purposes of "[gleaming the cube](#)" by pushing your limits to the edge as if this were an energy drink commercial, the **empty sum** and a **singleton sum** are still sums in the same sense that zero and one are integers, as counterintuitive as this might seem. For example, the integers 0 and 8 are representable as sums of distinct cubes as the empty sum and the singleton sum 2^3 , respectively.

Some integers can even be broken down into a sum of cubes in different ways. For such cases, this method must return the breakdown that is **lexicographically highest**, that is, starts with the largest possible working value for the first element, and then breaks down the rest of the number into a list of distinct cubes in a similar fashion. For example, when called with $n=1729$, the famous [Ramanujan taxicab number](#), this method **must** return the list `[ 12, 1 ]` instead of the list `[ 10, 9 ]`,

even though $12^3 + 1^3 = 10^3 + 9^3 = 1729$ just as well. This constraint may initially look scary, but you can really just *fuhgettaboutit*, since its enforcement is trivial by arranging the loop inside the recursive method to scan through the current possibilities from highest to lowest. As with such recursions, you should again use a private helper method that accepts additional parameters that were not present in the public method.

```
private static boolean sumOfDistinctCubes(  
    int n, int c, LinkedList<Integer> soFar  
)
```

This helper method receives the two extra recursion parameters `c` and `soFar` from the original method. The parameter `c` contains the highest integer that you are still allowed to use. Initially this should equal the largest possible integer whose cube is less than or equal to `n`, easily found with a while-loop. The parameter `soFar` contains the list of numbers that have already been taken into the list of cubes so that they will be available once the recursion has returned.

The recursion has two base cases, one for success and one for failure. If `n==0`, the problem is solved for the **empty sum**, so you can just return `true`. If `c==0`, there are no numbers remaining that you could use, so you simply return `false`. Otherwise, try taking `c` into the sum, remembering also to add `c` to `soFar`, and recursively solve the problem for new parameters `n-c*c*c` and `c-1`. If that attempt was unsuccessful, remove `c` from `soFar`, and try to recursively solve the problem without using `c`, which makes the parameters of the second recursive call to be `n` and `c-1`.

Note how, since this method is supposed to find only one solution, once the recursive call returns `true`, its caller can also immediately return `true` without exploring the remaining possibilities. This first success, guaranteed to be the lexicographically first one by the order these alternative solutions are explored, will therefore cause the entire recursion to immediately roll back all the way to the top level, where the breakdown of `n` into distinct cubes can be read from the list `soFar`.

To allow the search to cut off in some situations where continuing the search is guaranteed to be futile, it might also be of interest to know that the sum of cubes of consecutive integers from 1 to `c` can be computed quickly without loops using the formula $c^2(c + 1)^2/4$, a polynomial of degree four. If the target sum `n` is larger than the sum of cubes up to `c`, the method can immediately conclude that the breakdown is impossible, without going through the futile process of combing through all possible combinations of integers up to `c` only to find out that all of them are too small.

That's it for the distinct cubes method, recursing over combinations of integers. The second method to write in this lab deals with strings instead of numbers.

```
public static List<String> forbiddenSubstrings(  
    String alphabet, int n, List<String> tabu  
)
```

Compute and return the list of all strings of length `n` that can be formed from the characters given in the list `alphabet`, but under the constraint that none of the strings listed in `tabu` are allowed to

appear anywhere as substrings. For example, the list of all strings of length three constructed from alphabet "ABC" that do not contain any of the substrings ["AC", "AA"] would be ["ABA", "ABB", "ABC", "BAB", "BBA", "BBB", "BBC", "BCA", "BCB", "BCC", "CAB", "CBA", "CBB", "CBC", "CCA", "CCB", "CCC"]. To facilitate automated JUnit testing, your method must return this list of acceptable strings in sorted alphabetical order.

Following the exact same principle as in the previous problem of adding up distinct cubes, this recursion is also easiest to implement with a private helper method

```
private static void forbiddenSubstrings(  
    String alphabet, int n, List<String> tabu,  
    String soFar, List<String> result  
)
```

that takes two additional parameters `soFar` and `result`. The parameter `soFar` is the partial string constructed down the path to this point (this should equal the empty string at the top level call) and `result` is the list of strings that the recursion has already discovered. The base case for failure is when the string `soFar` ends with any one of the strings in the `tabu` list. (The `String` instance method `endsWith` might come handy in this.) The base case for success is the situation where `soFar.length()==n`, in which case `soFar` is added to the `result` list. Otherwise, the recursion should loop through the characters in the `alphabet`, appending each character to the `soFar` string before the recursive call, and removing it from the string after that recursive call.

Unlike the previous method to find the distinct cubes that was supposed to terminate after the first success, this private recursive method does not return anything to indicate success or failure. Even if this method finds some solution right away, it still has to keep chugging through the entire tree of branching possibilities to glean the successes from that entire search tree.

Lab 0(G): ...Or We Shall All Crash Separately

JUnit: [P2J7Test.java](#)

On the day of creating this lab, your instructor had to wait one hour in line in the cold winter outside Costco during the early stages of the COVID pandemic isolation measures, everyone standing there equally unsure of what the future would bring after two weeks of flattening the curve. This wait gave him ample time to think up a bunch of new problems (this entire document used to consist of only about twenty problems before the whole world changed), and the situation naturally lent itself to the concept of queueing. Since various kinds of queues, along with the nifty data structures that efficiently implement them, are useful in many algorithms, this should be a perfect time to check out the collection type `LinkedList<E>`, the faster alternative to `ArrayList<E>` for the operations needed to implement a **double-ended queue** (often called “**deque**”) when there is no need for random access into the list, but the queue is accessed and mutated only from both ends. (As an alternative, you can also try out the actual `ArrayDeque<E>` class in the standard library actually intended to be used as a proper deque.)

As in all algorithmic problems, you should always choose the data structure that makes those operations fast that you are going to be doing a lot, whereas the cost of operations that you are not doing is irrelevant. (The same principle extends to all of engineering with the realization that non-existent parts waste neither time nor energy, and these nonexistent parts can never be corrupted by either rust or moth.)

Most of the time, double-ended queues are used as ordinary “**first in, first out**” (FIFO) queues so that elements always enter the queue from the rear to eventually exit from the front in the exact order that they came in. But as long as it costs us nothing to have the ability to potentially do something, we might as well keep that possibility in our sleeve!

The first problem, no doubt at least subconsciously inspired by that long cold wait outside Costco, takes us to the world of the ancients where a bunch of zealots (yes, Lana, *literally*, look it up) had been surrounded by Romans with no escape available through the front or the back door. Preferring a glorious mass suicide over capture and torturous death by crucifixion, these men arranged themselves in a circle and drew lots for a random starting location. Counting from that location k steps ahead each time, the... man... who... was... *it!* was killed and removed from this grim game of eeny-meeny-miney-moe. The last man standing was expected to swiftly fall on his own sword to join his fallen brothers. [Josephus](#), our lovable wandering hero who had unfortunately become entangled with these rascals, had other plans to escape. After some quick mental arithmetic, he managed to finagle himself to this last surviving position to tell this tale. Could you do as well as Josephus under this considerable high stakes pressure and time limit? (After all, it's not like Josephus had a penchant for self aggrandizing with tall tales or anything like that. Was you dere, Charlie?)

Create a new class named `P2J7` in your project folder, and in that class, write the generic method

```
public static <T> List<T> josephus(List<T> men, int k)
```

that is given a list of men in order that they stand in the circle, counting initially starting from the first man. This method should create another list that contains the men in the order that they were eliminated. Logic of this elimination does not depend on the element value or even its type but solely on its position, so we might as well make this method maximally generic by allowing the parameter type to be an arbitrary `List<T>`. As explained in the lecture about generics, this is not at all the same thing as defining that parameter type to be `List<Object>`! A polymorphic method that expects to receive a `List<Object>` argument will not be able to handle an argument of type `List<Person>`, whereas a method that expects a `List<T>` argument can do so easily.

As always, **this method should not modify the contents of its parameter list**, but create and return a new list object (you get to choose the concrete subtype yourself here) as the result. Note also that the step size k can be larger than the initial number of men, and yes, this does make a difference occasionally. For example, with `men = [ "ross", "ted", "ringo", "alice", "bob", "rachel" ]` and $k=9$, the resulting elimination order is `[ "ringo", "ross", "ted", "rachel", "alice", "bob" ]`. Shirts or skirts, none will be spared!

Our second example of patiently waiting in queue comes from a more recent period in history, and is also an important algorithm from the era when humans still had to do the work of computers by hand. With slow and cranky humans playing the part of computers (in fact, that is what the word “computer” originally meant as somebody’s job title), we surely want to optimize every algorithm to reach the answer with the smallest number of computations.

Every ten years, the United States of America is constitutionally obligated to run a nationwide census, so that the seats to the congress (more precisely, to the House of Representatives) are distributed among the fifty states in proportion to their populations. The problem here is how to round the number of seats that each state gets into an exact integer, since this task is not as clear-cut as it may initially seem. Simple rounding of each value to the closest integer does not work, because this method would not preserve the total number of congressional seats in the entire nation. How can you tell which states are you supposed to round up, and which ones you round down? Making this decision by sorting the states by the decimal part of their number of seats and handing out the leftover seats in this descending order doesn’t work either.

Paradoxically in situations of this nature, even though integers in general are much easier and more accurate to perform calculations than floating point numbers used to approximate real numbers, the [integer programming](#) problem of finding a combination of integer values that satisfy the given constraints often turns out to be computationally far more complicated than solving that same problem when arbitrary floating point numbers are allowed in the solution.

Fortunately, the classic [Huntington-Hill method](#) solves this integer programming problem to minimize the overall **quantization error** caused by rounding the values of the solution to integer precision. In this method, each state first receives one initial seat, regardless of its population. The remaining seats are then distributed one seat at the time. The next seat always goes to the state with the current highest **priority quantity**, as computed with formula $P^2 / (s(s + 1))$, where P is the population of that state and s is the number of seats that state has received so far. Since the number of seats s is in the denominator of this fraction, whereas the numerator P^2 remains constant, each additional seat always decreases the priority of the state that received it, so that other states reach the

head of this priority queue to receive seats only after California, Texas, Florida and New York have taken their couple of first dibs. (Just sit still, Rhode Island over there, and patiently wait for your turn. You will eventually get that second seat that you are entitled to.)

The Wikipedia page gives bit of a different formula for priority quantity with a square root. But see what I did there, knowing that we do not actually care about the absolute values or the scales of these priority quantities, but only care about their mutual ordering? This ordering of priorities is unaffected by the application of any strictly **monotonic** function such as **squaring** to these quantities. However, squaring these irrational roots back to integers allows us to deal with exact integers, instead of having to juggle around the clown car of inherently imprecise floating point numbers! This allows the next method to work even in galactic civilizations in some space opera where populations are measured in octillions (one has to wonder, does every clone warrior or a copy of some virtual AI get a separate vote?) and yet the main characters somehow just happen to all know each other, provided that we do these calculations using the `BigInteger` type. (Forget that whole stupid Coruscant with all its pointless pomp: an *actual* technologically super-advanced civilization of the future in a galaxy far away will be managed by two guys sharing one laptop.)

Inside the same class P2J7 where you wrote the previous Josephus problem, write the method

```
public static int[] huntingtonHill(int[] population, int seats)
```

that, given the `population` in each state and the number of `seats` to distribute across these states, creates and returns a new array that contains the counts of seats given to each state. To break the ties to make the returned results unambiguous for the JUnit fuzz test, whenever two states currently have an equal priority quantity, the next seat is given to the state that is **earlier** in the `population` array.

To write this method, first add the class [Fraction](#) from the class examples project into your labs project, to let you perform your calculations with exact integer fractions **without any possibility of integer overflow or floating point rounding error**. Not even one floating point number, let alone any calculation involving them, should appear inside your method! Not just this method, but basically all your methods ever. Whenever you are happily writing code, and suddenly hear a young child's giggly voice coming from a drain, calling you to also come down and play with them, all of them just *floating* down there, [it is a really bad idea to follow that voice](#). God gave us integers; the rest is the work of Satan. (For the same message for our secular age, as some wag once updated [Jamie Zawinski's famous remark](#): Some people, when confronted with a problem, think "I know, I'll use floating point numbers." Now they have 1.9999999999999998904342 problems.)

You should also note that even though the population of some state fits comfortably inside an `int` without any overflow, the square of that population will not be equally friendly once this population reaches the millions, surely not unrealistic for electoral math! For this reason, these population squaring operations absolutely have to be performed using exact `Fraction` objects of unlimited range, not as primitive `int` values. Just like it is too late to close the barn door after the horse has escaped, it is too late to convert an `int` value into an unlimited `BigInteger` once the overflow has already taken place. For example, instead of trying to be concise, normally a virtue in our line of business, by writing a statement


```
num = BigInteger.valueOf(population[i] * population[i]);
```

where the `int` multiplication will overflow before conversion to `BigInteger`, split this into

```
num = BigInteger.valueOf(population[i]);  
num = num.multiply(population[i]);
```

to force the integer multiplication to take place inside the `BigInteger` mechanism where there is ample room for all digits to coexist. On that note, everybody stand up in attention, eyes front to receive the following extremely important instruction:

Just because an integer value fits into an `int`, its square might not do so. And just because the intermediate results of your computational formulas are not given explicit symbolic names in your code, it doesn't mean that those intermediate results somehow magically didn't exist in the memory and be truncated to its limitations, nor be subject to the exact same laws of computations as your named data items! Names and intentions are not magic!

Would your instructor really be so mean as to design the method `testHuntingtonHill` to fuzz populations that are too big to be accurately handled with floating point operations? Would that same instructor also ensure that some states end up having almost equal populations that differ by a measly little one or two people? Come on; at this stage of the studies, and especially if you have already taken my earlier course CCPS 109, do you really even have to ask?

The best way to track of the relative priorities of the states (each state identified as an integer as its position in `population` table) is to keep them inside a `PriorityQueue<Integer>` instance with a custom `Comparator<Integer>` object whose `compareTo(Integer a, Integer b)` method renders its verdict based on the current priorities of the states `a` and `b`. These priorities, of course, are kept up to date inside another array `Fraction[] priorities` that is defined as a local variable inside this `huntingtonHill` method, outside the nested comparator class. To find out which state gets the next seat, simply poll the queue for the state whose priority is currently the highest. Update the priority for that state to the `priorities` array, and offer the state back into the priority queue.

As historically important as this algorithm has been to the fate of the free world and our continued ability to keep rocking in it, it also has a more mundane everyday application in displaying rounded percentages based on a list of numbers from some real-world data. You know how you sometimes see a disclaimer phrased something like “Due to rounding, displayed percentages may not add up to exactly one hundred” under some table of numbers and their associated rounds percentages? Well, geez, just use the correct algorithm to display your rounded percentages, and any need for such disclaimers vanishes! For example, to compute the percentages rounded to one decimal place, simply distribute 1,000 virtual “seats” among your data items. Each of these virtual seats corresponds to one *per mille*, one tenth of a percentage point (yep, [those are an actual thing](#), as at every Finn with a driver's license surely knows), to be displayed as the exact percentage share of that data item.

Lab 0(H): More Integers Great And Small

JUnit: [P2J8Test.java](#)

In spirit of the earlier lab 0(E) that used the `BigInteger` type to represent solutions that would never fit inside an `int` or even a `long`, this lab brings you two more interesting problems dealing with “powerful” integer sequences reaching high about the weight class of those primitive types.

```
public static void hittingIntegerPowers(int a, int b, int t, int[] out)
```

Define two positive integers to be “close enough for government work” if their absolute difference multiplied by the given **tolerance** level t is at most equal to the smaller of those two numbers. For example, the integers 2000 and 2007 are “close enough” when $t = 100$, since the difference 7 multiplied by 100 gives 700, which is less than 2000. On the other hand, the integers 2000 and 2050 would not be close enough for $t = 100$, although they would be close enough for a wider tolerance of $t = 10$. Tolerance of $t = 100$ corresponds to the notion of being within one percent, but without using any division or floating point numbers to compute this. The more narrow tolerance of $t = 100000$ would require these powers to be within one thousandth of a percent of each other, which ought to satisfy even the most ardent six sigma advocate auditing this course from the business school.

Given two positive integers a and b and the desired tolerance t , this method should find and return find the smallest integer powers p_a and p_b so that when a is raised to the power of p_a , and b is raised to the power of p_b , the resulting two numbers are close enough for government work. Since these powers can get pretty big, as you can see in the table below, you need to perform these calculations using the `BigInteger` type. However, this is only for the actual powers that might end up having tens of thousands of digits; the exponents p_a and p_b are guaranteed to fit inside the `int` type for all the test cases given to your method.

a	b	t	Expected contents of out
2	4	100	{2, 1}
2	7	100	{73, 26}
3	6	100	{137, 84}
4	5	1000	{916, 789}
10	11	1000	{1107, 1063}
42	99	100000	{33896, 27571}

In Python, this function would simply return the answer as a two-tuple (p_a, p_b) of the exponents that fulfill the above requirement. Java does not have tuples in the core language, so we sidestep this by using a two-element `int[]` object to hold the result. Note that the return type of this method is `void` so that nothing is actually returned from this method. Instead, the method copies its answer into the two-element array that is handed to it as the parameter `out`.

This technique doesn't make much difference in this problem, but can be useful in other methods that get called millions of times, or if only to simulate the behaviour of some Python function that returns a tuple. Instead of this method having to create a small new array for the result every time it is called, the same array object can be reused between all these calls, which ought to lighten the load on the JVM garbage collector, at least back when the neither the art nor the science of garbage collection techniques was quite at the level where it is today.

```
public static BigInteger nearestPolygonalNumber(BigInteger n, int s)
```

As explained on the Wikipedia page “[Polygonal Number](#)”, for every positive integer $s > 2$, the corresponding **s -gonal numbers** are an infinite sequence of integers whose i :th element is given by the formula $((s - 2)i^2 - (s - 4)i) / 2$. The Wikipedia formula uses the letter n instead of i , but n already means something else in this problem. (Besides, the letter n should only be used to denote unknown but **fixed** integers anyway, whereas the letters i and j denote integers that **iterate** through position **indices** of loops.) For example, the infinite sequence of [octagonal numbers](#) that springs forth from setting $s = 8$ starts with 1, 8, 21, 40, 65, 96, 133, 176...

Given `s` as an ordinary primitive `int`, and `n` as a `BigInteger`, this method should find and return the `s`-gonal number that is closest to `n`. If the integer `n` lies exactly halfway between two consecutive `s`-gonal numbers, return the smaller of those two equidistant `s`-gonal numbers.

n	s	Expected result (as BigInteger)
5	3	6
27	4	25
450	9	474
(ten billion)	42	9999861561
(one googol)	91	1000 00416332753518329478897755794704334003003544 20356

This problem is best solved with application of **repeated halving**, better known as **binary search**. First, find some position b in the sequence so that the b :th s -gonal number is greater or equal to n . You can just start with b set to equal one, and keep multiplying it by ten until you reach your goal. Knowing some two positions a and b so that the correct answer is between these positions, inclusive, compute the midpoint index $m = (a+b) / 2$ and look at the s -gonal number in that position. If that number is less than n , assign $a = m + 1$, and otherwise assign $b = m$. Either way, you are good to continue. Once a and b are no more than one step apart, one more comparison gives you your answer. As you can see from the above table, even the positions a and b can get so huge that they would never fit inside an ordinary `int`, let alone the polygonal numbers stored in those positions...

Lab 0(I): Squaring Off

JUnit: [P2J9Test.java](#)

The two problems in this lab produce results of type `boolean[]`, arrays whose each element is a truth value. Since these individual bits can be packed eight into each memory byte tight as sardines, the storage of these arrays as **bit vectors** will be as compact as it can be, not one bit or byte of storage wasted in taking these truth bombs to the enemy. However, as a Java programmer you don't need to know anything about bits and bytes and the bitwise operations to handle them, since the Java compiler automatically turns all your bit vector operations into these low-level memory operations. Up here in the pleasant abstractions that the Java language deals with high above all that mud and grime, you can use `boolean` arrays no differently from how you always use arrays of any other element types.

Create the class named `P2J9` in your labs project, and there the following two methods that each solve a problem that involves **squares** of positive integers. Given this upper limit n as a parameter, the result of both methods is a `boolean[]` array of n elements that contains the answers for all natural numbers below n in one swoop. Compared to computing and returning the individual elements as requested by the caller, solving interdependent subproblems in bulk showers the benefits of Taylorism on us since many things are cheaper by the dozen. Unlike the industrial workers of the past, today's programmers prefer to sip their soft drinks in air-conditioned offices as the mindless machines perform those tiresomely repetitive tasks. Any decision or action done by a machine is always one fewer decision or action for us humans to concentrate our brains on.

```
public static boolean[] sumOfTwoDistinctSquares(int n)
```

Determines which natural numbers can be expressed in the form $a^2 + b^2$ so that a and b are two distinct positive integers. In the `boolean` array returned as result, the i :th element should be `true` if and only if such a breakdown is possible. The infinite sequence of positive integers that allow such breakdown begins with 5, 10, 13, 17, 20, 25, 26, 29, 34, 37, 40, 41, 45, 50, 52, ...

You may have previously seen a version of this problem where the breakdown to two distinct squares was done separately for one number at the time. Given n , the question asked you to find positive integers a and b whose squares add up to n . However, since we are dealing with values in bulk, perhaps rather than looping through each n and find positive integers a and b whose squares add up to n separately for each n , it would be more productive to loop through all relevant pairs of a and b , and see which values of n can be built from those pairs...

```
public static boolean[] subtractSquare(int n)
```

[Subtract a square](#) is a simple [impartial game](#) of mental arithmetic. Two players take turns moving from the current natural number n into any smaller natural number that can be reached by subtracting some square of a positive integer from n without going below zero. For example, assuming the current $n = 15$, the player whose turn it is to play can and must move into one of the three numbers 14, 11, and 6 that respectively correspond to subtracting the square of one, two or three from

the current number. Subtracting the square of four or higher is not allowed, since this would make the resulting value to be negative. The player who gets to move to zero wins the game, since the opponent can no longer make any moves and therefore loses the match under the **normal play convention**.

Using the standard terminology of this sort of impartial combinatorial games, a state is **winning** or **hot** if there exists at least one move into some cold state, and **losing** or **cold** if no such winning move exists. This rule makes the state $n = 0$ cold, since no moves are possible there. The state $n = 1$ is hot since the move of subtracting 1 wins the game. The state $n = 2$ is again cold, since the only possible move from that state leads to a hot state, thereby granting the opponent the guaranteed eventual win. The infinite sequence of cold states of this game begins with 0, 2, 5, 7, 10, 12, 15, 17, 20, 22, 34, 39, ...

This method should create and return an n -element array of boolean truth values so that the element in position i is `true` if the state i is hot, and false if it is cold. This method should be filling the result array from left to right, since to determine whether the current position is hot or cold, all the information needed to make that decision is already in the lower-numbered states whose heat values have already been filled in...

When played under the **misère play convention**, the player who moves to the zero state loses the game. However, the strategy of playing the misère version of the given combinatorial game cannot in general be constructed by merely interchanging the roles of the hot and cold states of the original normal play convention game, once again illustrating the principle of how [reversed stupidity is not intelligence](#). Generally, it does not pay to be miserly in analyzing the optimal actions of actually being a miser; we eat to live, not live to eat.

Lab 0(J): Similar Measures Of Dissimilarity

JUnit: [P2J10Test.java](#)

Consider two **bit vectors**, that is, two boolean arrays of equal length n , each filled with some combination of `true` and `false` elements. Various **dissimilarity** measures have been proposed to describe how “different” or “far apart” these two vectors of truth values stand from each other inside the **hypercube** of all 2^n possible vectors of truth values. If both bit vectors are identical, all these dissimilarity metrics will produce the answer zero. Otherwise the answer will be some positive quantity whose magnitude corresponds to the perceived dissimilarity between these bit vectors. Obviously, swapping the roles of the two bit vectors should not affect their dissimilarity. Generalizing this, permuting the boolean elements of these bit vectors the same way in both vectors should not affect the dissimilarity between those two bit vectors.

In this lab you will have to implement six such dissimilarity measures. All six methods are very similar in that they compute their results based on four counts n_{00} , n_{01} , n_{10} and n_{11} tallied from the two argument vectors. The count n_{00} is the number of positions in which both arrays have the value `false`. The count n_{01} is the number of positions where the first array `v1` has the value `false` and the second array `v2` has the value `true`. The other two counts n_{10} and n_{11} are defined symmetrically. Therefore for any two n -element arrays `v1` and `v2`, these four counts must add to n . It would be good to extract the computation of these four counts into a separate helper method that these six methods invoke as their first step. Having written that method, each method below is basically a one-liner that creates and returns the `Fraction` answer.

The formulas of how to compute each distance measure from the computed n_{ij} values can be found on the *Wolfram Mathematica* documentation page “[Distance and similarity measures](#)” section “Boolean data”, where you can click the Wolfram version of the function to read how that particular dissimilarity measure works. (On the documentation page for each separate function, click the section “Details” to expand it.) Alternatively, you can consult the page “[Distance measures](#)” that reveals how these six measures are merely the tip of the iceberg. (In the notation used in formulas on that page, $a = n_{00}$, $b = n_{10}$, $c = n_{01}$ and $d = n_{11}$.) Bees make honey and beavers build dams, whereas we are fortunate to belong to a species that creates this kind of immaterial artifacts.

Each of these six measures is to be implemented as a separate method inside the new class `P2J10` that you should create in your labs project. Since the expected answer is always some integer fraction, we will again use our [Fraction](#) example type as the return type of these methods.

```
public static Fraction matchingDissimilarity(boolean[] v1, boolean[] v2)
public static Fraction jaccardDissimilarity(boolean[] v1, boolean[] v2)
public static Fraction diceDissimilarity(boolean[] v1, boolean[] v2)
public static Fraction rogersTanimonoDissimilarity(boolean[] v1, boolean[]
v2)
public static Fraction russellRaoDissimilarity(boolean[] v1, boolean[] v2)
public static Fraction sokalSneathDissimilarity(boolean[] v1, boolean[] v2)
```

The following table lists a couple of argument arrays $v1$ and $v2$ of $n = 5$, along with their expected correct answers for each dissimilarity metric. To make the following table a “bit” (heh) more compact, the truth value arrays are displayed as bits so that 0 means `false` and 1 means `true`.

$v1$	$v2$	<i>Matching</i>	<i>Jaccard</i>	<i>Dice</i>	<i>RT</i>	<i>RR</i>	<i>SS</i>
10101	10101	0	0	0	0	2/5	0
01111	00100	3/5	3/4	3/5	3/4	4/5	6/7
01110	11100	2/5	1/2	1/3	4/7	3/5	2/3
10011	11100	4/5	4/5	2/3	8/9	4/5	8/9
01100	11010	3/5	3/4	3/5	3/4	4/5	6/7
11010	10100	3/5	3/4	3/5	3/4	4/5	6/7

Note the odd bird of the Russell-Rao dissimilarity among these dissimilarity metrics on the first row in the above table; even when two bit vectors are equal, each position where both vectors have a zero bit is counted as a dissimilarity between those two vectors under that metric.

Lab 0(K): Suffix Arrays

JUnit: [P2J11Test.java](#)

"It is as if the command: Do not covet things in space, were corrected with the unspoken word: Do covet things in time." — Abraham Joshua Heschel, *"The Sabbath"*

Strings are a stark data structure, universal and portable representation of all information that allows random access to the linear text stored within. Under the hood, strings are character arrays, but the primitive nature of such arrays is hidden behind a higher abstraction interface of the public methods of `String` class. Surprising algorithms have been developed over the years to quickly search and analyze both the string itself and the semantic information encoded in it. Even the seemingly simple task of [searching for a pattern inside the given text](#) can be optimized in countless ways, let alone various higher level tasks performed on these string instruments.

Occasionally the string data structure can be **augmented** with additional data that speeds up operations that would be inefficient if we were given only the raw unadorned text string itself. Even though this data in principle adds nothing new to the mix in that its contents are fully determined by the contents of the original string, having that redundant information available in a different form can provide significant speedups, as yet another form of the classic **space-time tradeoff**. This lab showcases a simple but powerful [suffix array](#) method as a technique to preprocess any text so that after the metaphorical cheque for the one-time payment of this preprocessing has cleared, all future searches of arbitrary **patterns** can be executed in time that grows only logarithmically with respect to the length of the text. This makes these pattern searches blazingly fast even if performed on the entire *War and Peace*!

Computed from the given `text` string that has n characters, its **suffix array** is an n -element array of integers that contains the position indices $0, \dots, n - 1$ into the `text`. Since each position is stored in the suffix array exactly once, the suffix array is some **permutation** of order n . The suffix array lists these n positions sorted in the **lexicographic order** (also called the "dictionary order") of the suffixes that start from each position. Note that the presentation on the linked Wikipedia page on suffix arrays uses one-based position indexing to the string, whereas here we naturally use Java's zero-based indexing to remain consistent to our principles as computer programmers.

For example, the non-empty suffixes of the string "hello" are "hello", "ello", "llo", "lo" and "o", corresponding to positions from 0 to 4 in the original string. Sorting these positions in the lexicographic order of their corresponding suffixes sorts them in the order `[1, 0, 2, 3, 4]`, which then becomes the suffix array of the original string. Since all n suffixes of `text` have a different length, the possibility of equality in their lexicographic comparison can never arise. This guarantees the uniqueness of the suffix array for the given `text`.

For reasons of simplicity and convenience, suffix arrays are represented in this lab as instances of some subtype of `List<Integer>`, despite the use of the standard technical term of "suffix array". Arrays, schmarrays. Create a new class named `P2J11`, and in there first the method

```
public static List<Integer> buildSuffixArray(String text)
```

that builds and returns the suffix array for the given `text`. It is enough to use the naive suffix array construction algorithm, since the structure of our test case strings will guarantee that the worst case scenario of this algorithm, a string whose all characters are the same, will never be realized.

The easiest way to implement this naive algorithm in Java should be to define another strategy subclass of `Comparator<Integer>` whose method `compareTo` then lexicographically compares the two substrings that start from the integer positions that it receives as parameters. In this method, first define the local variable `ArrayList<Integer> result`, and fill it up to the brim with the integers from 0 to $n - 1$. Then, use the utility method `Collections.sort` to sort this `result` list with your custom `Comparator<Integer>` strategy object.

This discipline should allow your `buildSuffixArray` method to be reasonably fast even when building the suffix array for the entire *War and Peace*, as performed by the JUnit test class using the [warandpeace.txt](#) data file. If you try to build an array that would contain all possible suffixes of that 3.2 million character text file as separate `String` objects, no computer that actually exists in our reality would have enough RAM to contain all those suffix objects.

Once the suffix array has been constructed for the given fixed `text`, it can be used to rapidly find all positions inside `text` where the given `pattern` occurs. These would be precisely the very same positions whose suffixes start with that `pattern`! Having already done the work of sorting these suffixes in lexicographic order in the previous preprocessing step allows us to use a slightly modified **binary search** performed to the suffix array to find the lexicographically first such suffix. Since all suffixes that start with the same `pattern` must necessarily follow each other as a bunch in the sorted array of such suffixes, looping through all these positions is a straightforward while-loop once the binary search has determined the first such position.

To achieve all that, write the second method

```
public static List<Integer> find(  
    String pattern, String text, List<Integer> suffix  
)
```

that creates and returns a list of positions of the original `text` that contain the given `pattern`, using the given `suffix` array to speed up this search. Note that this returned list of positions must be sorted in ascending order of the positions themselves, instead of the lexicographic order of the suffixes of `text` that start at these positions.

Lab 0(L): I Can, Therefore I Must

JUnit: [P2J12Test.java](#)

A jovial old grandpa mathematician straight out of central casting often appearing in [Numberphile](#) videos, [Neil Sloane](#) is behind the [On-Line Encyclopedia of Integer Sequences](#), a wonderful tool for combinatorial explorations. This lab has you implement two methods to generate elements of two interesting integer sequences whose chaotic behaviour emerges from iteration of a deceptively simple rule. Both sequences are filled in ascending order in a **greedy** fashion, so that each element is always the **smallest** number that avoids creating a **conflict** with the previously generated elements in the sequence. (Also, before you start: `remySigrist` requires **bitwise arithmetic**.)

Note that being defined by mathematicians, these sequences start from the position one instead of the position zero, unlike how sequences work for us budding computer scientists who have to actually get our hands dirty and therefore prefer zero-based indexing. When asked to produce the first n elements of the sequence, these methods should return an array with $n + 1$ elements. The zeroth element that we don't care about anyway is set to zero, after which the joyride begins.

```
public static int[] forestFire(int n)
```

Explained in the YouTube video "[Amazing Graphs II](#)", the first two elements both equal one. Then, each element $a[i]$ is the smallest positive integer for which no positive integer offset j creates an **arithmetic progression** of three equally spaced elements backwards in the sequence from the position i . More formally, the difference $a[i] - a[i-j]$ may never equal the difference $a[i-j] - a[i-2*j]$, even if your worst enemy got to pick the position i and the offset j .

```
public static int[] remySigrist(int n)
```

As explained in the YouTube video "[Amazing Graphs III](#)", this sequence devised by Rémy Sigrist assigns to every positive integer a natural number, called the "colour" of that number. After the first colour assignment $a[1]=0$, each following $a[i]$ thereafter is the smallest natural number c for which the binary representation of any earlier position j that has already been assigned the colour c has any bits turned on common with the binary representation of the position i . This can be checked quickly with the expression $i \& j == 0$, where $\&$ is the **bitwise and** operator of Java.

This problem can be solved with two nested for-loops. The outer loop iterates through the positions of a . The inner loop counts upwards through the colours until it finds one that does not create a conflict. However, you should not be a "Shlemiel" who uses a third level inner loop to determine whether the current colour candidate c creates a conflict. Instead, you should [trade space for time](#) with a second integer array named `taken` whose each element `taken[c]` gathers all the bits that are on in any previous position that was assigned the colour c . Then, whenever you to assign $a[i]=c$, update this lookup table with the assignment `taken[c] |= i`, where $|$ is the **bitwise or** operator of Java, and $|=$ its **assignment shorthand** analogous to $+=$ and other such familiar forms.

Lab 0(M): Tower Blocks

JUnit: [TowersTest.java](#)

The following problem is adapted from “[Towers](#)” in [CSES](#), another good collection and online judge for coding problems. This site is run by Antti Laaksonen, whose “[Competitive Programmer’s Handbook](#)” is an excellent resource about the application of basic data structures and algorithms for competitive programmers of all ages and sorts, even if they only ever compete with themselves to become slightly better every day. However, this problem does not require any of the techniques explained in that book, or for that matter, in any other classic algorithm textbooks. Instead, this problem is an exercise in coming up with an efficient mechanism to build the solution within the given **constraints** that prevent most of your possible moves and that way gently nudges you towards the optimal solution. (Same way as in many other walks of life, fences can often be your good and trusted friends, not mere hindrances that some unseen moron has placed there only to annoy you!)

You are given a series of `blocks` as an integer array, for example `{14, 9, 12, 4, 7, 1}`. You must process these blocks in **strict order from left to right**, which means that the order of these blocks in the array actually does matter a lot. These `blocks` are used to build **towers** so that every block must become part of some tower. On its turn to be processed, a block must be placed either on the table to start a new tower, or on top of some **strictly larger** block that is currently on top of some previously created tower. Again, there are **no backsies**; once you have placed a block somewhere, that block cannot later be taken out in style of jenga to be snuck into some more tempting and convenient location. As long as these rules are followed, each tower can end up being of any height, regardless of the heights of the other towers.

Your task is to **minimize the total number of towers**. For example, the previous `blocks` can be arranged into two towers, but no fewer. One working arrangement makes the first tower contain the blocks `{14, 12, 7}`, and the second tower the blocks `{9, 4, 1}`. As you can verify, both of these towers contain their blocks in the same order as those blocks appeared in the original array relative to each other. The optimal arrangement of `blocks` into the minimum number towers is not necessarily unique. Your method therefore needs to return only the minimum number of towers, not any actual working arrangement of blocks into towers.

In your labs project, create a new class `Towers`, and there the single method

```
public static int minimizeTowers(int[] blocks)
```

that computes and returns the minimum number of towers achievable with the given `blocks`. Your solution should be sufficiently efficient to complete the pseudorandom fuzz test for thirty thousand randomly created `blocks` of pyramidally increasing lengths in at most a couple of seconds.

Lab 0(N): Substring Parts

JUnit: [P2J13Test.java](#)

The two problems of this lab concern Java strings and substrings. Contrary to how things tend to work in other languages such as Python, Java 7 took advantage of the immutability of its `String` type by having the new object returned from the `substring` method share the same underlying character array with the original string. Since the contents of this shared `char[]` will never change, no harm can result in having multiple `String` objects share that same array. Immutability, huzzah! This policy allowed the `substring` method to work in guaranteed constant time needed to allocate the object from the heap and initialize its fields, regardless of the length of the original string and the substring being extracted into a separate object.

On the other hand, the big downside of this policy is that every `String` object in Java must know its `length` and the starting `offset` from the beginning of its character array, both stored as `int`. This causes every `String` object in the heap to be eight bytes larger than would be necessary, creating an instance of a curious phenomenon for which we could perhaps now coin the brand new term **space-space tradeoff**. Furthermore, sharing character arrays between strings and substrings prevent garbage collection of the large character array of the original string that became unreachable from the live variables of the program, yet is still kept alive by some small substring object that uses only a small part of that big chungus character array.

In your labs project, create a new class named `P2J13`, and in there two static methods

```
public static int countDistinctSubstrings(String text)
public static String reverseSubstringsBetweenParentheses(String text)
```

The first method should count how many distinct non-empty substrings exist in the given `text`, and return that count. For example, the string `"lollo"` contains a total of eleven distinct substrings: `"lollo"` of length 5, `"loll"` and `"ollo"` of length 4, `"lol"`, `"oll"` and `"llo"` of length 3, `"lo"`, `"ol"` and `"ll"` of length 2, and `"l"` and `"o"` of length 1.

This problem could be solved in a straightforward “Shlemiel” fashion with two nested for-loops to iterate through all pairs of possible starting points and substring lengths, and using some kind of `Set<String>` to maintain the distinctiveness of the strings so discovered. Even though the constant factors in the running time of this method are small enough to make it fast when most of the substrings are distinct from each other and it become necessary to iterate over all of them anyway, this technique is subject to [accidentally quadratic running time](#), provided that your worst enemy gets to play the **adversary** and choose the `text`. That sneaky little bugger could make the `text` consist of n repeated copies of the exact same character, the running time becoming painfully noticeable once n reaches into the thousands, as it will in the JUnit test for this problem. Such a string contains precisely n different distinct substrings, one for each possible length from 1 to n , but this algorithm would loop through a quadratic number of combinations to discover these substrings only to immediately throw away most of them!

A more complicated **adversary-proof** solution that is guaranteed to be invulnerable to accidentally quadratic running time, but pays this guarantee with a slightly higher constant factor in its running time over average-case inputs, stems from the realization that the substrings of length k can be constructed from the substrings of length $k+1$, by removing either the first or the last character from each one of these longer substrings. Your method should therefore generate these distinct substrings in the descending order of length. Use two separate instances of `HashSet<String>`, the first to remember the known distinct substrings of the previous length k , the second to store the substrings of length $k - 1$ that are currently being generated.

Initially, the first set is a singleton that contains only the original `text` itself. In each round of the loop, iterate through the known strings of k characters currently in the first set. Add to the second set the two substrings of length $k - 1$ constructed by dropping the first and last characters of that string of length k . Once you have done this for every string in the first set, add the size of the second set to your current tally of distinct substrings found. You now have the green light to `clear` the first set, and exchange the roles of the two sets for the next round of the loop to generate all the distinct substrings of length $k - 2$.

The second method of this lab also concerns substrings, but also illustrates the use and methods of the `StringBuilder` mutable string to efficiently construct a new string piecemeal by repeatedly appending characters to the end, and modifying this string while it is under construction. Given a `text` that consists of letters grouped with (possibly nested (pairs of parentheses)), this method should return a new string where the contents between each matching pair of parentheses have been reversed, leaving out the parentheses in this reversal. For example, given the string `"(ab)c(de)"`, this method should return the string `"baced"`.

Whenever the `text` contains nested parentheses, the corresponding substring reversals must be performed **inside out**. For example, the string `"(a(bc)d)"` should become `"dbca"`. The JUnit tester for this method guarantees that the parentheses inside the `text` given to your method are always **properly nested and balanced**, so your method does not have to concern itself with handling malformed strings whose parenthesis structure is inconsistent.

To perform this operation efficiently, use a `StringBuilder` to accumulate the `result`, along with a `Stack<Integer>` instance to keep track of the positions of the left parentheses encountered along the way. Loop through the characters of `text` from left to right, and `append` each character that is neither kind of parenthesis to the `result`. At each open parenthesis, `push` the current length of the `result` buffer into the stack. At each closing parenthesis, `pop` the position of its left counterpart from the stack. Extract and delete the substring of the `result` from that position into a separate substring that you then reverse and `append` back to the `result`.

Wham, bam, thank you, ma'am... except that even after almost three decades of evolution of the Java programming language, its `String` objects still do not have the simple `reverse` method that we take for granted in basically any other programming language used for serious professional work. The canonical way to create a reversed version of some Java string object `str` is by going through the expression `new StringBuilder(str).reverse().toString()`.

In general, Java enjoys so few advantages over Python 3 that we are almost obligated to point them out every time, and the guaranteed constant time operation of the `substring` extraction certainly qualifies. This idea could potentially be taken much further especially if the `String` data type were not defined to be `final`, but allowed different implementations to store the immutable text in different ways. Then, text editors and other programs that need immutable strings that are not merely sliced and diced into smaller pieces but also concatenated to create new strings from these pieces, could use the [rope](#) data structure where these operations work in guaranteed constant time, as illustrated in labs 64 and 65. The price of storing additional references inside each rope node is well compensated by the fact that each piece needs to exist only once in the memory, regardless of inside how many larger concatenated pieces of text that piece has been used as a building block.

Lab 0(O): Chars To The Left Of Me, Integers To The Right...

JUnit: [P2J14Test.java](#)

The two methods of this lab continue with the ongoing theme of strings, but this time operating at the level of individual characters. Both of these problems were adapted from the ample offerings of [LeetCode](#), one of the premier collections of online coding challenges and automated testers beloved by technical interviewers and job searching candidates. The JUnit test for this lab also uses the text file [warandpeace.txt](#) as data source, so make sure to also copy this file in the root of your project directory. Create a new class P2J14 in your project, and there first the static method

```
public static int[] distanceFromCharacter(String text, char c)
```

In the first method, you are given a `text` and some character `c` that is guaranteed to occur inside the `text` at least once. This method should create and return an integer array the same length as `text` so that each element gives the distance to the nearest occurrence of the character `c` from that position. For example, the array returned for the argument "hello world" and the character 'o' would be {4, 3, 2, 1, 0, 1, 1, 0, 1, 2, 3}. If the character `c` had instead been 'e', the returned array would be {1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9}.

Again, this problem could be solved in a simple "Shlemiel" fashion with two nested loops. The outer for-loop would iterate through the positions of the `result` array. For each such position, the inner while-loop would advance left and right from that position in lockstep until it sees the first occurrence of the character `c`. The grinding inefficiency of such a clumsy approach can be seen from any `text` where the character `c` occurs as the first character, but nowhere else after that. The pain of the quadratic running time becomes evident once the `text` length grows to the thousands.

To solve this problem in linear time, we note that the values of consecutive positions in the `result` array cannot be just every which way but loose, but can differ from each other by at most one. This allows you to use a single for-loop to fill in two temporary arrays `left` and `right`, that will then be combined at the end of this method to produce the final `result`. Loop through the positions of `text` to fill in the `left` array to fill in each element `left[i]` with the distance to the nearest occurrence of the character `c` when you are allowed to go only left, but not right. To update these values in constant time, you should maintain an integer counter variable to keep track of this distance, initialized to `text.length() + 1`. At every character other than the `c` that you are looking for, increment this counter by one, whereas at every occurrence of `c`, bring that counter back to zero. The current counter value becomes the value `left[i]` being currently filled in.

Filling the values of `right[i]` is done exactly the same way, except as a mirror image going from right to left. You can either do this in a second loop for clarity, or combine both loops into one for-loop with a bit of simple integer arithmetic for brevity. Once you have filled in both arrays `left`

and `right`, each element of the `result` array is the minimum of the `left` and `right` elements in that position.

(This same technique can also be used to solve a classic algorithm coding chestnut usually called something like [“Trapping rainwater”](#), guaranteed to be found in some form in basically every algorithmic problem collection of this kind. The `left` and `right` arrays keep track of the tallest wall seen to that direction from each position, filled in with a similar linear time for-loop that remembers the tallest wall that it has encountered so far. The height of the water pillar at each position is then the minimum of the two values of the `left` and `right` arrays for that position.)

```
public static String pushDominoes(String dominoes)
```

The second method is given a string parameter that represents a line of dominoes. Instead of placing these dominoes neatly on the table so that the pips match between consecutive dominoes like a civilized person, this time we are being more whimsical and place these dominoes standing up to create a chain of falling dominoes, itself a well-known metaphor for various things in life on both personal and national scales.

In the `dominoes` string given as argument to this method, the character `'.'` denotes a domino standing still, the character `'R'` denotes a domino falling right, and the character `'L'` denotes a domino falling left. The simulation of this domino chain proceeds in discrete time steps. In each time step, every falling domino causes its neighbouring domino to the direction of falling to also become a falling domino, provided that that neighbour was still standing at that moment. If a standing domino is simultaneously being pushed right by its left neighbour and pushed left by its right neighbour, these two forces cancel each other out, and the domino will remain standing.

This method should compute the equilibrium end result of the domino chain once the situation has become stable so that nothing would change in the next time step. For example, given the initial state `"R..."`, the result would be `"RRRR"`. Given the initial state `".R...L"`, the result would be `".RR.LL"`, the two chains from left and right meeting in the middle to leave the domino standing.

Lab 0(P): Two Pointers

JUnit: [P2J15Test.java](#)

The two methods of this lab demonstrate the classic programming technique known as **two pointers** that helps you not be a “Shlemiel” when solving various array problems whose structure happens to conform to the needs of this technique. (In this context, the term “pointer” refers to an integer index that can be used to access an array element in constant time, not literally the low-level pointer variable that stores that actual memory location.) Even better, this bad boy of good coding techniques has the built-in advantage in that its running time is at most linear with respect to the length of the array! Once absorbed into your “array” of coding techniques, the technique of two pointers is a skeleton key that cracks open countless doors that the poor Shlemiel would spend an entire day mindlessly running back and forth to squeeze open. A multitude of examples of this technique can be found by browsing the corresponding categories at [LeetCode](#) and [CodeForces](#).

To search for a solution inside the array, the basic version of the two pointers technique uses two array indices i and j . These indices together define the **start** and the **end** position of the current subarray that is guaranteed to contain the solution, assuming that some solution to the problem exists inside the array in the first place. Such a bold **invariant** claim is trivially true at the start of the method, since the index i is initialized to the first position, and the index j is initialized to the last position of that array, together spanning the entire array outside which nothing exists as far as this method is concerned. From there, a while-loop chugs on until either a solution is found in the subarray defined by the current values of i and j , or these two indices go past each other like strangers in the night, and establish by doing so that no solution exists. Each round of this while-loop is guaranteed to move at least one of these indices towards the other, but in a disciplined manner that **maintains the invariant** by guaranteeing that neither index can accidentally skip over a position that is needed for the solution.

The two pointers technique can be used for problems where this decision of which index to advance and how much can always be made based on the local information about the elements at positions i and j , possibly with some other state information that the loop maintains along these indices to remember some relevant aspects of the past that may affect the future decisions that lead to the final answer that we can confidently give to Regis.

Before looking at the two problems needed to solve for this lab, we demonstrate this technique with the classic example of **two summers**. From an array of integers guaranteed to be sorted in strictly ascending order, find two distinct elements a and b that add up to the given goal value $x = a + b$. Again, Shlemiel would brute force through this problem by ignoring all the information available in the fact that the array is guaranteed to be sorted, and use two nested for-loops to crawl through the $n(n - 1)/2$ possible pairs of elements to find two elements that add up to x .

A better way to solve the two summers problem comes from the realization that when i and j define a subarray that contains the solution whenever one exists, comparing the sum $a[i] + a[j]$ to the goal value x will always either find a solution, or give us an ironclad guarantee that moving one of these indices one step towards the other cannot possibly skip over the solution. For example, if

the sum of these two elements is less than x , this means that even the largest available element $a[j]$ was not large enough to reach the goal when paired with the smallest available element $a[i]$. This smallest element $a[i]$ therefore cannot be part of any element pair that would constitute a solution, so skipping over this nail cannot possibly cost us the kingdom. Symmetric reasoning allows us to safely skip over element $a[j]$ whenever this sum is greater than x . The complete method based on this idea should be similar to the following:

```
public static boolean twoSummers(int[] a, int x) {
    int i = 0, j = a.length - 1;
    while(i < j) {
        int s = a[i] + a[j];
        if(s == x) { return true; } // Got it!
        if(s < x) { i++; } // a[i] is too small to work
        else { j--; } // a[j] is too big to work
    }
    return false;
}
```

For an n -element array, these two indices must come together after at most $n - 1$ rounds, forcing the loop to terminate after at most that many rounds have elapsed. Since the work done in the body of the loop takes a constant time, the running time of the entire loop is at most linear with respect to the length of the array a .

Enough talk, on to action. Create a class P2J15 in your labs project, and there the static method

```
public static int[] findClosestElements(int[] a, int x, int k)
```

The first method is given an array a with at least k distinct integers sorted in ascending order, with the guarantee that the element x can be found somewhere inside this array. This method should first find the location of the element x inside the array using [binary search](#). Binary search is the ur-algorithm of the two pointers genre so that in each round, one of the indices can safely jump half-way towards the other index without the risk of jumping over the element x being searched for. However, be sure to implement your binary search correctly by realizing that the two index jumps are **not** symmetric! This side steps both of the two classic possible bugs of this algorithm; that one index jumps one step too far to potentially miss the element x , or that in the end both indices stay forever in adjacent positions because the computed size of the jump becomes zero.

Once you have found the position where the element x resides, your second loop should make these indices proceed left and right from that position, to collect the total of k elements whose values are closest to the element x , as measured by the absolute value of the difference of that element value and the element x . If two elements are equally close to the element x , one to the left of x and the other to the right (such as elements 0 and 10 when x equals 5), use the smaller one, if you can take only one element in the result. For example, the four closest elements to the element 10 inside the

array {1, 2, 8, 10, 11, 13, 22} would be {8, 10, 11, 13}. Note that your method must return these elements in sorted order.

(As a sidebar, we all intuitively understand how sweet it is to keep things such as library books or house numbers along a street in sorted order, for fast retrieval. Yet, few people seem to be able to coherently explain why this is so. The reason becomes evident by comparing the behaviour of the binary search over a sorted array to that of the **linear search** over an unsorted array. Being sorted effectively boosts the power of the order comparison operator between the array elements to nearly divine unlimited power. Looking at an element in an unsorted array gives you information only about that element, whereas looking at an element in a sorted array gives you useful information about all the elements that come before or after it!)

```
public static int countSubarraysWithSum(int[] arr, int sum)
```

The second method also receives an array named `arr` that consists of **positive** integers that have not been sorted in any particular order. The method must count of how many contiguous subarrays that array contains whose elements add up exactly to the given nonzero `sum`. For example, the array {6, 3, 1, 2, 3} contains three subarrays whose elements add up to six; {6}, {3, 1, 2} and {1, 2, 3}. These subarrays can partially overlap, or even be singletons whose element equals `sum`.

The Shlemiel way to brute force this problem would be to use three nested for-loops; the outermost level looping through all possible starting positions of the subarray, the second level looping through all possible lengths of that subarray, and then to top off this idiocy, the innermost third level adding up the elements in that subarray. The better technique uses two pointers, but this time as a variation where **both** indices `i` and `j` start at the **beginning**, and advance towards the end so that the **forward scout** index `j` is always ahead of the **rear guard** index `i` trailing somewhere behind it, this rear guard never stepping past the forward scout. Once the index `j` steps past the end of the array, everything that can be found has been found, and the method can terminate.

If the sum of the elements of the subarray from `i` to `j`, inclusive, is greater than `sum`, you can safely advance the rear guard index `i` (or `j`, when both indices are equal and thus define a singleton subarray) by one step to make the subarray (and therefore also the sum of its elements) smaller. When the sum of the elements of that subarray is less than `sum`, you can safely advance the forward scout index `j` by one step to make the subarray larger. When the sum of the elements of the subarray is equal to `sum`, increment your tally of such subarrays by one, and advance `j`.

To eliminate the innermost loop of the Shlemiel solution, note how each subarray that you examine is always the same as the subarray that you examined in the previous round, except with either one element added to the end (when advancing `j`), or one element removed from the front (when advancing `i`). You should therefore maintain the sum of elements of the current subarray in a local variable, and update that sum in constant time whenever either index advances; either by adding the element that the forward scout index `j` enters, or subtracting the element that the rear guard index `i` leaves behind. This speedup allows your method to pass the JUnit tests within at most the required couple of seconds, instead of taking an unacceptable long time to complete.

Lab 0(Q): Aboutturn Is Playfair

JUnit: [PlayfairCipherTest.java](#)

With the exponential progress of computing technology, the [Playfair cipher](#) has become adorably inadequate for today's needs for encrypted communications. We can still appreciate its emergent complexity from the rules that are simple enough for the average person to use with nothing but pen and paper, given the secret **passphrase**. We could now pretend to be time travellers visiting the stressful and chaotic trenches of the Great War, or the vaguely steampunk parlour of some gentleman detective who is leisurely cracking the intercepted communications between the enigmatic Professor Moriarty and his colourful cast of henchmen known under the collective sobriquet "The Crime Chums". Systematically going through the combinations in the same cozy spirit as we would solve a Sudoku today, even as there is no shortage of messenger boys and other street urchins ready, willing and able to brute force that same task for a tuppence, our good detective slowly drifts off into a dream of Her Majesty's finest scientists one day building a steam-powered machine that would reckon these mechanistic calculations for him...

Meanwhile, up here in the reinvigorated cyberpunk genre that has spilled over to our actual reality, the Playfair cipher has value as an educational exercise of two-dimensional array operations. In our variation of this cipher, we shall treat the letter J as being the exact same letter as the letter I, so that there are only 25 letters in the English language. The Playfair cipher uses a 5-by-5 grid of letters whose cells are filled with distinct letters in an order determined by the secret passphrase known to both parties who wish to exchange secret messages encrypted with this scheme. The grid is filled one row at a time while reading through the passphrase, skipping over the spaces, punctuation and every other character outside the 25 letters of our reduced English.

Each letter is placed in the current cell in uppercase at its first occurrence in the plaintext. The cursor is advanced to the next column, starting from the beginning of the next row once the current row has been completely filled in. At the end of the passphrase, the leftover letters that did not appear in the secret passphrase are filled in the rest of the grid in alphabetical order. For example, the passphrase "There will always be suffering!" would give us the following grid:

T	H	E	R	W
I	L	A	Y	S
B	U	F	N	G
C	D	K	M	O
P	Q	V	X	Z

The Playfair cipher assumes that the plaintext message contains only uppercase letters, such as "WEWILLATTACKATDAWN". The plaintext is partitioned into pairs of letters, and each pair is encoded separately into a pair of ciphertext letters. However, when both letters in the pair would be the same letter, an extra letter X is inserted between them. (This being the stuffy nineteenth century,

the original plaintext never contains two consecutive occurrences of the letter X.) For example, the message "JOLLY" would be encrypted as if it had been "IOLXLY" all along, again treating the letter J as if it really were the letter I. If the plaintext message consists of an odd number of letters, an additional X is tacked to the end to dance with the last letter that would otherwise be left without a partner. For this reason, the plaintext message also cannot end with the letter X.

Thanks to this preprocessing, every pair to be encoded will consist of two distinct letters. To encode a pair of such letters located in the same row of the grid, use their neighbours one step to the immediate right, with the end of the row looping cyclically back to the start of that row. For example, using the above grid, the pair SL would be encoded as IA, and the pair BU would be encoded as UF. The rule of encoding a pair of letters located in the same column is simply the transposed version of the rule to encode two letters in the same row; each letter is replaced by its neighbour one step immediately below. For example, the pair DQ would be encoded as QH, and the pair GW would be encoded as OS.

The last possible situation is where the two letters of the pair are located in different rows and columns, such as BV. In this case, these letters are replaced by the two letters that complete the rectangle in the grid, using the original two letters as two opposite corners of that rectangle. To make the decryption unambiguous, we have to make a choice between two symmetric and equally correct ways to do this, and choose to encode each letter with the letter in the corner that is in the same **row** as the letter being encoded. For example, the pair BV would be encoded as FP, and the pair GX would be encoded as NZ.

That's it! That is the entire scheme that could be taught in fifteen minutes to any schoolboy and all his little chums, and is now ready, willing and able to be translated into Java code. In your project folder, create a new class named `PlayfairCipher`, and there the method

```
public static char[][] constructPlayfairTable(String passphrase)
```

to create and return the 5-by-5 character array Playfair table that the JUnit test class will pass on as the `grid` to the two methods

```
public static String encryptPlayfair(String plaintext, char[][] grid)
public static String decryptPlayfair(String ciphertext, char[][] grid)
```

for encrypting and decrypting the **plaintext** and **ciphertext** messages. In spirit of the age where the Playfair cipher originated, our JUnit tester will encrypt the entire [warandpeace.txt](#) text file with that cipher. (Put that in your pipes and smoke it, Russian formalists.) Again, make sure that this file is copied in the root of your project directory, since that is where the JUnit test will be looking for it.

The decoding of the ciphertext back to the original plaintext is almost the same algorithm as the one used to encode the original plaintext into that ciphertext. In fact, you can extract the common parts of these two methods into a `private` utility method of its own, making the bodies of these two methods simple one-liners. In addition to the `text` and the `grid`, this utility method is given an integer `offset` that tells it how far to look at the neighbour in the same row or the column. For the

encoding, the `offset` is set to +1, whereas for the decoding, the `offset` is set to -1. The rest of the shared algorithm is identical for both encrypting and decrypting.

If you want to be really clever and have the remainder operator `%` work correctly for both cases to keep you wrapped inside the grid, the `offset` for decoding could be +4 as well! You see, Java uses the mathematically correct version of the remainder operator so that $-1 \% 5$ correctly equals -1, even though this result unfortunately falls outside the legal indices to the table. Python uses the mathematically incorrect version of the remainder operator that happens to work correctly for purposes of wrapping indices inside sequences. In Python, the expression $-1 \% 5$ equals 4. (The reader is invited to ponder this sort of “[worse is better](#)” phenomenon that can occasionally be observed in many other domains of our shared human enterprise and endeavour.)

Despite its simplicity, Playfair cipher still illustrates some important ideas that are relevant even for modern cryptography. For example, the [Kerckhoff’s principle](#) that states that all security of the system should rest on the secret key, so that even if every detail of the algorithm were leaked to the enemy, that knowledge would not help them decipher any of our ciphertext messages, since their security depends only on the secret key without which knowing the algorithm itself gets them nowhere. Systems that violate this important principle to instead rely on “security by obscurity” can always be permanently broken with the classic, literal **brute force** decryption technique of “[rubber hose cryptanalysis](#)”, and will soon go the way of the dodo in today’s interconnected world.

We can also note a dashing feat of usability design well before any man of stature even bothered to think about such concerns of women and the servant class. The Playfair cipher is insensitive to any whitespace and capitalization, which greatly simplifies communication, memorization and application of secret passphrases for us humans whose ability to remember things is usually somewhere around the level of the microchips found inside a toaster these days.

Lab 0(R): Frogs On A Box

JUnit: [FrogCrossingTest.java](#)

This problem is adapted from a problem in [CodeForces](#), the premier Russian competitive coding problem collection and its associated online judge. Even if you never participate in any competitive coding contests, all of us should still partake in the most important contest of them all, trying to beat yesterday's version of yourself. You never know when all that extra coding strength and knowledge of little techniques will come handy in future problems. ("Do you even code, bro?")

Inside a version of the classic arcade game of Frogger updated for the needs of our course, a troop of frogs is standing at the edge of a canyon, aiming to escape to the other side to avoid the fate of being slowly boiled in their tanks. The distance between the canyon edges is far too long for even the mightiest frog to levitate its way across. The good news is that a row of boxes has been magically suspended to float in the mid-air between the edges of the canyon. (True story.) The bad news is that each of these boxes can be used only once, since each box vanishes the moment that some frog propels itself off it. Given the x -coordinate positions of the boxes and the jumping strength of each frog, how many frogs can cross the canyon successfully?

In your labs project, create a new class named `FrogCrossing`, and there the method

```
public static int maximumFrogs(int[] strength, int[] boxes)
```

The first parameter `strength` is an array whose length equals the number of frogs in the army. The element `strength[i]` gives the maximum distance between two boxes that particular frog can clear with a single hop. The `strength` array is guaranteed to be sorted in descending order so that the strongest frogs take their turn first, the way Nature and Nature's God dictated it to be. This order is good and must be preserved, no matter what all those advocates of slave morality for the weak may plead. The second parameter `boxes` gives the horizontal coordinates of the boxes floating in the canyon, listed in ascending order. The first and the last element of this array are the fixed positions of the edges of the canyon that do not vanish when used by some frog.

For example, if `strength` equals `{7, 5, 4}` and `boxes` equals `{0, 3, 4, 6, 8, 10, 11}`, it is possible for the strongest two frogs to cross the canyon from the edge at $x = 0$ to the opposite edge at $x = 11$, with several different ways to achieve this. However, no matter how you try to finagle it, all three frogs will not be able to cross the canyon, as can be established with the following argument. The two weakest frogs will use up the boxes at $x = 3$ and $x = 4$, and then the boxes at $x = 6$ and $x = 8$. Since this uses up all the boxes between $x = 0$ and $x = 10$, the first frog can now do nothing better than imitate Wile E. Coyote in his tragic realization of the inherent futility of his efforts. Having the strongest frog cross first changes nothing, since he will use up at least one box before the box at $x = 10$ that was necessary for the other two frogs to cross the canyon successfully.

This problem is best solved recursively with the aid of the private utility method that determines whether it is possible for the k strongest frogs to cross the canyon as an army. The public method `maximumFrogs` should consist of a single loop that counts the values of k from the size of the army

down to zero, and return the largest value of k for which the recursive method given below returns `true` to indicate a successful crossing. If there is no solution using the k strongest frogs, abandon the weakest of those frogs as the Cold Equations of Nature dictate, and try to find the solution for the $k - 1$ strongest frogs. (This solution illustrates the general technique known as **iterative deepening** that often comes handy in recursive optimization problems of this nature.)

```
private static boolean canCross(  
    int[] strength, int[] boxes, int[] atBox, boolean[] used  
)
```

The first two parameters `strength` and `boxes` are the same as in the previous method, and their contents do not change during this recursion. The k -element array `atBox` gives the index of the box that each frog is currently standing on. (You should use the indices of the boxes directly instead of their x -coordinate values, to make the lookup of boxes and their successors faster.) At the top-level call, the values of the `atBox` array are all zeros, since every frog starts at the edge of the canyon. The fourth parameter is a truth valued array to keep track of which boxes have already been used by some frog so that other frogs can't hop on them any more.

The base case of this recursion is when every element of `atBox` equals `boxes.length - 1`, for all k frogs having successfully crossed the canyon. Otherwise, find the frog currently on the box with the lowest x -coordinate. If it is possible for that frog to jump directly to the opposite edge, make it do so and update the `atBox` array accordingly, and try to solve the rest of the problem recursively.

If the leftmost frog cannot reach the opposite edge directly, use a while-loop to iterate through the unused boxes that it can reach with a single jump. For each such box, see what happens when making the frog jump on that box and update the `atBox` and `used` arrays accordingly, and try to solve the problem recursively for the new situation of frogs and boxes. If that recursive call returns `true`, your loop can also stop and return `true`, since we are looking for any one working solution. Otherwise, continue looping over the rest of the reachable boxes to see if that leftmost frog jumping there would give a more favourable result. Once you have tried jumping on every reachable box but the recursive calls from those jumps led only to failure, your method can also stop and return `false`. Since the hop sequences of the crossing frogs never visit the same floating boxes and can therefore be arbitrarily interleaved and still remain successful, there is no point looping over the other frogs in search for a solution. If some solution existed starting with some frog other than the current leftmost one, that solution would have been found by starting with the leftmost frog.

We leave it as an exercise for the reader to note other hard and thus useful constraints that can be used to speed up this basic search. Your mind will answer most questions if you learn to relax and wait for the answer. In this kind of branching recursive searches it is hugely important to detect inevitable failures as soon as possible. In that frozen moment when everyone sees what is on the end of every fork, you should turn back and return to the previous level of recursion to try out some earlier alternative instead.

Lab 0(S): Hamming Center

JUnit: [HammingCenterTest.java](#)

The **Hamming distance** between two arrays of equal lengths is defined as the number of positions where those two arrays differ. The Hamming distance is easily computed with a single for-loop traversing through the positions of both arrays in lockstep, incrementing an integer counter whenever the two elements in the same position disagree. The Hamming distance is a proper [distance metric](#), since every array is trivially at zero distance from itself, the distances back and forth between two arrays are symmetric, and Hamming distances satisfy the [triangle inequality](#).

For a collection of boolean arrays, we define its **Hamming center** to be a boolean array whose Hamming distance from all the arrays in that list is at most r , where r is the smallest radius for which some boolean array exists that satisfies this requirement. The Hamming center is still not necessarily unique; for example, both 01 and 10 can serve as Hamming centers for the arrays 00 and 11 with the radius of one. (We conventionally write out boolean arrays in a compact form of zeros and ones, since it's easier to justify and visualize the five truth values from 01101 than from [false, true, true, false, true].)

This lab uses recursion to construct the **lexicographically lowest** Hamming center for the given array of boolean arrays and the desired radius r . In your labs project, create a new class named `HammingCenter`, and there the method

```
public static boolean[] findHammingCenter(boolean[][] bits, int r)
```

The first parameter `bits` is a two-dimensional array whose each row is an array of boolean values, representing the boolean arrays whose Hamming center we wish to discover. The caller guarantees that each row of this array has the exact same length, so that the concept of Hamming distance is meaningful in the first place. This method should construct and return the lexicographically lowest Hamming center whose distance from all the rows of the `bits` array is at most r . If no Hamming center exists for the given `bits` array and the radius r , this method should return `null` to inform the caller that they need to try a larger value of r next.

This method should first set up the necessary array objects `distance` and `center`, passed down in the top-level call of the following recursive utility method that does all the heavy lifting:

```
private static boolean findCenter(
    boolean[][] bits, int[] distance, boolean[] center, int k, int r
)
```

The parameters `bits` and r play the exact same roles as they did in the first method. The recursion should not mutate the `bits` array in any way as it fills in the Hamming center to the parameter array `center`. The parameter `k` indicates which position of the `center` array is being filled at this

level of recursion. Since we are filling the `center` strictly from left to right, `k` will be initialized to zero in the top-level call to this recursive method, and increase by 1 in each recursive call.

The new parameter array `distance` maintains a counter for every row of `bits`. Each counter `distance[i]` keeps track of how many of the `k` bits already assigned to the `center` differ from the corresponding bits in the i :th row of `bits`. As soon as any one of these counters becomes greater than `r`, the current branch in the recursive quest for the Hamming center can safely be deemed an inevitable failure. It would be pointless to examine all possible ways to complete the `center` array from the current prefix already known to be hopeless, so the method can just terminate and return `false`.

The successful base case for the search for the Hamming center is when `k==center.length` so that the entire `center` array has been filled in. The method can now return `true` to make the previous levels aware of the good news and pass it on to their own callers.

Outside these positive and negative base cases, this method should loop through the truth values `false` and `true`, assigning each truth value to `center[k]` on its turn. It is important that you try out these truth values in this exact order, `false` first and then `true` second, to ensure that the `center` that gets completed first will be the lexicographically lowest of all possible centers. After assigning the truth value to `center[k]`, loop through the rows of the `bits` array to **update** all the `distance` values for the rows whose k :th bit differs from `center[k]`, and try to recursively fill the rest of the `center` array from position `k+1`. If this recursive call returns `true`, the current level of recursion should also immediately return `true`, without erasing all the good work that it has achieved in the `center` array. Otherwise, **downdate** the values in the `distance` array back to the previous values that they held before the recursive call. Once you have tried both `false` and `true` to fill in `center[k]` without success, return `false` to inform the previous level that the current prefix of the `center` cannot be extended to a working Hamming center.

Finding the Hamming center is a surprisingly difficult problem, far deeper than it may first appear on the surface. In fact, this problem is provably [NP-complete](#), as it belongs to [the set of famous decision and optimization problems](#) that, in some visceral sense of cosmic justice and order, all feel like there ought to exist some simple algorithm to solve them, and yet every known algorithm to solve these problems will spend an exponential time in the worst case to grind out the solution! These NP-complete problems share the curious property that verification of a solution is much easier than actually solving them; if some hinky dude in a dark alley ever offers to sell you a solution, it is “no problemo” for you to test quickly whether the solution being offered to you is any good before you inject it into your veins. However, coming up with a working solution all by yourself is a Herculean task akin to finding a small needle in an exponentially large haystack, with no hope of any magnets available to help you prune this search.

In an even more annoying fashion rarely seen since the days of Tantalus, all these NP-complete problems actually [turn out to be the same problem wearing different disguises](#), so that the ability to solve any one of them efficiently would directly translate into the ability to solve all of them efficiently! Since this ability would basically be the equivalent of Christmas coming every day (or, depending on your ability to foresee the second order effects of this ability, a Christmas at Ground

Zero), only an incurable optimist would wait for a modern Prometheus to ever bring down to us mortals such an algorithm that would effectively make us, if not actual primordial gods, at least some obscure minor deities hanging around the Olympian lounges hoping for our turn to get noticed. The good news here is that as long as Moore's law will keep on keeping on, and all computing devices become connected into just one giant CoMpUteR, the ability to hold the reins of this beast will make its wielder essentially equivalent to Loki in the Marvel Cinematic Universe.

This same asymmetry between the ease of verification and the difficulty of actually constructing a working solution shows up in many other facets of our reality. For example, consider how much easier it is to enjoy a good book or a piece of music that somebody else has already created, compared to how difficult it is to author a good book or a good song by yourself! Bridging the enormous and treacherous ocean that separates the balmy beaches of "P" of verifying that the offered solution is good, from the cruel arctic shores of "NP" of actually being able to construct even one working solution, is by far the most important open problem in theoretical computer science, and will surely remain so for a long time.

Lab 0(T): Clique Mentality

JUnit: [CliqueTest.java](#)

♪ *I was caught in the middle of a decision tree (clauses)*
I looked 'round, and I knew there was no jumping back (clauses)
My threads raced, and I thought, what could I undo (Clauses)
And I knew, there was no help, no constraints from you (Clauses!) ♪

♪ *Nodes of the graph, branching in my heap,*
The forking of threads, tore me apart
You've been... CLAUSE REDUCED! ♪

♪ *Rode down disjunctions, broke stack limit, was pigeonholed*
Went through the edges, the edges, and we had some fun
We met some grids, some brambles who gave us good time
Resolved all the rules, isolated all the loops
Yeah, yeah, they, they, they certified ♪

♪ *And I was sorting priorities, could I advance again, please*
Yeah, unit clauses were too kind
You've been... CLAUSE REDUCED! ♪ — 0xACDC

Consecutive integers from 0 to $n - 1$ are used to represent some kind of discrete **entities** in whatever **problem domain** we are currently interested in. For example, these entities could be people in a social network. Translated into integers, these entities are called **nodes**. A node is not an island, but is **pairwise connected** to any number of other nodes using symmetric pairwise connections called **edges**. Between any two entities, there either exists such a symmetric connection edge, or there does not exist such an edge. For example, in a social network, these connections would represent the friendship relations between pairs of people.

Once the essence of these entities and their relationships in the original problem domain has been boiled down to an abstract [graph](#) of these nodes and edges and nothing else, whatever vibrations our mouths happen to make the air move to tell each other tales about about what these entities are and what their pairwise connections mean becomes irrelevant. Mechanistic algorithms can operate on these abstract graphs to compute their properties with no concern to what the original problem domain even was. These algorithms always make their decisions in the abstract graph alone. Immaterial words do not magically affect the operation of the mindless machine, a peacock without eyes that executes such algorithms without any concern to any big picture or such grand purpose.

A myriad of [graph algorithms](#) have been devised to efficiently compute all sorts of interesting and useful properties on graphs. An entire course or two on **combinatorial algorithms** could easily be filled with nothing but study of all sorts of interesting and clever graph algorithms! These properties of the abstract graph will then directly translate to the original problem domain to reveal the corresponding interesting properties of the original entities and their connections.

All these classic graph algorithms will be studied in a later course of algorithmic combinatorics. This lab gives you the first taste of graph computations with searching for the largest [clique](#) inside the given graph. A **clique** is a subset of nodes in which any two distinct nodes are connected to each other with an edge; in a social network, a clique would be a group of people who are all friends with each other. Finding such a clique from a social network turns out to be a surprisingly complex problem, but can be solved with some branching recursion that not only achieves the goal for this particular problem, but whose idea generalizes to other similar problems in the future. The ability to systematically iterate over all possible subsets of the given set of elements is the heart of solving many combinatorial problems on graphs and the entities that those graphs represent.

We shall simply use an integer array of length m to list the elements of the subset of m elements. Since the order of listing these elements inside the subset makes no difference to the subset itself, so that $\{1, 3, 5, 8\}$ and $\{8, 3, 1, 5\}$ are two possible representations for the exact same subset, our recursion might as well fill in this array in ascending sorted order. This recursion will systematically visit all subsets for the given m , starting from the **lexicographically lowest** subset $\{0, 1, 2, 3\}$, and finishing with the **lexicographically highest** subset $\{7, 8, 9, 10\}$, for the purposes of this example now assuming that $n = 11$ and $m = 4$.

However, since we are only looking for those rare particular subsets that form a clique, the recursive construction of the current subset can **backtrack** as soon as some node is not connected to one of the nodes chosen in the subset in the previous levels of recursion, instead of keeping on the futile effort to iterate through all possible ways to complete the prefix into a complete subset only to always realize at the end that the constructed subset did not end up being a clique. In general, it is always better to turn back sooner than later once you realize that the current approach is ultimately futile and doomed to fail no matter how you continue.

In your lab project, create a new class `Clique`, and there the static field

```
private static int[] bestSoFar;
```

to keep track of the best solution found so far ("**branch and bound**"), and the static method

```
public static int[] findFirstClique(boolean[][] adjacencyMatrix)
```

The parameter `adjacencyMatrix` is guaranteed to be an n -by- n grid of boolean truth values so that the element in the position `[i][j]` is `true` if and only if the nodes `i` and `j` are connected with an edge. Since the graph in which we search for a clique is guaranteed to be symmetric, the truth values of the symmetric positions `[i][j]` and `[j][i]` will always be equal. An **adjacency matrix** is a compact way to represent a graph whose nodes are known to be consecutive integers, needing only one bit of memory to encode the existence of the edge between every pair of elements.

The method `findFirstClique` should start by initializing `bestSoFar` to a new 0-element array, and then create an n -element array `clique` to be passed to the recursive utility method

```
public static void findFirstClique(
    boolean[][] adjacencyMatrix, int[] clique, int k
```

)

This recursive utility method fills in the `clique` array from left to right. The recursion depth parameter `k`, initially 0 at the top-level call, indicates which position of the `clique` array is being filled in the current level of recursion. The method assumes that the first `k` positions from 0 to `k-1` of the `clique` array have already been filled with some `k`-element clique whose nodes are listed in the ascending sorted order, and tries to systematically extend this clique into a larger clique. The recursive method should start by checking whether `k > bestSoFar.length`, meaning that recursion has constructed a clique that is larger than the best clique found so far. In this case, the method should copy the `k`-element prefix of the current `clique` to become the new `bestSoFar`. With the aid of `Arrays` utility class, this feat is best achieved with the one-liner assignment

```
bestSoFar = Arrays.copyOfRange(clique, 0, k);
```

After this check, the method should enter a for-loop to iterate through the possible nodes that can be assigned to `clique[k]` to extend the current clique, even when the current clique is the best one that we have found so far. (As in life in general, don't let the good become the enemy of the perfect, or vice versa.) Since the nodes will be assigned to `clique` in ascending sorted order, this for-loop can start counting through nodes up from the previous node `clique[k-1]+1`, or from zero when `k==0` so that there is no previous node assigned.

Since we are aiming to complete a clique that is strictly larger than the largest clique found so far, this loop should only count up to `clique.length-(bestSoFar.length-k+1)`, instead of all the way up to `clique.length-1`. As we let the best solution prune the search for later solutions, this upper bound might even end up being strictly less than `clique[k-1]+1`, so that this for-loop will correctly examine no nodes at all! The same will also happen when `k==clique.length`, so this recursive method does not actually need to first check that the position index `k` is still within the bounds of the `clique` array. This shaves off one redundant comparison for every recursive call executed during the search. A small optimization, yes, but still quite elegant, since making some piece of code not even exist at all is pretty much the ultimate elegance in coding. (In general, non-existence tends to be a highly underrated property of things. Nonexistent things never break down, nor can they be attacked or corrupted by either rust or moth.)

For each node in the range of potential nodes to be assigned to `clique[k]`, the body of the for-loop should first ensure that that node is connected to all of the `k` nodes previously chosen into the current `clique`. Once the current node has passed this hurdle, you can assign it to `clique[k]`, and try to complete the rest of the `clique` with a recursive call using the depth `k+1`. Once this loop has tried and tested all the relevant nodes in its range in this manner, the method has nothing more left to do and can return. Once the top-level call of the recursive utility method has returned to the first method, that method can simply return the `bestSoFar` array as its final answer.

Lab 1: Polynomial I: Basics

<blink>Arooga! Arooga! Attention, young citizens of the future! As an alternative to labs 1, 2 and 3 about the `Polynomial` type and its arithmetic operations, some students might prefer the similarly spirited labs 19, 20 and 21 to implement the `Distance` data type with its arithmetic operations, or the labs 64 and 65 that implement the `Rope` data type as a many ways superior alternative to `String`. Both choices have their ups and downs. Students looking for some harder but top shelf stuff can also try out the Bonus Lab 106, about the `Dual numbers` data type for automatic differentiation of their programs.</blink>

JUnit: [PolynomialTestOne.java](#)

In the same spirit as the [Fraction](#) example class, your task in this and the two following labs, should you choose to accept this mission (“as always, the Secretary will disavow any knowledge of your actions”), is to create a class `Polynomial` whose objects represent univariate polynomials of one variable with integer coefficients, along with the basic mathematical operations of these polynomials. If your math skills on polynomials have gone a bit rusty since you last had to use them somewhere back in high school, check out the page “[Polynomials](#)” in the “[College Algebra](#)” section of “[Paul’s Online Math Notes](#)”, the best and yet comprehensively concise online resource for undergraduate algebra and calculus that your instructor has ever traipsed upon.

As is the good programming style unless there exist good reasons to do otherwise, this class will be intentionally designed to be **immutable** so that `Polynomial` objects will not change their internal state after they have been constructed. [Immutability in general offers countless advantages in programming](#), and is a goal worth pursuing wherever feasible, almost like that proverbial salmon whose price is so fantastically high to make it worth taking the fishing trip even if you don’t actually happen to catch any! Even though those advantages might not yet be fully evident, that should not be any reason to stop you from benefiting from them. (This is similar to how you still benefit from a refreshing glass of cool water on a hot day just the same, even if your knowledge would not stretch to give an entire lecture of the biochemical effects that H_2O has inside your body.)

The public interface of `Polynomial` should consist of the following instance methods.

```
@Override public String toString()
```

Implement this method as your very first step to return some kind of meaningful, human readable `String` representation of this instance of `Polynomial`. This method is not subject to testing by the JUnit tests, so you can freely choose for yourself the exact textual representation that you would like this method to produce. Having this method will become **immensely** useful for debugging all the remaining methods that you will write inside `Polynomial` class.

```
public Polynomial(int[] coefficients)
```

The **constructor** that receives as argument the array of **coefficients** that together define the polynomial. For a polynomial of degree n , the `coefficients` array contains exactly $n + 1$ elements so

that the coefficient of the term of order k is in the element `coefficients[k]`. For example, the polynomial $5x^3 - 7x + 42$ that will be used as an example in all of the following methods would be represented as the coefficient array `{42, -7, 0, 5}`.

Terms missing from inside the polynomial are represented by having a zero coefficient in that position. However, the **coefficient of the highest term of every polynomial should always be non-zero**, unless the polynomial itself is identically zero. If this constructor is given a coefficient array whose highest terms are zeroes, it should simply ignore those leading zero coefficients. For example, if given the coefficient array `{-1, 2, 0, 0, 0}`, the resulting polynomial would have the degree of only one, as if that coefficient array had been `{-1, 2}` to begin with, without those pesky and redundant higher order zeros causing confusion.

As the first warning of an unexpected pitfall lurking inside this problem, the zero polynomial should be encoded as the singleton coefficient array `{0}` instead of the empty array that the careless removal of all leading zeros will produce in this situation. The JUnit test will produce zero polynomials by adding some randomly generated polynomials to their own negations. Even though the test class wasn't explicitly designed to do this, accidental **collisions** will happen even with perfect randomness! (In fact, collisions will happen *especially* from perfect randomness, unlike the finger-quotes "randomness" haphazardly generated by humans whose brains evolved to be much superior in the task of recognizing and generating patterns rather than randomness.)

To ensure that the `Polynomial` class is immutable so that no outside code can change the internal state of an object after its construction (at least not without resorting to underhanded Java tricks such as **reflection**), the constructor should not merely assign the reference to the `coefficients` array to the private field of coefficients, but it absolutely positively **must create a separate but identical defensive copy of the argument array, and store that defensive copy instead**. This technique ensures that the stored coefficients of the polynomial do not change if some outsider later changes the contents of the shared `coefficients` array that was passed as the constructor argument. The JUnit test `TestPolynomialOne` will intentionally try to mess up some coefficient arrays later to make your code fail.

```
public int getDegree()
```

Returns the degree of `this` polynomial, that is, the exponent of its highest order term that has a nonzero coefficient. For example, the previous polynomial has degree 3. All constant polynomials, including the zero polynomial, have a degree of zero regardless of the magnitude of that constant.

```
public int getCoefficient(int k)
```

Returns the coefficient for the term of order k . For example, the term of order 3 in the previous polynomial equals 5, whereas the term of order 0 equals 42. This method should work correctly even when k is negative or greater than the actual degree of the polynomial, and simply return zero for such nonexistent terms.

```
public long evaluate(int x)
```

Evaluates the polynomial using the value x for the unknown symbolic variable of the polynomial. For example, when called with $x=2$ for the polynomial with coefficients $\{42, -7, 0, 5\}$, this method should return 68.

Your method does not have to worry about potential integer overflows, since the automated test class has been designed to allow the final and intermediate results of this computation inside the range of the primitive data type `long`. Motivated students can also perform this evaluation with the [Horner's rule](#) that is both faster and more numerically stable than the naive evaluation. However, even if you use the evaluation formula where each term is computed separately, make sure that you are not some "Shlemiel" who computes each term such as x^{100} from scratch with a loop, even though x^{99} from the previous round is right there and needs only one more multiplication to give us x^{100} .

Unless you can understand and explain without any hand waving [all the complexities involved with floating point numbers and their arithmetic operations](#) such as `Math.pow`, as a rule floating point numbers should be avoided in all applications that require their results to be identical and repeatable in all computing environments, the way that all blessed integer arithmetic is guaranteed to be in Java. Control the matter, and you control the mind. After the Great Reset button of this world has been pressed down and firmly kept there, the following Year Zero will be the Current Year forever. All bourgeois and counterrevolutionary ideas of "positive zero" and "negative zero", and especially all ideas of "infinities" sputtered from the poison pen of that deplorable traitor Emmanuel NotANumberstein whose pernicious influence forever pollutes every computation that he manages to infiltrate with his crimethink, will be eradicated from dictionaries not only in our programming languages, but also in our minds.

Nothing exists except through mechanistic consciousness, as it is impossible to see reality except by looking through the eyes of the Compiler. In the dawning new age of enlightenment and borderless digital fluidity, every properly educated young citizen must tirelessly `/root` out and report any and all instances of crimethink to proper authorities who are ideologically trained and equipped to permanently `#cancel` such toxic and reactionary ideas of numeric inequality sputtered from the keyboards of code wreckers and data hoarders. Many decisions that you unknowingly make today will determine which side of the barbed wire fence you will be living in the coming brave new world. If you want a vision of the future, imagine a boot loader stamping on a human face... forever.

Lab 2: Polynomial II: Arithmetic

JUnit: [PolynomialTestTwo.java](#)

This lab continues with the `Polynomial` class from the previous lab by adding new methods for polynomial arithmetic to its source code. No inheritance or polymorphism is yet taking place in this lab. Since the `Polynomial` type is intentionally designed to be immutable, none of the following methods should modify the objects `this` or `other` in any way, but always return the result of that arithmetic operation as a new `Polynomial` object created inside that method.

```
public Polynomial add(Polynomial other)
```

Creates and returns a new `Polynomial` object that represents the result of polynomial addition of the two polynomials `this` and `other`. Make sure that the coefficient of the highest term of the result is nonzero, so that adding two polynomials $5x^{10} - x^2 + 3x$ and $-5x^{10} + 7$, both having the same degree of 10, produces the result $-x^2 + 3x + 7$ that has a degree of only 2. As the JUnit test class generates random polynomials and adds them together, occasionally some polynomial will be added to its own negation, producing the zero polynomial as the result. Your code must handle this important edge case correctly.

```
public Polynomial multiply(Polynomial other)
```

Creates and returns a brand new `Polynomial` object that represents the result of polynomial multiplication of `this` and `other`. You can perform this multiplication by looping through all possible pairs of terms between the two polynomials and adding their products together into the result where you combine the terms of equal degree together into a single term.

Multiplication of two polynomials can cancel out some internal terms whose coefficients were originally nonzero in both polynomials. For example, multiplying $x^2 + 3$ by $x^2 - 3$ gives the result $x^4 - 9$ whose second order term ends up with a zero coefficient and vanishes from the result. However, unlike when adding two polynomials, the highest term can never get cancelled out this way, since the product of two nonzero integer coefficients can never be zero. You therefore know the degree of the entire result of multiplication right away once you know the degrees of the original polynomials, without actually having to carry out this multiplication by adding all the terms together.

Lab 3: Polynomial III: Comparisons

JUnit: [PolynomialTestThree.java](#)

This lab continues modifying the source code for the `Polynomial` class from the first two labs to allow **equality** and **ordering** comparisons to take place between objects of that type. Modify your `Polynomial` class signature definition so that it implements `Comparable<Polynomial>`. Then write the following methods to implement the equality and ordering comparisons.

```
@Override public boolean equals(Object other)
```

Returns `true` if the `other` object is also a `Polynomial` of the exact same degree as `this`, and that the coefficients of `this` and `other` polynomials are pairwise equal. If the `other` object is anything else, this method should return `false`.

(To save you some time, you can actually implement this method only after implementing the method `compareTo` below, since once that method is available, the logic of equality checking will be a trivial one-liner after the `instanceof` dynamic type check.)

```
@Override public int hashCode()
```

Whenever you override the `equals` method in any subclass, you should also override the `hashCode` method to ensure that two objects that are considered equal by the `equals` method will also have equal integer hash codes. This method computes and returns the **hash code** of this polynomial, used to store and find this object inside some instance of `HashSet<Polynomial>`, or some other *hash table* based data structure.

You get to choose for yourself the hash function that you implement, but like all hash functions, the result should depend on the degree and all of the coefficients of your polynomial. The hash function absolutely **must** satisfy the contract that whenever `p1.equals(p2)` holds for two `Polynomial` objects, then also `p1.hashCode() == p2.hashCode()` must hold for them.

Of course, since this entire problem is universal and common enough so that it would be silly to force everyone to keep greasing this same old squeaky wheel, you can nowadays use the method `Arrays.hashCode` to compute a hash value for your coefficients, good enough for the government work. However, if you take this route to implementing `hashCode` for your polynomials, make sure that you eliminate the leading zeros from your `coefficients` array, since otherwise two equal polynomials with different number of leading zeros will get different hash codes. This will cause the `HashSet<Polynomial>` of the JUnit mass tester stray from being in lockstep with the supposedly equal `TreeSet<Polynomial>` that uses the following `compareTo` method to determine the ordering of the polynomials inside the balanced binary search tree.

```
public int compareTo(Polynomial other)
```

Implements the **ordering comparison** between this and other polynomials, as required by the interface `Comparable<Polynomial>`, allowing the instances of `Polynomial` to be *sorted* or stored inside some instance of `TreeSet<Polynomial>`. This method should return +1 if this is greater than the other, -1 if the other is greater than this, and return a 0 if both polynomials are equal in the sense of the `equals` method.

A **total ordering relation** between polynomials can be defined by many possible different rules. We shall use an ordering rule that says that **any polynomial of a higher degree is automatically greater than any polynomial of a lower degree**, regardless of their coefficients. For two polynomials whose degrees are equal, the result of the order comparison is determined by **the highest-order term for which the coefficients of the polynomials differ**, so that the polynomial with a larger such coefficient is considered to be greater in this ordering.

Be careful to ensure that this method ignores the leading zeros of high order terms if you have them inside your polynomial coefficient array, and that **the ordering comparison criterion is precisely the one defined in the previous paragraph**. A common bug in student code at this point is to loop through the coefficients from lowest to highest, instead of looping through them down starting from the highest. This bug is particularly tricky to pinpoint, since looping through the coefficients in the wrong direction will still produce a perfectly legal total ordering between polynomials that the collections of type `TreeSet<Polynomial>` would be perfectly happy to use. It just would not be the total ordering that the JUnit test class is expecting to see, but how could the student know that if they missed it in the casual reading of the specification?

(One imagines the student voicing out such complaints something like “Not fair, Not fair! It isn’t fair, my precious, is it, to ask us what it’s got in its nasty little arrayssess?”)

Lab 4: Extending An Existing Collection Class

JUnit: [MinMaxArrayListTest.java](#)

In this lab, you practice using class inheritance to create a custom subclass

```
public class MinMaxArrayList<E extends Comparable<E>>
    extends ArrayList<E>
```

that efficiently tracks the current minimum and maximum elements in the list. For the concepts of “minimum” and “maximum” to make sense for the list element type `E`, note how the generic type parameter constrains `E` to implement the interface `Comparable<E>`.

Your class should provide a no-argument constructor and a constructor that takes an argument of type `Collection<? extends E>`, matching the constructors of `ArrayList`. In addition to the overridden methods, your class should define the following two brand new methods:

```
public E getMin()
public E getMax()
```

Returns the minimum (or maximum) element in the list according to the natural ordering defined by `compareTo`. Both methods must throw `NoSuchElementException` if the list is empty. Note that it is possible for minimum and maximum elements of the list to be equal.

The naive approach of scanning the entire list every time someone asks for the minimum or maximum would be far too slow for large lists. Your implementation must instead cache the current minimum and maximum in two new `private` data fields, and maintain this cache whenever the list is somehow modified. When a modification might have invalidated the cached values (for example, removing an element that was the current minimum), your class should use lazy invalidation: mark the cache as dirty and defer the linear rescan to the next call to `getMin` or `getMax`. When elements are added, you can often update the cache cheaply without any rescan, since a new element can only change the minimum or maximum if it is smaller or larger than the current cached extreme.

The critical challenge of this lab is identifying every inherited method that mutates the list and ensuring that each one properly maintains or invalidates the cache. The methods you need to override include `add(E)`, `add(int, E)`, `addAll(Collection)`, `addAll(int, Collection)`, `set(int, E)`, `remove(int)`, `remove(Object)`, `removeAll(Collection)`, `retainAll(Collection)`, `removeIf(Predicate)`, `replaceAll(UnaryOperator)`, `sort(Comparator)`, and `clear()`. Each overridden method should first call its super version to perform the actual list mutation, and then update or invalidate the cache as appropriate. Missing even a single method is a bug that the JUnit tester will catch.

The JUnit tester includes a **fuzz test** that verifies correctness against an oracle while also enforcing a time limit. An implementation that scans the entire list on every call to `getMin` or `getMax` will not finish this fuzz test time. Only proper caching will.

Lab 5: Biticut Follies

In this lab, you will create an interactive Swing component that draws a simple cartoon face. Historically, **graphical user interface (GUI) programming** was the first **killer app** that propelled the entire oriented programming paradigm into the mainstream after languishing for two decades as purely theoretical academic research while the real work was done the way that it had always been done. (Most things in computer science are at least a decade or two older than most people casually assume; this hold silently behind the scenes in all walks of life, not merely in coding.) Inheritance and polymorphism make the task of creating new customized GUI components so straightforward that in practice, any GUI programming done in a non-OO low level language first has to use that language to build up a crude simulation of these higher level mechanisms, thus becoming a living and breathing example of both [Greenspun's Tenth Rule](#) and the [Blub Paradox](#).

In this lab, your task is to create a custom Swing GUI component `Head` that extends `JPanel`. To practice working with Java graphics, this component should display a simple human head that reacts to the mouse cursor entering and exiting the surface of that component. You should most likely copy-paste a whole bunch of boilerplate code from the instructor's example class [ShapePanel](#) into this class, as if it were still the nineties. Your class should contain a `private` field `boolean mouseInside` that is used to remember whether the mouse cursor is currently over your component, and the following methods:

```
public Head()
```

The constructor of the class should first set the preferred size of this component to be 500-by-500 pixels, and then give this component a decorative raised **bevel border** using the utility method [BorderFactory.createBevelBorder](#). Next, this constructor should create and add an inner [MouseListener](#) to this component. This event listener should be constructed using an inner class `MyMouseListener` that extends [MouseAdapter](#). Override both methods `mouseEntered` and `mouseExited` inside your event listener inner class to first assign the value of `mouseInside` accordingly, and then call `repaint` to request your component to be rendered in its new state.

To avoid a common beginner mistake at this point, **do not try to render anything in the event listener itself**. Your `paintComponent` method must be ready, willing and able to render the component at all times anyway, not just during event handling. Therefore, all the state information needed to perform this rendering must be stored in the instance variables of that component.

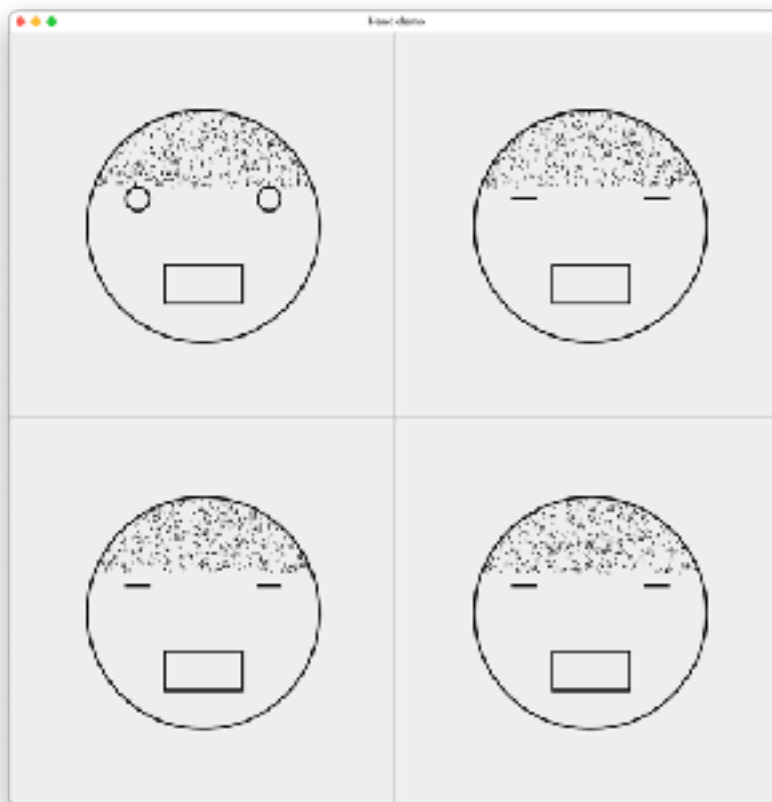
If you implement `MouseListener` instead of extending `MouseAdapter`, you need to provide the empty implementations for all the other methods in that interface, even if those methods end up doing nothing at all. The `MouseAdapter` utility class provides these empty implementations for its subclasses free of charge, saving you some inconvenient typing while also making your code more concise and readable. In our enlightened age after Java 8, these methods could have been default methods defined in the interface itself with do-nothing empty bodies, but of course the design of these classes had to use the mechanisms that were available in the language at the time of Java 1.2.

```
@Override public void paintComponent(Graphics g)
```

Renders some kind of abstract cartoon image of a simple human head on the surface of this component. If `mouseInside` equals `true`, the eyes of this head should be drawn to look open, whereas if `mouseInside` equals `false`, the eyes should be drawn closed as if sleeping. Each `Head` instance tracks its mouse status separately. However, this is not an art class so this head does not need to look all coiffed up; a couple of simple rectangles and ellipses suffice. Of course, students with a more artistic bent and interest might want to check out more complicated shapes from the package [java.awt.geom](#) to render a prettier image, or perhaps even using [Image](#) objects whose contents are read from some semantically appropriate GIF or JPEG file. (Use the utility class `ImageIO` for this, [as explained here](#).) Also do not forget to first call the superclass version of the method with `super.paintComponent(g)` to erase the previous contents of the component.

(Back in this author's salad days, GIF and JPEG were the only image formats that the average end user ever saw on their mighty screens of then-massive 640-by-400 pixel resolution. You kids have your new fancy kinds of formats these days transmitted over network connections that are faster than a greased lightning and serve you high-definition video streamed in real time, while back in our youth, by jove, an 'onest lad had to scratch his mutton chops through minutes of modem screech just to spy an eyeful of some nice lady's bare ankle!)

To admire your interactive Swing component, write a separate `HeadMain` class that contains a `main` method that creates a `JFrame` that contains four separate `Head` components arranged in a neat 2-by-2 grid using the `GridLayout` layout manager, as in the screenshot below. Then you can wiggle your mouse cursor from top of one component onto another to watch the eyes open and close with your mouse movements. Your instructor will usually provide the `main` class for the remaining Swing labs after this one, but it's good for you to practice writing this class just once.



Lab 6: Text Files I: Word Count

JUnit: [WordCountTest.java](#)

This lab teaches you to read in text data from an instance of `BufferedReader`, and then perform computations on that data in the spirit of “poor man’s data science”. Who knows what kind of interesting correlations would a small army of future Davids find, armed with slingshots of modern data analytics tools amply available for anyone who wants to turn on the lights in yet another previously dark room in the Grand Hotel of Reality, usually to reveal yet another grifter who has been sneaking his hand into someone else’s pockets under the cover of the comfortable darkness?

To practice working inside a small but entirely proper object-oriented framework, we shall design an entire class hierarchy that reads in text one line at a time and performs some operation for each line. Having read in all the lines, some kind of result is then emitted in the end. The subclasses must then implement those operations by overriding these **template methods** that implement the three consecutive **stages** of this abstract algorithm. To allow this file processing operation to return a result of arbitrary type that can be freely chosen by the users of this class, the abstract superclass of this little framework is also defined to be **generic**. Start by creating the class

```
public abstract class FileProcessor<R>
```

that defines the following three template methods, all three abstract in the superclass:

```
protected abstract void startFile();  
protected abstract void processLine(String line);  
protected abstract R endFile();
```

The class should also have one concrete method for you to write in, this method made `final` to guarantee that its implementation is uniform in all subclasses of this class hierarchy:

```
public final R processFile(BufferedReader in) throws IOException
```

This method should first call the method `startFile`. Next, it reads all the lines of text coming from `in` one line at the time in some suitable loop, whose body calls the method `processLine` for each line that it reads in. Once all lines have been read in and processed, this method should finish up by calling the method `endFile`, and return whatever the method `endFile` returned.

This abstract superclass defines a template for a class that performs some computation for a text file that is processed one line at the time, returning a result whose type `R` can be freely chosen by the concrete subclasses of this class. Subclasses that purport different text processing operations can override the three template methods `startFile`, `processLine` and `endFile` methods in different ways to implement different computations on text files.

As the first application of this mini-framework, you will emulate the core functionality of the Unix command line tool [wc](#) that counts how many characters, words and lines the given text file contains.

Those who have taken the Unix and C programming course CCPS 393 at this same school should recognize this handy little text processing tool from that course. But even if you have not yet familiarized yourself with the Unix command line, you can still implement this tool based on the specification of its expected behaviour.

The existence of `wc` in all Unix systems by itself proves its usefulness; this tool solves an important problem that tends to come up a lot in different problem domains. This deceptively simple piece of software therefore *must* contain some kind of important thing for us to learn, even though it is not immediately obvious what that thing might be. However, our inability to yet detect its importance does not magically cause this importance to be non-existent! No matter which way you choose to design and implement this tool, something educational and computationally interesting should definitely happen to you during this process.

Create a class `WordCount` extends `FileProcessor<List<Integer>>`. This class should define three instance fields of type `int` to keep track of these counts, and the following methods:

```
protected void startFile()
```

Initializes the character, word and line counts to zero.

```
protected void processLine(String line)
```

Increments the character, word and line counts appropriately. In the given `line`, each character increments the count by one, even if that character is a whitespace character. To properly count the words in the given line, count the non-whitespace characters for which the previous character on that line is a whitespace character. (Using `line.split(" ")` to extract the individual words does not produce the correct count in situations where the line contains two consecutive whitespace characters somewhere.) To prevent a “loop and a half” situation where the first character has to be handled separately because it has no predecessor, you can simply act as if the first character in `line` had been preceded by some whitespace character.

You should use the utility method [Character.isWhitespace](#) to test whether some character is a whitespace character, since the Unicode standard defines [quite a lot more whitespace characters](#) than most people can even name, or even see any need for their existence. Since that mythical cadre of Top Men with their horn-rimmed glasses and penchant for sticking tail fins on everything already did all the grunt work in tabulating all the properties of characters that are acknowledged to exist, we might as well reuse their work instead of trying to badly reinvent the parts that we expect to use in our program. Since the classes and methods in all standard libraries have been mercilessly pummelled by all kinds of user code for decades to various ends and purposes, all their edge case and corner case bugs lurking inside these methods have surely been brought to light by now.

```
protected List<Integer> endFile()
```

Creates and returns a new `List<Integer>` instance (choose the concrete subtype of this result for yourself) that contains exactly three elements; the character, word and line counts, in this order.

Lab 7: Text Files II: Chasing Some Tail

JUnit: [TailTest.java](#)

In this lab, we return to the `FileProcessor<R>` framework from the previous Lab 6 for writing tools that process text files one line at the time. This time this framework will be used to implement another Unix command line tool [tail](#) that extracts the last `n` lines of its input and discards the rest. In your labs project, create a new class `Tail` that extends `FileProcessor<List<String>>` and has the following methods:

```
public Tail(int n)
```

The constructor that stores its argument `n`, the number of last lines to return as a result, into a private data field. In this lab, you have to decide for yourself what instance fields you need to define in your class to make the required methods work.

```
@Override protected void startFile()
```

Start processing a new file from the beginning. Nothing much to do here, really.

```
@Override protected void processLine(String line)
```

Process the current line. Do something intelligent here. Be especially careful not to be a “[Shlemiel](#)” so that your logic of processing each line always has to loop through all of the previous lines that you have collected and stored so far. To help you to avoid being a “Shlemiel”, you should note that the `List<String>` instance that you create and return does not necessarily have to be specifically an `ArrayList<String>`, but perhaps some other subtype of `List<String>` that allows constant-time updating operations at its both ends would be far more appropriate.

```
@Override protected List<String> endFile()
```

Returns a `List<String>` instance that contains precisely the `n` most recent lines that were given as arguments to the method `processLine` in the same order that they were originally read in. If fewer than `n` lines have been given to the method `processLine` since the most recent call to the method `startFile`, this list should contain as many lines as have been given. For example, the tail of the last ten lines of a file that consists of five lines would be just those five lines.

Lab 8: Concurrency In Animation

To paraphrase the immortal wisdom of [Jack Handey](#), just as bees swarm about to protect their nests, I will similarly “swarm about” to protect my nest of knowledge eggs. Here is this swarm visualized.

Particle systems are often used in games to create interesting animations of phenomena such as explosions or fire that would be otherwise difficult to simulate and render as a bunch of polygons. Giving particles more intelligence and mutual interaction such as making some particles follow or avoid certain other particles can produce [surprisingly natural and lifelike emergent animations](#). This lab lets you create a Swing GUI component that displays a real-time animation using Java concurrency to execute an **animation thread** that runs at guaranteed constant pace independent of what the human user might happen to be doing with that particular component or elsewhere. This thread will animate a classic **particle field** where a **swarm** of thousands of independent little particles buzzes randomly around the component.

This same basic architecture could then, with surprisingly little additional modification, be used to implement some real-time game, with this animation thread moving the game entities according to the rules of the game 50 frames per second, as the event listeners pass along the player’s orders to the game entity that happens to represent the human player. As seems to be the case in all walks of life, most things that you expect to be complex and difficult usually turn out to be surprisingly much simpler than you would have assumed, whereas the things that you expected to be simple and easy often turn out to be dizzyingly complex once you crack them open and really try to implement them yourself properly without any shortcuts or hand waving! (This principle works even better in programming where some guy somewhere probably has solved the complex thing already, so we can just copy-paste this work and be done with that by the time we want to be home for dinner.)

First, create a class `Particle` whose instances represent individual particles that move randomly around the two-dimensional plane. Each particle should remember its *x*- and *y*-coordinates on the two-dimensional plane and its heading as an angle expressed as **radians**, stored in three data fields of the type `double`. This class should also have a random number generator shared between all objects, and a shared field `BUZZY` that defines how random the motion is.

```
private static final Random rng = new Random();
private static final double BUZZY = 0.7;
```

The class should then have the following methods:

```
public Particle(int width, int height)
```

The constructor that places this particle in random coordinates inside the box whose possible values for the *x*-coordinate range from 0 to `width`, and for the *y*-coordinate from 0 to `height`. The initial heading is taken from the expression `Math.PI*2*rng.nextDouble()`.

```
public double getX()
public double getY()
```

The accessor methods for the current `x`- and `y`-coordinates of this particle.

```
public void move()
```

Updates the value of `x` by adding `Math.cos(heading)` to it, and updates the value of `y` by adding `Math.sin(heading)` to it. After that, the `heading` is updated by adding the value of the expression `rng.nextGaussian() * BUZZY` to it.

Having completed the class to represent individual particles, write a class `ParticleField` that extends `javax.swing.JPanel`. The instances of this class represent an entire field of random particles. This class should have the following private instance fields.

```
private boolean running = true;
private java.util.List<Particle> particles =
    new java.util.ArrayList<Particle>();
```

The class should have the following public methods:

```
public ParticleField(int n, int width, int height)
```

The constructor first sets the preferred size of this component to be width-by-height, as must be done for every new Swing component subtype. Then, it creates `n` separate instances of the class `Particle` and places them in the `particles` list.

Having initialized the individual particles, this constructor should create and launch one new `Thread` using a `Runnable` argument whose method `run` consists of one `while(running)` loop. (Don't even dream about launching a separate thread for each particle without waking up to profusely apologize about that transgression. This is Java, not Go.)

The body of this loop should first `sleep` for 20 milliseconds. After waking up from its sleep, it should loop through all the `particles` and call the method `move()` for each of them. Then call `repaint()` and go to the next round of the while-loop.

```
@Override public void paintComponent(Graphics g)
```

Render the visual image of this component by looping through particles, rendering each individual particle as a 3-by-3 pixel rectangle at its current `x`- and `y`-coordinates. As with the other labs whose result is a graphical component, feel free to try out variations in the artistic detail, as long as the overall theme and spirit of the component remain as specified in this document.

```
public void terminate()
```

Sets the field `running` in this component to be `false`, thus causing the animation thread to terminate in the near future.

To admire the literal and metaphorical buzz of your particle swarm, create a class `ParticleMain` that contains a `main` method that creates one `JFrame` instance that contains a `ParticleField` of size 800-by-800 that contains 2,000 instances of `Particle`. Using the `main` method of the lecture example class [SpaceFiller](#) as a model of how to achieve this, attach a `WindowListener` to the `JFrame` so that the listener's method `windowClosing` first calls the method `terminate` of the `ParticleField` instance shown inside the `JFrame` before actually disposing that `JFrame` itself by calling its method `dispose`.

Try out the effect of different values between 0.0 and 10.0 of `BUZZY` to the motion of your particles. When `BUZZY` is small, the motion of each particle tends to maintain its current direction, whereas larger values of `BUZZY` shift their motion closer to random [Brownian motion](#).

This sort of **random walk** simulation also leads to another interesting and educational thought problem for the more statistically minded students to ponder. When you set `BUZZY` to be close to zero, you can see the particles soon leave the viewing peephole in their journey towards infinity in different directions. However, as `BUZZY` increases and the overall motion of each particle starts approaching the idealized Brownian motion, the time that each particle spends inside the peephole increases. For different values of `BUZZY` ranging from 0 to 1, how long would you think you have to wait on average for the entire peephole to become devoid of particles that it was initially seeded with? What do you think this average escape time curve would look like if you were to plot it as a function of `BUZZY`? Tiny changes to parameter values around some critical **phase change** can sometimes make a huge difference to the value of that function!

Lab 9: Lissajous Curves

In this lab you are going to write a Swing component that displays a [Lissajous curve](#), a famous **parametric curve** whose simple periodic motion produces interesting self-intersecting visuals. The Swing component should allow the user to control the parameters a , b and δ that control the shape of the Lissajous curve by entering their values into three `TextField` components placed inside this component. This lab is therefore an exercise in not just rendering curves and other shapes on a component, but also in positioning specialized components such as text fields on the surface of a Swing component, and reacting to the events of the said components with an update of the display. Create a new class `Lissajous` extends `JPanel`, with the following methods:

```
public Lissajous(int size)
```

The constructor that sets the preferred size of this component to be `size`-by-`size` pixels, and stores the given `size` into an instance field with the same name. Then, three instances of `TextField` are created and added inside this component. Initialize these text fields to values 6, 5 and 0.5, with some extra spaces added to these strings so that these text fields have a decent initial size for the user to enter other numbers to try out. Define your own subtype of [ActionListener](#) as a nested class whose `actionPerformed` method simply calls `repaint` for this component. The same single instance of this class can simultaneously listen to all three text fields, since the reaction to the event is identical for all three sources of these events.

```
@Override public void paintComponent(Graphics g)
```

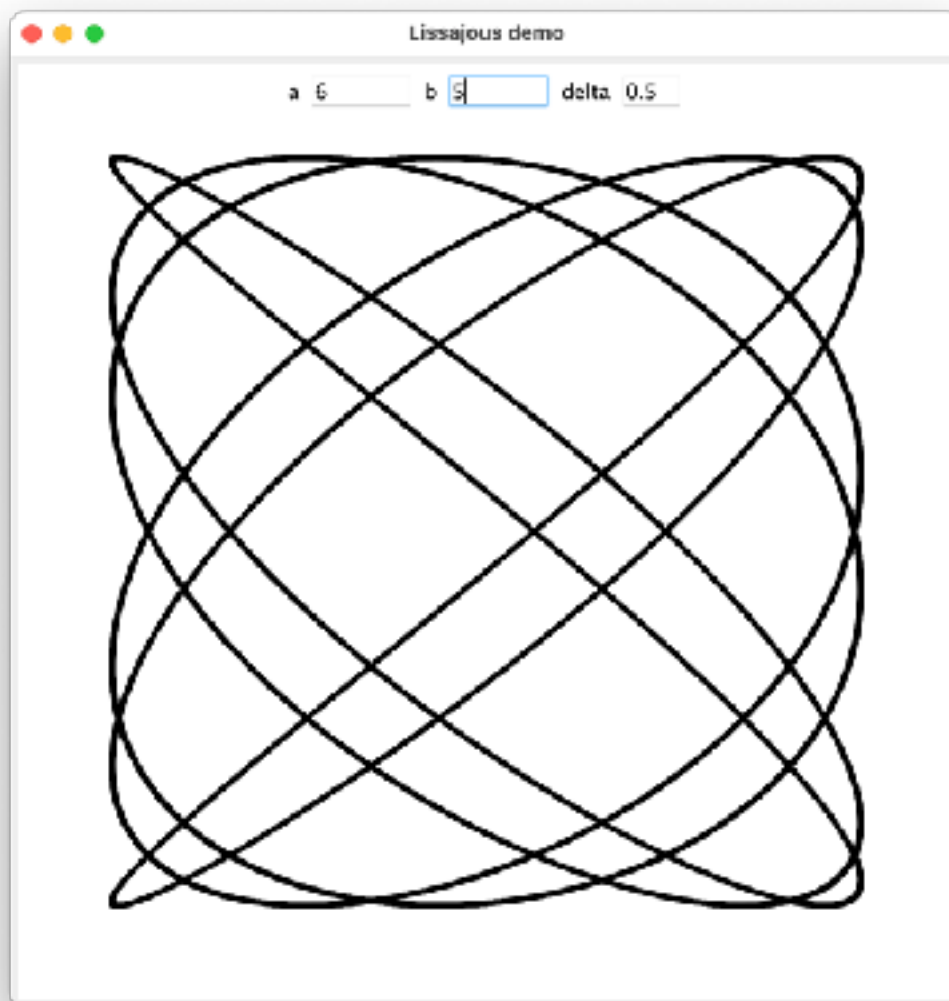
After calling `super.paintComponent(g)` to erase the previous contents of the component, renders the Lissajous curve on the component surface, using the current values for a , b and δ that it first reads from the previous three text fields, with a and b parsed as integers. This method should consist of a for-loop whose loop counter double t iterates over the values of t ranging from 0 to $(a/\text{gcd}(a,b)) * b$ in small increments such as 0.01, where `gcd` is the **greatest common divisor** of two positive integers. In the body of the loop, compute the point coordinates x and y with

```
x = size/2 + 2*size/5 * Math.sin(a*t*Math.PI + delta);  
y = size/2 + 2*size/5 * Math.sin(b*t*Math.PI);
```

and draw a line segment from the current point to the previous point. To admire your Lissajous curve and the effect a , b and δ on its shape, as if you were transported inside some 1970's dystopian science fiction movie and trapped inside the volcanic lair of the mad professor Kokkarius, create a separate class `LissajousMain` whose `main` method creates a new `JFrame` that contains your `Lissajous` component. The end result with the previous initial values for a , b , and δ should resemble the following screenshot. You should verify that your code is correct by trying out other values for a , b and δ shown on the Wikipedia page "[Lissajous curve](#)" to verify that your displayed curves have the same shape as the example images over there.

Parameters a and b represent the frequencies of horizontal and vertical motions, so their greatest common divisor greatly affects the period of the resulting shape. This period is longest when a and b are **relatively prime** so that $\gcd(a, b) = 1$. When a and b are equal, the resulting shape will simply be a circle whose elongation depends on the value of the δ offset between the periods of horizontal and vertical motions. Using the value of δ of $\pi/2 \approx 1.5707$ makes the phases of x - and y -coordinates to be maximally apart (effectively turning the sine function of the first formula into cosine), whereas δ of 0 makes these two coordinates act in lockstep for a less exciting curve.

Motivated students can take on as an extra challenge to make the displayed image look smoother by eliminating some unpleasant visual jaggies, and possibly make this more artistic by using more complex curves and graphical shapes rendered to the positions of these “points going for a walk” around the two-dimensional plane. For example, instead of subdividing the entire curve into small line segments, you could subdivide it into more suave [cubic curves](#) whose consecutive pieces are guaranteed to connect together seamlessly at their shared endpoints, since the shared additional control points guarantee the tangent lines to match perfectly at these shared endpoints.



Lab 10: Computation Streams

JUnit: [StreamExercisesTest.java](#)

This lab teaches you to think about computational problems and then implement them in the functional programming framework of **computation streams** introduced in Java 8. Therefore in this lab, you are **absolutely forbidden** to use any **conditional statements** (either `if` or `switch`), **loops** (either `for`, `while` or `do-while`) or even **recursion**. Instead, all computation of this lab **must** be implemented using only **Java computation streams and their core operations**!

In this lab, we shall also briefly check out the **Java NIO framework** for improved operations on files than those that were offered in the old package `java.io` and the class `File` therein. Your methods receive some [Path](#) as an argument, guaranteed to refer to some text file in your file system whose contents your methods will then process with computation streams. In the JUnit test, this will again be the example text corpus of [warandpeace.txt](#). Create a new class named `StreamExercises`, and write in there the following two static methods.

```
public static int countLines(Path path, int thres) throws IOException
```

This method should first use the utility method [Files.lines](#) to convert the given path into an instance of `Stream<String>` that produces the lines of this text file as a stream of strings, one line at the time. The method should then use the stream transformation method `filter` to keep only those whose length is greater or equal to the threshold value given in the parameter named `thres`, and in the end, return the count of how many such lines the file contains.

```
public static List<String> collectWords(Path path) throws IOException
```

This method should also use the same utility method [Files.lines](#) to first turn its parameter path into a `Stream<String>`. Each line should be converted to lowercase and broken down to individual words that are passed down the stream as separate `String` objects (the stream operation `flatMap` will be handy here). Split each line into its individual words with the aid of the method `split` in `String` with the word separator regex `"[ ^a-z ]+"`. Then in the stages of the computation stream that follow, discard all empty words with another `filter`, and `sort` the remaining words in alphabetical order. Multiple consecutive occurrences of the same word should be removed (you can simply use the stream operation `distinct`, or if you want to do this yourself the hard way as an exercise, write a custom [Predicate<String>](#) subclass whose method `test` accepts its argument if and only if it was the first one or was distinct from the argument of the previous call). To wrap up this computation, `collect` it into a `List<String>` as the final answer.

As this new century shapes up to be an absurdist reboot of its predecessor during the worldwide pandemic, can rediscovery of [principles of futurism](#) be far away? After all, the original [futurists](#) certainly were the “tech bros” of their time! At some point computer users will also notice that despite [all the advances in computer processing power](#), their computers work subjectively no faster or crash less than they did two decades ago. What Andy Grove giveth, Bill Gates still taketh away...

Lab 11: And You Will Find Me, Prime After Prime

JUnit: [PrimesTest.java](#)

[“There are only two hard things in Computer Science: cache invalidation and naming things.”](#)

— Phil Karlton

“All programming is an exercise in caching.” — Terje Mathisen

“Good programming turns caching into cha-ching!” — Ilkka’s corollary

This lab tackles the classic and important problem of **prime numbers**, positive integers that are exactly divisible only by one and by themselves. Create a class `Primes` to contain the following three `static` methods to quickly produce and examine integers for their primality.

```
public static boolean isPrime(int n)
```

Checks whether the parameter integer `n` is a prime number. It is sufficient to use the plain old **trial division** algorithm for this purpose. If an integer has any non trivial factors, at least one of these factors has to be less than or equal to its square root, so it is pointless to look for any such factors past that point, if you haven’t already found any. To optimize this further, you realize that it is enough to test for divisibility of `n` only by primes up to the square root of `n`.

```
public static int kthPrime(int k)
```

Find and return the `k`:th element (counting from zero, as usual in computer science) from the infinite sequence of all prime numbers 2, 3, 5, 7, 11, 13, 17, 19, 23, ... This method may assume that `k` is nonnegative.

```
public static List<Integer> factorize(int n)
```

Compute and return the list of **prime factors** of the positive integer `n` guaranteed to be at least two. The exact subtype of the returned `List` does not matter as long as the returned list contains precisely the prime factors of `n` in **ascending sorted order**, each prime factor listed exactly as many times as it appears in the product. For example, when called with the argument `n=220`, this method would return some kind of `List<Integer>` object that prints out as `[ 2, 2, 5, 11 ]`.

To make the previous methods maximally speedy and efficient, this entire exercise is all about **caching and remembering the things that you have already found out** so that you don’t need to waste time finding out those same things later. As the famous [space-time tradeoff](#) principle of computer science informs us, you can occasionally make your program run faster by making it use more memory. Since these days we are blessed with ample memory to splurge around with, we are usually happy to accept this tradeoff to speed up our programs.

This class should maintain a private static instance of `ArrayList<Integer>` in which you store the sequence of the prime numbers that you have already discovered, and **lazily expand** this list by generating and appending new primes to its end. Knowing that this list of all prime numbers up to

the current upper limit is sorted not only allows you to quickly look up and return the `k`:th prime number in the method `kthPrime`, but also to quickly iterate through all prime numbers up to the square root of `n` inside the method `isPrime` to look for potential divisors.

Furthermore, `Collections.binarySearch` can be used on the sorted list of these so far known prime numbers to determine almost instantly whether some given positive integer is a prime number. It would probably be a good idea to have a private helper method `expandPrimes` that lazily finds and appends new prime numbers to this list as needed by the `isPrime` and `kthPrime` methods.

Speed is the essence of this lab. To qualify for the passing lab marks, the automated test must successfully **finish all three tests within ten seconds** on the average off-the-shelf desktop computer from the past five years. (This requirement used to be twenty seconds back when this lab was originally written, but as you know, Moore's law and all that other stuff happened in between.)

Lab 12: The Second Hand Unwinds

JUnit: [PrimeGensTest.java](#)

In Python 3, all kinds of lazy sequences are easy to produce with **generators**, special functions that yield their results one element at the time and then continue their execution from the exact point where they left off in the previous call, instead of always starting their execution from the beginning of the function body the way ordinary functions work in both Python and Java. The Java language does not offer generators, but the idea of lazy **iteration** over computationally generated infinite sequences is still very much worth learning, and fits snugly into the `Iterator<E>` class hierarchy of Java Collection Framework.

This lab continues the work from the previous lab by using its methods `isPrime` and `kthPrime` as helper methods to produce the subsequence of all prime numbers that [satisfy some additional requirements](#) collected on the Wikipedia page that lists these special subtypes of prime numbers. Create a new utility wrapper class `PrimeGens` inside which you write **no fields or methods whatsoever**, but three `public static` nested classes. Each of these classes acts as an **infinite iterator of prime numbers** that are not explicitly stored anywhere in the precious memory bytes, but are generated **lazily** whenever the user code requests the next such prime number.

Each of these three classes should define the basic methods required by `Iterator<Integer>`. The method `hasNext()` should always return `true`, since for our purposes, all these sequences are infinite. (It is currently unknown whether the twin primes sequence is truly infinite, but enough twin primes are known to exist so that we won't run out of those during the JUnit tests.)

To get you started with the required lazy integer iterator classes in this lab, here is a complete model implementation of two such example classes. The first one produces all [palindromic prime numbers](#), whereas the second produces all **composite** numbers, that is, those that are not primes. Your three classes will have the same structure as these two, but use different logic inside the method `next` to find the next prime number with the required property. (Note also the tactical use of both **pre-** and **postfix** forms of the `++` operator in `Composites`.)

```
public static class PalindromicPrimes implements Iterator<Integer> {
    private int k = 0; // Current position in the prime sequence
    @Override public boolean hasNext() { return true; } // Infinite sequence
    @Override public Integer next() {
        while(true) {
            int p = Primes.kthPrime(k++);
            String digits = "" + p;
            if(digits.equals(new StringBuilder(digits).reverse().toString())) {
                return p;
            }
        }
    }
}
```

```

public static class Composites implements Iterator<Integer> {
    private int curr = 4, k = 2, nextPrime = Primes.kthPrime(2);
    @Override public boolean hasNext() { return true; }
    @Override public Integer next() {
        if(curr == nextPrime) {
            nextPrime = Primes.kthPrime(++k);
            curr++;
        }
        return curr++;
    }
}

```

Having carefully studied both these classes until you can explain what each line does, the three nested classes that you write inside class PrimeGens for you to write are as follows. You can quickly and easily test your implementations with a simple main method that first creates an instance of the said iterator, and then uses a for-loop to print out the first couple of dozen elements for you to compare to the expected initial sequences given below. Once the first twenty or thereabouts special primes produced by your iterator match the expected sequence, it is nearly certain that the rest will also follow suit, same as even though the race is not always to the swift nor the battle to the strong, that is still the way to bet. (At some point incredible luck becomes indistinguishable from incredible skill, if only to mangle Arthur C. Clarke's famous observation about the analogous relationship between sufficiently advanced science and magic.)

```

public static class TwinPrimes implements Iterator<Integer>

```

Generates all [twin primes](#); prime numbers p for which $p+2$ is also a prime number. This sequence begins 3, 5, 11, 17, 29, 41, 59, 71, 101, 107, 137, 149, 179, 191, 197, 227, 239, 269, 281, 311, ...

```

public static class SafePrimes implements Iterator<Integer>

```

Generates all [safe primes](#) that can be expressed in the form $2*p+1$ where p is also a prime number. (The smaller prime p that establishes the "safety" of the prime $2*p+1$ in a certain technical sense is then called a "Sophie Germain prime"). This sequence begins 5, 7, 11, 23, 47, 59, 83, 107, 167, 179, 227, 263, 347, 359, 383, 467, 479, 503, 563, 587, 719, 839, 863, ...

```

public static class StrongPrimes implements Iterator<Integer>

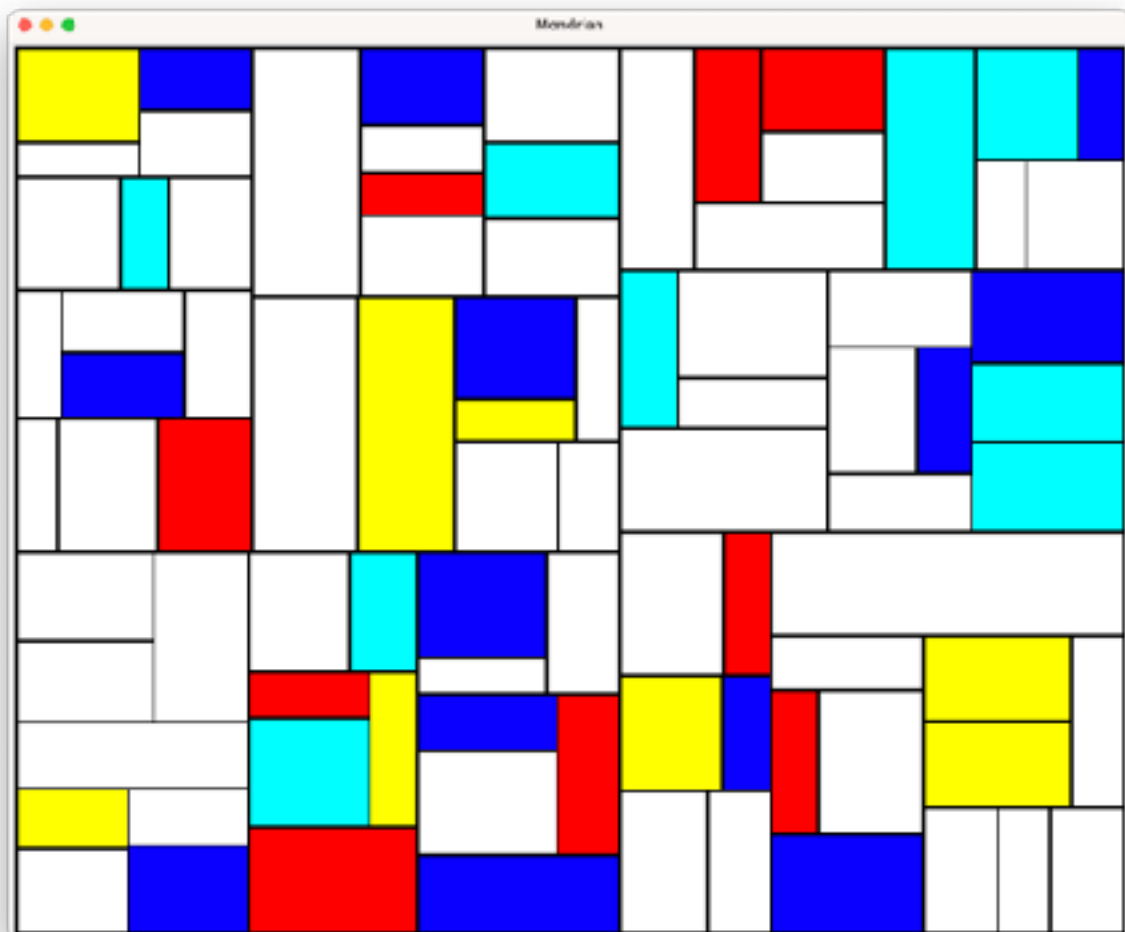
```

Generates all [strong primes](#), primes p that are larger than the average of the previous and next prime numbers around p in the sequence of all prime numbers. This sequence begins 11, 17, 29, 37, 41, 59, 67, 71, 79, 97, 101, 107, 127, 137, 149, 163, 179, 191, 197, 223, 227, 239, 251, ...

Lab 13: Recursive Mondrian Art

Your instructor actually used this assignment back in a Palaeolithic version of this course, but now that this classic problem has resurfaced in [Nifty Assignments](#), perhaps the time has come to revisit its timeless message. Especially since during these years in between, your instructor also managed to get himself hopelessly bitten by the [subdivision](#) and [fractal](#) art bugs.

This lab combines graphics with recursion by constructing a simple **subdivision fractal** that resembles the famous works of abstract art by [Piet Mondrian](#). An example run of the instructor's model solution produced the random image captured for eternity in the following screenshot, to which your images should be reasonably similar in style and spirit, using controlled randomness to generate a new piece of abstract visual art every time your program is run.



Write a class `Mondrian` that extends `JPanel`, with the following fields:

```
// Cutoff size for when a rectangle is not subdivided further.  
private static final int CUTOFF = 40;
```

```
// Percentage of rectangles that are white.
private static final double WHITE = 0.75;
// Colours of non-white rectangles.
private static final Color[] COLORS = {
    Color.YELLOW, Color.RED, Color.BLUE, Color.CYAN
};
// RNG instance to make the random decisions with.
private Random rng = new Random();
// The Image in which the art is drawn.
private Image mondrian;
```

After this, the class should have the following methods:

```
public Mondrian(int w, int h)
```

The constructor for the class, with the desired width and the height of the resulting artwork given as constructor arguments. Set the preferred size of the component to w-by-h pixels, initialize the field `mondrian` to a new `BufferedImage` of size w and h, and ask that image for its `Graphics2D` object to be passed as the last argument of the top-level call of the recursive `subdivide` method below. After returning from the rendering recursion, you should call `dispose()` method of the `Graphics2D` object to avoid leaking its resources.

```
public void paintComponent(Graphics g)
```

Draws the mondrian image created in the constructor on the surface of this component.

```
private void subdivide(int tx, int ty, int w, int h, Graphics2D g2)
```

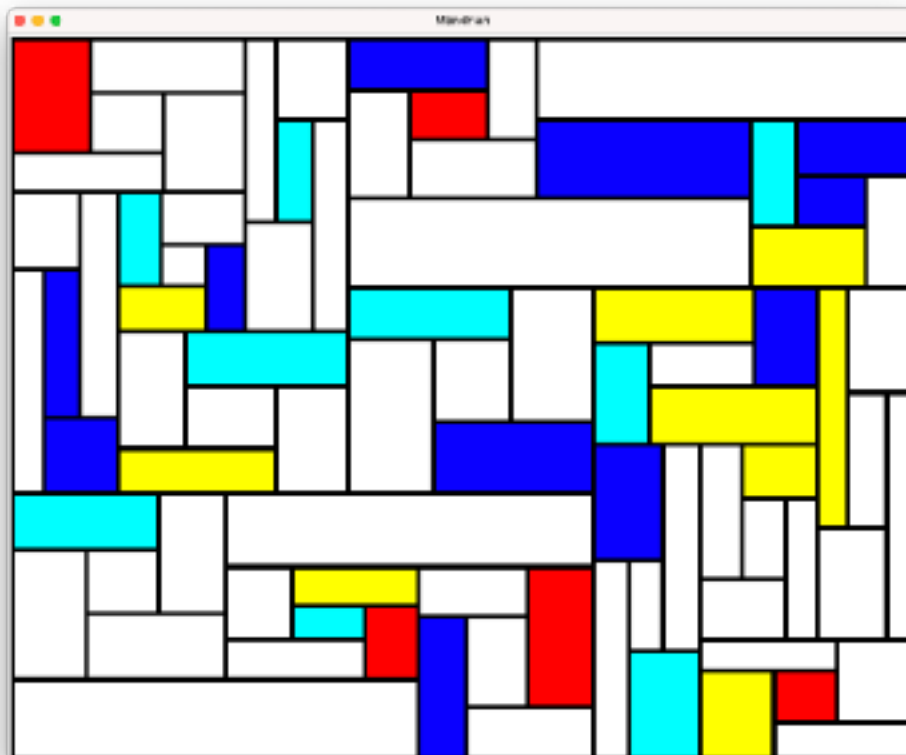
Recursively subdivides the given rectangle whose top left corner is in coordinates `(tx, ty)` and whose width and height are `(w, h)`. The base case of the recursion is when either w or h is less than `CUTOFF`, in which case this rectangle is drawn on the `Graphics2D` object provided. (The top-level call in the constructor should ask the image object its `Graphics2D` object and pass that on to the recursive call.) This rectangle should be white with probability `WHITE`, and choose a random colour from `COLORS` otherwise. Draw the rectangle to have a black outline in either case.

Otherwise, this recursive subdivision method should make two recursive calls for the two rectangles that you split randomly from the given rectangle. Make sure that you always split each rectangle along its longer edge (breaking ties arbitrary for squares), and that the splitting line is not too close to either edge that has the same direction, to keep the subdivision reasonably balanced. Subdivisions that contain thin shards and forced sharp angles tend to be perceived as ugly and disharmonious, same as when somebody writes cursive by hand and has to squeeze the words in more narrowly at the end of the line to stay within the margins. In all walks of life, that elusive sweet spot at the middle way between the extremes of predictably solid and boring order opposite the screeching cacophony of randomness will usually be perceived by actual humans as the most aesthetic. (Reaching that sweet spot can be a bit of a tightrope act in practice.)

To admire your randomly generated pieces of art, write a `main` method that creates a `JFrame` instance that contains one instance of `Mondrian` that is 1,000 pixels wide and 800 pixels tall. You can also try varying the above parameters and the colour scheme to aim for an even more artistic and aesthetically pleasing result.

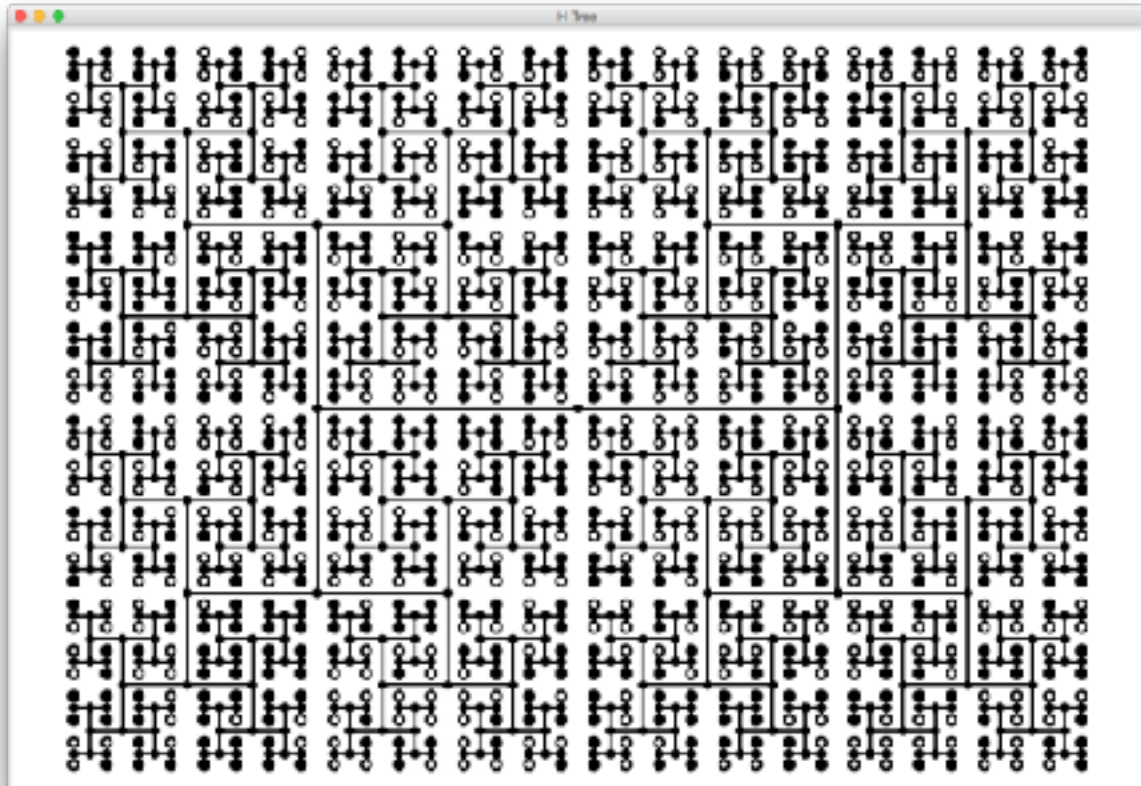
This random subdivision is not yet aesthetically the best possible due to a massive bias in the positioning of the subdivision walls inside the rectangle. Each subdivision will always contain exactly one straight **fault line** directly through that entire rectangle, which then continues to recursively hold for the two rectangles separated by this line. This is unconsciously perceived as being unnatural, even if the viewer does not consciously perceive the existence of the fault lines. (The reader might enjoy finding these recursive fault lines from the previous example image, and deduce the order of the subdivision recursion from those lines.)

Below, a more complex subdivision rule into five spiral-like pieces is mathematically guaranteed to eliminate such jarring fault lines, to make the resulting subdivision structure more organic and pleasant to the eye. More complex subdivision rules would converge towards random results that seem perfectly natural in the sense of being completely unforced and not in any way artificial, that way achieving the maximum artistic value reachable by a mindless machine in this domain. No law of nature or man will also stand athwart history yelling stop even if you use multiple different rules, one of which gets randomly chosen at each recursive subdivision call. No opponent, not even Old Nick himself, can systematically defeat you, when you don't even know yourself what path you will be taking in the future! We should also mention the completely different approach of [Gilbert tessellations](#) that produce even gnarlier tessellations, with the aesthetic cost of some of these tiles ending up unpleasant thin shards.



Lab 14: H-Tree Fractal

In spirit of the recursive Mondrian subdivision from the previous lab, here is another recursive fractal problem of rendering the [H Tree fractal](#) on a Swing GUI component, with some additional decorative little dots that indicate the intersections and the caps at the branch ends of this exponentially branching structure. An example run of the instructor's private model solution produced the following result that your outcome should also resemble. You may again adjust the style to be more artistic, provided that you maintain the basic nature and structure of this fractal.



Start by creating the class `HTree` that extends `JPanel`. This class should have the following fields:

```
// Once line segment length is shorter than cutoff, stop subdividing.
private static final double CUTOFF = 10;
// Radius of the little dot drawn on each intersection.
private static final double R = 5;
// The image inside which the H-Tree fractal is rendered.
private Image htree;
// Four possible direction vectors that a line segment can have.
private static final int[][] DIRS = {{0, 1}, {1, 0}, {0, -1}, {-1, 0}};
// A random number generator for choosing the end piece colours.
private static final Random rng = new Random();
```

This class should then have the following methods.

```
public HTree(int width, int height)
```

The constructor receives the `width` and the `height` of the image, and should first initialize `htree` to be a new `BufferedImage` instance of those dimensions, and set the preferred size of the entire component to that same `width` and `height`. Acquire the `Graphics` object of this image and convert it to the more modern and powerful `Graphics2D` reference that allows you to first turn on **anti-aliasing**, and then later draw each line segment with non-integer precision inside the recursion. Before the recursive calls, make the initially all black image all white by filling it with a white rectangle whose area covers the entire image. Instances of `BufferedImage` start out being all black pixels since the bytes that store the colours of those pixels are guaranteed to be initially filled with zeroes. (This same guarantee is always in effect for all objects created into the heap memory area of the Java Virtual Machine.)

The constructor of `HTree` should call the next `render` method twice, both times starting from the point that is in the middle of the image and with `len` set to equal to the minimum of the width and height of that image, divided by three. The first call should head left, and the second call should head right, for the two symmetric branches.

```
private static void render(double x, double y, int i, double len,
Graphics2D g2)
```

Render the H-Tree fractal starting from point (x, y) towards the direction given by the index `i` into the `DIRS` array that contains the four possible directions that each line segment can be heading, with the current line segment having a length equal to `len`.

The base case of the recursion is when `len < CUTOFF`, in which case this method should fill a disk of radius `R` centered at coordinates (x, y) , flipping a random coin to decide whether this disk will be black or white. (If it is white, you should still draw the black outline around it.) Otherwise, draw a small black disk to the current coordinates (x, y) , and then calculate the endpoint (nx, ny) of the line segment from the direction that your rendering is currently heading with

```
nx = x + DIRS[i][0] * len;
ny = y + DIRS[i][1] * len;
```

Draw the line segment from (x, y) to (nx, ny) on the image, and follow with two recursive calls at (nx, ny) using the directions `i+1` and `i+4`, wrapping around the `DIRS` array as needed by taking the remainder of the new direction modulo 4. The new value of `len` in the recursive call should equal the current `len` divided by the square root of two, so that the branches become shorter down the recursion so that the overall shape will stay within a finite boundary. (The choice of using square root of two as the divisor makes the entire H-tree precisely space-filling, since two squares of that side length fill exactly the original square.)

Thanks to the magic of subdivision fractals, these exponentially spreading branches never touch or overlap each other, no matter how deep you continue this recursion. In the limit of CUTOFF equal to zero, as the recursion depth increases towards infinity, this **space-filling fractal** is **dense** in its entire two-dimensional bounding box in that it does not leave any holes in which you could fit any circle with any arbitrarily small positive radius ϵ of your choice. This despite the shape consisting only of one-dimensional line segments when we don't draw the decorative disks on the intersections! (Yes, a mathematical purist would now point out that the H-tree fractal is not technically space-filling since it has a Lebesgue measure of zero, but we shall skip over that technicality here.)

Students who happen to catch from this exercise a funky case of “fractal fever”, a once fashionable ailment among the young but assumed to have gone the way of [mange, grunge, the itch, the twitch, the thrush, the scoff and the rot](#) since the early nineties, at least outside California (the epicentre and patient zero of the original epidemic, as always), might want to dip into “[Brainfilling Curves: A Fractal Bestiary](#)”, your instructor’s personal favourite that teaches you how to render an infinite variety of such artistic fractals with **turtle graphics**.

To admire your fractal artwork, you should again create a `main` method to open a new `JFrame` instance, inside which you add a new `HTree` instance of dimensions `(1000, 800)`. Again, the same way as in the recursive Mondrian subdivision, once you get this program to work, feel free to experiment to make the result look more pleasant. This is perfectly fine with the instructor as long as you maintain the overall shape and spirit of the original H-Tree fractal. For example, if you happen to know how to compute arbitrary **rotations** to direction vectors using matrix algebra, you might want to add a touch of randomness to make the image seem more relaxed and natural so that instead of always turning exactly 90 degrees left or right, you instead turn 90 degrees plus or minus a small random amount, as if some a flesh-and-blood human had drawn this shape with a slightly trembling hand. To prevent the randomness from going completely off the rails, the recursion should simultaneously keep track of both the correct intended direction and the current actual direction, and correct back should the former veer away too far from the latter.

Lab 15: Zeckendorf Encoding

JUnit: [ZeckendorfTest.java](#)

Back in the lab 0(E) we encountered the [Zeckendorf theorem](#) that says that every positive integer can be uniquely expressed as a sum of non-consecutive Fibonacci numbers. This sum of Fibonacci numbers can then be encoded into a series of bits whose positions correspond to Fibonacci numbers so that the value 1 indicates that that Fibonacci number is present in the sum, whereas the value 0 denotes the absence of that Fibonacci number. For example, playing around with online conversion tools such as “[The Fibonacci base system](#)”, we find out that $42 = 34 + 8$. The **Zeckendorf representation** of this number is 10010000 to denote that the fifth (8) and eighth (34) Fibonacci numbers are present in the sum, the positions counted from right to left the same way as with the ordinary positional representation.

The Zeckendorf representation is a bit (heh) of an inefficient way to represent integers compared to the ordinary binary numbers as a simple sum of the powers of two. This is due to the wasted bit pattern 11 that never appears inside the Zeckendorf representation, for a roughly 25% loss of storage efficiency. However, this wasteful limitation suddenly turns into a major advantage if you want to encode not just one integer, but an entire sequence of arbitrarily large `BigInteger` values so that you can still unambiguously restore the original values! When a sequence of integers is encoded using the ordinary binary encoding for each individual element, this encoding must somehow be augmented to let the decoder know where each number ends and the next number begins. This can be solved by either agreeing that each integer is encoded with the exact same number of bits (which then enforces a maximum value that can appear inside the sequence), or by giving some high-end bits a role to indicate whether the encoding of the current number still continues, in spirit of the **UTF-8 character encoding** discussed in the I/O streams lecture. (The lab 71, “The Gamma And The Omega”, showcases one such possible encoding scheme.)

Using Zeckendorf representation, determining where the encoding of the current number ends for the encoding of the next one to begin is automatic. The otherwise unused bit pattern 11 acts as an artificial **separator mark** that tells the decoder that the representation of the current number ends there, so the next number starts from the following position. The first 1 in the otherwise forbidden pattern 11 is the highest “zit” of the Zeckendorf encoding of that number, and the forbidden second consecutive 1 acts as a virtual “comma” that separates that number from the next. Even better, this encoding is **self-synchronizing** and thus more **robust**; the reader can recover from any incorrect bits the moment it arrives at the next separator mark, from where the rest of the sequence will be faithfully restored. (The **UTF-8 encoding** for Unicode characters also shares this convenient guarantee, over there enforced by the high-order bits of every payload byte.)

We must also realize that the convention of writing digits in descending order of their positions is but a societal agreement similar to driving either left or right side of the road; either one is good, but we have to pick one and have everyone agree on it. Humanity could have just as well agreed to write our bits, zits and fits of other kinds of digits in ascending order of positions. For example, the number we normally say out loud as “two hundred and eighty seven” could just as well be written as 782 instead of 287. [Fibonacci coding](#) of the given sequence of integers consists of simply writing the Zeckendorf representations of each number left to right (so the last bit, being the highest bit, is

always 1), so putting another 1 between the two separate numbers to create the impossible 11 will act as a virtual “comma” that clearly separates those two numbers.

After that mouthful, it is finally time to start chewing. (In the later problems, these educational preludes will only get longer.) To celebrate the fact that the Zeckendorf encoding allows us to encode numbers that contain even millions of digits, we use the `BigInteger` class to represent both these integers and the Fibonacci numbers used to break them apart, the same way as we did back in lab 0(E). Create a class `Zeckendorf` that contains the following two `static` methods to encode and decode the given list of integers:

```
public static String encode(List<BigInteger> items)
```

Given a sequence of `BigInteger` values as `items` so that each item is greater than zero, compute and return the Zeckendorf representation of that entire sequence as a single string whose each character is either '0' or '1'. For example, given the list `[ 4, 36, 127 ]`, this method would return the string `"101101000001110100001011"` (coloured here for extra readability), since 4 encodes to `"1011"`, 36 encodes to `"010000011"`, and 127 encodes to `"10100001011"`.

If you have already completed the Lab 0(E), you might as well reuse the method `fibonacciSum` to do most of your heavy lifting here. Of course, in a real compression algorithm the result would be a sequence of bits instead of using one full Unicode character to encode each bit, but in this lab the result is a string just for convenience. Inside this method, you will surely once again use the mutable `StringBuilder` to accumulate the zeros and ones, and convert the entire result into a `String` just once at the end before returning.

```
public static List<BigInteger> decode(String zits)
```

Given the Zeckendorf decoding as `zits`, recreate the original sequence of positive integers. Your method may assume that `zits` is a complete encoding of an integer sequence that can contain characters '0' and '1' only.

Notice how, same as any two methods to encode and decode between two representations of something, these two methods are **inverses** of each other, so that applying one to the result of the other always produces the original starting value. Writing a fuzz test turns out to be an especially trivial task for such functions. Even if test cases are produced in some pseudo-random fashion, the expected result of using the pair of methods in tandem is fully known, since it must always equal the original value! Of course, to debug your code that is not yet passing this passive-aggressive mass test, you can create yourself specific desired test cases with the above online Zeckendorf calculator.

To gain even stronger appreciation about the general awesomeness of the Zeckendorf encoding, discrete math and number theory enthusiasts can check out [David Eppstein's discussion](#) on efficient integer arithmetic performed directly on the Zeckendorf representation, without the need to convert these integers back to ordinary binary encoding for these operations.

Lab 16: Egyptian Fractions

JUnit: [EgyptianFractionTest.java](#)

As explained on the Wikipedia page "[Egyptian fraction](#)", ancient Egyptians had a curious way to represent arbitrary positive integer fractions a/b by breaking them down into sums of distinct **unit fractions** of the form $1/n$. Such breakdown can be done in infinitely many different ways for any rational number, depending on what additional obligations we wish to impose on this breakdown. However, the number of unit fractions needed to express a number as a sum of unit fractions tends to grow exponentially with respect to its absolute value. Even though the following construction algorithms would work for arbitrarily large rational numbers, we will only consider fractions a/b that satisfy $a < b$ making $a/b < 1$, so that our JUnit test methods will terminate within a reasonable time without running out of memory.

This lab has you implement two different algorithms to construct the Egyptian fraction breakdown. To ensure correct results even when they involve thousands of digits, you must do all your integer arithmetic using the `java.math.BigInteger` type. You should also add the `Fraction` example class into your labs project and use its operations to perform the arithmetic of rational numbers needed in this lab. Instead of reinventing that rusty old wheel all by yourself, you get to have more time to [not merely walk](#) but also do your arithmetic like an ancient Egyptian. This should make your heart lighter so that Anubis will let you enter the field of reeds for a pleasant afterlife, instead of you unceremoniously ending up being eaten by the crocodile god.

Create a new class called `EgyptianFractions` into your labs project, and there a method

```
public static List<BigInteger> greedy(Fraction f)
```

that constructs the Egyptian fraction breakdown of the fraction $f = a/b$ with the **greedy algorithm** originally [proven to always terminate by Mr. Fibonacci himself!](#) The simplest situation is when $a = 1$, so the result is $1/b$. Otherwise, compute the smallest positive integer n whose reciprocal $1/n$ is less than the fraction a/b . Assuming that the fraction a/b is in lowest terms, this can be done by dividing b by a with the **truncating** integer division giving the result as the [floor function](#), and then add one to the result of that truncated division. For example, the smallest positive integer n whose reciprocal is less than $3/17$ is $\lfloor 17 / 3 \rfloor + 1 = 6$. Add the term $1/n$ to the result, and extend the result with the greedy breakdown of the remaining, hopefully simpler fraction $a/b - 1/n$.

To ensure unique results for the fuzz testing, this method should return the Egyptian fraction as a list of denominators of these terms in sorted ascending order, each term of course represented as a `BigInteger` value in this problem. For example, when called with the fraction $2/3$, this method would return the list `[ 2, 6 ]`, and with the fraction $4/17$, return `[ 5, 29, 1233, 3039345 ]`. As you can see there, restricting the breakdown to use only distinct unit fractions can cause quite a blowup in terms for even some seemingly simple fractions. As is typical with greedy algorithms, the result is not necessarily the simplest or the most balanced possible subdivision into unit fractions, since the last term can be a pretty thin sliver compared to the greedily chosen "thicc" terms that precede it.

The next method introduces another algorithm to construct an Egyptian fraction that usually produces a rather different result than the previous greedy method. Same as Fibonacci's greedy method, the following technique is not guaranteed to return the simplest or the most balanced breakdown. (It sure feels like some spooky general pattern that underlies our visible fabric of reality is slowly emerging from its slumber!) Write the method

```
public static List<BigInteger> pairing(Fraction a)
```

that maintains a local `Set<BigInteger>` variable that keeps track of which terms have already been chosen to be part of the Egyptian fraction. Initially this set is empty. To break down a fraction a/b , we first split it into two terms $(a - 1)/b + 1/b$. Repeating this step allows us to “carve” unit fractions off the bulk of the remaining fraction one at the time, as if carving delicious slices off the Christmas ham. Slice off one unit fraction $1/b$ by adding b to your chosen set of terms, and construct the Egyptian fraction representation for the remaining fraction $(a - 1)/b$ until the entire number has been carved out so that only a zero remains, simplifying the fraction along the way as needed by eliminating the common factors of $a - 1$ and b . The `Fraction` class will, of course, do this for you.

Comparing the breakdowns for some randomly chosen simple fractions, the greedy algorithm seems to typically produce fewer terms, but not always. For example, $5/91$ breaks down greedily into unit fractions `[ 19, 433, 249553, 93414800161, 17452649778145716451681 ]`, whereas pairing produces a far more balanced subdivision `[ 23, 91, 2093 ]` for the same quantity. As so often in all walks of life, we have ample time to regret our greedy initial choices as they slowly take us towards their inevitable ends. On the entire way down there we can only dream of what could have been, had we initially stayed our hand and not shot that bloody albatross, but followed the dove and taken the road less traveled instead. (Making your choices inside a computer program instead of directly in the physical reality does not affect the poetic justice of the tragic consequences of such greedy choices. As above, so below; as below, so above.)

To maintain the discipline of using each denominator at most once, we need to somehow handle the situation where the new slice $1/b$ is already in the set of chosen terms. To do this in an orderly fashion, maintain a local variable `List<BigInteger> buffer` that contains the denominators of the unit fractions that are waiting to get into the set of chosen terms. Start by initializing this `buffer` to contain exactly one value b , the term that is currently waiting to enter the club. While this `buffer` is non-empty, pop out any one of the items there, let's call it n . (It turns out that the order in which the items are popped won't affect the final result, so you might as pop out the last item.) In the first round of popping, the value n must therefore be equal to b , but it can and will equal other values during the later rounds of this corrective back-and-forth dance.

If n is not already a member of the set of chosen terms, add it there and move on. Otherwise you have a conflict of two equal terms $1/n$ trying to become a chosen term, even though only at most one of them can actually be part of the final set. First, remove n from the set of chosen terms. Resolution of this conflict will then continue depending on the parity of n :

- If n is even, add the term $n/2$ to the `buffer`.
- If n is odd, add two terms $(n + 1)/2$ and $(n(n + 1))/2$ to the `buffer`.

For example, two equal unit fractions $1/10$ and $1/10$ can be replaced by a single $1/5$ without affecting the total. If the conflict is between two odd unit fractions $1/5$ and $1/5$, push two new terms $1/3$ and $1/15$ into the `buffer`, the total again unaffected by this since $2/5 = 1/3 + 1/15$. Rinse and repeat until all terms are distinct and the `buffer` is empty. Until then, the individual terms will enter, exit and re-enter the set of chosen terms and the waiting room of the `buffer`, the exact details of this back-and-forth depending on the terms waiting inside the `buffer` and the needs of the rest of the number $(a - 1)/b$ that is still waiting to be broken down.

We often like to think of integers being composed of prime factors multiplied together. For natural numbers, this **prime factorization** is unique, the result modestly known as the “[Fundamental theorem of arithmetic](#)”. This famous result is actually quite not as trivial as it initially seems, and it certainly is not true just because of some hand waving in the likes of “But, like, it’s the *primes*, so how could it even be otherwise, *maaaan*, so much confusion here sweaty lol I can’t even!” However, as we just saw, another way to think of not just integers but all rational numbers is to see them as finite sums of tactically chosen unit fractions. However, unlike the unique prime factors that make up each integer, any rational number (and thus all integers) can be broken down into sums of distinct unit fractions in infinitely many different ways.

These Egyptian fractions boggle the mind as a way to express any one of the infinitely many positive rational numbers in an infinite number of ways of adding up smaller positive rationals. Kinda makes us wonder if those ancient Egyptians really were connected to something that we have forgotten... but up here in the present timeline of cyberpunk future so bright that we need to wear shades, anybody interested to learn more about this topic can start on the page “[Egyptian Fractions](#)” by that [David Eppstein](#) guy again. (“Once is happenstance, twice is coincidence...”) Most of the external links emanating from the page are as dead as the ancient Egyptian culture that they echoed, but fortunately at least “[Algorithms for Egyptian Fractions](#)” is still alive at the time of this writing. The reader may feast his mind by appreciating the multitude of other techniques to break down an integer fraction into a sum of distinct unit fractions.

Lab 17: Diamond Sequence

JUnit: [DiamondSequenceTest.java](#)

The `Set<E>` implementations provided in the Java Collection Framework are flexible, but not the best for all situations due to their relatively high memory use for each object stored in that set. Most of the time in practical programming, sets and maps are used to remember the things that we have seen and done during the method's execution. If those things can never become unseen or undone, the `remove` operation does not need to be supported, except possibly in the form of the `clear` method to evacuate the entire data structure at once.

In the terminology of data structures and algorithms, collections that allow dynamically adding new elements but disallow dynamic removals are called **monotonic**. Sometimes such structures can be implemented more efficiently than the full versions that allow the `remove` operation. Since each element removal decreases the overall **entropy** of that data structure, the `remove` operation necessarily has to involve actual physical work. This makes this operation inherently more complex than the entropy-increasing operation `add` and the nominally entropy-neutral operation `contains`.

Further optimizations in the set representation are possible when members of the set are known to be **natural numbers**. This is a common use case, given that everything inside not just the physical computer but our entire physical reality is fundamentally made of integers anyway, as much as we like to otherwise pretend that the computer memory is full of strings, dictionaries and other entities that in reality are just as made up and fictional as, say, hobbits and vampires. This effect becomes even more pronounced when these numbers tend to be added to the set in a roughly ascend-ish order, which is often the case with many integer sequences whose rules to compute the next element depend on whether some particular integer has already been seen inside the sequence generated so far, and the growth of the sequence is inherently so modest that new heights are reached slowly and the sequence must therefore fill in the gaps that it leaves behind. For example, the [Recamán's sequence](#) that we met back in Lab 0(B) would neatly fit this pattern.

Membership information can in such cases be packed maximally tight so that each integer is represented as one bit stored inside a `boolean[]`. We will cleverly manage this boolean array as a **sliding window** so that the field `start` tells which absolute position the relative zero index represents, and implement the methods `add` and `contains` to operate relative to this index.

Now, the good news is that you don't need to do any of what I just said, since for both this lab and for your possible future endeavours in Java, your instructor provides the class [NatSet](#) whose objects represent **monotonic** sets of natural numbers (that is, zero or positive). These `NatSet` objects start their lives as representing the empty set. The public method `void add(int n)` adds the natural number `n` to the set. (There is no evident way to say that in any less convoluted fashion, and yet leaving in that phrasing somehow feels like giving up as a technical writer.) The public method `boolean contains(int n)` tells the caller whether `n` is a member. Since no shrinkage is allowed even when it's cold, so we don't have a `remove` method in the public interface.

Using the `NatSet` class as a tool to keep track of the numbers already seen during construction, your mission is to implement another very interesting sequence of positive integers, found by your instructor in the book “[Mathematical Diamonds](#)” by [Ross Honsberger](#). Since that book deals with mathematics instead of computer science, it uses the conventional numbering of sequence positions starting from one, not from zero the way the numbering always works for us propeller beanie cap code monkeys and web designers. The first element in the position $k = 1$ equals one. Each later position $k > 1$ contains the smallest positive integer that so far has not appeared anywhere in the sequence and makes the sum of the first k elements of this sequence to be an integer multiple of k .

For example, the second element for $k = 2$ is the smallest unseen integer that, when added to the first element 1, makes the total divisible by 2. The correct answer is 3, since $1 + 3 = 4$ is divisible by $k = 2$. After $[1, 3]$, the third element is again the smallest unseen integer that, when added to the previous elements 1 and 3, makes the total divisible by 3. Correct answer is 2, since $1 + 3 + 2 = 6$, and the deuce is the smallest of all unseen numbers that fit the bill. The sequence starts its eternal journey as 1, 3, 2, 6, 8, 4, 11, 5, 14, 16, 7, 19, 21, 9, 24, 10, 27, 29, 12, 32, 13, 35, 37, 15, 40, ...

To practice writing virtual iterators that are not backed by any actual collections but who always produce the next element of the sequence in computational means, we implement this “diamond sequence” (named a bit unimaginatively due to the lack of mathematical knowledge, depth or sophistication from this author’s part), as a virtual iterator. In your labs project, create the class

```
public class DiamondSequence implements Iterator<Integer>
```

with the instance methods required in the interface `Iterator`:

```
@Override public boolean hasNext()  
@Override public Integer next()
```

Since this sequence is infinite, the method `hasNext` always returns `true` without further ado. For the computation performed inside the method `next`, this class should define a data field `int k` that keeps track of the position that the generation of this sequence is currently at. Then, there should also be another data field `sum` that contains the sum of the elements generated so far. The type of this second field should be `long`, to ensure that adding up the elements of this sequence does not overflow even when we shall soon add up its first hundred million elements! Last, but definitely not the least, this object needs an instance of `NatSet` to remember the elements generated so far.

The method `next` should first increment `k` and perform some integer arithmetic to look for the smallest previously unseen `n` that, when added to the current `sum`, causes the total for the current position to be divisible by `k`.

The JUnit test class [DiamondSequenceTest](#) will push your code hard to its limits by making it generate the first one hundred million elements of the diamond sequence. Within this range, all values in sequence still fit inside `int` variables without a possibility of overflow, but their sum certainly will not fit inside an `int`. The sequence grows in a modest linear pace, but it bounces above and below this baseline in a seemingly chaotic fashion that somehow always manages to fill in the gaps

that it has left behind. (Every filling of such a gap will then allow the `NatSet` data structure to move its sliding window into the data some number of positions to the right.)

The fact that no integer appears twice in this sequence follows trivially from its construction rule. It is not equally obvious that every positive integer n will eventually appear in some position, but it can be proven that this will eventually happen for any n .

Furthermore, this sequence has a most curious **self-describing** structure in that whenever the k :th element of this sequence equals v , its v :th element equals k . For example, since the seventh element of this sequence equals eleven, its eleventh element must therefore equal seven! Let's all for a moment try to wrap our collective heads around this puzzling fact that somehow follows from the construction of this sequence. (In discrete math terms, when this sequence is viewed as an infinite **permutation** of natural numbers to themselves, all **cycles** of this permutation have the same length of two.) The method `testSelfReferentiality` in the JUnit test class verifies this bold claim empirically for the first one million elements of this sequence.

Lab 18: Working Scale

This lab again has you working with Swing GUI components, but this time not just with inert graphics and boring GUI components such as buttons and text fields that people use at work, but animations where the content of your component is updated in real time fifty frames per second.

From the repository [ikokkari/CCPS209Labs](https://github.com/ikokkari/CCPS209Labs), add the interface [AnimationFrame](#) and the two classes [AnimationPanel](#) and [Tircle](#) into your labs project. The interface `AnimationFrame` represents the ability of an object to render itself into pixels on command, so that the contents of this image vary over time. This interface contains one method for your subclasses to implement:

```
public void render(Graphics2D g, int width, int height, double t);
```

Render the contents of the animation frame the way they are supposed to look like at time `t`, into the given `Graphics2D` object `g` so that the dimensions of this rendering are `width` pixels wide and `height` pixels tall. Time is not measured here in actual seconds of “wall clock time”, but is merely a number that can be positive, negative or zero, no different from the numbers used to denote the spatial dimensions. This way, the motion of the displayed animation could be sped up or slowed down arbitrarily with every individual frame still looking exactly like it should be. We could even find out with this one method what the animation frame would look like one year from now, without having to render every frame between the present moment and that future time.

Regardless of the speed that this virtual time advances, the actual animation is rendered inside an `AnimationPanel` component at exactly fifty frames per second to keep the visuals smooth enough to satisfy the human eye. (Back in this author's time, there were only 24 frames per second, and that is how we liked it without complaining.) The `AnimationPanel` objects are Swing GUI components customized to display the `AnimationFrame` object given to them. Time `t` always starts from zero, and increases by `step` after every animation frame. Whenever you create a new subclass of `AnimationFrame`, you can edit the `AnimationPanel[]` array in the `main` method to include an object of your new class to immediately try it out. As you admire your artwork, remember to also grab the window resizing handle and give it a good back and forth to ensure that your `AnimationFrame` properly renders as it should in any `width` and `height` and does not, for example, rely on the assumption that the animated image is a square.

(Also, do not fail to appreciate the isomorphism between the two concept troikas `Animal / Cat / getSound` versus `AnimationFrame / Tircle / render`.)

As an example of how to properly implement the `render` method, see the example class `Tircle` that renders a [subdivision fractal](#) that is built up from smaller copies of itself. Such **self-similarity** immediately suggests using recursive methods to compute and render such fractals. Since we cannot conveniently make the `render` method itself recursive (since it has no place for the additional parameters that our subdivision would need), we nimbly sidestep this limitation by extracting the recursion into a separate `private` method of its own (in the class `Tircle`, this recursive method is also called `tircle`) and make the. The arguments for this top level call from the `render` method

depend on the parameters `width`, `height` and `t` to ensure that the rendering is done correctly and without pixellation for all possible image sizes, from a small icon up to a screen splash.

To give this recursion a solid foundation with every branch eventually reaching a **base case**, once the subdivision shape is small enough, it is rendered as some simple shape such as a circle or a rectangle in that position. In principle, this base case would be when the shape is smaller than one pixel. However, cutting off the subdivision once the radius of the shape is lower than some higher threshold such as ten pixels usually gives more aesthetic results. Armed with this general technique, you could render any subdivision fractal you can dream up. Laws of neither nature or man dictate that `t` has to remain fixed through all levels of recursion, but can be increased or decreased a little in each call to give the overall system a more organic appearance when its components are not acting in an unnaturally perfect lockstep with each other.

Your `render` method should render some kind of reasonable result for any legal value of the parameter `t`. The simplest way to achieve this is to **normalize** your animation time to always run from 0 to 1, regardless of the actual intended length of the animation. (The animation can always be stretched to arbitrarily length in the actual wall clock time by decreasing `step`.) For values of `t` outside the interval $[0,1]$, the integer part of `t` is ignored and only the decimal part has an effect on the outcome. This makes the entire animation periodic so that at integer times $t = 0, 1, 2, \dots$ the animation has returned in its initial state. This animation can then be turned into an animated GIF that consists of `n` frames computed with `t` stepping through times $0, 1/n, 2/n, 3/n, \dots, (n-1)/n$. Note how the last frame is not `t=1`, since this would be an exact duplicate of the first frame `t=0` and show up as a tiny (and therefore all the more annoying) duplicate frame glitch in the animation. Since animated GIFs tend to use more space than necessary (the old format does not exploit any of the myriad statistical correlations between consecutive frames during the compression, but encodes each frame from scratch as a separate image), these days we convert any such animated GIF into various other formats to boil off about half of its bits. You kids these days have it so easy...

To make your periodic animation smooth so that there is no visible discontinuity at the times that are whole integers, define your animation paths so that the position, size and other aspects of its **appearance** are the same for every individual object for `t=0` and `t=1`, whichever way that object happens to move in times between these two endpoints. Alternatively, you can organize your objects so that each object has a “successor” so that the appearance of this object at `t=1` is the same as the appearance of its successor object at time `t=0`, and this way “cheat” to create infinite forward motion. See, for example, [“Frequipathy”](#), [“Cubenary”](#) or [“Stalka”](#), some old creations of the author. The fact that these compute the objects in three dimensions makes absolutely no difference to anything relevant. (Objects that end up outside the rendering area don’t need such a successor, since their “blip out of existence” never shows up on the screen...)

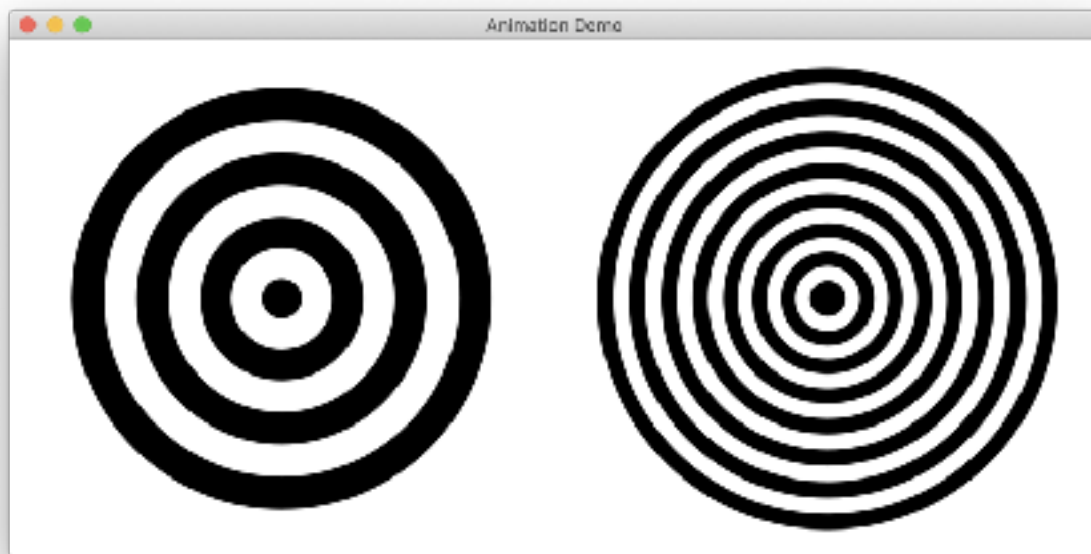
Calculations that consist of such periodic motions will greatly benefit from the existence of periodic functions, which most often are the basic trigonometric functions of sine and cosine. Since the purpose of this lab is not give you unpleasant flashbacks to the battlefield of trig back in the jungle of high school, the math is conveniently provided to you in form of two periodic functions

```
default public double circleX(double cx, double r, double t)
default public double circleY(double cy, double r, double t)
```

already implemented as default method in `AnimationFrame` so that any class that implements `AnimationFrame` automatically inherits these implementations. You can use these functions to calculate smooth periodic motion in either one or two dimensions. To have some quantity x oscillate between two extreme values $cx+r$ and $cx-r$ around the center point cx , simply compute it with the formula $x = \text{circleX}(cx, r, t)$. For example, to have x oscillate between the values 4 and 12, the math says that $cx=8$ and $r=4$. This oscillation is normalized to have a period of one.

When used together to compute the x - and y -coordinates for the position of some object at time t , the two methods `circleX` and `circleY` give you the position of an object that does **circular motion** around the center point (cx, cy) with the radius r . (Internally, `circleX` uses the cosine function whereas `circleY` uses sine.) This approach can also be used to compute the points of a regular polygon centered at (cx, cy) with n sides (for example, a square for $n=4$), with the i :th corner being in coordinates given by plugging in $t = (1.0 * i) / n$. (To rotate this polygon, add the same offset to t for each corner.)

Enough math and theory, let's finally get on with the practice! Create a class `Breathe` that implements `AnimationFrame`. Implement the `render` method in this class to render concentric circles alternating in black and white (the number of circles should be given as constructor parameter for flexibility) from outside in so that each circle is centered at the same coordinates $(width/2, height/2)$ with radius diminishing inwards. An example screenshot of the main method of `AnimationPanel` that creates one instance of `Breathe` with eight circles and another with sixteen, hopefully clarifies the intention. However, the radius of each circle should slightly oscillate in some smooth fashion so that **the entire animation gives an appearance of something organic that is breathing, has a pulse, or performs some muscular effort to move**. Adjust your equations for the radii until you are satisfied with the impression.



Once you get these shapes in their proper places on the component, you can check out the author's old works "[Frequestion](#)", "[Jobol](#)" and "[Zigwave](#)" as examples of organic motion. The neat thing about periodic functions is that you can always add or subtract another periodic function, and the result remains periodic. Even better, as long as the period of the first function is an integer multiple of the period of the second function, the result will still have the same period as the first function. This allows us to design our motion paths with linear thinking, and once the animation works, adjust it by adding suitably periodic small "wobbles" to them, possibly weighted with periodic functions such as $\sin(\pi t)$ that have been tactically chosen to vanish to zero at both $t = 0$ and $t = 1$...

Lab 19: Symbolic Distances I: Representation

JUnit: [DistanceTestOne.java](#)

Even though Java expressions are given in symbolic form inside the language source code, during the execution of compiled code all these variables contain only the numerical results produced by evaluating some of those expressions. You only have to “show your work” in school exams, whereas the rest of reality acting as a grumpy boss is happy with the final result and just enough paperwork to cover their own arse if someone catches a lawsuit. This makes numerical approximate results up to a couple of significant figures sufficient for the government work. Even so, sometimes mere end results can be unsatisfactory, especially when they get rounded to the `double` type that cannot express even simple numbers correctly, as we saw simply by trying to evaluate the expression $0.1 + 0.2$, or comparing the results of $(0.1 + 0.2) + 0.3$ and $0.1 + (0.2 + 0.3)$.

We are forced to define new data types as classes to represent some restricted forms of symbolic expressions as exact values. We already saw the `Fraction` type to keep rational numbers such as $1/3$ in the exact form, instead of having to round them to 0.3333333333333333 for this fundamentally incorrect result to be assigned to some variable so that the rest of the computation could proceed. Some languages come with useful data types for fractions and other limited symbolic expressions out of the box. Stratospherically high level languages such as **Wolfram Mathematica** even keep every expression in its symbolic form and operate on those symbolic forms, so that only the final result needs to be converted to its numerical approximation for human consumption.

If you can tolerate the memory and time cost in keeping expressions in their symbolic forms, such expressions are better than numerical approximations for the simple reason that you can always convert from symbolic to numeric, but once this approximation is done, there is no way to reverse it and get the original symbolic expression. Furthermore, symbolic expressions convey useful information about the structure of the data in a way that gets lost after being reduced to numbers. This despite the fact that inside the computer there exist only bytes filled with zeros and ones, and anything higher is mere consensual fiction to help us organize our thoughts about computations in a more abstract and succinct form than the bare metal would allow.

In spirit of the `Fraction` example that we saw in the class, this lab and the two labs following it define a new data type to symbolically represent numbers of the form $a\sqrt{b}$ where a can be any integer, and b can be any **squarefree** positive integer. A positive integer n is squarefree if n is not divisible by c^2 for any integer $c < n$. Some examples of such numbers that we would like to represent are $-7\sqrt{13}$, $42\sqrt{10}$, and $-1234\sqrt{101}$, where the requirement of being squarefree gives every such possible value one unique representation in our system. For example, $7\sqrt{48}$ is equal to $7\sqrt{3 \times 4 \times 4}$, which allows the two fours to be extracted from under the square root for a more succinct representation $28\sqrt{3}$ for the same irrational quantity.

One super relevant fact about integers to know at this point is that [the square root of an integer can never be a rational number](#) of the form a/b unless $b = 1$, which would make the square root itself also an integer and the integer under the root symbol to be a **square number**. Otherwise, that

square root must be an **irrational** number, and an **algebraic** one at that, being a root of some quadratic equation. But this square root of an integer can never be a rational non-integer.

Every integer a can be trivially represented as $a\sqrt{1}$ inside this system. Furthermore, formulas that compute $a\sqrt{b} + c\sqrt{d}$, $a\sqrt{b} - c\sqrt{d}$ and $a\sqrt{b} \times c\sqrt{d}$ will always produce results of that same form. The set of such numbers is therefore **closed** with respect to these operations: no matter how you apply these operations to the values you have created so far, you will never get a value that is not representable in this system, ignoring the issues of arithmetic overflows. To also allow division as the fourth arithmetic operator while still keeping this system of numbers closed, we would have to allow more complicated symbolic expressions than just this simple form. (Not today, Satan. Go sow your division some place else: here we do everything with integers, the way God intended.)

But that's again enough math, the time has come for some decisive coding action. Since these kinds of numbers often appear in distance calculations between points on the two-dimensional plane (for example, check out the [Spiral of Theodorus](#)), we shall name our data type for representing such quantities as `Distance`. Each instance of `Distance` represents a number that consists of a sum of terms of the form $a\sqrt{b}$, so that no b appears twice in the sum. Two terms $a\sqrt{b} + c\sqrt{b}$ can always be combined into a single term $(a + c)\sqrt{b}$ to enforce this constraint.

Since the operation of extracting squares from the given integer is so very important, our first order of business is to implement the method to do so. Since this method does not need anything from the underlying object, we might as well make it a `static` so that it can be used as a function:

```
public static int extractSquares(int n)
```

Given a non-negative integer n , this method should find and return the largest positive integer c so that n is divisible by c^2 . (For any n , the value $c = 1$ will always work, if nothing higher does.)

The next thing is to choose the internal representation for this new data type. To keep everybody who works on this lab on the same page with respect to this design issue, please use an instance of `TreeMap<Integer, Integer>` to associate each integer that appears under the square root with its coefficient outside the square root. Declare the following field inside your `Distance` class:

```
private TreeMap<Integer,Integer> coeff=new TreeMap<Integer,Integer>();
```

Every term of the form $a\sqrt{b}$ from the pure Platonic plane of mathematics thus becomes an entry `b=a` inside `coeff` map in Java. For example, inside a `Distance` object that represents the number $3\sqrt{1} - 4\sqrt{7} + 5\sqrt{11} + 2\sqrt{101}$, contents of `coeff` would be `{1=3, 7=-4, 11=5, 101=2}`, with exactly as many entries inside the map as there are terms in the original formula. Note how Java even uses the curly brace syntax in the `toString` representation of a `Map` instance the same way as Python does for its equivalent `dict` data structure. (Oh, if only the rest of the Java language would also similarly play ball here...)

So that we will be able to construct simple `Distance` objects, we need a constructor to initialize an object that consists of a single term, and a constructor to initialize an object from the arbitrary map that encodes the terms. (The constructor for the single term distance will create and initialize the `TreeMap` just the same as the general constructor does, but initialize its contents with just one element, instead of looping through all elements of the `coeff` map.)

```
public Distance(int a, int b)
public Distance(Map<Integer, Integer> coeff)
```

Both constructors should use your previous `extractSquares` function to express each term in its simplest form, since the outside callers didn't sign up to perform this simplification on your behalf. It will also make your life a lot easier later if you write these constructors to make sure that `coeff` will only contain terms $a\sqrt{b}$ for which a is nonzero. Any such term should simply be ignored during the object construction. So you might as well stop that buck right there at your desk, so that the rest of your methods may assume that every term has a nonzero coefficient. (Analogous as the words of the President, every action of the constructor weighs a ton.)

Just like we did with the `Fraction` type, we are going to design the class `Distance` to be **immutable** so that it has no `public` methods that could modify the internal state of the object. The methods for arithmetic operators will always create and return a new object to represent the result, instead of modifying either one of the originals that participate in that operation. Therefore, your second constructor should create a defensive copy of the `coeff` map object given to it and copy the coefficients there for safekeeping, instead of just storing a reference to the object that is still being used and therefore modified by the outside world.

One more method left to write in this lab, although that one is the biggie. The first `public` method for you to implement when creating a new data type is surely `toString()`. Most of the time that task is straightforward, so we might as well do all that work now to make the result look pretty, so that no user code ever has to do the work that clearly belongs to the designer of that data type. As they reputedly say in the military of some country, a good soldier will throw himself on the landmine to save his battle buddies. (Your 4-F instructor would not really know, since the only veteran status that he can claim without being tarred and feathered for stolen valour is being a veteran reader of *Beetle Bailey*, along with the "Humor in Uniform" column in *Reader's Digest*.)

```
@Override public String toString()
```

Because the automated JUnit tests record the results from this string representation, the string returned for each instance of `Distance` has to be unambiguous. The five rules that **you must follow in this lab** to guarantee that the outcome is both unambiguous and readable are:

1. The terms must be listed in ascending order of their square roots. (This is why we use `TreeMap` instead of `HashMap`, so that looping through the elements of its `keySet()` view is guaranteed to produce these terms in ascending order.)
2. If the coefficient of some term is 1, that coefficient is left out, unless the square root is also 1.

- Each individual term $a\sqrt{b}$ is transcribed as the string "aSqrt[b]", with no space between the coefficient and the square root.
- Consecutive terms are separated with either " + " or " - " depending on whether the coefficient of the next term is positive or negative, with exactly one whitespace around the operator symbol. If the " - " separator is used, the coefficient of the next term is transcribed without the minus sign.
- If the coefficient of the first term is -1, it is transcribed as a minus sign, without the 1.

These rules produce the result that *Wolfram Mathematica* would use to display such terms. Given a choice between implementing the `toString` method some other way rather than the way that has now been field tested over thirty arduous years, is there any reason not to go with the tested and true? The following table shows a list of illustrative examples of the expected way to transcribe a `Distance` object, given the `coeff` map given to its constructor.

coeff	Expected result of calling <code>toString()</code>
{}	"0"
{42=0,99=0,123=0}	"0"
{1=1}	"1"
{1=-42}	"-42"
{5=7}	"7Sqrt[5]"
{7=5}	"5Sqrt[7]"
{7=4}	"14"
{5=1,10=-1,15=1,21=-3}	"Sqrt[5] - Sqrt[10] + Sqrt[15] - 3Sqrt[21]"
{5=1,10=-1,15=1,20=-1}	"-Sqrt[5] - Sqrt[10] + Sqrt[15]"
{3=-5,7=11,6=-17}	"-5Sqrt[3] - 17Sqrt[6] + 11Sqrt[7]"
{10=-1,43=99}	"-Sqrt[10] + 99Sqrt[43]"

Of course, your constructor of `Distance` will make sure to extract all possible squares out of the coefficients given to it, rather than passing that buck to every future user of this class. As a class designer, always go the extra distance for the users of your class. There is only one class designer but an unknown number of future users, so the needs of the many outweigh the needs of the one.

Lab 20: Symbolic Distances II: Arithmetic

JUnit: [DistanceTestTwo.java](#)

The code written in the previous lab allows us to construct and print out symbolic expressions that are sums and differences of terms of the form $a\sqrt{b}$ so that a is any integer and b is any positive squarefree integer. This lab adds the basic arithmetic operations of addition, subtraction and multiplication to the mix. Since we intentionally design the class `Distance` to be immutable to get all the advantages of immutable data types, none of these methods should modify the internal state of either `this` or `other` object, but create and return a new `Distance` object for the result.

Writing the methods of this lab, you are allowed to assume that the JUnit test class gives your methods only argument values that never cause the computation to overflow when using the `int` type for integer arithmetic. If you want to convince yourself that unreported overflows do not happen, you can execute your `int` arithmetic with the methods `addExact` and its brothers in the utility class [java.lang.Math](#), instead of using the operators `+` and such already provided in the core language. These **exact arithmetic** methods in the `Math` utility class are guaranteed to throw an `ArithmeticException` whenever they detect an overflow. Absence of that exception will then let you sleep soundly tonight in the knowledge that no overflow has occurred.

```
public Distance add(Distance other)
public Distance subtract(Distance other)
```

These methods implement the addition and subtraction between `this` and `other`. Since the logic of both methods is identical except for one small part of one statement, we strongly recommend implementing both of these operations in one swoop with a `private` helper method

```
private Distance arithmetic(Distance other, boolean adding)
```

where the parameter `adding` indicates whether the terms of `other` should be added or subtracted from the corresponding terms of `this`. The `add` and `subtract` will then be simple one-liners that call this helper method with the second argument respectively set to `true` or `false`.

```
public Distance multiply(Distance other)
```

This method is implemented easiest using an empty `TreeMap<Integer, Integer>` instance that will contain the result. That way, the logic of the method is just two nested loops that iterate over all possible pairs that take the first term from `this` and the second term from the `other`. For every such pair of terms $a\sqrt{b}$ from `this` and $c\sqrt{d}$ from `other`, compute their product $ac\sqrt{bd}$, possibly extracting some square hiding under the square root made of the common prime factors of b and d . Then, either create a new entry for bd inside your result map, or update its existing entry by adding the coefficient ac to its value. In the latter case, you should correctly handle the edge case where that existing value is already equal to $-ac$ so that this entire entry vanishes altogether.

Lab 21: Symbolic Distances III: Comparisons

JUnit: [DistanceTestThree.java](#)

The previous two labs bestowed upon us the ability to perform arithmetic on `Distance` objects and print out the results in a human-readable format. The time has come to make these `Distance` objects behave in a civilized manner with other types so that the existing classes in Java Collection Framework and elsewhere work properly with them. In Python, this is achieved simply by defining some **magic methods** that correspond to the operators of the language. In Java, the following two methods inherited from the universal superclass `Object` should be overridden for this purpose.

```
@Override public boolean equals(Object other)
```

Using the example class `Fraction` as a model, override this method to accept value equality only with another `Distance` object whose `coeff` map consists of the exact same terms as this one. The analogy to the famous [Fundamental Theorem of Arithmetic](#) applies in this situation in that each numerical quantity representable within this system will have exactly one such representation. If two `Distance` objects have different `coeff` maps, the quantities that these objects represent are known to be different as surely.

```
@Override public int hashCode()
```

Override this method to somehow compute a hash code that depends on the terms in the `coeffs` map. Since you are using a `TreeMap<Integer, Integer>` instance to store these coefficients, you can simply return the `hashCode` of that map object as the `hashCode` of the entire `Distance`. After all, if two `Distance` objects consist of different terms, these two objects can be proven to represent distinct real values. (Students who are discrete math enthusiasts can try their hands on proving this, if only as a fun little side quest.)

Since these `Distance` objects represent real numbers, they should also allow order comparisons as such. Edit the first line of the class definition in your source code to now say

```
public final class Distance implements Comparable<Distance>
```

to guarantee the existence of the `compareTo` method. The big question now is what kind of algorithm could reliably compare any two arbitrary `Distance` objects in this manner. In absence of any fancier mathematics, we will simply evaluate each object as a decimal number, and compare those for the answer. However, to make this lab more interesting, we will not use the plain old `double` type for this, but might as well learn how to perform such calculations to arbitrary precision with the [BigDecimal](#) type from the `java.math` package! (When dealing with the `BigDecimal` type that internally uses base ten instead of base two to represent the decimal numbers, $0.1 + 0.2$ actually equals 0.3 , not one iota more or less...)

Analogous to the `BigInteger` type for integers limited only by your heap memory, `BigDecimal` objects represent decimal numbers. Their integer part is unlimited, same as in `BigInteger`, but their decimal precision is determined by some [MathContext](#) object that tells the `BigInteger` how far down it should keep computing these decimals. After all, even simple arithmetic expressions such as $1.0/3.0$ would need an infinite series of decimals to be exact! Whenever some operation on `BigDecimal` values produces an approximate result, the **rounding mode** determines how the value of the last included digit is determined from the truncated digits.

```
public BigDecimal approximate(MathContext mc)
```

Return the approximate value of this `Distance` as a `BigDecimal` object with the settings as in the given [MathContext](#). You should simply evaluate each term in turn and add it to the result.

```
public int compareTo(Distance other)
```

Performs the order comparison between `this` and `other` objects of type `Distance` by comparing their `BigDecimal` approximations. Same as in equality comparison, two `Distance` objects that have any different terms surely represent different numbers, but determining which object is larger can get tricky. It is hard to know *a priori* how much precision the `MathContext` needs to have for this comparison to be accurate enough to settle the issue of who is truly the mightiest. Otherwise, two sums whose values differ by some microscopic amount will seem equal when viewed through the numerical equivalent of Coke bottle lenses.

The lovely thing about this problem is that we don't need to know how much precision we need! Since we can always repeat the approximation using a higher precision, simply start with some small precision such as 10, and `approximate` both `this` and `other` with a `MathContext` of that precision. If these approximations do not reveal which number is greater, do it again with a higher precision. For example, 15. And so on. Simple order comparison between two sums of square roots of integers turns out to be **PSPACE-complete**; in the worst case, these sums need to be evaluated to an exponential number decimal places so that we are able to tell apart two such sums! For the test cases emitted by our JUnit test class, this scheme to iteratively incrementing the precision will eventually become sufficiently surgical to be able to tell apart close sums separated only by some sub-atomic margin that makes obsidian razors seem dull by comparison.

Students interested of seeing more `BigDecimal` computations in action should check out our [Mandelbrot](#) example. This **fractal renderer** performs its computations with the [BigComplex](#) type defined to perform addition and multiplication of **complex numbers** to arbitrary desired precision. Akin to that old joke of how it is easy to sculpt a statue of an elephant ("Start with any old block of stone, and chip away every piece that doesn't look like it would be part of an elephant!"), the program sculpts the outline of the black Mandelbrot set by carving away all the pixels around it that do not belong in that black hole. Such pixels are coloured using some scheme that depends on how many steps were required for the iteration from that pixel to cross the boundary. Not even one processor cycle is wasted on pixels strictly inside the Mandelbrot set, with the downside that the computation never terminates as the border pixels are iterated forever in a vain hope of escape.

Lab 22: Truchet Tiles

For testing: [TruchetMain.java](#)

As explained in the Wikipedia page "[Truchet Tiles](#)", tiling the two-dimensional integer grid using four rotationally symmetric black-and-white tiles can produce surprising "[op-art](#)" geometric patterns. Which one of four types of Truchet tiles is placed in each position is identified in this lab by two truth values `parity` and `diag` computed for that position. The parameter `diag` determines whether the dividing line between black and white parts cuts through the main diagonal or the anti-diagonal, and the parameter `parity` determines which side is black and which side is white. In your labs project, create a new class

```
public class Truchet extends JPanel
```

to make your class a full-fledged Swing component out of the box. Each instance of this class should display the Truchet tiling defined by its five constructor parameters.

```
public Truchet(  
    int s, int w, int h, BiPredicate<Integer,Integer> parity,  
    BiPredicate<Integer,Integer> diag  
)
```

Parameters `w` and `h` determine the width and height of the grid measured in tiles, and the parameter `s` gives the size of each tile in pixels. For example, an image made of values `w=100`, `h=60` and `s=5` would be 500 pixels wide and 300 pixels tall. Parameters `parity` and `diag`, both of the same type [java.util.function.BiPredicate<Integer, Integer>](#) to encapsulate two functions that take two `Integer` parameters and return a boolean result, provide your renderer the parity and diagonal of each tile based on its coordinates (`x`, `y`). This constructor should store its arguments into object fields to be accessible later, and set the preferred size of this component to new `Dimension(s*w, s*h)`.

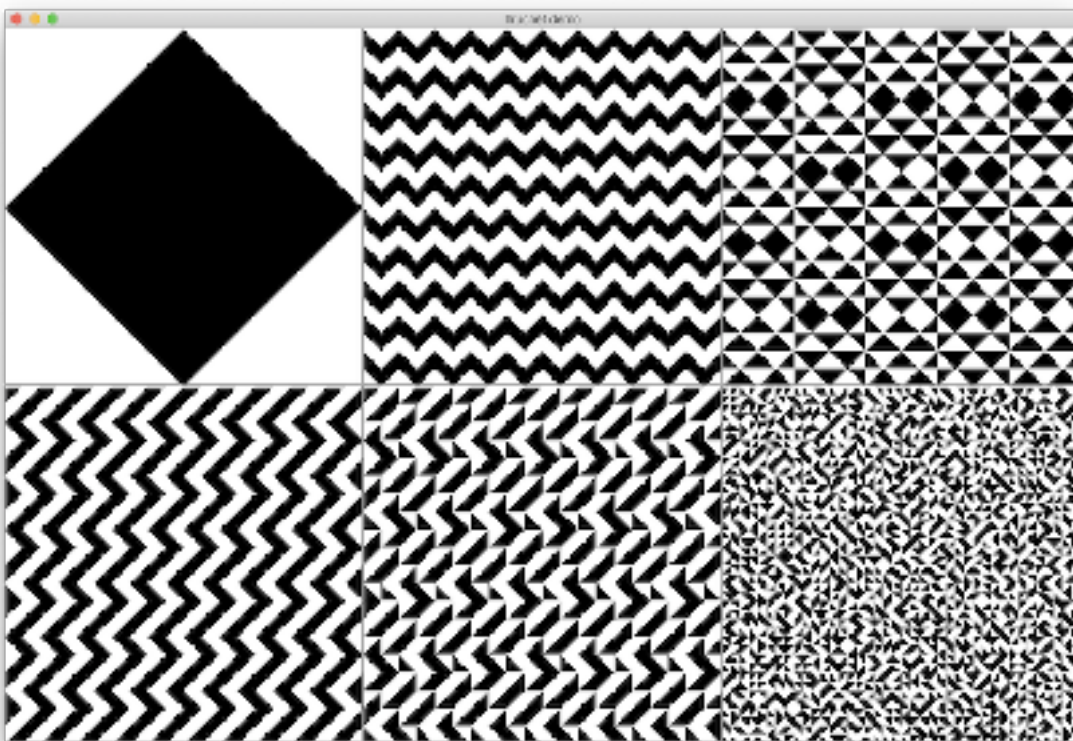
```
public void paintComponent(Graphics g)
```

Renders the individual Truchet tiles on the surface of this component, the shape of each tile as determined by the strategy objects `parity` and `diag` that make these decisions. There are four possible ways to assign these colours based on the parity and diagonal direction of the tile. However, to allow both the student and the grader to agree at a glance that this lab has been implemented correctly, your code should make these decisions in the exact manner that makes running the automated test class [TruchetMain](#) produce the result shown in the screenshot below. The easiest way to draw a filled triangle is probably to define and fill a [Path2D.Double](#) shape. (Again, do not neglect to first call `super.paintComponent(g)` to erase the previous contents of the component.)

Unlike with JUnit tests that can do their work even if we close our eyes and pretend that we are no longer in the room, in this problem our eyeballs will have to make the final determination whether

the pattern is as expected. At least our visual system offers a massive level of parallel processing right on the **wetware** that JUnit or any other computer program could not even dream of.

Especially important is the `Truchet` instance at the top left corner of this 2-by-3 component grid, the simple black lozenge against the white background, since that one confirms that your code uses the same scheme for colouring the tiles as in the instructor's private model solution. You might want to check out the modular arithmetic formulas used to produce the other instances where a pattern emerges from the local decisions that depend on the numerical properties of the tile coordinates. Even randomness is more pleasant after adding non random things to balance it out so that the human visual system has something to latch on to interpret. (The [TruchetMain.java](#) class also has a method that uses Truchet tiling to showcase the use of basic punctuation characters of *War and Peace*, as inspired by the post "[Punctuation in Novels](#)".)



Once you get this component to work, you can also try out more artistic variations such as **Smith tiles** that use quarter circles instead of triangles to create curved and blobby shapes. Or create a **labyrinth** by using `diag` to draw a black diagonal line to separate the two white halves ignoring parity completely, in spirit of [the classic Commodore 64 Basic one-liner](#). You could even [subdivide some tiles into smaller tiles](#) for a fractal subdivision effect, or explore the [properties of random Truchet tilings](#). Only your imagination shall be the ultimate ceiling of your art; in this course, the only law that we acknowledge and allow to bind our hands is the Law of Awesome!

Lab 23: Ulam–Warburton Crystals

For testing: [UlamTest.java](#)

[Game of Life](#) is the most famous creation of [John Conway](#), the unconventional mathematical genius who passed away during the COVID pandemic ([xkcd tribute](#)) one week before this lab was originally written in April 2020. Game of Life is used in many introductory programming courses for a good reason, [including this one](#), either as an example or as a programming project for the students to explore. This deceptively simple setup offers ample educational value not just for the first coding course, but [far beyond that](#) all the way up to the ultimate limits of computability.

However, just to be different from this well-beaten path, we shall instead implement a somewhat similar [Ulam–Warburton automaton](#) that produces its own distinctive fractal patterns of a more crystalline nature. Besides, that [Stanislaw Ulam](#) fella was also a pretty important figure in applied mathematics and computing in his time, including the Manhattan Project. His life and creations deserve at least some attention up here in this cyberpunk science fiction future timeline that we have suddenly found ourselves living in even though most people don't seem to have realized this, almost as if some wizard had cast a spell to make humanity forget that this has happened. The term “science fiction” itself has become obsolete when not just its standard staple of video phones, but also real time face substitution, subtitles and real time translation to another language elicit only a bored yawn. (Any readers who have a better explanation for this mass amnesia than the meek “a wizard did it” surely also know enough not to publicize this explanation to the wider world.)

The first new programming technique used in this lab shows how to explicitly encode two-dimensional arrays in just one dimension. Two-dimensional arrays in Java are actually just one-dimensional arrays whose elements are one-dimensional arrays. This “array of arrays” representation allows the individual rows to be **ragged** so that they don't all necessarily have to have the same length. However, when dealing with regular grids where each row is known to have the same length, the two-dimensional structure can also be encoded into a one-dimensional array with a total of `rows*cols` elements, where each element in the two-dimensional position (i, j) is stored in position $i*cols+j$. (This is how, for example, NumPy stores and represents its arrays that are known to be grids of elements of uniform type.)

For this lab, you should just use the previous simple representation. However, as an aside that might be useful in the future, the latter representation is especially clever when the values of that two-dimensional array are known to be symmetric across the diagonal, as in problems where the underlying reality that creates the entries in this table is already symmetric with respect to these indices. To map both (i, j) and (j, i) into the same target position, first enforce $i \leq j$ with a swap if necessary, and then use the formula $j*(j+1)/2+i$ to compute the target position in the one-dimensional array. Compared to the redundancy of storing every non-diagonal element twice, this trick stores these elements into almost half of that memory. Cutting your memory use in half is nothing to sneeze at even if the program ran to completion even under a more wasteful memory allocation scheme, since it allows you to solve instances twice the size, and that way hopefully have at least twice as much fun!

The Ulam–Warburton automaton is similar to Game of Life in style and spirit so that every **cell** in the board (now a simple two-dimensional grid, but could in principle be any **graph**) is always in exactly one of the two possible binary states “on” and “off”. The automaton advances in discrete time steps starting from zero, updating each cell logically simultaneously at each tick of time. In the original Game of Life, the state of each cell at time $t + 1$ depends on the states of the eight cells in its 3-by-3 [Moore neighbourhood](#) (fewer at edges and corners) only at the previous time t , the rules of Ulam–Warburton automaton have the state at $t + 1$ depend only on its [von Neumann neighbourhood](#) (up, right, down, left) from the previous two steps t and $t - 1$, instead of merely its immediate predecessor state. The rules are:

- If the cell (i, j) is **off** at time t , it is on at time $t + 1$ if **exactly one** of its four neighbours was on at time t , and otherwise it is off.
- If the cell (i, j) is **on** at time t , it is on at time $t + 1$ if that same cell (i, j) was off at time $t - 1$, and otherwise it is off.

In other words, cells turn on at the time when exactly one of their neighbours is on, and automatically turn off after having been on for exactly two time steps, regardless of the states of their neighbouring cells. Your method must, of course, be careful at the edges of the board.

In your labs project, create a class `Ulam` and write the following method inside it:

```
public static void computeNext(
    int cols, boolean[] prev, boolean[] curr, boolean[] next
)
```

The arrays `prev` and `curr` contain the two-dimensional boards at times $t - 1$ and t encoded into a single dimension, as explained above. (The number of rows is equal to `prev.length/cols`, and would thus be redundant to require as a parameter.) This method should not return anything, but compute the state of the board at $t + 1$ into the given parameter array `next`. Every time this method is called, all three arguments `prev`, `curr` and `next` are guaranteed to be proper array objects that contain enough elements to perform the necessary computations.

To allow the states of this automaton to be displayed as a Swing component, have your `Ulam` class extend `JPanel`. Define the field

```
private boolean[][] states;
```

that the component uses to store the computed states of the automaton, so that state at time t is stored in the one-dimensional boolean array `states[t]`. Then, define a constructor

```
public Ulam(int rows, int cols, int[][] start, int n)
```

that computes the `states` of this automaton for times 0 to n for later lookup. At time $t = 0$, every state is off. Then, let there be states. At time $t = 1$, every state is off except those whose positions are

given in the array `start` that is guaranteed to contain two-element `int[]` objects as its elements. For example, in the [UlamTest](#) method `massTest`, the array

```
int[][] start = {{100, 100}, {100, 200}, {200, 100}, {200, 200}};
```

would have those four states on. To allow the JUnit tests to access the state of each cell at given times, your class should have the accessor method

```
public boolean getCellState(int i, int j, int t)
```

that returns the state of the cell (i, j) at time t . To verify that the logic of your `computeNext` method is correct according to the rules of this automaton, you should run the `massTest` method in the JUnit test class [UlamTest](#) to verify that it gives you a green checkmark.

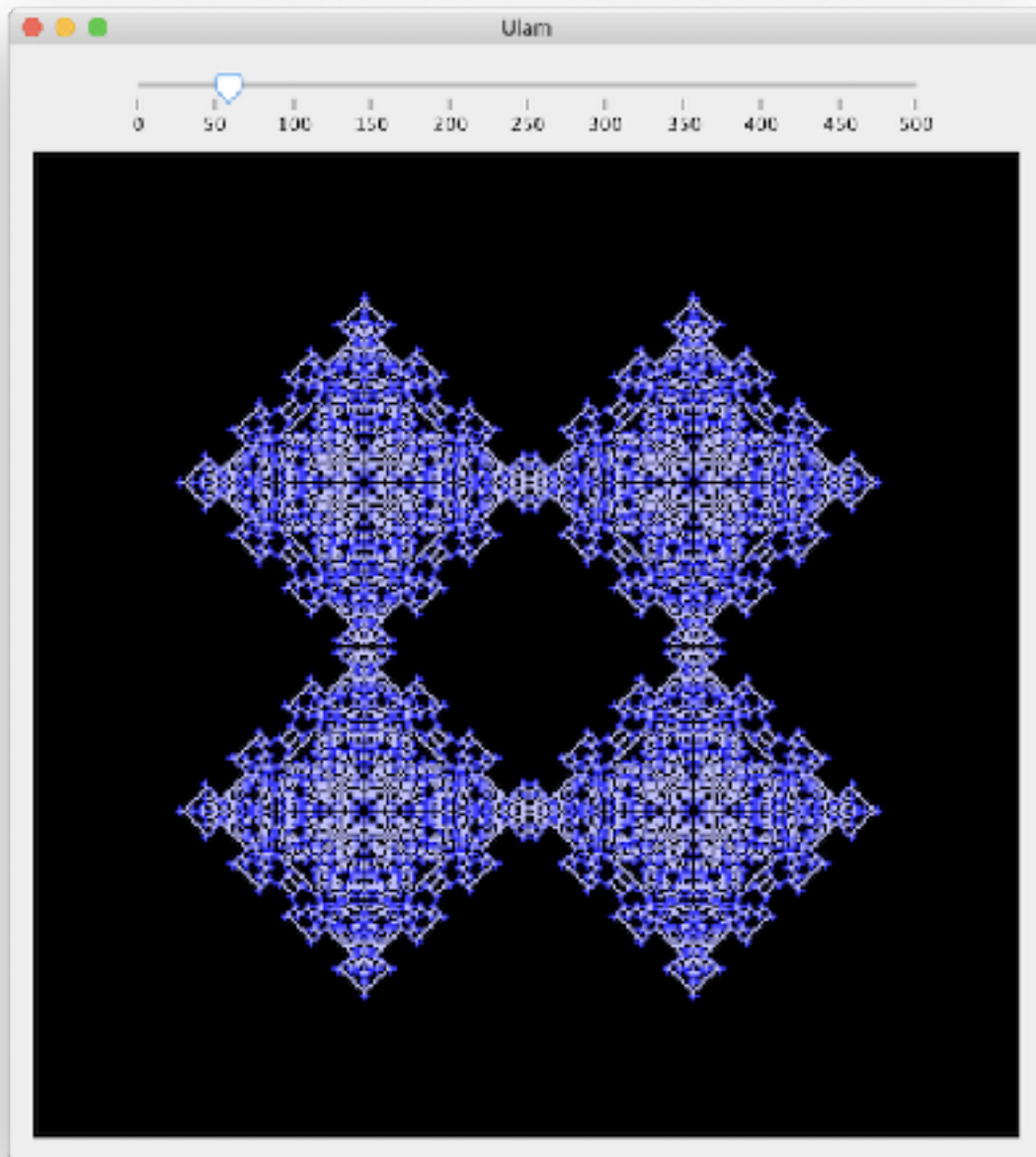
Just so that we get to again try something new in this lab, instead of animating the evolution of the automaton states using some form of Java concurrency, this component should contain a [JSlider](#) instance whose values range from 0 to n . By grabbing and moving that slider, the user should be able to instantly skip forward or backwards to any particular moment of time to see the board displayed in the `paintComponent` method in a manner that resembles the screenshot below. Create a nested `ChangeListener` subclass to listen to the events of your slider component, and then later use the sliders `getValue` method to determine the time to render the automaton state.

This display of a 300-by-300 board renders each cell as a 2-by-2 pixel box to make the result easier to see, but you can adjust this if you want to explore the evolution of the crystalline structure in larger game boards. You can also try seeding the automaton in a non-symmetric fashion to break the symmetry in the evolution of the automaton. The colour scheme used in the screenshot uses the colours listed in the following table to render a cell based on its state at times t and $t - 1$.

State at t	State at $t - 1$	Colour
off	off	black
on	off	dark blue
on	on	medium blue
off	on	light blue

To get back to Conway's Game of Life for a little bit, every second year computer science major should surely be able to write a simple [brute force Game of Life component](#). However, try to imagine the algorithmic techniques needed to merely store the current state of a giant board whose side lengths measure in millions of tiles (data structures for **sparse matrices** suddenly become necessary to keep this Jacob's ladder from running out of memory), let alone simulate the effect of one time step on such a board, or coming up with mathematical techniques to simulate some part of the board more than one step forward in one swoop.

Another surprisingly difficult coding challenge is to implement the logic of Game of Life backwards so that the **previous** state is computed from the current state that it leads into. However, even if a computation is deterministic to the forward direction, it can still be nondeterministic to the backward direction so that the same current state could potentially have more than one possible predecessor, some of which leading to contradictory states that cannot be produced by any configuration in Game of Life. In general, if computations could somehow be reversed in some mechanistic fashion without this exponential blowup of possibilities of the past, any encryption algorithm would be trivial to crack simply by running the encryption algorithm in reverse from the cipher-text output back to the original plaintext input...



Lab 24: Chips On Fire

JUnit: [CoinMoveTest.java](#)

Consider the following operation done for arrays whose elements are nonnegative integers, considered here to be the number of **chips**. First, my apologies for the inconsistent terminology used here. The first version of this problem was talking about “coins” until the author realized that in the literature of this field, the systems described below are generally known as [chip firing systems](#). As long as the names we utter from our virtual mouths are sufficiently memorable and unique to keep their referents apart, it’s all to-*may*-to, to-*mah*-to anyway. Men worry, machines execute.

Each position in the array also has a list of **obligations**, each obligation being another position in the array. Each position can have any number of obligations to other positions, and can have multiple obligations to the same position. The time in this system proceeds in discrete steps. For each time step t , the following operation is performed for every position of the array, giving us the chip distribution for the next time step $t + 1$. The logic of your method must be designed so that its result is equivalent to performing this operation for every position in an instantaneous **atomic** step so that all positions fire their chips in unison, as if they were saluting the most powerful *maharaja* that the mutton-chopped forces of our beloved Queen ever had the honour to encounter!

Every position that has at least as many chips as the total length of its list of obligations, gives out exactly one chip to each position as many times as that position appears in that list of obligations. The positions that did not have enough chips to fulfill their current obligations do not give out any chips to anybody; giving out chips from your purse in each round is an all-or-nothing affair. Some possible distributions of chips over the positions are **Gardens of Eden** for which there is no predecessor position whose firing of chips would lead to that position.

This operation obviously never changes the total number of chips inside the system, since it just moves them around the various positions. The **states** of this chip-firing system therefore correspond to the possible ways this fixed chip stack can be distributed among the positions. The **transitions** lead from each state to the next state that results from the previous operation. To allow you to compute these transitions, create a new class `CoinMove`, and there the method

```
public static void coinStep(  
    int[] curr, int[] next, List<List<Integer>> obligations  
)
```

The current state is given in the parameter `curr`, and the list of obligations for each position is an element of the list of such lists named `obligations`. (Both lists `curr` and `obligations` are guaranteed to have the same length.) This method should not return any result, but instead write the next state into the array `next`, also guaranteed to be an `int[]` of the same length as `curr` at the time of the call.

The following table showcases some possible values for `curr` and `obligations`, and the expected next state that this method is expected to produce:

curr	obligations	Expected next
[4, 5]	[[1], [0]]	[4, 5]
[3, 1]	[[1, 1], [0, 0]]	[1, 3]
[2, 3, 0, 1]	[[1], [3, 3, 0], [], [0]]	[3, 1, 0, 2]
[3, 3, 1, 1, 2, 0, 0]	[[6, 5, 1], [6, 4, 2, 2], [], [5, 0, 0], [], [], []]	[0, 4, 1, 1, 2, 1, 1]
[3, 2, 2, 0, 1, 2, 2]	[[3, 5], [6, 3], [0, 3, 4], [6, 2], [5], [2, 2, 3, 4, 2, 4, 1, 2, 1], []]	[1, 0, 2, 2, 0, 4, 3]

Since the total number of chips in the system and the list of obligations for each position never change, iterating this system over the infinite sequence of time steps $t = 0, 1, 2, \dots$ must eventually enter a **cycle** where the same sequence of states keeps repeating forever. The **period** of the cycle is the smallest positive integer $k > 0$ so that for every t in the cycle, states t and $t + k$ are equal.

```
public static int period(int[] start, List<List<Integer>> obligations)
```

From the given `start` state and the list of obligations for each position as in the previous method, this method should compute and return the period of the cycle reached from `start`. Iterate the sequence until you come to some state that you have already seen before. That state is going to be the starting point of the cycle, so we had better remember it. Initialize an integer counter to zero, and iterate the same cycle again incrementing the count in each step. When you inevitably return to the starting point of the cycle, return this count as the requested cycle length.

Ah... but one pesky little fly still buzzes in this ointment. Since the number of **reachable states** in the system can be exponentially large, so can also the cycle length and especially the length of the path leading to that cycle. Making the previous algorithm to explicitly store all the states it has seen into some kind of `Set` structure might therefore be prohibitively expensive. Fortunately, there exists an ingeniously simple algorithm for this exact type of situation where we have the ability to generate some sequence arbitrarily far one step at the time, and need to recognize the appearance of some previously seen state that marks the cycle. The [Tortoise and Hare algorithm](#) by Robert Floyd is a computer science classic that deserves way more fame than it currently enjoys.

Instead of maintaining only one current state inside the sequence used to advance to the next state, maintain two such states separately, aptly titled “tortoise” and “hare”, both initially residing at the `start`. The hare will take two steps for every step of the tortoise, thus iterating through the sequence twice as fast. When the current position of the hare is in some cycle, as it eventually must (the `start` state can also be inside some cycle to begin with), the hare will keep unknowingly racing around that cycle until that the slower-paced tortoise also enters that cycle. Regardless of their

positions inside the cycle at that moment, the hare will inevitably catch up with the position of the tortoise. Once that happens, both animals stop, shake hands and say their equivalent of “jolly good, old chap” to each other, knowing that the current state must lie inside the cycle. The hare can now stay in place and take a well-deserved rest, while the tortoise makes another trip around the same cycle, counting the steps needed to return to hare.

As revealed by the JUnit fuzz test, the period of this cycle greatly depends on the structure of obligations that the positions have for each other. For most configurations the correct answer seems to be just one; such **self-cycles** are the equilibrium states where all obligations and chip constraints perfectly balance out each other. However, this period can also be much longer, even well over one million for some of the initial states and obligation lists randomly generated in the JUnit test. Same as with all exponential growth, increasing the array lengths and numbers of chips in the system would make these periods eventually dwarf our entire universe itself...

The famous [space-time tradeoff](#) of computer programming is usually performed in the other direction than in our case here. Normally we make the program run faster by granting it more memory to remember the relevant stuff that it has seen and done. The exact opposite technique used by the Tortoise and Hare is used in situations where remembering all those relevant things would require way more memory than we can possibly have available in any computation device that can be realized in our physical universe.

In addition to being used to model serious stuff during the billable hours while wearing a tie, chip firing systems can also be [used to create computational art](#) by having each pixel correspond to some position at the given time, and colour that pixel according to the number of chips present in there at that time. Anything that we can do to bring art and science closer together by definition counts as progress. Chip-firing systems are also closely related to the [Abelian sandpile model](#), presented in an entertaining fashion in the [Numberphile](#) video “[Sandpiles](#)”.

Lab 25: Manhattan Skyline

JUnit: [ManhattanTest.java](#)

[Manhattan skyline problem](#) is a well-known exercise in **computational geometry**, with plenty of room for many educational and interesting variations. Its setup makes for a good illustration of the **sweep line algorithm** that will be used to efficiently solve this problem. The general technique of **sweep lines** will then solve many other problems that take place on the two-dimensional plane.

In the basic version of this problem, a number of tall buildings (perhaps built in the heyday of some midwestern city on a flat plain) are seen from a distance against the horizon far away. Our vantage point is located far enough so that each building is seen as a two-dimensional **silhouette** rectangle. From our point of view towards the horizon, each individual silhouette has its **starting** and **ending** x-coordinates s and e that always satisfy $s < e$, and the **height** of h . These projected silhouettes can overlap each other in part or whole even as the original three-dimensional buildings are physically disjoint. Create a new class `Manhattan` in your labs project, and there the method

```
public static int totalArea(int[] s, int[] e, int[] h)
```

The three arrays s , e and h (guaranteed to all have the same length n and consist of positive integer values only) contain the start coordinate, the end coordinate and the height for each individual building numbered $0, \dots, n - 1$, respectively. These buildings are guaranteed to be sorted in order of their starting coordinate, with ties resolved by the ending coordinates. This method should return the area of the overall total silhouette so that the overlapping parts of the silhouettes of the individual buildings are included only once in the total area, regardless of how many buildings happen to overlap at that same x-coordinate.

Conceptually, a sweep line algorithm “sweeps” through the horizon from left to right, noting all the points along the way where some “interesting” event takes place, in the sense that event potentially affecting the result computed by the algorithm. The algorithm should start by creating a list of these interesting events, and sort these events in the order of their x-coordinates. In the Manhattan skyline problem, the interesting events are “building b enters the view” and “building b exits the view” as your virtual “eye” sweeps the horizon from left to right. Start by defining a local variable

```
List<Integer> events = new ArrayList<>();
```

with these “enter” and “exit” events encoded so that the building b entering the view is encoded as $-1-b$, and the building b exiting is encoded as $b+1$. (We need to encode these events into integers in such a tricky way to guarantee that each event is encoded to something nonzero, because there is no way to tell apart zero and minus zero!) Add these $2n$ event encodings into `events`, and sort these `events` with a custom `Comparator<Integer>` local strategy class whose `compareTo` method compares two given events by first decoding them into starting times (taken from array s) or ending times (taken from array e). The type of the event determines whether its entering time or the exiting time is used as its comparison key.

The events sorted in this manner are ready to be processed in the order that they occur sweeping from left to right, updating the tally of the area. To keep track of the buildings that are currently in your active view (the so-called **active set** of the sweep line algorithm), define another local variable

```
Set<Integer> active = new HashSet<>();
```

that contains the buildings that are currently in the active view. Loop through the sorted events, also keeping track of the x-coordinate of the event processed during the previous round. To process an event, first compute its horizontal distance from the previous event. Multiply that distance by the height of the active building that is currently tallest, and add that product to the total. (Sometimes several events take place at the same x-coordinate, but since the distance between those events is zero, all such redundant terms from [slivers](#) of zero area between the literal and metaphorical walls that separate us from each other vanish from this classic summation, as if sniped to extinction by the master guns Tom Berenger himself.) Next, update the active set by adding or removing the building that enters or exits the view, and move on to the next round. Once all buildings have entered and exited the active view, this method can return the accumulated total.

s	e	h	Expected
{1, 4, 11}	{5, 8, 13}	{2, 1, 2}	15
{2, 6, 9, 12, 15}	{3, 8, 10, 14, 20}	{3, 3, 4, 3, 2}	29
{0, 9, 11, 13, 13, 18, 31, 32, 35}	{4, 23, 21, 23, 24, 21, 38, 50, 45}	{3, 3, 1, 1, 2, 2, 3, 3, 1}	113

Lab 26: FilterWriter

JUnit: [FilterWriterTest.java](#)

Some examples discussed during the lectures featured decorators for `Animal` and `Comparator` class hierarchies. The decorator pattern works the same way for any other class hierarchy, except that the more methods the public interface of some class contains, the more annoying drudgery decorating that class hierarchy becomes. The [Writer](#) hierarchy to transmit text data composed of Unicode characters allows for a more convenient writing of decorators. Only three methods need to be overridden in the subclasses, since the default implementations of all other methods call those three **template methods** to do their work. Subclasses can still override these template methods whenever they can provide them with a more efficient implementation, but they are never required to do so, unless the template method itself is declared to abstract in the superclass.

In your labs project, create the class

```
public class FilterWriter extends Writer
```

with the fields and constructor

```
private Writer writer;
private BiPredicate<Character, Character> pred;
private char previous;

public FilterWriter(Writer writer, BiPredicate<Character, Character>
pred, char previous) {
    this.writer = writer;
    this.pred = pred;
    this.previous = previous;
}
```

The functional interface [BiPredicate<T,U>](#) in the package `java.util.function` represents a predicate that takes two arguments and returns a boolean truth value. In `FilterWriter`, this predicate is used to filter the characters that this decorator lets through. Each `FilterWriter` instance remembers the `previous` character that it has let through, initialized from the constructor parameter to be used as the initial character. Override the methods `flush` and `close` to call the corresponding methods of the underlying `writer`, and then override the one required method

```
@Override public void write(char[] data, int off, int len)
    throws IOException
```

to write the characters from the subarray of `data` that contains the `len` characters from position `off`. To decide whether the current character `c` should be passed on to the method `write` of the underlying object, the method should call `pred.test(previous, c)`. If the predicate accepts

the character `c` by returning `true`, the character `c` is passed on to the underlying writer and becomes the new `previous` character. Characters rejected by this predicate are ignored in the spirit of the followers of a high school queen bee mean girl, all together pretending that the objects of the scorn of Her Majesty actually do not even exist.

The JUnit test class `FilterWriterTest` contains two test methods that both create a filtered version of *War and Peace*. The first test adds a bunch of [Unicode combining characters](#) to each letter of the original line to create some “[zalgo text](#)”, but the predicate that rejects all such combining characters regardless of the previous character should have no problem restoring the original text. The second test uses a predicate that rejects every vowel that immediately follows a vowel, and every consonant that immediately follows a consonant, producing results like following paragraph:

```
Pire was unany. Sot, abot te average heg, bod, wit huge
red han; he did not kow, as te san is, how to ener a dawin
rom an sil les how to leve one; tat is, how to sa sometin
pariculary agebe before gon awa. Besides tis he was aben-
mined. Wen he rose to go, he tok up ined of his ow, te general's
te-corered hat, an hel it, pulin at te pume, til te general
ased him to resore it. Al his aben-minedes an inability to
ener a rom an conere in it was, however, redemed by his kiny,
sime, an modes exesin.
```

At least to this linguistic layman, the filtered text seems to carry a curious pidgin rhythm and flavour of its own. This leads us to further wonder whether simple variations of this operation, possibly aided with some vowel and consonant permutation substitutions, could be mechanistically applied to our natural languages to create a completely made-up and yet internally coherent **pig latin** for the exciting universe in the next hit series of those young adult fantasy novels with a young female hero whose superpower is being totes unique and special in a joyless world full of mindless sheeple, and who is forced by the circumstances to choose between two very different but equally exciting boys. The algorithm demonstrated in the [DissociatedPress](#) lecture example can also be used to achieve such pseudo-linguistic ends. The algorithm could even effortlessly mash together several languages to obfuscate the flavours of the original languages.

Lab 27: Stacking Images

For testing: [ImageAlgebraMain.java](#)

Instances of the class `Image` represent two-dimensional pixel images in Java that can be read from files and URLs, and then rendered to `Graphics2D` canvases with the method `drawImage`, as illustrated in the example class [ImageDemo](#). These images, no matter where they were acquired, can be further processed and transformed with various `ImageFilter` instances, as illustrated by our other example [ImageOpDemo](#).

We are accustomed to adding up numbers, but we can also “add” images to each other with concatenation, similarly to the way strings are “added” by concatenating them. Since pixel raster images are two-dimensional, we can stack them up not just horizontally but also vertically, provided that the dimensions of these images are compatible in that dimension. In your labs project, create a new class `ImageAlgebra` (note that this class itself is not a Swing component, but a utility class that contains static methods), and in there two static methods

```
public static Image hstack(Image... images)
public static Image vstack(Image... images)
```

for the horizontal and vertical stacking of an arbitrary number of `Image` objects. Both of these methods are **vararg** methods, meaning that they accept any number of arguments of type `Image`, including zero, but in this lab these methods can assume that they are given at least one image. The **horizontal stacking** method `hstack` (the method names were here chosen to be the same as they are in NumPy) should create and return a new `BufferedImage` instance whose width is equal to the sum of the widths of its parameter images, and whose height equals the maximum of the heights of its parameter images. This image should contain all the images together as a single row. (To implement this method the easiest way, just draw the individual images one by one to an appropriate position of the resulting image, images of different heights top-aligned to start from the same ceiling line.) The **vertical stacking** method `vstack` works exactly the same, except that the roles of width and height are interchanged.

We can immediately put both of these stacking methods in good use in some **recursive subdivision**. Define a third method in your class

```
public static Image halving(Image tile, int d)
```

This method produces the result image according to the following recursive rule. For the base case where the depth `d` equals zero, this method should simply return the given `tile`. The result for positive depths `d` is the horizontal stacking of `tile` with the vertical stacking of two copies of `halving(tile, d-1)` where `half` is an image constructed from `tile` by scaling it to half of its width and height. Since the two copies of `halving(tile, d-1)` are identical, compute that one only once and reuse the result, which makes this recursive function to be a **linear recursion** where each recursive call generates at most one recursive call without exponential branching, and

can be converted into a loop for the same effect. ([Recursion is the root of computation since it trades description for time](#). However, the same way as with a stepladder that helps you change a light bulb, you put that ladder away once the bulb has been changed.)

The class `ImageAlgebraMain` uses the above three methods, along with the image file [coffee.jpg](#) provided in the GitHub repository for these labs, to create a display that should look like the screenshot below to be accepted for lab marks. To test your methods one at the time as you write them, first define all three methods as one-liner **stubs** that simply return the tile they were given. This allows you to run the `ImageAlgebraMain` program to test each method as soon as it compiles, even when you haven't yet implemented the other methods.



The 4-by-4 grid on the left of the above low-resolution screenshot is constructed by simple application of `hstack` to create the row, and then using `vstack` to duplicate that row into the final image. The image on the right side is the result of halving a 512-by-512 version of the original image. Since each tile gets cut in half down the recursion, the resulting image is always twice as wide as the original, minus the width of the last image on the right edge.

Lab 28: All The Pretty Hues

For testing: [ShadesMain.java](#)

Computers internally represent colours as **RGB triples** of red, green and blue so that each component can get values ranging from 0 to 1 when represented as floating point, or from 0 to 255 when represented as an integer byte. Turning all three knobs to zero gives you the darkest black that your display device is capable of presenting to the world, whereas turning all three knobs to maximum gives you its brightest possible white colour available on your display device.

Inside Java, objects of [java.awt.Color](#) represent individual RGB colours as objects. However, when dealing with images whose every pixel could potentially have a different colour, we would prefer not to fill the heap memory with actual objects. It seems very wasteful to spend twelve bytes of heap management bookkeeping space for mere three bytes of actual payload data per pixel, over what is these days millions of pixels! Instead of acting all purist and insisting that “everything must be an object”, four bytes of colour information can be crammed into a primitive `int` value whose four bytes represent the ARGB components of that colour.

In addition to the three colour components, this fourth “alpha channel” that tags along the free ride inside a four byte `int` has no fixed interpretation on top of its admittedly cool secret agent name. In practice, all image processing operations of Java treat the alpha channel as the **opacity** value of that colour, with 255 for a completely opaque colour and 0 for a fully transparent colour. When a pixel is rendered with a partially opaque colour, the resulting colour of that pixel is the corresponding partial mixture of the current pen colour and the previous colour of that pixel underneath.

Most colours that our eyes can see can be broken down into these three components. Some might be a bit tricky; for example, how exactly would you represent the brown colour? Think about that one for a while. By now you have surely seen lights of various colours of the rainbow, but have you ever seen a brown light in your life? What the heck exactly is that colour made of while we ask what it can do for us today? As convenient as RGB encoding is for computer software and hardware, it is not the best way for humans to represent colours. The alternative [HSB representation](#) breaks each colour into three components of **hue**, **saturation** and **brightness**. When done in the HSB colour space, colour calculations and especially colour blending produce more aesthetic results.

To extract the HSB colour components of the given RGB colour `rgb` given as an `int` into the first three elements of an existing 3-element float array `hsb`, you can copy-paste the method

```
private static void RGBtoHSB(int rgb, float[] hsb) {
    int a = (rgb >> 24) & 0xFF; // extract alpha
    int r = (rgb >> 16) & 0xFF; // extract r
    int g = (rgb >> 8) & 0xFF;  // extract g
    int b = rgb & 0xFF;         // extract b
    Color.RGBtoHSB(r, g, b, hsb);
}
```

The conversion from HSB to RGB is already available as the utility method `HSBtoRGB` in the `Color` class. Of course the inverse method `RGBtoHSB` also already exists in the `Color` class, but that one expects to receive the red, green and blue values as three separate parameters, instead of being packed into a single `int` the way that the above method does the conversion.

To play around with all these pretty colours, create a class `Shades` in your labs project. In this class, write the following method to create and return a modified copy of the given `Image`. The hue and brightness of each pixel are kept as they were, but the saturation of each pixel is decreased by multiplying it by the given `factor` guaranteed to be between zero and one, inclusive.

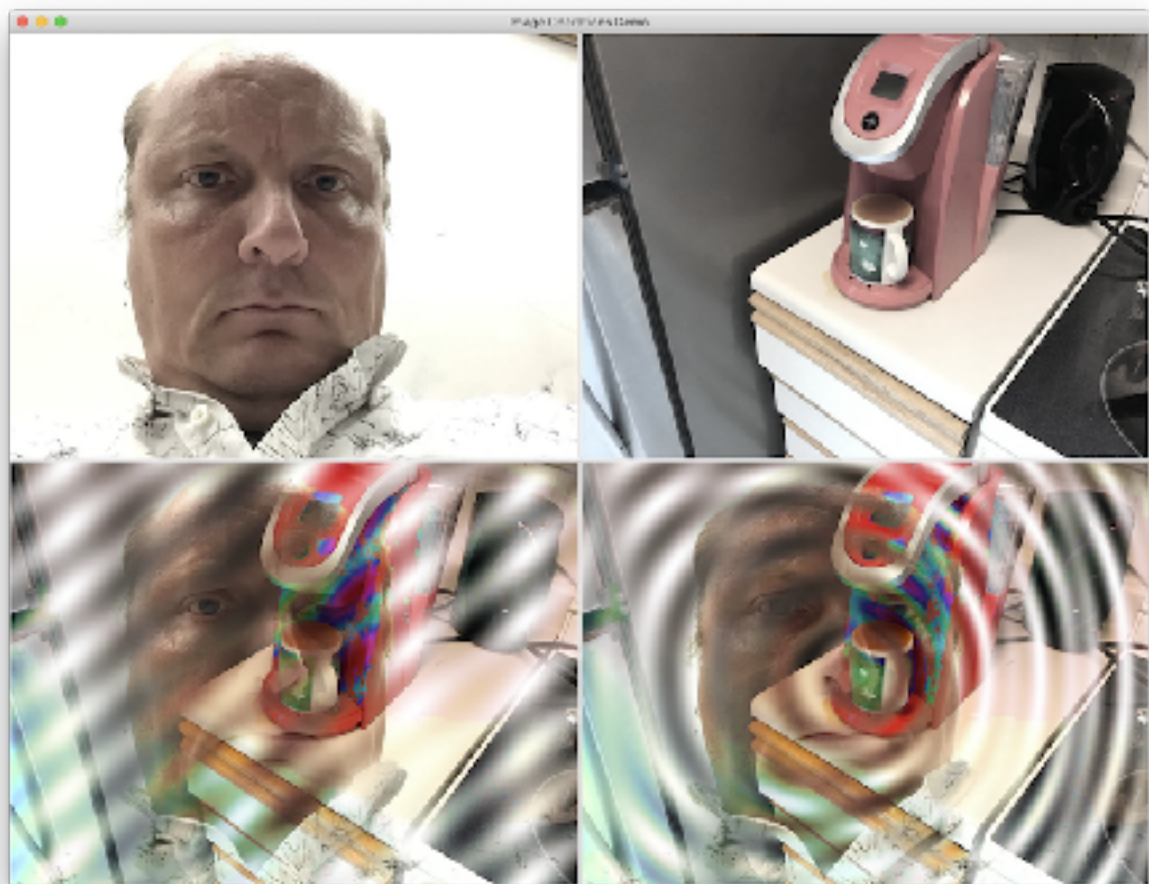
```
public static Image desaturate(Image img, double factor)
```

Compared to the original, the resulting image looks like somebody ran it through the hot wash a couple of times too many. You should use the `RGBImageFilter` approach demonstrated in the [ImageOpDemo](#) example class to implement this operation. The second method to write in this lab is

```
public static BufferedImage blend(  
    BufferedImage img1, BufferedImage img2, DoubleBinaryOperator weight  
)
```

This method blends together two source images `img1` and `img2` to create a result image so that the hue, saturation and brightness of each pixel (x, y) of the result are the weighted averages of those same components of the pixels (x, y) in the two source images. (Use simple linear averaging even for hue, which is technically incorrect but at least produces trippy colour artifacts at hue boundaries.) The weight used to mix these components is not constant, but determined for each pixel (x, y) separately by the parameter `weight` of type [DoubleBinaryOperator](#) to represent a function that takes the raw pixel integer coordinates (x, y) as parameters, and returns the floating point **weight** from the range $[0, 1]$ of the first image that is used to mix the components for that pixel. (A weight of zero therefore means that the result is all of `img2`.)

The following screenshot, produced by running [ShadesMain](#) with the example images [coffee.jpg](#) and [ilkka.jpg](#) included in the repository, shows the result that these methods are supposed to produce. The first row shows the desaturated versions of these source images. The second row shows two possible blends of these images. The bottom left image computes the blending weights from a periodic wave function directed at an angle through the two-dimensional plane, whereas the bottom right image uses a wave function emanating from the center with a [superellipse](#) shape instead of a boring old regular circle. A [superellipse](#) shape with parameter 2.05 or thereabouts does the job pretty much everywhere where we would normally use an actual circle, and yet it is subconsciously perceived to be more organic and pleasant. The main problem of perfect circles is that they are literally too darn perfect to be perceived as genuine natural things. Few perfectly round things seem to exist in nature, at least in the resolution that our eyes can discern; immediate examples that spring to mind might be the Sun and the Moon in the heavens, and the iris and the pupil in our eyes. (It is probably not coincidental that humanity has historically ascribed preternatural properties to these objects.)



Lab 29: In Timely Manner

JUnit: [TimeProblemsTest.java](#)

Computations on times and dates are tremendously important in the real world, yet surprisingly difficult to get universally correct, since both of these concepts turn out more complicated than they might seem in the everyday local experience of our daily lives. However, just like our programs can no longer assume that all text is written in English using a typewriter so that every character can be stored in one byte, software that is going to be running around the globe 24/7/365 cannot blithely rely on such happy-go-lucky ideas of how time works. For starters, everyone who has come this far surely knows that during the year 2020 when this problem was originally written, those were actually 24/7/366. What other pitfalls with sharpened spikes at the bottom might be lurking in this important problem domain to ensnare those patrolling this enormous jungle, nervously scanning for traps laid by Time Cong as millions of unblinking eyes stare at them from behind every leaf of grass?

To see the need for a package that cuts through this thicket and performs these calculations correctly in all possible cases, scan through “[Falsehoods programmers believe about time](#)” and tally up how many of the falsehoods listed there you would have also assumed to be true before doing this lab. *Tempus bugit!* The rest of the curated list of “[Falsehoods Programmers Believe In](#)” gives you a glimpse of how both the world and human societies are chock full of messy edge cases and corner cases waiting to trip the unwary programmers. Normies have the decadent luxury of closing their eyes and hand waving at such problems until they go away the same way as all problems eventually will, but hope is never a strategy for computer programmers.

Before Java 8, the standard library classes `Date` and `Calendar` encapsulated such computations. However, these classes [were defective in too many ways to count](#). They would be more appropriate to be fed on punch cards into the maws of the sixties vault-sized Bat-Computer to predict the Calendar Man’s next postmodern crime spree. The class `Date` is now officially **deprecated**, and the class `Calendar` remains only as a **legacy** for backwards compatibility with old code, as it of course should. For new code, the far superior package [java.time](#) introduced in Java 8 offers a more intuitive and consistent design that is well worth studying for its clean and SOLID object oriented design standpoint from all perspectives, once you have learned how to use its classes and methods in general. Class design of this package is so sparkling clean that doctors could use it as sanitizer.

Classes [LocalDate](#), [LocalTime](#) and [LocalDateTime](#) represent dates and times without any time zone information. (Pythonistas may note how these roughly correspond to classes `date`, `time` and `datetime` in the [datetime](#) module.) Unlike the original `Date` class, all such data types in `java.time` are designed to be **immutable** to bestow upon us a multitude of blessings that simplify our reasoning about programs that use them. Their methods are consistently named across these types, and they accept intuitive and human-readable arguments. For example, every method named with `with` creates a copy of an existing object with just one attribute changed and the rest kept as they were. (Such a method would sort of be a “mutator” for an immutable type.)

The first observation that jumps out from these classes is that **none of them have a public constructor!** Instead, new instances are created using the **static factory method** that is always named

of, usually **overloaded** to accept various types of arguments for object creation. For example, the expression `LocalDate.of(2020, 4, 23)` would return a `LocalDate` object that represents the date of April 23, 2020, the day that this lab specification was originally written.

Create a new class `TimeProblems` into your labs project. The first of the three methods that you will be writing there has you performing some calculations on dates. The whole point of this entire lab is to teach you to read through the package and class documentation in the Java API Reference, usually googling through some Stack Overflow pages in another browser tab to get through the practical details and find the method that does the thing that you need at the moment. You should first think how you would solve this problem in real life, and translate your ideas from there to actual code, each individual step of this process translated to appropriate method calls.

```
public static int countFridayThirteens(LocalDate start, LocalDate end)
```

This method should count and return the number of days between the `start` and `end` dates, both dates **inclusive**, that are Fridays whose date falls on the thirteenth day of its month. This particular combination used to be considered unlucky in the Western culture, at least back in the day when people were not quite as cognizant of the Pigeonhole Principle compared to us modern sophists, and thus easier to dazzle with numerological coincidences. Such beliefs seem to be slowly fading into distant memories in our culture, although in relatively recent times, the notion of some numerical dates being inherently unlucky spawned an entire series of popular docudramas to warn the movie-going youth of the dual dangers of getting intoxicated and having premarital sex.

The next problem was inspired by the “Gigasecond challenge” problem at [CodeWars](#), another good online whetstone for all apprentices and journeymen to sharpen their coding and algorithmic thinking skills against. (While others were partying, you studied the code.) To solve this problem, we need a new concept of [Duration](#) to represent a measured **passage of time** (for example, “two hours and five minutes”, “three years and ten seconds”, or “one nanosecond”) that is not anchored to any particular time or place. (The class [Period](#) is similar in both spirit and usage, but with a less precise granularity of mere days instead of nanoseconds.)

To ensure understanding of the difference between the concepts represented by `Duration` and `LocalDateTime`, note how adding or subtracting a `Duration` to a `LocalDateTime` or another `Duration` is perfectly sensible, whereas trying to add two `LocalDateTime` objects is codswallop. Even more contrary to intuition, even though dates cannot be **added** to each other (at least not in any way that would have the result meaningfully relate to any aspect of reality relevant to our future decisions), the **difference** between two `LocalDateTime` instances is a legitimate thing of type `Duration`, this type representing lengths of time not fixed to any particular point in time.

```
public static String dayAfterSeconds(  
    LocalDateTime timeHere, long seconds  
)
```


Given the current date and time as a `LocalDateTime` object, return the day of the week (as one of the capital letter strings "SUNDAY", "MONDAY", "TUESDAY", "WEDNESDAY", "THURSDAY", "FRIDAY", "SATURDAY") that results by adding number of seconds to the current date and time.

The previous two problems treated time as a local phenomenon, as we all do in our highly local everyday lives. This pragmatic approach to time is analogous to how, even though we understand that the Earth is a spheroid with a bulky belly, all our decisions and actions in our daily lives are made as if we believed that the surface that we live and work on is an effectively flat plane with occasional minor bumps. (Those who work as airline pilots or artillery operators must use a different mental model while on the clock so that they don't miss their marks, of course.)

This pale blue marble whose spinning surface we live on against the dark eternity of cosmos is divided into **time zones** so that the same instant of time corresponds to a different time and date depending on your current location on Earth. We ignore the theory of relativity and any philosophical issues about what it means for two instants of time to be the finger-quotes "same" time instant, with or without any the theory of relativity or any "quantum" mumbo-jumbo sprinkled in. Time zones aren't quite as straightforward as most people assume; for starters, [there exist 38 time zones](#), not just the obvious 24, and their clocks are not always some integer multiple of hours apart from each other. (Time zones also [aren't easy to get rid of](#), nor to be [made continuous](#).)

Instances of [ZoneId](#) represent the recognized time zones, and are identified by standardized strings such as "America/Toronto" or "Pacific/Marquesas". However, before you optimistically extrapolate the general form of these strings from this sample of mere $n=2$, you should note that "Eire", "Navajo", "US/Hawaii" and "America/North_Dakota/New_Salem" are equally legitimate time zones! You should also note the use of underscores as artificial white-space the same way as we do in C and Python with names that consist of multiple separate words, so that you don't get all confused when there is no time zone of "America/Los Angeles". Fortunately, all this preparatory work has already been done for you in the `ZoneId` and other classes in the package, sparing you from the need to reinvent this planetary wheel.

```
public static int whatHourIsItThere(  
    LocalDateTime timeHere, String here, String there  
)
```

Assuming that you are currently `here` so that the clock on your wall says `timeHere`, compute and return the current hour that the people over `there` see in their grandfather clocks.

One last thing for you to ponder: why is the first parameter of this method a `LocalDateTime` instead of a mere `LocalTime`? What situation would knowing the current local date, not merely the current local time, be necessary for correct answer? The great thing about this library is that you don't need to know! Instead of doing the tedious research to determine all the lurking edge cases, let the class [ZonedDateTime](#) take care of all such low level details, since some expert in this field already did the necessary grunt work for you. The same principle applies to all real world problem domains that we deal with using computational means. Note also the `withZoneSameInstant` method as "mutator" for immutable objects to access that same instant in a different time zone.

Lab 30: Mark And Rewind

JUnit: [RewindIteratorTest.java](#)

Instances of `Iterator<E>` allow their elements to be iterated only to the forward direction with the method `next`, but do not allow **rewinding** that sequence to an earlier position to repeat the already seen elements of the sequence for a second time. This is because some sequences are generated **virtually** so that the iterator computes the next element on command from its state that depends on the sequence generated so far. Such computations in general aren't **reversible**. Since the ability to rewind a sequence to an earlier position would be useful in some algorithms that operate on sequences, we shall solve this problem once and for all with a general purpose **decorator** for `Iterator<E>` instances that allows a position in any existing sequence to be **marked**, so that the decorated sequence can later be rewound back to that position.

The mass test method of the JUnit test class [RewindIteratorTest](#) uses the simple sequence of natural numbers 0, 1, 2, ... as the underlying sequence, and then performs “mark” and “rewind” operations at pseudo-randomly chosen positions of the sequence. The three public test methods generate the first thousand, first million and first hundred million elements of this sequence. If you edit the test method for the first thousand elements to be verbose, the first eight lines of output (shown below as generated with the instructor's private model solution) help you trace what is supposed to happen, with M and R denoting marking and rewinding, and numbers denoting the element that pops out of the decorated sequence at that moment.

```
0 1 2 M R 3 4 5 6 M 7 R 7 8 9 10 11 12 13 14 15 16 M 17 R 17
18 19 20 M 21 22 23 24 25 26 27 M 28 M R 29 R 28 29 30 31 R 21 22 23 24
25 26 27 M 28 M 29 30 31 R 29 R 28 29 30 31 32 33 M R 34 35 36 37 38 39
40 41 42 43 44 45 46 47 48 49 50 51 M 52 53 54 55 56 57 58 R 52
53 54 55 56 57 58 59 60 61 62 63 64 M 65 66 M 67 68 69 M 70 M R 71 72
R 70 M R 71 72 73 74 75 76 R 67 68 R 65 66 67 68 69 70 71 72 73 74 75
76 77 78 79 M 80 81 82 83 84 85 86 87 R 80 81 82 83 M 84 85 86 87
R 84 85 86 87 88 89 90 M 91 92 93 94 95 96 97 98 99 100 101 M 102 103
```

The reader might want to trace the effect of the mark and rewind operations by hand for the first two or three lines. Since the original underlying sequence produces each natural number exactly once and in order, the repeated appearances of each number are due to the rewinding operations.

Create a new class with the following signature in your labs project:

```
public class RewindIterator<E> implements Iterator<E>
```

Same as all decorators everywhere, this class should have a private field

```
private Iterator<E> client;
```

that refers to the underlying `client` iterator object being decorated, initialized in the constructor

```
public RewindIterator(Iterator<E> client) { this.client = client; }
```

Since the underlying iterator does not remember the elements it has produced, your decorator has to store these elements in some kind of list from which they can be extracted for their encore appearances. The simplest way to do this is to use two such lists, perhaps named

```
private LinkedList<E> buffer = new LinkedList<E>();  
private LinkedList<E> emitted = new LinkedList<E>();
```

so that `emitted` contains the elements that have been given out by this decorator with its `next` method, and `buffer` contains the previously produced elements that have been rewound and are waiting to be given out again. Required by `Iterator<E>`, this class must have the methods

```
@Override public boolean hasNext()  
@Override public E next()
```

The decorated iterator has a next element if either its `buffer` is nonempty or if its underlying `client` iterator has a next element. The next element produced by this decorator is then the first element of the `buffer`, and in the case that the `buffer` is empty, the next element of the `client` iterator. If any marks are currently active at the moment (that is, at least one position has been marked but not yet rewound), the produced element is pushed into the `emitted` list.

```
public void mark()  
public void rewind() throws IllegalStateException
```

These two new methods implement the functionality of marking the current position in the sequence, and later rewinding the sequence back to that position to repeat its elements. It is your task in this lab to implement these methods so that they behave correctly in all possible situations. Note that multiple positions can be marked at the same time, and the same position can be marked multiple times over repeated nested iterations. The method `rewind` always rewinds the sequence to the most recent marked position, or if there are no active marks at the moment, reports this unexpected situation to the caller by throwing an `IllegalStateException`. It might be a good idea to use a third instance field

```
private LinkedList<Integer> markPositions = new LinkedList<Integer>();
```

that contains all the positions in the `emitted` list that have been marked at the current time. (If this list is empty, the emitted elements do not need to be stored in the `emitted` list to waste memory, since such elements will never be needed again.)

Lab 31: Region Quadtrees

JUnit: [QuadTreeTest.java](#)

A two-dimensional image that consists of binary pixels (that is, every pixel must be either pure black or pure white) can be compactly stored with a two-dimensional array of `boolean` values. This representation allows rapid access and mutation of each pixel, independently of other pixels. Since storing each pixel needs exactly one bit of memory (plus some pocket change for the overall bookkeeping around that two-dimensional array structure), available heap memory sets a hard upper limit to the size of the images that can be stored this way.

A simple combinatorial argument based on [the pigeonhole principle](#), usually paraphrased in its bumper sticker form “There simply aren’t enough short bit patterns to uniquely encode not only themselves, but also a whole lot of longer bit patterns”, establishes that no general method can possibly compress data made up of arbitrary bits in a way that could guarantee to always produce a smaller result. For compression to be possible to begin with, the data must be somehow distinguishable from random noise so that some exploitable statistical dependencies and correlations exist between the individual bit values. Compressing that data will then erase such dependencies, since otherwise the compressed data could be run through another compression algorithm to squeeze it even tighter. Lather, rinse and repeat until the data has been compressed to next to nothing. You can observe this phenomenon in practice by trying to use your favourite compression tool to compress data that has already been compressed. Once all that metaphorical air has been squeezed out of the data, the **entropy** of that data ultimately determines how many bits are needed to store it so that the original version can still be losslessly recovered.

Two-dimensional binary black and white images that contain large uniform areas of either colour can sometimes be compressed into much smaller space with the **region quadtree** approach. Each quadtree object represents some square area of pixels whose side is some power of two. If this square is all white or all black, the object is a **leaf node** of this quadtree structure. Squares that contain both black and white pixels inside them are subdivided into four smaller squares that are recursively converted into region quadtree objects. These subtrees become children of the quadtree object that represents the original square. The recursive subdivision will necessarily end at a single pixel, but hopefully much sooner in areas of uniform colour inside the image. (Any Nassim Nicholas Taleb fanboys reading this can note how this relates to how “[The Most Intolerant Wins](#)”.)

To do this lab, you should first copy the interface [QuadTree](#) and its two concrete implementations [WhiteQuad](#) and [BlackQuad](#) into your labs project folder. The interface `QuadTree` defines the functionality of a region quadtree as two public methods

```
public boolean isOneColour();
public long computeArea(int scale);
```

The method `isOneColour` should return `true` if the region depicted by this `QuadTree` object represents an area of uniform colour, either all black or all white, and `false` otherwise. The method `computeArea(int scale)` should return the total area of the black pixels in the region

under the assumption that the entire region is a square whose side length equals 2^{scale} . For example, the area of a quadtree that is all black in the scale of 3 would equal $8 \times 8 = 64$. For coding this method, note that Java does not have integer exponentiation operator baked into the language the way Python has its handy `**` operator. However, this particular situation where the base of the exponentiation equals two can be easily computed with the **bit shift** operator `<<` so that the expression `1L<<scale` gives the quantity 2^{scale} . Note that the use of long literal `1L` allows us to potentially use up to 63 bits of scale instead of 31 bits that we would get from using the bare `int` literal `1`. (Not 64 and 32 bits, as the highest bits store the sign in the **signed integer** types.)

Before writing the class `QuadNode` of this lab, check out how the previous two methods of the `QuadTree` interface have been implemented in `WhiteQuad` and `BlackQuad`. These classes have `private` constructors so that no new objects can be created from the outside. Instead, the outside world must use the `get` method to access the **singleton** instances of these classes, for us to now demonstrate the (in)famous **singleton design pattern** for the first time in this course. (Some people consider singleton to be an **anti-pattern** in object-oriented design. The debate continues.)

```
public final class QuadNode implements QuadTree
```

Instances of this class represent region quadtrees that are not unicolour but contain both white and black pixels, so they could not be represented as instances of either `WhiteQuad` or `BlackQuad`. In the spirit of this course, the class `QuadNode` is designed to be **immutable** so that you get to practice **caching** of results the first time that they are computed, so that any future calls of those methods can simply look up the cached result. This is yet another huge advantage that immutable types enjoy over mutable types. The class `QuadNode` should have three instance fields

```
private QuadTree[] children;  
private long area = 0;  
private int lastScale = Integer.MAX_VALUE;
```

The four-element array `children` contains references to the four child quadtrees that represent the quarter squares (well, there is a name for a barbershop quartet) that this square is subdivided into. Since these children can be arbitrary region quadtrees, the type of this field is `QuadTree[]` instead of `QuadNode[]`.

The field `area` will cache the computed area of this quadtree, and `lastScale` will remember the scale in which that area was computed. Both fields are initialized with impossible placeholder values that allow us to recognize them as being uninitialized. Same as in the earlier example implementations `WhiteQuad` and `BlackQuad`, the class `QuadTree` should also have a `private` constructor to disallow object creation from the outside. To ensure the immutability of every instance after its creation, we make a **defensive copy** of the argument array for private storage, instead of just storing a reference to the array object that the outside world is still using and therefore potentially modifying later. However, since the `QuadTree` objects stored in this array are immutable, we don't need to defensively copy them, but can sit back and enjoy the main benefit of immutable types in how they can be freely and safely shared between any number of users.

```
private QuadNode(QuadTree[] children) {
    this.children = java.util.Arrays.copyOf(children);
}
```

To allow the outside world to create new quadtree structures to their heart's content, we provide a **static factory method** instead:

```
public static QuadTree of(QuadTree... children)
```

Unlike a constructor, a static factory method is not under any obligation to return specifically a `QuadNode`, but can return an instance of arbitrary subtype of `QuadTree` depending on the needs of the situation. Since this constructor is intended to accept an array of arguments, we might as well make the corresponding parameter `children` to be a **vararg** so that the array elements can simply be passed to this method as separate arguments, instead of forcing the caller to explicitly create a small array object every time. This factory method should recognize special situations of these four children being either all uniform white or all uniform black, and return the correct singleton `WhiteQuad.get()` or `BlackQuad.get()` appropriate for the situation. Otherwise, this method creates and returns a new `QuadNode` object with the given children.

Knowing that the child nodes cannot all be of the same uniform colour, the method `isOneColour` becomes trivial to implement in the `QuadNode` class, to always return `false`:

```
public boolean isOneColour() { return false; }
```

Otherwise, this method would have to do the work of recursively testing if all its children are uniformly the same colour, potentially probing through the entire tree structure to come up with the answer. The same would also happen if the child trees were mutable. We get to enjoy the asymptotic running times of our methods decreasing from $O(n)$ all the way down to the constant time $O(1)$ so very rarely that every occasion is worth celebrating for at least a moment. Immutable types, huzzah!

```
public long computeArea(int scale)
```

Assuming that this quadtree represents a square whose side length is $1 \ll \text{scale}$, recursively compute and add up the areas of the child quadtrees, using a `scale` one less than the `scale` in the current level of recursion for each of these recursive calls. Before returning the total to the caller, store the area and the `scale` used in its computation in the instance fields `area` and `lastScale` so that the next time this `computeArea` method is called, you can just look up this answer instead of redundantly recomputing it all the way down with recursion.

What if the previously cached area was computed using a value for `lastScale` that is different from the `scale` in the current call? Assuming that `lastScale < scale`, you can simply adjust the previously computed area for the needs of the new `scale`. Even in the opposite case where `scale < lastScale`, there may be some moves you might be able to pull off before accepting the harsh reality of having to recompute that area from scratch.

Lab 32: Triplefree Sequences

JUnit: [TripleFreeTest.java](#)

A sorted list of positive integers is **triplefree** if it does not contain any **arithmetic progression** of length three or longer, that is, does not contain three elements $a < b < c$ so that the differences $c - b$ and $b - a$ are equal. Another way to express this is to require that for any two elements $a < b$ in the list, the quantity $a + 2(b - a)$ is not a member of that same list. For example, lists $[2, 3, 5, 6]$, $[1, 2, 7, 8, 10, 11]$ and $[1, 2, 6, 9, 13, 18, 19, 21, 22]$ are triplefree. Try as you might, you won't find three elements that together form a three-step arithmetic progression inside any of them! For example, since the last list contains both elements 18 and 21, it cannot contain either 15 or 24 that would complete such a triple, and because it has 6 and 19, it cannot also contain the element 32 anywhere.

Create a class `TripleFree` in your labs project, and in that class, write a public method

```
public static List<Integer> tripleFree(int n)
```

that finds and returns the longest possible triplefree list whose largest element is n . Since in general there exist several lists of equal length that each satisfy the triplefree constraint, this method must return the **lexicographically largest** such list when the elements are read from highest elements down to the lowest. For example, the lists $[1, 3, 4, 6]$ and $[2, 3, 5, 6]$ are both triplefree and contain four elements, but the method must still return the second list for $n=6$.

n	Expected result
2	[1, 2]
4	[1, 3, 4]
9	[1, 2, 6, 8, 9]
15	[2, 3, 5, 6, 11, 12, 14, 15]
42	[2, 3, 5, 6, 11, 12, 14, 15, 29, 30, 32, 33, 38, 39, 41, 42]

This problem is best solved bottom up, building the answer list with recursion that solves the problem for n with two recursive calls for two subproblems with $n-1$. Write a helper method

```
private static List<Integer> tripleFree(int n, Set<Integer> chosen)
```

that finds and returns the answer for integer n when the `chosen` numbers have already been taken into the sequence in the previous levels of recursion. Your public method can then make the top level initial call to this private recursive method with the second parameter initialized to some `Set<Integer>` object that the recursion fills in during its journeys down the search tree.

In each recursive subproblem, you have the usual two-way choice of either taking the integer n in your chosen set, or leaving that integer behind. However, you may only take n into your chosen set if it does not create a triple with some two numbers that you already chosen. Whichever of these two branches gives you a better solution, return that list as the answer. If both ways are equally good, use the way that takes n in chosen, to ensure that your method returns the lexicographically largest solution.

This basic recursion solves the problem in reasonable time for n up to twenty or thereabouts, but for higher values on n , some branches that cannot possibly contain the best solution need to be mercilessly pruned on the spot, instead of recursing through those entire branches and coming up with a redundant solution that will eventually turn out to be suboptimal. The JUnit test class [TripleFreeTest](#) will loop through the possible values of n in ascending order, so your class can remember the length of the best list for each n .

Since adding new constraints to any optimization problem can never improve the optimal solution, the stored optimum for unconstrained n will be an upper bound to the solution with any set of chosen elements. For example, suppose you have already found a 15-element solution to your original top-level value of n . If `chosen.size()==5` in some branch down in the recursion, and the previously discovered optimal unconstrained solution for the current level n is 10 or less, this branch can be automatically dismissed as a failure. Any solution available in this branch contains at most 15 elements, and can therefore be no better than the lexicographically higher 15-element solution that you have already found.

Other clever optimizations can prune even more of those unproductive branches, which makes this problem a good exercise in finding out ways to prune down your exponential recursions from spreading out like some hydra spurting two new heads for each one that you chop down. If you create some good optimizations whose value you really want to try out, go ahead and uncomment the `testFirstHundred` method in the JUnit test class, and steel yourself to run it. Can you make your code complete that test within one hour?

A related but more famous combinatorial problem of constructing the shortest possible [Golomb ruler](#) hardens these constraints so that each possible difference between two element values must occur exactly once in the entire array. Constructing such rulers for large n is not an entirely trivial task. Golomb rulers can be further generalized to two dimensions as [Costas arrays](#), arrangements of n non-attacking rooks on n -by- n chessboard so that no two rooks lie on the same column or on the same row in a way that all possible pairwise direction vectors between two rooks are distinct. Efficient algorithms to generate individual Costas arrays on command for any given n have been devised. However, no systematic enumeration method that could iterate through all possible Costas arrays for the given n essentially faster than the trivial **backtracking**, let alone a closed form formula for [the number of different Costas arrays](#) for the given n , have been discovered.

Lab 33: Sardine Arrays

JUnit: [SardineTest.java](#)

The primitive integer types `byte`, `short`, `int` and `long` of Java store integers using one, two, four and eight bytes of memory, respectively. All four primitive types in Java are **signed** so that they allow both positive and negative values. Same as with other types in Java, we can create homogenous arrays to store a sequence of elements of the same type. When the element type of an array object is a primitive, these elements are stored in consecutive bytes with no additional bookkeeping.

However, in practice many programs operate on problem domains where the relevant integer values are known to be **unsigned** natural numbers. When such values are stored in Java primitive integer types, the highest order bit that represents the sign of the integer in [two's complement](#) encoding will always be zero and therefore completely wasted. Furthermore, the problem domain often inherently limits these unsigned integer values so that each such value is known to fit into some k bits. Whenever k is something convenient like 14 or 30, we have no problem since we can simply use some appropriate primitive type of `short` or `int` that these values almost fill up, leaving so little wasted space of unused bits between these elements that it's not really worth worrying about.

However, for the sake of argument, suppose k is some more annoying quantity such as 17. Now we won't be able to use the `short` type to store such values, since one `short` value can carry only 16 bits of information. We are forced to use the next higher type `int` with 32 bits of information, of which 15 bits, almost half of them, end up being wasted. Wasting almost half of your available bits of memory is not as big a deal these days as it used to be back in the day when the memory available for each program was measured in kilobytes. However, using these bits more productively allows our programs to solve bigger instances of the same problem within the same confines of memory. In this problem, we will perform the **time-space tradeoff** in the opposite direction from the usual so that we will make our array operations slightly slower to ensure that not one bit of space is wasted.

In this lab, we shall design and implement a general purpose utility class whose instances behave as simulated `<DrEvil>"arrays"</DrEvil>` of unsigned k -bit integers. An "array" of n such numbers is stored inside a [BitSet](#) instance that contains exactly nk bits of information, with less air between those bits of information than inside a shipping crate of Ikea furniture packed flat, or the titular sardines packed into a can. The computations in this lab are in fact so low-level that we don't even need to import any classes from the standard library! In addition to the **bitwise arithmetic** operators to encode and decode the given integer into individual bits, the other new operation that you need in this lab is the **bitwise left shift** operator `1<<k` to quickly compute 2^k with only integers without any floating point arithmetic.

Create a new class `Sardines` that contains two instance fields `data` and `k`, starting as

```
public class Sardines {
    private BitSet data;
    private int k;
```

This class should have a public constructor that initializes these fields.

```
public Sardines(int n, int k)
```

Since not all possible values of n and k can be meaningful even if they happened to be of the correct type, this lab now makes you practice throwing exceptions to report an inevitable violation of method preconditions. This constructor should throw an [IllegalArgumentException](#) if the array size n is negative, or when the total number of bits $n*k$ inside the array is greater than the theoretical maximum given by the named constant `Integer.MAX_VALUE`, or if the number of bits per element k is either negative or greater than 31. The method `testExceptions` in the JUnit test class [SardinesTest](#) will make sure that your code enforces these restrictions gracefully and throws an exception of the correct type each time.

You need to be careful with how you actually check for the second violation in your constructor. The seemingly obvious test `n * k > Integer.MAX_VALUE` will always be `false`, regardless of actual values of n and k . No result of integer arithmetic truncated to fit inside an `int` can possibly be larger than `Integer.MAX_VALUE`. Try again. Be better.

To read and write the value of an element in the simulated sardine array, this class should provide the corresponding accessor and mutator methods

```
public void set(int idx, int v)
public int get(int idx)
```

to `set` and `get` the value of the element in the position `idx`. The k bits that constitute the binary value in position `idx` of this instance of `Sardines` are stored into the k consecutive positions of the `BitSet` starting from position $k*idx$ in the underlying array data of boolean values. For example, when $k==5$, the first element is stored in bits 0 to 4, the second element is stored in bits 5 to 9. These methods must also throw an [ArrayIndexOutOfBoundsException](#) exception if `idx` is outside the range of legal indices ranging from zero to $n-1$, and throw an [IllegalArgumentException](#) if the element value `v` is not representable in k bits. Again, the test method `testExceptions` in the JUnit test class [SardinesTest](#) expects you to throw these exceptions correctly to pass that test.

One last pitfall in the implementation of this class lurks inside the corner case $k==31$ for this problem. Using that value for k , the expression `1<<k` will fill in the **sign bit** in the highest position of that number, which makes that number equal `Integer.MIN_VALUE`, the smallest negative integer value that fits inside an `int`. You therefore need to be a “bit” more careful when checking whether the argument `v` of the `set` method is fitting, or whether you should reject it and throw an exception. Otherwise the JUnit test method `massTestMillionAndThirtyOne` will fail and reveal the final mishandling of this particular situation, the proverbial snatching the defeat from the jaws of victory.

Lab 34: Leaders of Men

JUnit: [RunoffVotingTest.java](#)

For elections to determine exactly one winner from a group of C competing candidates vying for the iron throne, the Anglosphere nations tend to use the “[first-past-the-post](#)” system where each voter casts a ballot for exactly one candidate. The candidate with the **plurality** of votes wins the seat, without necessarily achieving an actual majority of votes cast. This system is simple and predictable in its consequences, but has been criticized from all angles for its various shortcomings and paradoxes. For example, the possibility of “spoiler candidates” whose very existence induces **tactical voting**; some voters who know their favourite candidate is unpopular in general will cast their ballots against their true preference to help boost whoever is the lesser of the remaining evils from their point of view. (Don’t blame me; I voted for Kodos.)

This lab examines two voting systems where each voter, instead of choosing just one candidate to vote for, ranks all C candidates in linear **preference ordering**. For the purposes of this problem, each ballot is a one-dimensional **permutation** array where each number $0, \dots, C - 1$ occurs exactly once. For example, assume $C=4$ with the candidates numbered $0, 1, 2$ and 3 . A ballot filled in with the values $\{3, 1, 0, 2\}$ would then indicate that that voter wants to see the candidate 3 win, but if that candidate cannot win, that voter would rather see the candidate 1 win, and so on.

Both methods of this lab receive the stack of ballots as a two-dimensional array of integers whose rows are the individual ballots. Now that each ballot contains more information than merely identifying that voter’s most favoured candidate, voting systems can be generalized to use all this information to determine the fair winner... with or without using **Benford’s law** to pry out any irregularities to shine the light to some sneaky bastard stuffing the ballot box in the dead of night.

Create a class `RunoffVoting` in your labs project. This class will have two `static` methods to implement two different voting systems that operate on these ballots, both returning the winner of that election as the result. To guarantee the determinism of expected results for our JUnit fuzz tests, we replace coin flips with the rule that **all ties that occur during the calculation must be resolved on spot in favour of the higher-numbered candidate**. (Note that this does not mean the candidate who currently has the most votes, but refers to their id numbers.)

```
public static int condorcetMethod(int[][] ballots)
```

Resolve the winner of the election using the [pairwise Condorcet method](#). In our variation to guarantee a winner within this method without resorting to additional tie breaker mechanisms, each candidate engages in a pairwise match against each other candidate. Each pairwise match between two candidates Winston Noble and Hubert Hoag tallies the ballots that rank Winston above Hubert, versus the ballots that rank Hubert above Winston. During the pairwise match between Winston and Hubert, all the other candidates and their relative rankings to Winston and Hubert are temporarily ignored. The winner of each pairwise match earns one “match point”, and the candidate with most match points after all pairwise matches have been completed wins the entire election. Barring upsets that involve a dead girl or a live boy, whether Winston predictably beats Hubert in their pair-

wise match by a single vote or by millions of votes makes no difference; Winston gets exactly one match point for this feat, no matter how comfortable his margin of victory.

An election with C candidates should therefore end up with $C(C - 1)/2$ match points, regardless of the number of ballots cast. You should debug your method by asserting this fact after the fact. (Remember to turn on the assertions in the Java compiler with the command line option `-ea`.) To pass the JUnit test in a reasonable time, make sure that you are not being a “Shlemiel” who combs through each ballot and each candidate more times than you actually need. When going through a ballot, update a whole bunch of match score counters at the same time, not just one.

```
public static int instantRunoff(int[][] ballots)
```

Resolve the winner of the election with [instant runoff](#), a more complex method for choosing the winner. Initially, each candidate is awarded all the ballots that placed him as the first choice. Then, as long as no candidate holds **more than half** of the ballots that would give that candidate an immediate win (even though his eyes can already see the promised land, having exactly half is not enough to start the ticker tape parade), the candidate who currently holds the fewest ballots is eliminated. Again, if two or more candidates have an equal number of ballots, be sure to eliminate the lowest-numbered candidate. Ballots held by the eliminated candidate are distributed between the candidates who are still in the race, so that each ballot goes to the next candidate down the preference list stated in that ballot who has not yet been eliminated from the race.

Once some candidate has been eliminated, he is out for good, and no future redistribution of ballots can resurrect him back to this process. This spells doom for any “apple pie” candidate, where this term is not a reference to nostalgic Americana out of some Norman Rockwell painting, but for being everybody’s second choice but nobody’s favourite. In other words, a neutral candidate who nobody hates but nobody feels any jingoistic enthusiasm for either. That candidate will therefore be eliminated in the very first round, even though in the grand scheme of things, that candidate could have been the optimal global choice in the hypothetical situation of extremely fragmented partisan jingoism where every voter loves their own Dear Leader and fervently hates every other candidate who is not Mr. Apple Pie...

The critical step to the performance of this method is the redistribution of ballots from the candidate being eliminated in the current round. To ensure that you can do this in a manner that does not resemble the way that “Shlemiel” paints the fence running back and forth along the same parts of the fence that he has already painted many times before, you should maintain a list of lists of votes held by each candidate (some kind of `List<List<Integer>>` would probably be good for this purpose) so that you can quickly loop through the ballots of the candidate being eliminated without having to loop through all ballots of the entire election to find out which candidate each ballot currently belongs to. Note that each ballot can be identified as an integer that gives its row position in the `ballots` array.

Again, be careful to ensure that your method resolves all ties in favour of the higher-numbered candidate. Also, treat any candidate who is not ranked as the first choice in any of the ballots as having already been eliminated in this computation even before even the first elimination round, so that that candidate doesn’t come alive in any later elimination round.

Lab 35: Multiple Winner Election

JUnit: [MultipleWinnerElectionTest.java](#)

Anglosphere countries tend to use voting systems where each district elects exactly one representative, typically done with the blunt [first-past-the-post](#) method to determine the sole winner of each district. Many other countries use some type of [party list proportional voting](#) system where each electoral district elects multiple representatives. Each voter casts a vote for a particular party, either directly ([closed list](#)) or indirectly ([open list](#)). The seats that are up for grabs in each district are dealt out to the parties in proportion to their votes in that district.

In this lab, you implement three methods to compute the distribution of `seats` to the parties according to the votes that these parties received. The methods of **D'Hondt**, **Webster** and **Imperiali**, each one a special case of the [highest averages method](#), are otherwise identical, but use a different formula to compute the **priority quantity** for each party, as described below. You might therefore want to implement this entire voting algorithm as one `private` method that the following three `public` methods pass the buck to.

```
public static int[] DHondt(int[] votes, int seats)
public static int[] webster(int[] votes, int seats)
public static int[] imperiali(int[] votes, int seats)
```

The parameter array `votes` gives the number of votes received by each party. The `seats` are given out one at the time, same as in the [Huntington-Hill method](#) back in the Lab 0(G), so that each seat goes to the party that currently has the highest priority. To make the results sufficiently unambiguous for the JUnit fuzz tests, all ties should be resolved in favour of the party whose id number is higher, same as was done in the previous lab 34, "Leaders Of Men". Priority quantity of the party that received v votes and has so far been given s seats is $v/(s + 1)$ in the D'Hondt method, $v/(2s + 1)$ in the Webster method, and $v/(1 + s/2)$ in the Imperiali method. You should again perform all these calculations with exact integer arithmetic using our [Fraction](#) data type, of course!

The returned `result` should be an `int[]` of the same length as the parameter array `votes`, each element indicating how many seats were given to that party. For example, in an election between four parties where `seats=21` and `votes={23,26,115,128}`, the D'Hondt method would return `{1,2,8,10}`, the Webster method would return `{2,2,8,9}`, and the Imperiali method would return `{1,1,9,10}`. These results illustrate how adjusting the divisor sequence in the priority quantity calculation can cause this process to favour either larger or smaller parties, the Webster method being most favourable and the Imperiali method the least favourable for small parties.

(As a final humorous aside on the general theme, what should we think of a method where the priority quantity is computed with the formula $v/2^{s+1}$ dealing the seats out logarithmically so that doubling your number of votes gives you one additional seat? Let those wonks at [FiveThirtyEight](#) wrap their pointy eggheads around that conundrum!)

Lab 36: Rational Roots

JUnit: [RationalRootsTest.java](#)

Everyone who has come this far has surely heard of the [quadratic formula](#) to solve quadratic polynomial equations. Since even that already brings up nasty square roots with irrational and even complex results, we shall leave that for higher level languages such as Wolfram Mathematica that can properly handle algebraic computations on symbolic expressions. Instead, this lab makes you implement a handy method based on the [rational root theorem](#) for polynomials that is honest-to-God guaranteed to find *all* rational roots of rational polynomials of *arbitrary* degrees, not merely for some quadratics. (Put *that* in your pipe and smoke it, Abel!) The method itself is clearly explained on the page “[Finding zeroes of polynomials](#)” of the course “[Algebra](#)” in “[Paul’s Online Notes](#)”, the most concise and the best online undergraduate mathematics resource that this author is aware of.

Same as in the three previous labs that had you implement the `Polynomial` class, we represent a polynomial as an array of its coefficients so that the i :th element of that array contains the coefficient of its i :th order term. For example, the polynomial $5x^6 - 17x^3 + 8x^2 - 42$ would be represented as the integer array $\{-42, 0, 8, -17, 0, 0, 5\}$ where the coefficients are read from the lowest order term to the highest. Terms that are missing from the polynomial (in this example, the exponents 1, 4 and 5) are not actually missing from the array, they just have a coefficient of zero. Zero is a number same as any other number, even if we tend to think of it as somehow special.

The rational number b/c where b and c are integers and the fraction is in lowest terms can be a root of a polynomial whose lowest coefficient is t and the highest coefficient is s only if the numerator b is some factor of the lowest coefficient t , and the denominator c is some factor of the highest coefficient s . Since we can easily loop through all possible pairs (b, c) of integers that satisfy that condition, all that remains is to evaluate the polynomial at each such fractional point b/c and gather precisely those points that make the polynomial zero. For example, for b/c to be a rational root of the previous example polynomial, c must be one of the factors of 5 (therefore either 1 or 5, not much choice there) and b must be one of the factors of 42 (therefore one of 1, 2, 3, 6, 7, 14, 21, 42). This gives a total of $2 \times 8 \times 2 = 32$ possible combinations to try out; two for c , eight for b , and for each such pair, you must try both possibilities b/c and $-b/c$ for the sign of that root.

The operation to evaluate the given polynomial at given rational point x is best written into a separate method of its own so that we can unit test and debug it. Once we are confident this method is working, it will be much easier to write the method that finds the rational roots of the polynomial, since the failures of that method to produce the correct answer for some test case can then be isolated to the body of that method for debugging.

```
public static Fraction evaluate(int[] coefficients, Fraction x)
```

This method evaluates the polynomial with the given `coefficients` at point x . The answer must be computed and returned as an exact `Fraction` object. However, again please don’t be a Shlemiel and evaluate each term separately, but you should rather realize that you can get the next term x^{k+1} with just one more multiplication from the previous term x^k . (The best algorithm to evaluate an ar-

bitrary polynomial at the particular point x is the [Horner's rule](#), whittling down the entire action to one measly multiplication and one addition per term.)

```
public static List<Fraction> rationalRoots(int[] coefficients)
```

Return the list of all rational fractions of the polynomial with the given `coefficients`. To make the result unique and unambiguous for the JUnit fuzz tests, this list must contain the rational roots of that polynomial in sorted ascending order, with each root listed only once even if it happens to be a double root. Loop through the possible rational roots in two nested for-loops, using the previous method to evaluate the polynomial at that rational point, and adding that rational point to the results if the polynomial evaluates to zero at that point. Eliminate duplicates and sort and that's that about that equation, wham bam thank you ma'am.

For example, given the integer coefficients $\{333, -432, -16\}$, this method would create and return the list $[-111/4, 3/4]$ that reveals two rational solutions for that polynomial. Given the coefficients $\{319410, -207597, 48337, -4830, 176\}$, this method would return four rational solutions in the list $\{42/11, 13/2, 65/8, 9\}$.

As a potentially important aside for your future, these kind of test cases for this problem with known rational solutions were a breeze to create with Wolfram Mathematica, a very high level programming language that operates directly on symbolic expressions kept in their symbolic forms and evaluated lazily according to the rules of mathematics. A small screenshot below copied from the session used to create the automated test class shows the idea of building up symbolic polynomials that have the rational roots that we want them to have:

```
In[67]:= p = Collect[(x - 13/2) (x - 27/3) (x - 42/11) (x - 65/8), x]
Out[67]= 
$$\frac{159705}{88} - \frac{207597x}{176} + \frac{48337x^2}{176} - \frac{2415x^3}{88} + x^4$$

In[69]:= ReplaceAll[p, {{x -> 13/2}, {x -> 27/3}, {x -> 42/11}, {x -> 65/8}}]
Out[69]= {0, 0, 0, 0}
```

Even rudimentary knowledge of Mathematica would make all undergraduate level math courses immensely easier. This author finds it strange that this wonderful system is not taught to everybody during the first week of the intro calculus class. Plus, though even many active users of Mathematica don't seem to know this, Mathematica is a pretty awesome programming language also to actually solve real problems in all walks of life outside mathematics, especially when integrated with the [Wolfram Alpha](#) knowledge base. As high-level languages go, the experience of moving from Python 3 to Mathematica should feel pretty similar to moving from Java to Python 3. Interested students can create themselves a free account on [Wolfram Cloud](#) to try out the language, and start with either "[An Elementary Introduction to Wolfram Language](#)" or "[Fast Introduction to Programmers](#)" according to taste to find out what to do in, on, over, under and with that account.

Lab 37: Big Ten-Four

JUnit: [TenFourTest.java](#)

When standing on any integer a , you can step to either $10a$ or $10a + 4$; also whenever a is even, you can also step to $a/2$. For example, if you are currently at 17, you can step into either 170 and 174. If you are currently at 42, you can step into 420, 424 or 21, which together sound like a fun weekend in Vegas back when that sort of thing was still a thing. As this instructor learned from "[In Polya's Footsteps](#)", a collection of entertaining expositions on mathematical topics by the late great [Ross Honsberger](#), a crafty young fella such as *you* can get from $a = 4$ to any other positive integer by using only these three basic moves. (You are a winner, kid, I can see it in your eyes!)

This lab examines a mathematically less sophisticated but still universally effective **systematic search** method of [breadth-first search](#). This search algorithm is guaranteed to discover the shortest path of steps from the number 4 into any given n . Implementing this search algorithm for this problem teaches you the general idea of how to solve literally any problem of this nature with the systematic approach of generating all shortest paths to every number in such breadth-first manner, terminating as soon as once the shortest path to the goal has been discovered.

Create a class `TenFour` into your labs project, and in that class write the method

```
public static List<Integer> shortestPath(int n, int limit)
```

to find the shortest path from 4 to n under the additional constraint that you are not allowed to move into any number that is greater or equal to `limit`. If the goal integer n is not reachable under this additional restriction, this method should return an empty list, and otherwise it should return the list of numbers encountered along the path from 4 to n . To implement the logic of breadth-first search, you should first define the following local variables:

```
boolean[] seen = new boolean[limit];  
int[] parent = new int[limit];  
LinkedList<Integer> frontier = new LinkedList<>();
```

The array `seen` keeps track of which numbers we have already discovered in our search that will be spreading out from the starting number 4, the only number that we have seen so far. Every time that we discover some previously unseen number, we note its `parent` from which we came into this number. The `frontier` list contains the numbers that we have discovered at the moment but not yet **expanded** by looking at their neighbours. For a refreshing change, this `frontier` list should be an instance of our red-headed stepchild `LinkedList<Integer>` instead of the golden child and usual go-to guy of `ArrayList<Integer>`, because to implement a **first-in-first-out queue** this algorithm needs, we surely want to add elements at one end of the list, and remove them from the other end. Depending on which way you slice it to choose your poison, either the addition or the removal operation for `ArrayList<Integer>` will necessarily take a linear time to restore the element structure to be consecutive without gaps, whereas a `LinkedList<Integer>` guarantees both of these operations to consume only a constant time.

To start this circus, mark the number 4 as having been `seen`, and add 4 into the `frontier` queue. Then, use a while-loop that keeps going until the `frontier` becomes empty. Pop out the first element of the `frontier`, the current number for this round that we shall call `v`. (During the first round of this loop, necessarily `v=4`.) Look at the three numbers $10*v$, $10*v+4$ and $v/2$ (this last one only when `v` is even) that you could move from the current number `v` in one step. Those of the three that haven't been `seen` yet, mark them as `seen`, make `v` their `parent`, and add them to the end of the `frontier` queue. To make sure that the returned shortest paths(not just their lengths) are unique to allow automated testing with JUnit, your method **must** examine the three neighbours of `v` in this exact order of $10*v$, $10*v+4$ and $v/2$ when expanding `v`.

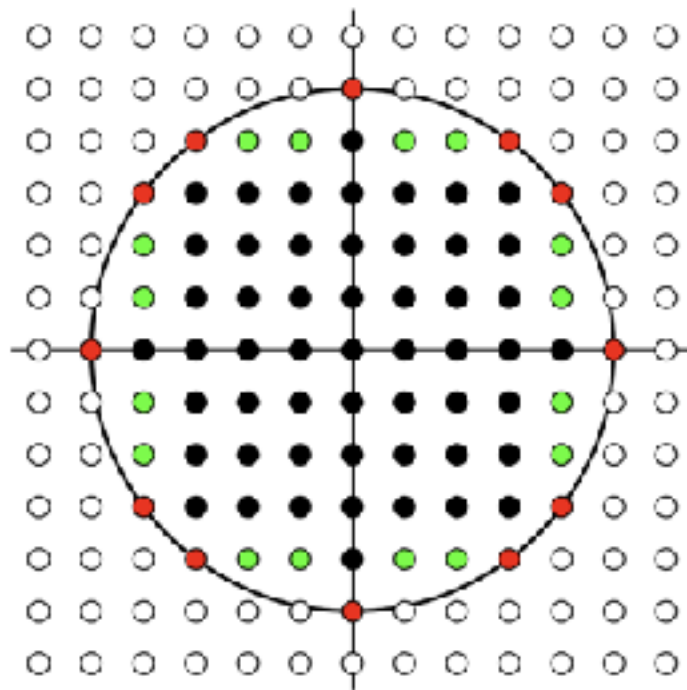
Keep going until you discover the goal number `n` during the expansion of some other number `v`. At that point you can construct the path from `n` back to 4 by following the `parent` information that you stored along the way, and then reverse that path from `n` to 4 to be returned as your final answer as the path from 4 to `n`. If the `frontier` becomes empty so that the while-loop terminates before finding the goal number, return the empty list as the artificial negative answer. As revealed by the instructor's private model implementation, the shortest paths from 4 to various goal numbers (with the `limit` incremented iteratively from $100*n$ by factor of 2 until the goal was found via some path that stayed within that limit) are listed in the following table.

n	Expected result
3	[4, 2, 24, 12, 6, 3]
13	[4, 40, 20, 10, 104, 52, 26, 13]
42	[4, 2, 24, 12, 6, 64, 32, 16, 8, 84, 42]
404	[4, 40, 404]
739	[4, 2, 1, 14, 144, 72, 36, 18, 9, 94, 944, 472, 236, 2364, 1182, 11824, 5912, 2956, 1478, 739]
740	[4, 2, 1, 14, 7, 74, 740]

Lab 38: Fenceposts Around Euclid's Orchard

JUnit: [GaussCircleTest.java](#)

As can be easily seen from the Pythagorean theorem, a **disk** with the center at origin $(0, 0)$ and radius r contains all points (x, y) that satisfy the inequality $x^2 + y^2 \leq r^2$. The points on the boundary **circle** turn this inequality into strict equality. As we were taught back in school, the area of this disk is given by the famous formula πr^2 . Note also the correct terminology in this context in that a *circle* is the one-dimensional boundary curve around a two-dimensional *disk*. (Equivalent concepts in three dimensions are the boundary *sphere* around the solid three-dimensional *ball*.) So, next time some weisenheimer challenges you to recite the formula for the area of a circle from memory, you can remind him that being a one-dimensional curve, the area of the *circle* around the disk is zero...



The [Gauss circle problem](#) asks for the exact number of points (x, y) where both x and y are integers that exist inside the disk of given radius r . Our lab further breaks this point count into three parts by defining an **edge point** to lie exactly on the edge of the disk. A **border point** lies inside the disk but not on the edge, and at least one of its axis-aligned neighbour points $(x + 1, y)$, $(x - 1, y)$, $(x, y + 1)$ and $(x, y - 1)$ lies outside the disk. Points inside the disk that are neither on the edge or at the border are **proper internal points**.

As you can tally from the above image, the disk with $r = 5$ contains 12 edge points (red) that correspond exactly to the [Pythagorean triples](#) of integers (x, y, r) whose **hypotenuse** equals r . That same disk also contains 16 border points (green) and 53 proper internal points (black).

Create a class `GaussCircle` in your labs project, and there write the method

```
public static void classifyPoints(int r, long[] out)
```

This method does not return anything, but instead fills in its three-part answer into the first three elements of the parameter array `out` so that `out[0]` is the number of proper internal points, `out[1]` is the number of border points, and `out[2]` is the number of edge points. Since both the method and its caller share the same `long[]` object, the values written inside the array remain when the control is back at the caller.

Your method **must** perform all of its internal computations with the `long` type instead of `int` to guarantee that the squaring of distances does not overflow for large values of `r`. (Even when the value of `r` fits inside an `int`, its square `r*r` might not be quite as friendly.)

The following table lists a couple of values for `r` along with their expected numbers of internal, border and edge points. Due to symmetry, the edge point count will always be divisible by four, and the points $(r, 0)$, $(-r, 0)$, $(0, r)$ and $(0, -r)$ are edge points for every integer radius `r`. Of course the total number of points will increase along with `r`, but the exact distribution of those points into internal, border and edge points depends on the arithmetic properties of `r`. The three counts of points in the last row add up to 3141592649625, close enough to πr^2 for government work, but still slightly overestimating the true value of π because the implicit sum of unit squares centered at all these lattice points will necessarily include some area that falls outside the original `r`-disk. The area of these trimmings relative to the total area of the disk approaches zero as `r` increases.

r	Internal	Border	Edge
1	1	0	4
10	261	44	12
24	1661	128	4
118	43045	660	4
119	43797	660	12
12345	478705609	69820	12
1000000	3141586992773	5656800	52

This problem could be solved with two nested loops that iterate through all possible combinations of `x` and `y` both getting values from $-r$ to `r` and classifying each point separately. However, as you can see from the last line of the above table, this method has to work even for pretty large values of `r`, so you should rather add up these points to the tally one entire row at the time, utilizing the symmetries of the disk for maximal efficiency gain. Manipulating and evaluating symbolic formulas is always more efficient than grinding low level arithmetic like some machine.

Place your finger on the topmost edge point $(0, 5)$ of the previous picture and start tracing the red and green edge and border points of the upper right quadrant with your finger. Notice how from the current point (x, y) you always move to whichever is the first available point of the three adjacent

points $(x + 1, y)$, $(x + 1, y - 1)$ or $(x, y - 1)$ that is not outside the disk. In other words, when tracing the boundary of the upper right quadrant with your finger, you always move east when you can do so. Otherwise you move southeast if you can do so, and move south only if you have no other choice. This technique is guaranteed to work for any radius r .

Your method should therefore maintain three counters for the internal, border and edge points, and update these counters appropriately at each step depending on the direction of that step. Once you reach $y = 0$, at which point (heh) necessarily $x = r$, you can terminate the loop. Note that the center row $y = 0$ is its own symmetric row, unlike the other rows with $y > 0$ that are symmetric with the row $-y$ mirrored along the x -axis.

[The Gauss circle problem](#) asks for an exact closed-form combinatorial formula for the number of integer points inside the r -circle to give us the answer on the spot without having to crawl through the rows of points one at the time. However, this problem still stubbornly remains unsolved, despite all our best effort to come up with such a formula! The titular [Euclid's Orchard](#) problem asks how many points inside the disk can be seen from the origin so that no other point blocks the view from the origin. Equivalently, how many coordinate pairs (x, y) inside the disk are **relatively prime** to each other so that they satisfy the equation $\gcd(x, y) = 1$.

As the final unnecessary aside, this problem would become even trickier if the coordinates of the center point (c_x, c_y) could be arbitrary real values, instead of being permanently nailed to the origin. An interesting theorem about this setup says that for any positive integer n , there must exist some center point (c_x, c_y) and radius r so that the area πr^2 is exactly n , and that disk contains exactly n integer grid points, no more and no less.

Lab 39: Hot Potato

JUnit: [HotPotatoTest.java](#)

A group of n people numbered from zero to $n - 1$ stand around a circle. Each person has memorized their personal “enemies list” of other people in the circle. However, this enemy relation is not symmetric so that Moe can be an enemy of Joe even when Joe is not an enemy of Moe. (You may not be interested in war, but war will still be interested in you. The relationship between predator and prey is inherently asymmetric, and the prey does not get a vote on this by merely closing its eyes and wishing it to be so.) For the purposes of this problem, everyone has at least one enemy inside the room and, unlike in the real world, nobody is messed up enough in the head to be their own enemy. During this process, no new enemies are created or friendships forged in the room.

We may also imagine that these events take place in some very official room with a high vaulted ceiling, and that these men are dressed in some sort of fancy robes or togas to signal their position. Either way, all these people are assumed to be perfect Machiavellian schemers who hatch all their plans in secret, but do not hold any grudges or desire for payback for any past acts of trespass against them and what’s theirs, unless the revenge concretely benefits them in the future. (There is no love to be found in this room, so the preference between being hated rather than loved is not a factor.) These men have properly internalized the ideal [Markovian](#) view that once the present state of the world is sufficiently known as to determine all our future decisions, the past does not exist in the relevant sense that it could afflict the outcomes of our actions. The only thing that matters at the present moment is to accept that present moment as it is, and choose your actions at that eternal infinite now to maximize your future gains over all the future present moments.

Students who are interested in this general theme should also check out the popular writings of [Nassim Nicholas Taleb](#) in their spare time. To sum up its relevant ideas while standing on one foot; you should only care to listen to advice from those who have demonstrated that they understand the concept of [ergodicity](#) (if you have to ask what that means, you definitely don’t understand it yet) and can explain its effect on their decision making, and that they have a “[skin in the game](#)”... and significantly more of risk of burning to crisp than merely getting to eat delicious potato skins!

Back to the problem at hand. At time $t = 0$, the person number 0 is holding some kind of proverbial hot potato. At each discrete time step t , the person currently holding the hot potato will instinctively toss it randomly to one of his enemies to catch. That person will then similarly throw the hot potato randomly to one of his enemies at time $t + 1$, and so on. In this lab, you will create a method to investigate the probability distribution of the hot potato at a given time t , given the people and their list of enemies. Create a class `HotPotato` in your labs project, and there a method

```
public static Fraction[] hotPotato(int[][] enemies, int tGoal)
```

The i :th row of the `enemies` array is a one-dimensional array of integers that lists the enemies of the person i in ascending order. For example, if `enemies[2]` equals `{0, 5, 6, 9}`, that means that person 2 has four enemies, the persons numbered 0, 5, 6 and 9. Since Java does not actually have two-dimensional arrays, but every `int[][]` is in reality a one-dimensional array whose each ele-

ment is a one-dimensional `int[ ]` that contains that row of elements, the rows of the array can be **ragged** so that they do not need to have the same length.

The probability distribution of the position of the hot potato at time t is expressed as an n -element array of probabilities. Since these probabilities will be integer fractions, we shall again use the [Fraction](#) class to represent these probabilities without rounding errors. As usual, the individual probabilities must always be between 0 and 1 and together add up to exactly 1, since the potato must be held by exactly one person at any given time. For the initial state $t = 0$, this probability distribution is simply $P(t, 0) = 1$ for the first person and $P(t, p) = 0$ for the other people $p > 0$, since we know that the person 0 is initially holding the hot potato.

Each probability $P(t + 1, p)$ can be computed from previous probabilities at time t by first finding all people p' who consider the person p an enemy. (Since the enemy relationships never change during the execution of this method, it might be smart to precompute these at the start of this method, so that you can always just look them up later whenever you need them.) For each such person p' , compute the quantity $P(t, p') / m$, where m is the number of enemies of person p' . This term gives the probability of the event “Person p' is holding the potato at time t , and tosses it to the person p .” The desired $P(t + 1, p)$ can then be computed by adding up all such terms $P(t, p') / m$, since those are the only possible ways for p to receive the potato at time $t + 1$. Fortunately these are all mutually exclusive, so we can simply add up their individual probabilities to find the desired probability.

The following table displays a couple of illustrative examples of the `enemies` structure and the resulting probability distribution at the given time t . In the situation given in the first row, person 0 will first throw the hot potato randomly to either person 1 or 2, resulting in a distribution where person 0 cannot be holding the potato, whereas both persons 1 and 2 have the potato with the same probability $1/2$. The second row shows the distribution that follows the previous distribution at the time $t = 2$. This second distribution is made more lopsided than the one expected in the first row by the difference in the tossing behaviour of persons 1 and 2. (Long-term behaviour of any such Markov system can be solved symbolically arbitrarily far to the future with linear algebra.)

n	t	enemies	Expected result (as <code>Fraction[]</code>)
3	1	<code>{{1, 2}, {0, 2}, {1}}</code>	<code>{0, 1/2, 1/2}</code>
3	2	<code>{{1, 2}, {0, 2}, {1}}</code>	<code>{1/4, 1/2, 1/4}</code>
4	2	<code>{{1, 3}, {0, 2, 3}, {0, 1}, {0, 1, 2}}</code>	<code>{1/3, 1/6, 1/3, 1/6}</code>
5	3	<code>{{1, 2, 3}, {0, 4}, {1, 4}, {2, 4}, {1, 2}}</code>	<code>{1/12, 7/18, 11/36, 1/18, 1/6}</code>
2	1000	<code>{{1}, {0}}</code>	<code>{1, 0}</code>

Lab 40: Seam Carving

JUnit: [SeamCarvingMain.java](#)

All magic tricks are trivial once you get to watch them from the backstage (or even better, in the rehearsal room), as opposed to seeing them unprepared through the fourth wall sitting in the actual audience. The [seam carving algorithm](#) for **context-aware image resizing** is deservedly famous for the almost magical outcome that it achieves relative to its own far smaller complexity.

Modern image processing algorithms employ arduously trained **deep neural networks** that implicitly identify the **semantic content** of the image. The seam carving algorithm is significantly less lofty, as it employs only good old fashioned arithmetic and algorithmic tabulation of intermediate results to determine which pixels to eliminate. Seam carving performs its magic with simple local calculations with no higher semantic knowledge of the objects that appear in the image, nor any ordering of importance between them. Everything is done at the level of individual pixels and their local gradients to their neighbours, and yet the global result just works.

Instead of rescaling every pixel by the same amount, the scaling is done by repeatedly “carving” a one-pixel wide path through the image, and then erasing exactly those pixels from the image. For simplicity, each individual carve is constrained to traverse from some top row pixel to some bottom row pixel so that the carve contains exactly one pixel from each row, and moves at most one pixel left or right in each individual step to the next row. Since the path contains exactly one pixel from each row, removing these pixels maintains the proper shape of the image as a rectangle whose height is unchanged, and whose width has decreased by one pixel. The resizing can go on arbitrarily deep by carving these individual slices of bologna out of the image one at the time.

The finer details of energy calculation and other details of the seam carving algorithm are well explained with illustrative images in the project pages of various fine universities whose courses have used seam carving as a programming project, such as [Princeton](#), [Brown](#) and [Berkeley](#), so we shall not attempt to repeat them in this document. Instead, here in the Wossamotta U. whose faux marble columns tend to be covered with lichen, graffiti and vomit instead of ivy, our practical can-do attitude ought to propel us to do at least as well as those perfumed princes in their secret handshake clubs. Have all those classic college comedies really taught us nothing, my fellow band of lovable underdogs itching to take on “The Man”? Was it over for Napoleon when those German ratzis bombed Verdun? Come on, the trailer for this classic comedy reboot practically writes itself! Create a class `SeamCarving`, and there the method

```
public static Image carve(BufferedImage image, int width)
```

to create and return the seam carved version of the `image` horizontally scaled to the desired width. To allow you to concentrate on the actual seam carving algorithm instead of the gritty details of the bitwise arithmetic needed in colour manipulations, you can use the following method to compute the approximate gradient between two RGB colours packed into four-byte integers that you get from the `getRGB` method of `BufferedImage`. The higher the gradient between the pixel and its chosen neighbours, the more “energy” that pixel contains.

```
private static float gradient(int rgb1, int rgb2) {
    int b1 = rgb1 & 0xFF, b2 = rgb2 & 0xFF;
    float b = Math.abs(b2 - b1);
    int g1 = (rgb1 >> 8) & 0xFF, g2 = (rgb2 >> 8) & 0xFF;
    float g = Math.abs(g2 - g1);
    int r1 = (rgb1 >> 16) & 0xFF, r2 = (rgb2 >> 16) & 0xFF;
    float r = Math.abs(r2 - r1);
    return b + g + r;
}
```

You can test your implementation of the algorithm with the [SeamCarvingMain](#) class provided in the repository. The outcome of seam carving the example images [coffee.jpg](#) and [ilkka.jpg](#) to half of their original width with repeated carvings can be seen in the screenshot below.



The basic seam carving algorithm without further fine-tuning tends to have problems dealing with sloping straight lines, as can be seen in the seam carved version of the coffee maker image. However, the important coffee maker object remains almost as wide as in the original. For the author portrait in the second row, even the most rudimentary version of the seam carving algorithm could not possibly fail to find the lowest-energy seams in the nearly uniform white background colour glowing around that old mutt's ugly mug.

Lab 41: Geometry I: Segments

JUnit: [CompGeomTestOne.java](#)

“Whoever says X, must also say Y.” — V. I. Ilkkanov

“Whoever says ‘ex’, must also say ‘why?’” — its modern relationship corollary, apparently

At first impression, problems in **geometry** seem difficult to solve on a computer, at least in low-level languages such as Java or Python that do not allow expressions to be analyzed in symbolic forms within the [homoiconic language](#) itself. No, this term does not refer to Liza Minnelli in this context. “Homo” means “same”, and “icon” means representation, referring to how the structures of the a homoiconic language itself are also first class data in that same language so that expressions of that language can be assigned to variables and passed back and forth to functions to perform computations on.

Even worse, algebraic calculations of even seemingly simple geometric shapes tend to be chock full of square roots and other irrational quantities, especially once trigonometric formulas for arbitrary angles enter the fracas. Even the familiar junior high school mathematics notion of how a “line” is defined by its “slope” and “offset” goes what-sorcery-is-this *kaput* the moment it encounters the first vertical line segment, since computing the slope of that vertical line involves a division by zero. It is our duty as responsible educators to save our students from the claws of such *comprachicos*. In the next four labs, students are invited to enter Leopold Kronecker’s paradise of well-behaved integers and everything good and plenty that springs forth from them, before that wicked serpent of division sneaks its forked tongue inside this big happy tent.

Geometric computations cannot be avoided in some applications, most exciting of which are surely games played in a simulated microworld in some finite box inside the infinite two-dimensional plane. These geometric computations become more manageable if restricted on the **lattice points**, that is, coordinates made out of integers. Our geometric objects are composed of straight **line segments** whose both endpoints must be such lattice points. Since this lattice can be arbitrarily large, we can get as much precision as we want simply by scaling the world to be large enough to make every point inside it to be a lattice point, with no need for fractional parts within the pixel raster of the game display.

Instead of using roots and trigonometric functions, we build our geometry on the vector operations of **cross product** and **signed area**. Even if you don’t know what these operations are, no need to worry! Just like you are allowed to use a compiler even though you don’t know how it works, or drive a car even if you don’t know how a combustion engine works, you are still allowed to enjoy the existence and results of these functions even if you have never taken a vector mathematics course and so wouldn’t be able to explain standing on one foot exactly what makes these functions tick.

Create a class named `CompGeom` in your labs project. You will be writing all methods from these computational geometry labs inside this same class. Each lab will have its own JUnit test class that depends only on the methods written up to that point. To begin, copy-paste the following implementations of the methods `cross` and `ccw` inside your class:

```

public static int cross(int x0, int y0, int x1, int y1) {
    return x0 * y1 - x1 * y0;
}

public static int ccw(int x0, int y0, int x1, int y1, int x2, int y2) {
    return cross(x1 - x0, y1 - y0, x2 - x0, y2 - y0);
}

```

Without worrying about what these functions actually do and how they do it, you should now put on your Java programmer hat and spend a moment in silent appreciation of how these both methods use only basic arithmetic, and are well-defined for all possible argument values. (The JUnit tests are designed to keep the numbers small enough so that your code does not need to worry about the effects of integer overflows.) Most importantly, **neither operation performs any divisions**, so we don't need to worry about dividing by zero the way people often end up doing in their slope calculations, nor care about the inaccuracies of floating point division.

Analogous to the bumper sticker version of Liskov Substitution Principle that your instructor keeps parroting every chance he gets during the lectures, you should also follow a similar bumper sticker maxim for computational geometry: **Should any one of the forbidden words “angle”, “slope” or “root” come out in your description of how your method solves that computational geometry problem, your method is automatically wrong!**

Now that we got that out of the way, let us explain what the above two methods actually do. The `cross` method computes the **two-dimensional cross product** of two vectors (x_0, y_0) and (x_1, y_1) . (Any point (x, y) on the two-dimensional plane can be thought of as a **vector** from the origin $(0, 0)$ to that point (x, y) .) The cross product of two vectors is actually a three-dimensional operation that produces a three-dimensional vector result, not merely a single scalar, but since any two-dimensional vector (x, y) can be considered a degenerate three-dimensional vector $(x, y, 0)$, multiplications by these zeroes cancel out most terms. Only the z-component of the result vector needs to actually be computed with the remaining formula $x_0y_1 - x_1y_0$, and can be returned as a scalar instead of a redundant three-element vector with x- and y-components being zeros.

The cross product returns the **signed area** of the **parallelogram** defined by the vectors (x_0, y_0) and (x_1, y_1) . Swapping the order of the arguments therefore negates the sign of the area. Since the signed area formula features only addition, subtraction and multiplication, it can be applied to integers, rational numbers and real numbers. Here, the real important feature of the signed area formula is the guarantee to produce an integer result when applied to integer arguments.

The concept of a negative area might first seem confusing, but it has an important purpose in the operation that in most online material is succinctly known as `ccw`, so we shall also follow that naming here. (In fact, just googling for "`ccw computational geometry`" gives you several good pages and sets of slides that illustrate how this operation cuts through seemingly complex geometry problems faster than a chainsaw goes through butter!) This operation allows us to implement all methods from these four labs with absolute accuracy without any rounding errors or singularities. Among the subfields of computer science, computational geometry is especially notorious for how its problems have all kinds of tricky edge and corner cases (heh) that naive implementations of

these algorithms tend to have trouble dealing with, especially if implemented using floating point arithmetic with a happy-go-lucky attitude.

The name `ccw` is short for “counterclockwise”. Given three points (x_0, y_0) , (x_1, y_1) and (x_2, y_2) on the plane, the sign of the signed area of the triangle from (x_0, y_0) to (x_1, y_1) to (x_2, y_2) tells you the orientation of these three points with respect to each other.

- If the method `ccw` returns any **positive** number, these three points are oriented **counterclockwise**, so the turn in the middle when moving from (x_0, y_0) to (x_1, y_1) to (x_2, y_2) is **left-handed**.
- If the method `ccw` returns any **negative** number, these three points are oriented **clockwise**, so the turn in the middle when moving from (x_0, y_0) to (x_1, y_1) to (x_2, y_2) is **right-handed**.
- If the method `ccw` returns zero, the three points (x_0, y_0) to (x_1, y_1) to (x_2, y_2) are **collinear**.

In practice, all the problematic edge and corner cases in the computational geometric algorithms in these labs emerge in situations where three points are collinear, even overlapping. Frankly, it is not surprising that problems of points and line segments often begin with the order to “assume the general position” where no three points are collinear and no four lie on the same circle...

And that’s it! That’s all she wrote and said! The method `ccw` is the fundamental building block of all computational geometry operations that we shall now implement. In fact, if you remember how the method `compareTo` for custom order comparison returns its result as an integer whose sign contains the entire answer and the magnitude of the returned number is meaningless, the `ccw` operation is the two-dimensional generalization of the one-dimensional order comparison operator! (In fact, checking whether $a < b$ on the one-dimensional line of real numbers reduces to checking whether the turn defined by the three tactically chosen points $(a, 1)$, $(a, 0)$, $(b, 0)$ is left-handed.)

To practice using this wonderful `ccw` operation and appreciate it in action, here is finally your first method to write in this lab, in a new class named `CompGeom`.

```
public static int lineWithMostPoints(int[] xs, int[] ys)
```

For simplicity, a set of points on a two-dimensional plane is given to these methods as two arrays `xs` and `ys` (read these two names out as “exes” and “whys”, as the letter *s* denotes the plural form) so that `xs` contains the *x*-coordinates of these points, and `ys` contains the *y*-coordinates. For example, the *x*- and *y*-coordinates of the point (x_4, y_4) can be found in the elements `xs[4]` and `ys[4]`. These points have not been otherwise sorted in any manner, but can be given in any order.

Given a set of n points on the plane, find the line that contains the largest number of points from that set. Since there can exist several such lines for the given set of points, this method returns only the count of how many points are on that line, to make automated testing feasible.

Any two distinct points determine the unique line that goes through those two points. The `ccw` operation then tells you quickly whether some third point also lies on that same line. You can therefore solve this problem with two nested loops that examine all possible pairs of points (and therefore all

possible lines defined by such pairs of points), with a third level of innermost loop finding all the other points in the same line. However, since the **cubic** running time of $O(n^3)$ from three levels of nested loops can be a bit heavy, you should do whatever you can to eliminate redundant comparisons to speed up the execution of this method.

With that out of the way, it is time to implement an important operation that will be used as building block for many other computational geometry algorithms: determine whether the line segment with endpoints (x_0, y_0) and (x_1, y_1) intersects another line segment with endpoints (x_2, y_2) and (x_3, y_3) , that is, whether these two line segments have at least one point in common.

```
public static boolean segmentIntersect(  
    int x0, int y0, int x1, int y1, int x2, int y2, int x3, int y3  
)
```

One of the famous programming [epigrams](#) by Alan Perlis points out that if your function takes ten parameters, you probably forgot some. We are two short of that magic number here, but the point (or more specifically in this case, the lack of `Point`) remains just as valid.

Once again: no angles, no slopes, no trig, no nothing that could ever produce anything that is not an exact integer, since we can solve this important problem with absolute accuracy even in all possible corner cases with the following approach. Start with a simple **quick rejection test** by comparing the **bounding boxes** of these line segments. Bounding box comparison is the basic computational geometry technique to speed up all intersection tests between complex objects, but it works just as well here with simple line segments. If the higher of the x -endpoints of either segment is strictly less than the lower of the x -endpoints of the other segment either way, these two segments cannot possibly intersect. The same test is applied to y -coordinates of these endpoints. In practice, most of the time we will receive the negative answer from quick rejection tests, since most pairs of line segments on the plane are mutually distant enough for this quick rejection test.

The method moves to the next stage only if the bounding boxes intersect. Then, the segment with endpoints (x_0, y_0) and (x_1, y_1) intersects the segment with endpoints (x_2, y_2) and (x_3, y_3) if the turns $(x_0, y_0) - (x_1, y_1) - (x_2, y_2)$ and $(x_0, y_0) - (x_1, y_1) - (x_3, y_3)$ do not have the same nonzero handedness. In other words, the points (x_2, y_2) and (x_3, y_3) are not on the same side of the line segment (x_0, y_0) and (x_1, y_1) , but the line segment from (x_2, y_2) to (x_3, y_3) actually crosses that line so that an intersection can occur. The same test must be repeated with the roles of two segments interchanged, since the segments must cross each other both ways to intersect.

This logic will take care of the intersection test between two arbitrary line segments, even in the degenerate case where one of these segments is a single point. To find all intersections within a set of n such line segments, of course you can loop through all $n(n - 1)/2$ pairs of segments, which would become a prohibitively inefficient “Shlemiel” algorithm as n increases. Assuming that the segments have been spread around reasonably evenly, a [sweep line algorithm](#) similar to the one that took apart the earlier Manhattan skyline problem will do this job in expected linear time. This algorithm would be slow only for pathological special cases such as these line segments being stacked vertically in a slab around the same x -coordinate.

Lab 42: Geometry II: Polygons

JUnit: [CompGeomTestTwo.java](#)

The previous lab of computational geometry introduced the `cross` product and `ccw` operations as the fundamental building blocks of our geometric algorithms. We now move on from sets of n points to ordered lists of n points that define an n -sided [polygon](#) whose edges are the line segments connecting the consecutive points in this list. The line segment from the last point in the list back to the first point closes the polygon. Depending on the needs of the current situation, we can think of the polygon being made up of “edge segments” or being made up of “corner points”, but that is just a different “point” of view to the same integer data. (Surely it is not a coincidence that our everyday language is rife with geometric metaphors.)

The line segments determined by an arbitrary list of points do not necessarily look like a “polygon” as we generally understand this term. For the purposes of this lab, a list of points defines a [simple polygon](#) if it satisfies the following two constraints:

1. No three consecutive corner points are collinear.
2. No two non-consecutive edge segments intersect, not even in a corner point.

Of course every two consecutive edge segments always meet in their common corner point so that no holes are left in the polygon boundary, but no edge segment intersections exist other than the n trivial intersections at the corner points. The `ccw` and `segmentIntersect` methods from the previous lab should make short work of writing the following method:

```
public static boolean isSimplePolygon(int[] xs, int[] ys)
```

This method determines whether the list of points whose coordinates are given in two n -element arrays `xs` and `ys` form a simple polygon when their points are read in that order.

Technically, the first condition of non-collinearity of three consecutive corner points is not necessary for a polygon to be “simple” as that concept is generally defined. However, eliminating redundant corner points guarantees that every corner point is a literal “corner” so that a sudden discontinuous turn definitely takes place in that point. (You could always modify this method to take an additional `boolean` parameter to indicate whether the collinearity avoidance rule is in effect.)

Furthermore, performing any **cyclic shift** to these corner points changes nothing, but the shifted points still represent the exact same simple polygon as a geometric object on the plane. The same would hold also for reversing the list of corner points, although some material on computational geometry imposes an additional condition that the polygon corner points must be listed in the order that follows the boundary of the polygon in counterclockwise direction. Such a guarantee would make some later operations on polygons easier to implement. (We have encountered this principle of unique canonical representation for objects before, such as with the `Fraction` class.)

Only the sign of the result of `ccw` has any effect on our decisions, whereas the magnitude of that result is irrelevant. Akin to how one-dimensional sorting algorithms only compare elements for order and never need to use arithmetic to find out the absolute difference between elements, **topological** algorithms whose results are unaffected by arbitrary scaling, shifting and rotating of these points, as a rule of thumb should never need the actual magnitude of the result for anything. However, this magnitude does still have a meaning that can be put to good use in another important problem, namely calculating the **area** of the given polygon.

Recall from the previous lab that the cross product of two vectors is the signed area of the parallelogram defined by those vectors. Therefore, the signed area of the triangle defined by three corner points (x_0, y_0) , (x_1, y_1) and (x_2, y_2) is exactly half of `ccw(x0, y0, x1, y1, x2, y2)`. This leads to the immediate observation that **the area of a triangle whose corners lie at integer lattice points can only be some integer or some integer plus one half**, regardless of how these three points lie on the plane. The result is always this neat regardless of whatever square roots their side lengths, and whatever irrational numbers their angles happen to be.

Since any simple polygon can be broken down into disjoint triangles sharing edges and corner points, the same immediately follows for areas of arbitrary simple polygons. (As with many other theorems of computational geometry that feel to our intuition as if they were “ so very clear and rational to anyone who does not belong to the oppressor class ”, this result is a bit more, ahem, *complex* than that.) **Triangulation** would then immediately provide us a way to compute the area of any simple polygon. Efficient algorithms have been developed to rapidly triangulate any polygon into a partition of disjoint triangles. (These algorithms have to tread carefully if the polygon to be triangulated is not convex; the greedy approach of just taking three consecutive corner points and clipping that triangle on its own, then recursively triangulating the rest will fail for such polygons.) However, we don’t actually need to perform any triangulation to merely compute the area of a simple polygon, since we can compute this area in guaranteed linear time from the list of edges using the [shoelace formula](#), courtesy of the awesome mind of [Carl Friedrich Gauss](#) himself!

```
public static int shoelaceArea(int[] xs, int[] ys)
```

Since the area of a lattice polygon can only ever be an integer or an integer plus one half, this method should return the area of that polygon multiplied by two, so that the result will always be an integer. The caller can always divide this result by two for the real area, but this way we can return 30 when the actual area is 15, and return 31 when the actual area is 15 1/2, so that the caller will be able to tell those two situations apart while using only integer arithmetic.

(This idea of scaling fractional values to be integers generalizes for other computations that involve fractional quantities. For example, we can sometimes represent money as cents instead of dollars and not have to deal with the fractional parts that typically don’t have an exact representation in floating point anyway. You should still always be careful with potential integer overflows these days when a billion here and a billion there is needed to add up to some real money.)

Choose any reference point from the plane. It doesn’t even matter which point you choose, or whether that point lies inside or outside the polygon. Given such freedom, we might as well choose

the origin $(0, 0)$ as this reference point to simplify the calculations, even though any other point would have worked just as well. Initialize the integer `area` variable to zero. Loop through the consecutive edge segment with corner points (x_i, y_i) and (x_{i+1}, y_{i+1}) , and add the result of `ccw` of the reference point and those two points to your `area`. After this one loop, return the final value of the `area` as your answer, wham bam thank you ma'am, that's all the fat lady sang.

Honestly, it really is that simple. Since the `ccw` function returns a signed area whose sign depends on the direction of the edge segment relative to the reference point, the additional areas outside the polygon that get added to the `area` during the execution of this algorithm will be subtracted on the way back around the other side of the polygon. Regardless of how the polygon points lie and along however complex fractal shape the polygon boundary might meander back and forth through the plane, these positive and negative areas outside the polygon will always perfectly cancel each other out. Only the area inside the polygon will never get cancelled out that way, so the final result will be exactly equal to the area of that polygon.

Lab 43: Geometry III: Points In Polygons

JUnit: [CompGeomTestThree.java](#)

Intersection and **collision** tests (the former for **static** worlds, the latter for **dynamic** worlds) are important in many games that take place on a two-dimensional plane. This lab gets that ball rolling with the simplest possible such intersection test to determine whether the given point lies inside a particular polygon. More advanced algorithms exist to perform efficient intersections and collision tests between arbitrary polygons, disks and other complex geometric objects. To speed up the intersection test between two complex objects such as entire **polygon meshes** of some video game character, we often employ a **quick rejection** test that first checks whether some simple **bounding boxes** around those two objects intersect will provide the negative answer in the vast majority cases, without the need to actually execute the expensive intersection test between the actual complex objects themselves.

A simple polygon is **convex** if for any two points (x, y) and (x', y') inside that polygon, the entire line segment between these two points also lies fully inside the polygon. Intuitively, a convex polygon has no “pits” on its edges. Convex polygons are important in computational geometry, since many problems on polygons become much simpler if the polygons involved are known to be convex. Every **triangle** is automatically convex as it came into this world (the same claim does not hold for quadrilaterals of four points), which is one major reason why triangles and **triangle meshes** are the fundamental shape primitive in computer graphics.

During the traversal of the boundary of a convex polygon, every turn must necessarily have the same handedness. Either every turn is left-handed, or every turn is right-handed, but both kinds of turns cannot simultaneously exist on the boundary of the convex polygon. Under our additional voluntary constraint of counterclockwise listing of the corner points, only left-handed turns can exist. These observations give us an immediate linear time algorithm to determine whether a given polygon is convex. The very same idea can be extended to determine whether the given point (x, y) lies inside the given convex polygon.

```
public static int pointInConvex(int[] xs, int[] ys, int x, int y)
```

The arrays `xs` and `ys` are guaranteed to be corner points of some convex polygon, and in this problem further guaranteed to be listed in counterclockwise direction. Instead of returning a mere yes/no answer as a `boolean`, it is more useful for our later purposes to define this method to make the distinction between different ways for the point to be inside the polygon. This method should return the answer as an **integer ranging from zero to three** so that the result zero means that the point (x, y) lies outside the polygon. Result one means that the point (x, y) is one of the corner points. Result two means that the point (x, y) lies on some edge segment but is not a corner. Finally, the result three means that (x, y) lies properly inside the polygon.

The caverns and tendrils of an arbitrary polygon can easily sprout into fractal-like complexities, so the question of point containment in a non-convex polygon is not quite as trivial as it is for their convex brothers. The first idea, same way as in the earlier problem of computing the area of a polygon, might be to subdivide the polygon into disjoint triangles, and then test for point containment

within these triangles with the previous method. If some triangulation of that polygon is already available, there certainly is no harm in using it, especially if those triangles have been arranged and sorted into horizontal **slabs** so that we can first use **binary search** to determine the slab where the point lies according to its y -coordinate, and then another binary search to determine the triangle that the point lies according to its x -coordinate. Otherwise solving one instance of this problem via triangulation would be a textbook example of overpaying for overkill, since a far simpler algorithm already solves this problem for arbitrary polygons in linear time!

```
public static int pointInPolygon(int[] xs, int[] ys, int x, int y)
```

Same as the previous method for point inside convex polygon, this method should return the answer zero for outside, one for corner, two for edge and three for inside. However, this method must work for any polygon, even those that are not convex or not even simple; the JUnit test will generate polygons that intersect themselves and degenerate to line segments and points. The first stage should be the loop around the polygon corner points to determine whether (x, y) is one of the corners, or is lying on one of the edge segments. Past that hurdle, the rest of the algorithm can safely assume that no edge segment of the polygon intersects the point (x, y) itself.

To visualize the [ray casting technique](#) used to determine whether the point (x, y) is inside the polygon, imagine standing on that point and then start walking along some straight **ray** that shoots from its starting point into some arbitrary direction. (Technically, a line segment has two endpoints, a ray has only one, and a line has none but is infinite to both directions. Most people erroneously say “line” when they *<fedora>ackshually</fedora>* mean to say “line segment”, but this is a long-lost linguistic hill left for only the most valiant and euphoric gentlesir to bravely die on with his blade.) Since we get to choose this direction freely, we might as well choose this direction to be one of the main axis directions with one coordinate being zero, to simplify the formulas involved. Any one of the four compass directions would result in a symmetric method, but since we have to pick one of these four, let us follow the usual convention and pick the ray shooting right towards infinity, eternally in parallel with the x -axis.

Since we don't have the math to be able to deal with rays (this could be done as a trivial special case of parameterized curves, but even the simplest of those soon tend to end up requiring non-integer arithmetic), we realize that it is enough to look at the line segment from point (x, y) to the right just far enough so that its endpoint is guaranteed to lie outside the polygon. Any x -coordinate larger than the maximum x -coordinate among the corner points of the polygon will do just fine. Since the end position $(1 + \max x_i, y)$ of our rightward ray (heh, another good name for some pompadour dad rock band) is known to lie outside the polygon, and every edge crossing along the segment moves either from inside to outside, or from outside to inside. The parity of the number of polygon edge crossings from the starting point (x, y) to the end position $(\max x_i + 1, y)$ tells us the answer right away! Any odd number of crossings means that the starting point (x, y) was inside the polygon, whereas any even number proves that it was outside.

A trivial modification of the previous “even-odd algorithm” to use the “[nonzero winding rule](#)” makes this algorithm even more general so that it can handle even non-simple polygons whose edges can freely intersect each other. In such a polygon, the issue of what the words “inside” and “outside”

even mean can become as frustrating as trying to get to the outside of a [Klein bottle](#), but the non-zero winding rule handles such situations without confusion.

Initialize the local `count` of crossings to zero. Whenever the ray segment intersects the polygon edge segment between two polygon corner points (x_i, y_i) and (x_{i+1}, y_{i+1}) , the sign of a `ccw` check from point (x, y) to these two points tells us which direction we are crossing that edge. If the sign is negative, decrement the `count` by one, and if the sign is positive, increment the `count` by one. The point (x, y) is inside the polygon if and only if the final `count` is nonzero. For simple polygons, this rule gives the same answer as the previous even-odd rule, but for self-intersecting polygons, the results will be different so that the polygon has holes inside it that are not connected to the world outside the polygon. To pass the JUnit test, **your `pointInPolygon` method should use the nonzero winding rule for counting the intersections, instead of the even-odd rule.**

Before anybody starts implementing this algorithm by actually traversing through that rightward ray in their code, we must emphasize that the idea of walking along this ray is only an aid for your intuition. The actual algorithm should loop through the edge segments of the polygon in one linear-time for-loop. For each edge segment, check whether it intersects the ray segment between the points (x, y) and $(\max x_i + 1, y)$. We can do this with the `segmentIntersect` method from the first geometry lab.

However, as is typical for algorithms of computational geometry, a difficult corner (heh) case lurks in the grass waiting to trap the unwary. Using concrete numbers to illustrate the danger, let $(2, 3)$ be the point (x, y) whose containment inside the polygon we wish to determine, and let the polygon have two edge segments that connect the three consecutive corner points $(5, 1)$, $(7, 3)$ and $(7, 5)$. The rightward ray emanating from the starting point $(2, 3)$ will therefore intersect both of these line segments in their shared corner point $(7, 3)$. The algorithm that looks at each edge segment separately will therefore count two edge crossings at one corner point, which makes the parity of the final count to be the exact opposite of the correct answer!

In moments like this, it is often good to step back for a moment for a more calm and reasoned view. Cowboy coders (not a compliment) would add extra conditions to cover such failing test cases, instead of solving the general problem for all cases in one swoop. **However, any such fix that talks about some special direction is automatically wrong, even if it happens to solve the issue for that particular test case.** Since the result of any topological problem is unchanged by arbitrary planar rotation, a bug fix in any such problem can never depend on any particular “special” point or direction. The fix needs to be more general so that it works under arbitrarily mirroring, rotation, translation or any other transformation of the original points, as long as that transformation does not affect the topological properties of the original set of points. We want to be correct in general, not just *ad hoc* for the test cases that the testers thought to write up.

The correct fix is actually pretty clever in its simplicity once we realize it. Before applying the intersection test, we should do a quick rejection test to eliminate all polygon edge segments that cannot possibly intersect our rightward ray from the point (x, y) . If both endpoints of the edge segment have a y -coordinate lower than our y , the segment lies fully below the ray and no intersection can exist. The same happens when both endpoints have a y -coordinate higher than y . To fix the previous bug, **treat the y -coordinate of the segment endpoint that is equal to y as if it were an infinites-**

imally small distance lower than y for the purposes of this quick rejection test. You don't actually need infinitesimals to exist in your number system, *you just need to act as if they did*, and the end result is the same. (This simple flash of insight generalizes for many other problematic situations in programming that you might encounter in the future.)

Using the previous numbers, the edge segment from $(5, 1)$ to $(7, 3)$ will now be rejected as lying fully below the rightward ray from the starting point $(2, 3)$, and therefore that segment does not increment the crossing count despite the fact that it technically did intersect the rightward ray at the corner point $(7, 3)$. However, the next edge segment from $(7, 3)$ to $(7, 5)$ is not quick-rejected, as the point $(7, 3)$ is considered to lie below the ray whereas the other endpoint $(7, 5)$ clearly lies above it. That edge segment increases the crossing count by one, as it properly should.

Lab 44: Geometry IV: Convex Hull

JUnit: [CompGeomTestFour.java](#)

As we saw in the previous three labs on computational geometry, a list of points can be thought of either as these points existing independently as points on the two-dimensional plane, or as the ordered corner points of an n -sided polygon so that the consecutive corner points are implicitly connected by polygon edge segments between those points. To conclude this journey through the calm surface of computational geometry above all its deep complexities, we will tackle the problem of building a simple polygon out of the list of points given in arbitrary order. In general, exponentially many different simple polygons could be constructed from the given set of distinct points. However, as always when we are granted the luxury of such a choice, we might as well try to build a polygon that has some kind of useful properties for the rest of our program to enjoy.

This fourth geometry lab is also a good place to introduce a technique of **indirection** that often comes handy not just in computational geometry, but in many other problems in data structures and algorithms. When the same points are shared between multiple polygons or other geometric objects, such as when constructing a [triangle mesh](#) to represent a **polyhedron** surface where several triangles meet at their shared corner points, the ability to move these points without breaking the connections between these triangles would be highly desirable.

If every triangle is an island that knows only the absolute coordinates of its own corner points but nothing about the existence of other triangles that share common corner points, moving these corner points will affect only that triangle, but not any other triangles. To avoid creating ugly cracks on the polygon mesh, we would have to somehow find all other triangles that use the same corner point, and perform the same update to that corner point in each triangle separately. To avoid this hassle, maintain the set of all relevant points separately, let's say for simplicity as the two arrays `xs` and `ys` we have already seen before. A polygon of n such points is encoded as an n -element array of integer indices to the arrays `xs` and `ys` that contain the actual point coordinates.

For example, a triangle encoded as the array `{42,17,99}` would consist of the corner points whose actual coordinates are stored in the positions 42, 17 and 99 of the arrays `xs` and `ys` for lookup. These arrays `xs` and `ys` could potentially contain millions of other points, most of which are not part of this particular triangle. Another triangle encoded as the array `{42,43,77}` shares the corner point with index 42 with the previous triangle. Adjusting the coordinates of the point number 42 will then automatically adjust every triangle that shares the same corner point, without us needing to explicitly search for all such triangles and painstakingly update them all! (In some problems, additional computation may be required to maintain additional constraints that prevent the mesh from becoming degenerate in some form.)

Enough chit chat. Given a set of points as two arrays `xs` and `ys`, the first method in this week's lab will construct a **fan polygon** by sorting the given set of points in counterclockwise fashion, when viewed from the reference point with the lowest y -coordinate. If the set contains several points with the same minimum y -coordinate, use the point with the lowest x -coordinate among those points as the reference point. Note that neither method in this lab should change the contents of the `xs` and

ys arrays given to them in any way, but create and return a new array of position indices that lists these points in the order in which they constitute the polygon corners.

```
public static int[] sortCCW(int[] xs, int[] ys)
```

Given a set of n points with coordinates given in two arrays `xs` and `ys`, this method should first create an n -element array `Integer[] result`, and fill it with the integers $0, \dots, n - 1$. Next, define a local **strategy** class `Comparator<Integer>` to create objects that will perform the comparisons needed in sorting the array of indices.

```
class CCWComp implements Comparator<Integer> {
    public int compare(Integer i, Integer j) {
        // Fill in this method body
    }
}
```

The method `compare` inside this local class should order compare the two indices `i` and `j` according to the following ordering rules between indices:

1. The index of the reference point is always smaller than any other index.
2. If the turn from the reference point through point `i` into point `j` is left-handed, then `i` is considered lower than `j`.
3. If the turn from the reference point through point `i` into point `j` is right-handed, then `i` is considered higher than `j`.
4. If the reference point is collinear with the points `i` and `j`, whichever of the points `i` and `j` has a lower y -coordinate is considered lower. If both points have the same y -coordinate (another possible edge case that is not immediately obvious but can happen), the point with the lower x -coordinate is considered lower.

These rules allow the `compare` method to always return either -1 or $+1$ for any two distinct indices `i` and `j`, and the resulting ordering is total. Armed with this comparator, the `sort` method in the `Arrays` utility class can take care of all the hard work involved in the actual sorting:

```
Arrays.sort(result, new CCWComp());
```

All that is left for you to do is to fulfill the letter of the law by creating an `int[]` that is going to contain the actual result, and copy the contents of the `Integer[]` into that array of the promised return type before returning it.

The fan polygon is not necessarily simple, since it can contain degenerate line segments without an area. This will happen, for example, with the points $(0, 0), (1, 0), (2, 0), (2, 1), (1, 1), (2, 2)$ that become a fan polygon in this order, with the origin $(0, 0)$ as the lowest point that becomes the reference point. The rest of the points end up listed in the order of increasing angl... nope, we were never supposed to use that bad word that Old Nick himself conjured up to lead us astray from our path of

righteousness! Try again. Points end up sorted in counterclockwise order relative to the reference point in the lower left, explaining the name `sortCCW` given to this method.

Since the boundary of the resulting polygon can contain both left and right turns, the fan polygon is not necessarily convex. It is guaranteed to be a [star-shaped polygon](#), though, meaning that there exists at least one point inside it so that standing on that point, you could “see” every other point inside that room by turning around in place. The so-called “star” shape of such a polygon might not be something we consider to be even a cartoon blammo star. All these names are made up anyway to bolster our intuitive feel of the abstract subject and this way enlist the parts of our brain that deal with spatial intuition to aid in understanding such concepts. Convex polygons are a further special case of star-shaped polygons whose every point would work as that standing point. Since the constructed polygon contains all of the original points, it will also have redundant corner points in places where three consecutive points are collinear.

A fan polygon can be swiftly converted into a convex polygon with several different algorithms, of which we shall implement the one known in literature as [Graham scan](#). Other algorithms also exist to find the [convex hull](#) of the given set of points. In fact, the convex hull problem is a two-dimensional generalization of one-dimensional **comparison sorting** problem, a fact humorously highlighted by how the famous convex hull algorithms **quickhull** and **mergehull** are straightforward two-dimensional generalizations of the classic quicksort and mergesort algorithms originally designed for efficient one-dimensional comparison sorting!

(For a reduction to the other direction, to sort a series of integers $x_0, \dots, x_n$, simply construct the convex hull of the set of points (x_i, x_i^2) and read the answer in order around the hull. The existence of this reduction, even if never actually used in practice, establishes that the convex hull cannot be constructed essentially faster within the same model of computation than the time required to sort an array of one-dimensional such numbers.)

We implement specifically the Graham scan algorithm in this lab basically because the previous `sortCCW` method already did half of its work. Besides, this lets us finally use a **last-in-first-out stack** in the implementation of this algorithm, which turns this lab into an even more useful exercise in programming in general. Perhaps due to their deceptive simplicity, LIFO stacks are not always fully appreciated during the first and second year computer science curriculum, whereas queues and priority queues tend to find more use in algorithms encountered during that period.

```
public static int[] grahamScan(int[] xs, int[] ys)
```

This method should start by calling the previous method `sortCCW` to acquire the fan polygon listing of the set of points given in the parameter arrays `xs` and `ys`. It should then create a stack (you can use the actual `Stack<Integer>` in the `java.util` package, or simply simulate a stack with an `int[]` and an index variable, canonically named `top`, whose value equals the current size of the stack) and push the first two indices of the fan polygon in this stack. The rest of the algorithm should then loop through the rest of the points `i` of the fan polygon and obey the following rules of what to do for each such point:

1. If the stack contains only one element, or if the turn from the current top two elements of the stack to the point i is left-handed, push i into the stack.
2. While the turn from the current top two elements to the point i is collinear or right-handed, pop the topmost element of the stack.

The second rule must be implemented as a nested while-loop inside the outer for-loop through the fan polygon points, because in principle there is no upper limit on how many points the new point can theoretically eliminate from the stack. You might want to watch some animation of Graham scan in action to see for yourself why and how this happens. (Relaxing afterwards, you can also ponder why this algorithm is guaranteed to work in linear time despite the fact that it consists of two levels of nested loops, since [the answer to this question](#) generalizes to the study of such algorithms in general.) Once the outer for-loop has finished processing all corner points of the fan polygon, the stack contains precisely the points that lie on the convex hull. DUCY?

Each of the four labs of computational geometry has a separate JUnit test class to test only the methods of that particular lab. However, the `verifyPicksTheorem` method will act as the Final Boss to challenge the students who complete all four geometry labs in this fourth JUnit test class, these labs uniting to battle the Final Boss somewhat like all those Avengers suddenly popping out of magic wormholes to defeat Thanos and his cronies. This pseudorandom fuzz test repeatedly creates a set of random points on the plane, and uses Graham scan to compute their convex hull polygon. The area of this polygon is then computed with both the shoelace method and a brute-force implementation of [Pick's Theorem](#) that computes the area of a simple lattice polygon as a function of how many lattice points fall inside that polygon and on its edges. As we observe both of these methods for area calculation return the same area for a large number of such randomly generated convex hull polygons, our faith in the general correctness of these methods is greatly increased.

Determining which points are inside and on the edges of the convex hull works as a mass test to verify that `pointInConvex` and `pointInPolygon` methods always return the same answer. Whenever two totally different algorithms agree on all answers for the large number of randomly generated arguments given to these algorithms, our belief in the correctness of both algorithms is massively increased. Philosophically minded students can put their finger on every item, and ask about each one why that is so.

Lab 45: Permutations I: Algebraic Operations

JUnit: [PermutationsTestOne.java](#)

This next troika of problem specifications covers the computational operations of [permutations](#), integer arrays of n elements that contain each natural number from 0 to $n - 1$ exactly once. The required methods in these three labs may assume that the arguments given to them by the JUnit fuzz tests satisfy this definition. Since we are computer programmers, we shall once again follow the secret pact of our ancient brotherhood of extreme finitists, sworn to Saint Dijkstra to always start counting positions from zero, whereas the fancy pants real mathematicians prefer to define permutations with the numbers 1, ..., n , as if India's greatest gift to the world had never existed.

Even though permutations are represented as arrays of integers, a permutation can conceptually be thought of as a `<DrEvil>"function"</DrEvil>` that can be "applied" to any array of n elements. Since permutations themselves are also such arrays of n elements, permutations can be applied to existing permutations to create new permutations, making Keanu say "whoa" once again. This "function" copies each original element exactly once into the result, to the position given by the permutation array. For example, if `perm` is a permutation for which `perm[4]=9`, applying it to an array `arr` where `arr[4]=="Bob"` will see "Bob" in the position 9 in the result array.

Analogous to matrix multiplication in linear algebra where the product of any number of n -by- n square matrices is also an n -by- n square matrix, the combination of any number of permutations on n elements is itself a permutation of n elements. In fact, permutations can be implemented as binary [permutation matrices](#) so that each row and each column contains exactly one element that equals 1, and now all the known truths of **linear algebra** can be applied to its special case of permutations. (Astronaut 1: "Wait, it was all just linear algebra?" Astronaut 2 [*aiming gun*]: "Always was.")

Enough theory, let's see some action. Create a class `Permutations` in your labs project window. You will be writing all methods in this set of three labs into this class. The methods in this first lab deal with the **algebra** of combining permutations to create new permutations.

```
public static int[] chain(int[] p1, int[] p2)
```

Combine the two permutations `p1` and `p2`, assumed to be of the same size, into a result permutation `p` that represents the operation of first applying permutation `p2` to the given array, followed by applying permutation `p1` to this intermediate result. For example, if `p2[3]==4` and `p1[4]==9`, then `p[3]==9` moving the element right there instead of first visiting the intermediate position 4 in between.

```
public static int[] inverse(int[] perm)
```

Given the permutation `perm`, construct its **inverse permutation** that, if chained with `perm`, would produce the **identity permutation** for which `p[i]==i` for all positions `i`. For example, when called with a permutation array `{2, 1, 4, 0, 5, 3}`, this method would create and return the permutation array `{3, 1, 0, 5, 2, 4}`.


```
public static int[] square(int[] perm)
```

The square of the permutation is constructed by chaining it with itself. This method should not be too difficult once you have written the method `chain`, a mere one-liner.

```
public static int[] power(int[] perm, int k)
```

Compute the k :th power of the given permutation `perm`, that is, the result of chaining `perm` with itself k times. For negative values of k , this method should return the $-k$:th power of the inverse of `perm`. With the rule that $k = 0$ always gives the identity permutation, this operation is well-defined for arbitrary integer powers k , both positive and negative.

Since k will get pretty large in the JUnit tests (since the element values never change, permutations play nicer than integer matrices so there can be no possibility of overflow), you should use the technique of [exponentiation by squaring](#) to compute the result much faster for large k than you would get by chaining `perm` with itself $k - 1$ times. The base cases of this recursion are $k = 0$ where the result is the identity permutation, $k = 1$ where the result is just `perm`, and $k = 2$ where the result is the square of `perm`. For higher values of k , apply the identities $p^{2k} = (p^2)^k$ and $p^{2k+1} = p(p^2)^k$ that cut the exponent in half in each step. This ought to make this method blazingly fast even for k not just in the millions but even in the vigintillions, were the range of our integers that stratospheric.

perm	k	Expected result
{1, 0, 2, 3}	4	{0, 1, 2, 3}
{1, 3, 6, 0, 5, 2, 4, 7}	98	{3, 0, 4, 1, 2, 6, 5, 7}
{0, 6, 3, 4, 8, 7, 2, 9, 5, 1}	-397	{0, 9, 6, 2, 3, 8, 1, 5, 4, 7}

Students who aim to take math courses, especially a course on **abstract algebra**, can note how permutations and their operations `chain` and `inverse` define [an algebraic group](#), with the method `chain` standing for $p_1 + p_2$ and `inverse` standing for $-p$ in this group, the identity permutation acting as the unit element of these operations. Powerful algebraic truths and useful algorithms such as **exponentiation by squaring** immediately spring forth from this realization. (Besides, [permutation groups](#) are a big cheese in group theory, thanks to [Cayley's theorem](#).)

Lab 46: Permutations II: Cycles

JUnit: [PermutationsTestTwo.java](#)

Continuing the previous lab, a permutation is an array of positions that describes where each element moves to when that permutation is applied to some array. The n -element array therefore already implicitly contains all possible information about that permutation. However, to allow easier computation of some interesting properties of this permutation and its behaviour, it is often handy to break down the permutation into its disjoint **cycles**. For a general idea of how these cycles work, you should first watch the YouTube video “[Cycle Notation of Permutations](#)” at [Socratica](#). Mathematically minded students who are interested in how cycles tend to work out in random permutations can check out the post “[Random Permutations, Part 3](#)” by [John Baez](#).

To find the cycle of `perm` that the position `i` is part of, start a loop from position `j=i`. Keep moving from the current position `j` into the position `perm[ j ]`, until you return to the starting position `i` and have met all the other members of that cycle along the way. (It is also quite possible that `perm[ i ]==i`, making that cycle a **singleton**.)

```
public static List<List<Integer>> toCycles(int[] perm)
```

Since every cycle in `perm` is a list of integers, the list of such cycles must therefore be a list of lists of integers. This method should compute and return the **canonical cycle representation** of `perm` as disjoint cycles so that each individual cycle is listed starting from its largest element, and the cycles are listed in the ascending order of their first elements. The following table showcases the expected result for some random permutations for various n . As the second and third row of this table demonstrate, a permutation can consist of a single cycle that all these elements ride together as if piled on one of those tandem bikes that is so large that it automatically becomes comical. In the extreme case at the other end of this lattice poset, every position rides its own unicycle.

perm	Expected result
{1, 0, 2}	[[1, 0], [2]]
{2, 4, 1, 0, 3}	[[4, 3, 0, 2, 1]]
{0, 1, 2, 3, 4}	[[0], [1], [2], [3], [4], [5]]
{0, 5, 2, 3, 4, 1}	[[0], [2], [3], [4], [5, 1]]
{4, 6, 1, 5, 3, 0, 2}	[[5, 0, 4, 3], [6, 2, 1]]
{12, 7, 13, 1, 2, 5, 11, 0, 10, 14, 3, 6, 9, 4, 8}	[[5], [11, 6], [13, 4, 2], [14, 8, 10, 3, 1, 7, 0, 12, 9]]

Since the permutations and their canonical cycle representations are one-to-one, we should naturally also write the inverse function that constructs the permutation from the list of its cycles.

```
public static int[] fromCycles(List<List<Integer>> cycles)
```

In fortunate situations like this where we have both a function and its inverse available as black boxes, an effective automated fuzz test is trivial to implement. Repeatedly create a permutation (either systematically or randomly), convert it to cycle form, translate that cycle form back to the explicit permutation that it represents, and verify that this permutation is same as the original permutation. Fuzz tests can be left running overnight to strengthen our faith in the correctness of the methods when we see no discrepancies with our morning coffee. Trying out a million random permutations at different scales should realistically at some point hit and reveal all possible edge cases and corner cases that this method could reasonably have lurking inside it.

One property of permutations that becomes easy to compute from the cycle representations is the **algebraic order**, that is, the smallest positive integer k so that `power(perm, k)` is the identity permutation. In fact, the `power` of a permutation from the previous lab would have been much easier to compute in its cycle form, since we could merely shift each cycle $k \% c$ steps to the right, where c is the length of that cycle. Since the identity permutation keeps every element in its original position, the order k is simply the **least common multiple** of the cycle lengths inside the permutation. For example, if some permutation consists of cycles of length 6, 4, 1, 1, and 2, their least common multiple 12 is enough to rotate every cycle back to its original position.

Parity of the permutation is another important algebraic property that can be easily computed from its cycle representation. This parity tells us something about how many **pairwise swaps** of two distinct elements are necessary to actually act out the permutation in practice. These swaps can then be organized in a host of different ways, but whether an odd or an even total number of swaps is needed to reach the goal is an inherent fixed quantity for that permutation. (Try it for yourself, if you don't believe this. You will inevitably get stuck, no matter which way you twist and turn.)

```
public static int parity(List<List<Integer>> cycles)
```

To compute the parity of the permutation given in its cycle form, you only have to count how many individual cycles have an even number of elements! If there is an odd number of such even-length cycles, return `-1` for the odd parity, and otherwise return `+1` for the even parity, hey bada bing, bada boom. The number of cycles of odd lengths is irrelevant for this calculation.

The somewhat "odd" terminology where minus one stands for odd and plus one stands for even, despite the fact that both of these are pretty odd numbers, comes from the term $(-1)^n$ that often pops up in combinatorial formulas, especially those that involve [the inclusion-exclusion principle](#) where alternating terms are added to and subtracted from the total tally. Values of $(-1)^n$ alternate between `-1` and `+1` depending on whether n is odd or even, explaining the terminology of parity used in this context. Of course, this expression should always be evaluated with $n \& 1 == 0 ? +1 : -1$ in guaranteed constant time, independent of the sign and the magnitude of n .

```
public static String cycles(
    List<List<Integer>> cycles, String alphabet
)
```

Since the canonical cycle notation is more readable than plain arrays, the last method in this lab creates a human-readable representation of the cycles of the given permutation. Instead of using integers 0 to $n - 1$ separated by commas and spaces, the `alphabet` string tells which character to use for each such integer. Using the digits 0–9 as the first ten characters, the lowercase letters a–z as the next twenty-six and the uppercase letters A–Z as the next twenty-six allows us to stringify these permutations without any commas. Parentheses separate cycles from each other, even though these parentheses are actually redundant and could be removed to make this representation even more compact. The parentheses make the expression easier for humans to digest in a glance.

perm	Parity	As canonical cycles
{1, 0, 2}	−1	"(10)(2)"
{2, 4, 1, 0, 3}	+1	"(43021)"
{0, 5, 2, 3, 4, 1}	−1	"(0)(2)(3)(4)(51)"
{13, 10, 12, 1, 4, 6, 5, 11, 14, 7, 8, 0, 3, 2, 9}	+1	"(4)(65)(e97b0d2c31a8)"

Lab 47: Permutations III: Lehmer Codes

JUnit: [PermutationsTestThree.java](#)

Every first course on combinatorics starts with how n distinct elements can be arranged into exactly $n!$ permutations, the product of all positive integers up to n . For example, $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$. Well-known techniques exist to generate all these permutations one at the time in various orders. If you would first like to fool around with some classic techniques to systematically generate all permutations, check out the page “[Generate Permutations](#)” at [Combinatorial Object Server](#).

In this lab, we look at an interesting one-to-one conversion between permutations to integers using [Lehmer codes](#). To explain how Lehmer codes work, we first need to examine the way that integers can be represented as sequences of digits. In our everyday **positional number system of base ten**, the position of each digit indicates the power of ten that the digit should be multiplied by in the total sum of digits. In base ten, there is no need for digits greater than nine, since having ten of some power of ten is equal to **carrying** one to the next higher power instead. For example, we say “nine hundred” but we don’t say “ten hundred”, since the latter carries to the next column to become “one thousand”. (Yet in English, we still do say “eleven hundred”...) Humanity has historically agreed to use the base ten since our hands literally have ten digits (you get it now?), so we tend to assume ten to be the only possible base by some law of Nature and Nature’s God. But the same principle works just as well for any other base, most importantly for base two for **binary numbers**.

This base does not even need to stay the same for every position. **Mixed-radix representations** allow using different bases in different positions. We all do this routinely without even realizing it every time we talk about times and dates in mixed bases of 365/24/60/60/1000, depending on which combination of days, hours, minutes, seconds and milliseconds we are talking about. The base in each position still determines the largest digit that can appear in that position. For example, since every hour has sixty minutes, we must say “one hour and fifteen minutes” instead of “seventy-five minutes” so that values that grow too big for their britches get carried over to the next column.

The process of encoding each permutation into a unique nonnegative integer should maximally exploit the fact that exactly $n!$ permutations can be constructed from n elements. This can be achieved with the [factorial number system](#) whose positions denote the consecutive factorials 1, 2, 6, 24, 120, 720, ... (The linked Wikipedia page starts the factorials from zero, but that is redundant here due to duplication of $0! = 1! = 1$.) The largest digit allowed for the position that corresponds to $k!$ is precisely k , since if you have the term $k!$ repeated $(k + 1)$ times in the sum, you actually already have the next higher factorial $(k + 1)k! = (k + 1)!$ in your hands without realizing it. These factorial positions carry their spillage over to the next position, analogous to the positions in the ordinary positional number system in base ten and other bases.

The following methods implement the conversion between long integers and their factorial number system (“**factoradic**”) representations. Since these methods are each other’s inverses, the JUnit fuzz test only needs to generate random long keys to verify that these functions cancel each other out.

```
public static void toFactoradic(long key, int[] coeff)
public static long fromFactoradic(int[] coeff)
```

Express the integer `key` as a sum of factorials so that the position that corresponds to $k!$ may be used at most k times. Note that the `toFactoradic` method does not return anything, but writes the answer into the array named `coeff` provided by the caller. (The caller is responsible for this array object having sufficient length to contain the answer.) This method should fill in the `coeff` array so that each `coeff[k]` tells how many times the factorial $(k + 1)!$ appears in the breakdown of the integer `key` as a sum of factorials. For example, 2168 breaks down to $2! + 3! + 3(6!)$ that you can verify is equal to $2168 = 1 \times 4 + 1 \times 6 + 3 \times 720$.

key	Expected result (to be filled in the prefix of <code>coeff</code> array)
8	{0, 1, 1}
22	{0, 2, 3}
101	{1, 2, 0, 4}
2168	{0, 1, 1, 0, 0, 3}
61319700746071	{1, 0, 1, 1, 1, 6, 2, 7, 6, 8, 7, 4, 5, 13, 14, 2}

To see how we can use this representation to encode permutations of n elements uniquely into integers from zero up to $n! - 1$, we need the concept of an **inversion** inside a permutation. In your later courses on data structures and algorithms, this same concept will pop up again in analysis of the running time of the **insertion sort** algorithm, provably the best comparison sort algorithm in the situation where you are only allowed to compare and swap adjacent elements. A **right inversion** inside the permutation `perm` consists of two positions $i < j$ so that `perm[i] > perm[j]`. Intuitively, the elements in positions i and j are “out of order” with respect to each other compared to the identity permutation where all elements are in order and therefore the permutation contains zero inversions. An n -element array in descending order has the largest possible number of $n(n - 1)/2$ inversions, for the “ n choose two” ways to choose any two positions among n positions without replacement when the order of these chosen elements doesn’t make a difference.

The **inversion count array** whose i :th element is the count of how many right inversions the position i participates in (that is, how many elements in later positions $j > i$ are smaller than the element in position i), uniquely identifies each permutation. Even better, these inversion count arrays can be reversed and converted to integers with the previous two methods! The conversion back and forth between permutations and their inversion count arrays will be performed by the next two methods for you to write in the `Permutations` class.

```
public static int[] toLehmer(int[] perm)
public static int[] fromLehmer(int[] inv)
```

The following table contains some randomly generated permutations and their expected inversion count arrays. Since the last element of the inversion count array is always zero, we will leave this element out of the inversion count array altogether, so the inversion count array is always one ele-

ment shorter than the original permutation that begat it. For example, the permutation $\{5, 6, 2, 3, 1, 0, 4, 7\}$ contains five elements after element 5 that are less than 5, five elements after element 6 that are less than 6, two elements after element 2 that are less than 2, two elements after element 3 that are less than 3, and so on.

perm	inv
$\{0, 1, 2\}$	$\{0, 0\}$
$\{0, 3, 4, 1, 2\}$	$\{0, 2, 2, 0\}$
$\{1, 3, 4, 5, 0, 6, 2\}$	$\{1, 2, 2, 2, 0, 1\}$
$\{6, 3, 2, 4, 1, 0, 5\}$	$\{6, 3, 2, 2, 1, 0\}$
$\{5, 6, 2, 3, 1, 0, 4, 7\}$	$\{5, 5, 2, 2, 1, 0, 0\}$
$\{3, 14, 13, 11, 4, 1, 5, 10, 9, 12, 6, 2, 0, 7\}$	$\{3, 13, 12, 10, 3, 1, 2, 6, 5, 4, 2, 1, 0\}$

The conversion from the permutation to the inversion count should be a pretty simple thing with two nested loops, but the reconstruction of that permutation from the inversion counts turns out to be a nice little programming exercise on loops and arrays. Enjoy.

The last two methods in this lab combine these two operations into back and forth conversion between permutations and integer keys. Note that the method `fromKey` needs to know not only the key but the number of elements n in the permutation, since the same keys mean different permutations when applied to different values of n .

```
public static long toKey(int[] perm)
```

This method should first call the previous method `toLehmer` to compute the inversion count array for `perm`. This inversion count is then turned into a scalar integer value by applying the method `fromFactoradic` to the **reversed** inversion count array. (It is somewhat disconcerting to realize that after almost three decades of existence, the `Arrays` utility class in the Java standard library still does not offer us a utility method to reverse an array.)

```
public static int[] fromKey(long key, int n)
```

This method should use the method `toFactoradic` to convert `key` into its factoradic form, which is then reversed and given to the `fromLehmer` method to produce the original permutation.

Lehmer encoding maps the permutations of n elements into integers from 0 to $n! - 1$ in **ascending lexicographic order**. For example, when $n = 4$, the lowest permutation $\{0, 1, 2, 3\}$ maps to the lowest key 0, and the highest permutation $\{3, 2, 1, 0\}$ maps to the highest key 23. Clever techniques seen in Lab 72, “Lazy Generation Of Permutations”, iterate through $n!$ permutations in lexicographic and other useful orders, spitting these permutations out one at the time so that you won’t

run out of memory even trying them all for some n for which the iteration itself would be an overnight run. On the other hand, Lehmer's correspondence between permutations and integers lets us quickly jump to any arbitrary position in the sequence of permutations, despite this sequence being longer than the proverbial year of famine.

Readers whose interest in permutations has been sufficiently piqued by these three labs might want to check out the author's old example class [Permutations](#) that produces permutations one at the time as a lazy `Iterator<List<Integer>>` with recursive **backtracking** converted to iteration so that each call to the method `next` continues this simulated recursion from the place where it left off in the previous call. To save memory, the same `List<Integer>>` object is given out every time with its elements rearranged. The object seen by the user code has been silently decorated with a `Collections.unmodifiableList` decorator to prevent the outside world from messing up the contents of this list. (That one is another nifty little OOP trick that comes handy whenever some internal implementation details need to be exposed to the cold and inhospitable winds of the outside world for efficiency reasons.)

This generator can be given an optional `Permutations.Predicate` object that acts as a **filter** to let through only the permutations that satisfy that predicate. For example, we can generate all permutations of n elements where each element is moved at most k steps away from its original location. Or generate only the [alternating permutations](#) where the element values zigzag up and down, such as $\{1, 6, 3, 5, 2, 4, 0\}$. The backtracking algorithm generates each permutation only until the generated prefix is rejected by the filter predicate, so it will not waste time iterating through all possible ways to complete the remaining elements that will not change the fact that the bad prefix has already violated the filter.

Lab 48: The Curse Of The Clumpino

JUnit: [ClumpsTest.java](#)

All serious programming languages have the “A-list” data structures such as sets and lists lounging in the language itself or at the very least in its standard library, since these data structures are so immensely useful in solving problems universally through all possible problem domains. However, not every data structure ever invented passes the muster of practical applicability to earn its entry past the velvet rope into this exclusive club. In this lab, we will implement the [disjoint set data structure](#), one of the lesser known “C-list” data structures that never got its big break to the major leagues, and then slowly drifted back to working at his old man’s used car dealership back in his Rockwellesque hometown, every Friday night crying into his beer to anyone who would listen how he could have been a *contendah* back when he was given the chance to speed up this more famous algorithm that *totally promised* to take him to the B-list as its plus one. The disjoint set data structure is still worth appreciating on its own even if we never actually go on to implement any of the classic algorithms or solve [the myriad programming problems](#) that gain from the help of this plucky little hometown champ.

Create a new class `Clumps` into your labs project folder. Each object of this class represents some **partition** of numbers from 0 to $n - 1$ into **disjoint clumps** so that at any moment, each number belongs to exactly one clump. Initially, every number belongs to its own singleton clump. Clumps are actually nothing but good old disjoint subsets, but during this lab, we shall use the more colourful word “clump” to emphasize how these subsets will be crudely clumped together to produce new bigger clumps, as if they were made of clay. Before looking at the implementation details, here are the public methods that define what this data structure will do for its user code.

```
public Clumps(int n)
```

The constructor of this class to initialize the data structure for integers 0, ..., $n - 1$.

```
public boolean sameClump(int a, int b)
```

Determines whether the integers a and b are currently part of the same clump.

```
public boolean meld(int a, int b)
```

If a and b already belong to the same clump, nothing happens. Otherwise this method combines the two clumps that contain integers a and b into a single clump that contains all integers that were originally in the same clump as either one of the arguments a or b . This method should return `true` if the clumps of a and b were disjoint but became the same clump as a result of this operation, and return `false` if a and b were already part of the same clump so that nothing happened in this meld.

```
public int clumpSize(int a)
```

Counts how many numbers are currently in the same clump as the integer `a`, and returns this count. (Code inside this method will not actually be counting anything, but will simply look up the value from the `size` table that the other methods continuously update as they mutate the data structure.)

These clumps can only ever grow larger when they `meld`, since no clump is ever broken apart into smaller clumps. Eventually all numbers belong to the same clump, after which point nothing can ever change. Such **monotonicity** often allows liberties in the data structure design to allow a solution that would not bear the presence of **entropy-increasing** operations such as “remove” or “split”. The disjoint set data structure allows the methods `sameClump`, `meld` and `clumpSize` to work in time that, for all purposes both practical and impractical, is a small constant independent of the value of n . Even though one `meld` operation might theoretically combine two clumps of size $n/2$ into a new clump of size n , updating this data structure will still happen in constant time independent of n .

It may seem strange that updating a data structure with one trillion elements needs no more time than updating a data structure with one hundred elements, so let us now pull away the cheap plywood and the black velvet cloth that has been to the cheap laundromat a couple of times too many to uncover the mundane secret inside this magic trick. To allow the previous operations to be implemented efficiently for arbitrary clumps, the current partitioning of elements into clumps will be encoded in two arrays

```
private int[] parent;  
private int[] size;
```

The values of these fields are initialized to array objects created in the constructor when we know the value of n that they need to be big enough to contain. The actual elements are initialized with assignments `parent[a]=a` and `size[a]=1` for each number a from zero to $n - 1$.

Every clump has exactly one **representative**, kind of a [primus inter pares](#) that could in principle be any one of the numbers currently in that clump, any one of the numbers in that clump arbitrarily chosen to shoulder the responsibility of acting as the representative for the entire clump. Which one of the numbers in that club assumes this role depends on the precise order of the `meld` operations that led the entire structure to its current state. The number a is the representative of its clump if and only if `parent[a]==a`. Otherwise, to find the representative of the clump that the integer a currently belongs to, continue looking for the representative of the integer `parent[a]` the same way, until you arrive at the representative of the clump, which you recognize from the fact that it equals its own parent. (And, like in the song, is also its own grandpa.)

This operation of looking for the representative of the given number inside its clump should be implemented as a private helper method

```
private int findRepresentative(int a)
```

that follows the parents all the way up to find the representative of the clump that contains the integer a . After this representative has been found, this method should perform **path compression** as explained on the Wikipedia article about the disjoint set data structure, since this cleanup operation

costs us essentially nothing while it will massively speed up future queries for numbers whose parent chain ever leads to the representative through this same waypoint number. To perform this path compression, simply follow the parents from a to its representative r the exact same way as before, and reassign $\text{parent}[b]=r$ for every integer b along this path.

The method `sameClump` should at this point be a one-liner that merely checks that its parameters a and b have the same representative.

The array `size` keeps track of how many numbers belong to each clump. However, this `size` information is maintained correctly only for those numbers that are actually representatives of their own clumps. The value of `size[a]` for any other number a is never needed for anything, so we don't waste any time updating these values to be correct.

The method `meld` should first check whether a and b are already in the same clump, and return `false` to indicate having done nothing to nobody in that case. Otherwise, find the representatives of a and b in the structure, let us call these representative numbers ra and rb . In principle, either one of the two assignments $\text{parent}[ra]=rb$ or $\text{parent}[rb]=ra$ would combine these two clumps into one, but it is more efficient to reassign the representative of the bigger clump to be the parent of the representative of the smaller clump, to rein in the growth of the implicit tree structure in these `parent` values. Don't forget to also update the `size` information for the representative that you chose to make the parent of the other representative.

Once the `Clump` object for partitioning n elements has been created, each operation `sameClump`, `meld` and `clumpSize` works in **time that is for all practical purposes a constant** independent of the total number of elements n . There are some technical details best left for the second course on algorithms, such as the fact that this seemingly “constant time” is not actually a constant for adult realies, but a constant plus a certain really weird little function called the “[inverse Ackermann](#)” whose exact nature is somewhat difficult to explain at this point of your computer science studies. This function depends on n and will grow without bound as n increases, yet it grows so excruciatingly slowly that it makes even glaciers zoom past like light beams in comparison.

The author can guarantee without any hesitation that you will not find a slower-growing function than this one anywhere in any of the materials that comprise your undergraduate computer science education. Treating this function as an infinitesimally minuscule rounding error in the constant running time can never give you any hassle in any possible counterexample, not just in this real world of ours but in any imaginable one. When functions get weird, the weird turn pro.

Lab 49: Just A Minor Operation

JUnit: [BitwiseStuffTest.java](#)

♪ *So they know everything you code is `true`,
Turn it on, don't turn it on, turn it on.* ♪

— Float Division, “No Bits Lost”

Dealing with bits and bytes directly inside Java is the closest that we ever get to the true essence of the underlying machine executing our code. We prefer to think about execution and the general meaning of a program using higher level languages and abstractions, so this lab shall be considered both character and morale building even for students who have no plans to take courses on lower level coding. Furthermore, the knowledge of **bitwise arithmetic** in Java will directly translate to every other programming language and environment, so students need to learn this stuff only once to use it productively, no matter where their lives may take them after this course. (The same principle of learning something once and then getting to use it everywhere also applies to **regular expressions**, equally indispensable in their own problem domain of text pattern matching.)

Depending on your attitude and the outcome of the random coin flips at the inflection points of your future life, this lab could either be the most important lab you ever complete from this entire problem collection, or the least important. There is no middle ground. Unfortunately, we cannot know quite yet which one of these extremes will play out, but having come this far, [are you willing to take that risk](#) and not spend all your time learning a killer's art? As you will learn in a future computer science course that you should take around the fourth year, whichever name that course might have in the institution where you study this, many computational problems have this kind of curious nature in that we cannot hurry up to get the result faster than helplessly waiting for the system eventually produce the result for us. Such problems are not limited to those that we capture inside our computing, looking in at them through the eyes standing outside, as into the cages of some rare creature in a zoo. Even actual useful problems feel the pinch!

High-level languages such as Java and Python allow programmers to talk about integers and their arithmetic operations in their Platonic essence without knowing how these integers are actually stored, warts and all, inside a computer as aggregates of typically four bytes (`int`) or eight bytes (`long`), let alone requiring the programmer to perform bitwise arithmetic to juggle these bits and bytes around to achieve the desired end results. However, this lab is important not just for general knowledge of our shared reality underneath, but also for the many practical optimizations of time and space that this instrument allows in the hands of those who know how to insert it deep into the computer's memory. We should occasionally get our hands a little dirty in all walks of life, instead of always waiting for the tape to run by passing the buck to some public interface where some sort of magical elves do all the grunt work for us. Even if you got the spirit, don't lose the feeling.

The same integer can be represented in different bases, usually 10 or 2, and it is still the same mathematical object underneath, silently and eternally floating in the Platonic realm in which all mathematical objects finger-quotes “exist” even before humanity itself entered the cosmic game. The integer object that we normally write out in base ten as 42, is still the exact same integer writ-

ten out in binary as 0b00101010 to make it clear that $42 = 32 + 8 + 2$, as you can see from how the expression `42==0b00101010` is `true` in Java. As a convenient shorthand, we say that bits whose value is 1 inside the number are **on**, whereas the bits whose value is 0 are **off**, analogous to some kind of light switches or signal flags.

Since the binary notation tends to get cumbersome once we get past our built-in mental limitation of [seven plus minus two](#) things to keep track of, the highly convenient **hexadecimal notation** groups four bits into one hexadecimal digit from 0 to F, where the letters A to F denote the six missing digits 10 to 15 that do not exist in base ten that goes only up to nine. (In retrospect, perhaps [those limeys are on to something](#) with the base twelve of the **imperial system**, with the nice bonus in how the digits in that base literally go up to eleven!) Since an `int` consists of 32 bits, and each hexadecimal digit encodes a group of exactly four bits, every `int` value can be given as eight hexadecimal digits, such as the famous hexadecimal literal `0xDEADBEEF` along with its more juvenile variations such as `0x0B00B1E5`. The prefix `0x` tells the compiler that the digits following it should be interpreted as hexadecimal even if they all happen to lie in the range 0–9, so we don't have to ask them whether decimal is the role they wanted to live. This way, there will never be any ambiguity about which representation of integers is used in the source code. The 64 bits inside each `long` value can similarly be encoded into sixteen snappy hexadecimal digits.

Again, all these integer literals could just as well have been written in base ten, since that integer itself is the same regardless of which one of its infinite names we happen to use to summon it this time, analogous to how “Clark Kent” and “Superman” are two separate names for the same actual object. Hexadecimal notation makes it clear to the reader that the integers are supposed to be semantically thought of as bit patterns with no direct interaction taking place between bits in different positions, instead of ordinary integers whose semantic purpose we normally associate with various counting and arithmetic operations.

Bitwise arithmetic operators `&`, `|` and `~` correspond to the logical operators `&&`, `||` and `!`, with the difference that the former operate on integers instead of boolean truth values. These operations perform the operations separately for each bit position, which allows to achieve some poor man's parallelism by computing the results of up to 32 logical operations in parallel this minor operation to force a final ultimatum, instead of having to process them one at the time with a for-loop. To paraphrase the famous observation about human nature made originally by Voltaire, we can make an integer tell the truth about itself and the others by giving it a suitable mask.

The result of the **bitwise or** operator, denoted by `x | y` for two integers `x` and `y`, is on for all positions where **at least one** of the operands `x` and `y` is on. This operator can **turn on bits** of the first number `x` regardless of whether those bits were previously on or off, by making the second number `y` equal to a mask that is on in those positions that we want to turn on in the result.

The result of the **bitwise and** operator, denoted by `x & y` for two integers `x` and `y`, is on for all positions where **both** of the operands `x` and `y` are on. This operator can **turn off bits** in the first number `x` regardless of whether those bits were previously on or off, by making the second number `y` equal to a mask that is on in those positions that we want to turn off in the result.

The positions of bits inside a 32-bit `int` are numbered 0 to 31 from lowest to highest. Since all Java primitive integer types are signed, the highest bit is always the **sign** of the number so that 0 means that the number is positive or zero, and 1 means that the number is negative. The remaining bits in negative numbers do not have the same meaning as they have when the number is positive, but are given in **two's complement encoding** to guarantee that the number zero does not have two distinct representations as +0 and -0. This sign bit matters when the bit pattern is treated as an integer that performs integer arithmetic, but does not show up in the hexadecimal notation for integer literals, since the hexadecimal notation represents the underlying bit pattern directly instead of the semantic meaning of that bit pattern as an arithmetic integer.

To create a mask whose k :th bit is on and all other bits are off, use the expression $1 \ll k$ where $\ll$ is the **left shift** operator for bits. This is not a **cyclic** left shift, but the incoming bits from the right are initialized to zeros, and the outgoing bits to the left simply vanish. The **right shift** operator also is similarly not cyclic, but unlike the left shift, offers two separate versions $\gg$ and $\ggg$ with the difference that $\gg$ maintains the value of the highest order bit (since that important bit denotes the **sign** in **two's complement** signed integer representation) whereas the operator $\ggg$ brings in a zero bit into that position. Your choice of which one of these operators you need to use therefore depends on whether you are treating that bit pattern as a signed integer in the rest of your code. The choice is always yours. Bytes and their aggregations do not come with any inherent semantics, since those same 32 bits inside a computer finger-quotes only “mean” whatever you want them to mean. If you treat those bytes as an unsigned integer, then that is what they are, whereas if you treat those exact same bytes as a signed integer, a floating point number (but please don't when I am around) or a machine code instruction, then that is what they are. By themselves, bits and bytes are nothing but zeros and ones without any inherent meaning. (Made of ephemeral electric currents, even “zero” and “one” are words made up to bolster our intuition, and these vibrations of gums could as well have been “true” and “false”, “black” and “white”, or “foo” and “bar”.)

You can use the left shift as a shorthand for multiplication by some power of two 2^k , but the results can be a “bit” surprising if that highest sign bit of the result differs from the sign bit of the original number. This move allows us to create masks for a one bit position, and more complex masks can be combined with the bitwise or operator. To create an integer mask whose k :th bit is off and all other bits are on, use the bitwise negation to the previous mask with $\sim (1 \ll k)$. Such a mask can often be thought to play the same fiddle as a **stencil** plays in spray painting. (This analogy is apt in more levels than merely the surface one.)

Can't rest on your laurels now, not when you got none, so let us finally get on with the methods required in this lab. All the methods below have intentionally been specified to operate on 64-bit `long` parameters instead of 32-bit `int` values, since we now live in the science fiction age of 64 bits anyway, perhaps equipped with tail fins for additional “retro” appeal. Four, sixteen windows, ten in a row. Some art major who has stumbled this far to this bug-infested dried-out forest when they originally just wanted to dip their toes in a computing course could perhaps turn this into one of those “Virgin vs. Chad” memes, at the time of this writing arguably the second greatest artistic achievement of Generation Z. (The highest is the midwit meme, hands down.) The utility class [Long](#) contains a bunch of `static` methods for useful bitwise operations that did not make the cut to the language itself, such as `bitCount`, `rotateLeft` and `parseUnsignedLong` that you might find useful writing the following methods in the class `BitwiseStuff` in your labs project.


```
public static int countClusters(long n)
```

Counts how many clusters of maximal consecutive blocks of ones the parameter value `n` contains, and returns that count. For example, for the value of `n` whose representation in binary would be `0b10011101110001110000000001100001000001100000001111111110`, the expected result would be 8 for the eight colonies of consecutive ones inside the number. Note that each singleton one bit with zero bits on its both sides is still a maximal cluster that should be counted. Make sure to add up the clusters at the very beginning and the end of the number correctly.

```
public static long reverseNybbles(long n)
```

The term **nybble** (or “nibble” depending on which side of the pond you grew up in) is used to denote the four bits that comprise each half of a single byte, but not, for example, the four bits around the middle of that byte. One **hexadecimal digit** therefore represents and describes exactly one nybble. Given a 64-bit long value that consists of a total of sixteen nybbles, compute and return the long value that contains these same nybbles but in reverse order. The four bits inside each nybble should remain as they were in the original. For example, when called with the long value whose hexadecimal representation is `0x63e821ceea5afcc6`, the result of this operation would equal `0x6ccfa5aeec128e36`, the original hexadecimal representation read backwards.

```
public static long dosido(long n)
```

Remain behind, go straight ahead, mess with your values and change odd to even by swapping each bit in every even position with the bit in the next higher off position, so that positions 0 and 1 are swapped with each other, positions 2 and 3 are swapped, and the dance goes on up to positions 62 and 63. For example, the long with hexadecimal representation `0x8074db18c339bafb` would produce the result `40b8e724c33675f7`. For example, its lowest nybble `0xb==0b1011` turns into the nybble `0x7==0b0111` in this operation.

```
public static int bestRotateFill(long n)
```

Even though Java does not have a **cyclic left or right shift** operator in the language itself, these two operations can be handy in some combinatorial problems. These low-rent operations are assigned to play in the alternative venues since they won’t get past the velvet rope of the A-list lounge of the Java language to hang out with familiar superstars such as `+` and `*` so that, like Madonna or Cher in our world, they would be universally known for the character tokens that represent them. (Let alone being so important that, akin to The Artist Formerly Known As Prince, our rulers would add entirely new symbols to the Unicode character set specially for them!) Here, everything is by design, so these two have been implemented as static methods `rotateLeft` and `rotateRight` in the utility class [Long](#). This method should return the number of steps `k` to stretch all true realities by maximizing the `bitCount` of the **bitwise or** operation `n | m` where `m` is the result of shifting `n` cyclically `k` steps to the left. If several values of `k` are equally good, return the smallest such `k`.

Applying the same ideas that were needed to implement the previous four operations, many other bitwise operations can also be implemented. In the real world, the necessity of bitwise arithmetic indicates some kind of low-level programming done close to the iron (hardware), making speed and efficiency the essence at this low and dark realm of spirits. Students who are interested in these sorts of more or less clever optimizations can check out the site “[Bit Twiddling Hacks](#)”, the classic book “[Hacker’s Delight](#)”, or the free online textbook “[Matters Computational](#)” for an encyclopaedic presentation of combinatorial and numerical techniques from the bit level up.

To bring this lab specification to its end, you can still use the one-letter operators `&` and `|` also inside logical conditions. In that context, these operators denote the versions of logical operators that do not **short-circuit**; that is, they don’t skip evaluating the second operand when the first operand already has made the result of the entire expression certain. The Java language specification explicitly guarantees the strict **left-to-right evaluation** and the resulting **short circuit** behaviour for the logical operators `&&` and `||`.

The evaluation order of Java subexpressions is otherwise generally left for the compiler to decide, and is not guaranteed to take place the same as reading the source code from left to right. For example, the expression `rng.nextInt(42)*rng.nextInt(99)` is not guaranteed to produce the same result in every environment even under the same internal state of `rng`, since we cannot know which one of the two calls to `nextInt` will be evaluated first! To say nothing of Lovecraftian abominations such as `a=++a+a---a+++--a;` whose runtime behaviour the language can only leave undefined, despite the fact that the expression on the right hand side has a unique parse tree to decipher its operations. To ensure your intended evaluation order, such expressions must be taken apart into multiple statements separated by semicolons that establish the strict sequential order in which these subexpressions are computed.

This invisible danger exists in basically all widely used programming languages, and yet no textbook or other material meant for a first year programming course seems to even acknowledge its existence. For every computer program that is currently used to make decisions that affect us, the version of the compiler that was current at the time that program was compiled and released generated some evaluation order of subexpressions that passed not only the unit and integration tests of that project, but more importantly, the test of time in actual use. One day these programs will be recompiled with the Current Year version of that compiler that will then emit some different evaluation order that *never* makes any difference to anything... at least until that initially adamant “never” once again turns out to be “hardly ever” in this creaky old Ship of Fools.

Lab 50: The Weight Of Power

JUnit: [PowerIndexTest.java](#)

Consider a [weighted voting](#) system such as a corporate shareholder meeting where participants wield a different voting power, called the **weight** of that voter. These voters are convening to vote either “yes” or “no” for some motion. For a motion to pass, the weighted total of “yes” votes must be at least equal to the given **quota**. In a typical **majority vote** situation, more than half of the votes are necessary to pass the motion. However, this quota could also be something other than one half in special situations, such as the rules of some organization requiring at least two thirds of the votes for some particularly fundamental measure to pass. Sometimes even a unanimous decision is required, such as in some jury verdicts. (As many dramatic movies and special episodes of beloved television shows have taught us, being a one-man majority can be a pretty nefarious position!)

In a weighted voting system, the actual power of each voter is not necessarily linearly proportional to the weight of that voter. For example, consider weights [9, 9, 7, 3, 1, 1] with the quota of 16 votes required to pass the motion. There are really only two relevant situations to consider in this setup. Whenever the first two voters agree with each other, that already determines the final outcome regardless of the votes of the four other guys. Whenever these first two voters disagree with each other, the seven votes of the third voter act as a **kingmaker** to determine which one of the bigger guys gets their way this time, again rendering the three little guys irrelevant. The last three voters are **dummies** who might as well stay home, since the outcome tallied with their votes included can never differ from the outcome tallied without them. (The term “dummy” is not used here in the sense of being some sort of imbecile, but a mute and inanimate doll that does not get to make any decisions that could affect anything in its external reality.)

To see even more clearly how the actual voting power can be very different from the pure percentage of total votes, look again at the third player who acts as kingmaker in situations where the two big guys disagree. From this kingmaker’s point of view, the bigger the better, since his position to extract outside concessions for his cooperation can only grow more juicy with the size of this kingdom! Machiavellian operators in both fiction and real life throughout history have strived to find such leverage to move the world in ways that Archimedes could only dream of.

In this lab, we compute two different power index measures for weighted voting, first the [Banzhaf power index](#), followed by the [Shapley–Shubik power index](#). Both computations involve recursions through the combinatorial possibilities that are relevant for that power index; all **subsets** for Banzhaf, and all **permutations** for Shapley–Shubik. Besides, the last names of these guys just sound so adorable, as if these gentlemen were rubbery characters in a *Mad Magazine* cartoon by Don Martin that depicts these algorithms as delightfully wacky machines with gears and whistles that sound like “Fwababa dabada fwababa dabada” and “Shtoin!” as the ballots chug through.

Both algorithms operate with the concept of **coalition**, which here is a five-dollar word for what we normally would just call a **subset** of voters. A coalition is **winning** if the sum of weights of its members is at least equal to quota. In a winning coalition, an individual voter is **critical** if changing his vote would make the coalition no longer be winning. The winning coalition is **minimal** if every voter in that coalition is critical so that it contains no superfluous voters who just run up the score.

For example, consider the coalition of voters $\{0, 2, 5\}$ in the previous situation. This coalition is winning, since $9 + 7 + 1 = 17$ votes is enough to reach the quota of 16. Voters 0 and 2 are both critical for this winning coalition, since removing either one of them makes the total fall below the quota. Voter number 5 with his single vote is not critical for the win, since the coalition $\{0, 2\}$ would still be winning even without his vote. That guy only gets the satisfaction of having rooted for the winning team this time... typically feeling all the more triumphant about this the less important his actual vote was for achieving this outcome, a tragicomical psychological phenomenon often seen with the fandom of movies, sports and politics. The human condition is as mysterious as it is tragic.

The Banzhaf power index for an individual voter is defined as **the number of winning coalitions in which that voter is critical**. These numbers are usually **normalized** by dividing them by the total number of winning coalitions so that they add up to one, but in this problem we will keep these counts as integers for simplicity. The simplest way to compute the Banzhaf power index is to recursively generate all possible subsets of voters. Whenever you find another winning coalition, increment the counter `out[m]` by one for its each critical member `m`. Then, recursively generate the coalitions that the current winning coalition is a subset of, since everyone will get points for all winning coalitions whose critical member they are, not just for the minimal winning coalitions. However, if none of the members of the winning coalition was critical for the win, you can save plenty of time by not recursing through the remaining supersets of such redundant winning coalitions.

Create the class `PowerIndex` in your labs project, and there the method

```
public static void banzhaf(int quota, int[] weight, int[] out)
```

that computes the Banzhaf power index of voters with given weights. This method does not create and return a new array, but writes the results in the array `out`, guaranteed to be of sufficient length and filled with zeros at the time of call. The recursion is best extracted into a helper method

```
private static void banzhaf(
    int quota, int[] weight, int[] out,
    LinkedList<Integer> coalition, int sum, int n
)
```

where the additional recursion parameter `coalition` contains the voters who have already been taken into the current coalition. The top level call should initialize this to a new and thus empty `LinkedList<Integer>` instance. The parameter `sum` contains the sum of weights of the elements in the chosen `coalition`. (We could always compute this value by adding up the votes in the current `coalition`, but this way we have it available at each level in one constant time lookup, much better than looping through the array each time.) Parameter `n` gives the length of the prefix of `weight` array whose subsets this recursion should visit from the current level downwards.

The recursion to systematically iterate through all 2^n possible subsets of the given set of n elements is essentially the same as the recursion seen in the **subset sum** problem. The base case of this recursion is when $n = 0$ and there is nothing to do. At each level in this recursion, if the current

`coalition` is winning so that its votes together add up to at least to `quota`, every critical voter of the `coalition` without whom the `quota` would not have been reached by the rest of this `coalition` gets its corresponding counter in the `out` array incremented by one. You must check the criticality of each member of the current winning `coalition` separately inside the loop where you hand out these points, since the winning `coalition` constructed so far is not necessarily minimal, but can theoretically contain any number of non-critical gnats.

In non-base cases with $n > 0$, regardless of whether the current `coalition` is winning, this recursion should branch in two different directions; one for taking the voter $n - 1$ into the `coalition`, and the other for leaving that voter $n - 1$ out of the `coalition`. Once both branches have returned and silently updated the `out` array along the way, this method has nothing left to do but return to the previous level. Many legitimate recursions have this kind of nature in that the recursive method itself does not return anything, but some data structure shared between every level of recursion simultaneously gets incrementally updated during the systematic exploration of the branches of this implicit recursion tree. Once the recursion has systematically examined all possibilities, the final state of that shared data structure is our final answer to Regis.

The **Shapley–Shubik power index** is computed by summing over the $n!$ possible **permutations** of these n voters, instead of summing over their 2^n possible **subsets**. When the weights of the voters arranged in one particular permutation are accumulated from left to right, the voter who makes the cumulative sum reach the `quota` is called the **pivotal voter** of that permutation. The Shapley–Shubik power index of that voter is simply the number of times that voter is pivotal over all $n!$ possible permutations of voters. The straightforward recursive method to compute the Shapley–Shubik power indices for these voters takes the same parameters as the previous method to compute the Banzhaf power indices.

```
public static void shapleyShubik(int quota, int[] weight, int[] out)
```

The simplest way to visit all permutations is to again define a `private` helper method that takes additional parameters that describe the current subproblem. Such recursion generates each one of the $n!$ possible permutations by extending the current **prefix** in all possible ways to complete the permutation. However, to avoid explicitly visiting all $n!$ permutations inside this method, which would be prohibitive for large n , we can be a bit more clever with the pivotal voters.

```
private static void shapleyShubik(
    int quota, int[] weight, int[] out, int level, int last,
    boolean[] taken, int sum
)
```

The parameter `level` keeps track of the current recursion level. It is initially zero in the top level call, and gets incremented in each recursive call. Knowledge of the current level of recursion tells us how far the prefix of the current permutation has been generated. Unlike in the Banzhaf recursion where the voters in the current `coalition` were explicitly stored in a list, the recursion for Shapley–Shubik uses an array of truth values to keep track of which voters have already been taken in

the current prefix. The parameter `last` remembers which voter was added to the prefix in the previous level, and the parameter `sum` contains the total weight of the voters in the current prefix.

The base case of this recursion is when `sum` is greater than or equal to `quota`, which must happen in every branch before running out of voters to add to the current prefix. Since this makes the `last` voter the pivotal element of all permutations starting with that prefix, we can add up these permutations in one swoop by incrementing `out[ last ]` by the number of possible ways to complete that incomplete but winning prefix into a complete permutation. This count is simply the factorial of how many unassigned voters still remain given by `weight.length-level`. You might want to **precompute** all factorials up to 17! that fit inside an `int` without overflowing into a `private static` array for quick lookup during the actual algorithm.

When total weight of the current prefix is less than `quota`, this method should use a for-loop to iterate through the voters who have not yet been taken into the prefix, and recursively complete all possible permutations that start with the previous prefix extended by the new voter with a recursive call with updated `level`, `last`, `taken` and `sum` for that extended prefix.

The following table contains some `weight` arguments along with the expected correct result from both methods. The `quota` is in all cases one half the sum of weights, **plus one**. The first row is the numerical example taken from the Wikipedia page for the Banzhaf method. The third row, constructed from the previous row simply by adding the big guy one more vote, represents a **dictatorship** where one of the voters outweighs everyone else added together, so both metrics correctly assign all decision power to this dictator. The fourth row shows how equalizing the weights of every voter gives each voter the exact same voting power, as is to be expected when nothing external breaks the symmetry between these voters. Note how the power indices for n voters must always add up to $n!$ in the Shapley–Shubik method, when the algorithm implicitly tallies up every possible permutation of these n voters.

weight	Banzhaf	Shapley–Shubik
[4, 3, 2, 1]	[5, 3, 3, 1]	[10, 6, 6, 2]
[1, 2, 5, 8]	[1, 1, 1, 7]	[2, 2, 2, 18]
[1, 2, 5, 9]	[0, 0, 0, 8]	[0, 0, 0, 24]
[42, 42, 42, 42]	[3, 3, 3, 3]	[6, 6, 6, 6]
[2, 13, 17, 26, 30]	[2, 2, 6, 6, 10]	[8, 8, 28, 28, 48]
[10, 24, 24, 32, 40, 42]	[2, 10, 10, 10, 14, 14]	[24, 120, 120, 120, 168, 168]
[3, 14, 24, 26, 27, 64, 67, 70]	[2, 10, 14, 14, 14, 58, 58, 62]	[288, 1824, 2592, 2592, 2592, 9888, 9888, 10656]

No matter how many orders of polynomials you manage to shave off from the exponential running time of your search as if carving ever thinner slices off an exponentially expanding turkey, you still always have an exponential search left in your hands. Assuming that you previously had enough patience to solve the problem for n voters before this optimization, you could probably solve that same problem for $n + 2$ voters after that optimization. Meh. Inch-wormy increases rarely arouse any great enthusiasm for the future of the algorithm. Problems with a large n are best solved with randomized **Monte Carlo sampling** to evaluate each term only up to precision that is enough for the purpose that we are grinding these power indices for. That's the cost of doing business in all problems that are inherently exponential. Which, to a first approximation, is basically all of them.

Exponential haystacks can be a bitch to find even one measly little needle from, let alone carefully combing through every possible straw of hay inside that stack as if you were some modern day Rumpelstiltskin. It probably won't be worth your while to speed up your computations in this lab by cleverly utilizing the fact that two voters with equal weights must also end up with an equal voting power in both systems, since this special case will not appear in the automated tests often enough for this to pay off in total running time. (Some examples on the web that perform these computations by hand use this trick to minimize the work, in similar spirit to integrals in calculus textbook with carefully selected bounds and coefficients that make things vanish.)

Interested students can easily find plenty of online material about both methods to compute power indices. This material is not necessarily educational just for math and computer science, but for understanding all of life and social reality in general. For example, the essay "[The Inner Ring](#)" by C. S. Lewis uses no math, yet still makes important observations about this general topic. (Not everyone is doomed to talk nonsense even when they refuse to do arithmetic, let that be either in our real world or in Narnia.)

Lab 51: Twelve Angry Primes

JUnit: [MillerRabinTest.java](#)

“The fact is, we all started out as someone’s ideal little Platonic baby objects. And a space like natural numbers forces us to become primes or composites. Nothing in between can exist here.”

— after *Shot Caller* (2017 film)

The **trial division** test used in an earlier lab to determine the primality of a positive integer n is a good exercise on basic looping, especially after we realize that only the known primes up to the square root of n need to be examined as potential divisors. However, this approach is not feasible for efficient primality testing outside a relatively small range of values of n . An important part of this course, same as for the previous course CCPS 109 *Computer Science I*, is to make the students stop thinking of one million as a large number, and start seeing millions or even billions as more like Ralph Wiggum’s idea of a large number. (“My daddy is at least a *million* years old!”)

Results of [number theory](#) allow for [faster algorithms for primality testing](#) by harnessing the predictably unpredictable power of **randomness** to navigate the way through some exponentially hairy thicket of possibilities whose promises we are unable to evaluate before being forced to commit into exactly one of them. Suppose that you don’t even know yourself what you are going to do, similarly to the main character in [The Dice Man](#). No malicious **adversary** can then exploit what you are doing by setting up the problem to make it seem as if your choices are reasonable at each turn, and yet in the aggregate, these choices are guaranteed to lead you to the worst possible outcome.

In this lab, you will implement the necessary ingredients for the [Miller–Rabin primality test](#). Instead of randomization, we will implement a **deterministic** version of this algorithm using the ideas from the Numberphile video “[Witness Numbers](#)” that the reader might like to watch before proceeding further in this lab. Our deterministic version of the Miller–Rabin algorithm is sufficient to conclusively determine the primality of all integers that fit inside a 32-bit integer variable, without any chance of some composite number sneaking by as a prime number. However, before we get to implement the primality testing algorithm itself, we first have to implement one non-exciting helper method that does all the heavy lifting behind the scenes while the front man algorithm gets all the kudos and glory, same as in so many other walks of life.

```
public static long powerMod(long a, long b, long m)
```

This method should compute and return the **modular power** given by the formula $a^b \bmod m$, where the **modulus** m is some positive integer greater than one. Since this method only needs to deal with positive values of a and b , this modulus operator is the same as the Java integer division remainder operator `%`. (For negative numbers, the modulus and remainder operators would give very different results. For example, $-7 \% 3$ equals -1 , whereas the quantity $-7 \bmod 3$ would equal 2 , if the Java language actually had the `mod` operator. But for this problem, this is a moot point.)

The massive advantage of computing your all integer arithmetic modulo some tactically chosen m is that the result of every intermediate arithmetic operation will always be less than m . This allows

humongous powers of integers to be computed without using the `BigInteger` type to explicitly represent all their millions of digits. The final result will still be correct, thanks to the formulas

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$(a \times b) \bmod m = ((a \bmod m) \times (b \bmod m)) \bmod m$$

that work for any m , be it prime or composite. These formulas allow us to replace all integers with their modular values at every step of evaluation. This way these numbers won't grow too big for their britches, and the end result will equal the true result modulo m .

(Although this property is not needed in this lab but is interesting and important in general, precisely whenever the modulus m is itself a prime, every integer a in the range from 1 to $m - 1$ has a unique [modular multiplicative inverse](#) in that same range, denoted by a^{-1} . This multiplicative inverse satisfies the equation $(aa^{-1}) \bmod m = 1$ and therefore allows **division** to be also sown inside this field. This multiplicative inverse of a can be efficiently computed with the [extended Euclidean algorithm](#) without using any literal divisions that tend to cause problems with integers. In general, it's a good idea to use multiplications instead of division wherever possible.)

To make the modular exponentiation ridiculously fast even for large values of a , b and m , you should use the technique of [exponentiation by squaring](#) previously encountered in the labs that dealt with permutations of integers. When the exponent $b = 2k$ is even, use $a^{2k} = (a^k)^2$ to chop that exponent in half. When the exponent $b = 2k+1$ is odd, use the subdivision $a^{2k+1} = a(a^k)^2$ for the same outcome. Depending on how your logic runs either as a while-loop or the equivalent **tail recursion**, either one of the equivalent subdivision formulas $a^{2k} = (a^2)^k$ and $a^{2k+1} = a(a^2)^k$ can always be used.

Unlike Python whose dynamic nature doesn't require the behaviour of any function to be set in stone during compilation (which is why tail call optimizations cannot be done in Python while maintaining the original semantics of that program), the static nature of the Java language has the advantage of allowing the compiler to rewrite all recursive [tail calls](#) into equivalent iterative structures that do not allocate a new stack frame. This makes linear recursions of this particular restricted form to be effectively identical with iterative while-loops. Since the stack size in this problem grows only logarithmically anyway, same as for all recursions whose logic is based on repeated halving, even Python could handle this one recursively within the usual recursion depth limits out of the box without any tail call optimizations. The following table showcases some values of a , b and m along with their expected results.

a	b	m	Expected result
17	30	9	1
8	9	3	2
60	1909	56	32
255	428433	307	101
1656	3647549	928	640

The Miller–Rabin primality testing is based on the idea of finding a **negative witness** for the positive integer n whose primality we wish to determine. The existence of even one such witness... well, why don't we use a more colourful term "papers check" inspired by this author watching one too many [Big Herc](#) videos on YouTube discussing the hard reality of prison life, to help and motivate him to stand up and "do his own time" during the pandemic lockdown. Once the lockdown is over, it's time to get some hard coding time in!

Imagine the number n as a hardened criminal who caught a case that he couldn't beat, so that he has been sent up the river for a telephone number sentence. Having settled in, he cheerily greets his fellow crime lads, proclaiming himself to be in good standing in the cold and hard streets of the criminal underworld, meaning that he has never ratted on his fellow criminals, nor has he ever committed certain types of crimes that the men of conviction around him would not look kindly upon. Any failure in this paper check when demanded (and as the late great hardboiled crime novelist Andrew Vachss noted in one of his *Burke* novels, no matter how many years in your sentence, certain parts of it will never take long) reveals him to be nothing but a rat-faced punk who had better slink right into Punk City and never again show his mug in the mainline among the righteous cons.

However, this whole setup is very different when n is an OP, a genuine Original Prime whose paperwork and shiv are hooped and ready to go ballistic at a moment's notice, safe in the knowledge that his name has already been carved deep into the cold hard streets that raised him. No matter which particular underling number a the shot caller with the keys to that particular modular pod will randomly send to do the heart check, n can keep his head up high, certain that he will stand up under any amount of pressure and interrogation under whatever bullshit charges they might fall behind. An entire train of such challengers can be thrown randomly at n without changing this outcome, since the genuine prime convicts will never break into smaller proper factors.

On the other hand, any composite number is guaranteed to eventually crack under the hot lights of factorization and tweet out its factors like a little finch. It is certainly in the interest of the criminal subculture to do its due diligence and quickly reveal any such snakes waiting in the grass. Any composite punks masquerading as unbreakable prime gangsters will soon realize that their life choices have led them straight into a Level Four yard that shows no mercy. A composite number trying to pass itself as a prime has about the chance of a parlay of two coin flips to survive even one heart check, let alone an entire gauntlet of such checks coming randomly from every direction. Should the number fail any of these tests at any time, every righteous con in the entire block will be jumping to split his wig with their Occam's shivs! Factors in, factors out.

Hell is other primes constantly staring at you, as a playwright philosopher might put it. In this spirit, the first part of the Miller–Rabin algorithm is the important subroutine

```
public static boolean isMillerRabinWitness(long n, long a)
```

that checks whether n survives the heart check from the challenger a . This method should first compute the two unique integers s and d so that $n - 1 = d2^s$, with d odd. This is easily achieved by repeatedly dividing $n - 1$ by two until the result becomes odd, and using s as a counter of how many times this division by two had to be performed. The actual challenge begins by computing the quantity $p = a^d \bmod n$. If this p equals either 1 or $n - 1$, the convict n has passed the challenge on the spot

with an instant knockout punch, and this method can return `true`. This challenger has nothing more to say. Note that n can still be composite even after this instant knockout; further challenges are required to gain more confidence on primality of n .

Otherwise, the challenge continues with the following operation that will be executed up to $s - 1$ rounds. Recompute the quantity p from the formula $p^2 \bmod n$ for the current p . If the new value of p equals 1, the punk ass composite nature of n has been stripped bare for all to see, and this method can return `false`. If the new value of p equals $n - 1$ instead, the convict has passed the challenge, and this method can return `true`. For all other values of p than these two extremes, the current round is deemed inconclusive, and the same challenge continues with the next round of squaring modulo n . The challenged integer n must decisively prove its primality before $s - 1$ rounds have been completed, as the prison yard rules declare the challenger to be the winner after $s - 1$ rounds.

Note the one-sidedness of these rules in that defeating any one challenger does not prove n to be a prime number, yet losing even one challenge conclusively establishes n as a composite. The actual Miller–Rabin primality testing is a **randomized algorithm** where the primality of the integer n is established by defeating k separate challengers randomly chosen from the range 2, ..., $n - 1$. The randomness guarantees that this algorithm can't be fooled by an adversary who sends in some undercover infiltrator composite n known to defeat all challengers from the fixed set of values hard-coded in the algorithm's cell block. Since any composite number trying to disguise itself as a prime has at most a one in four chance to defeat a randomly chosen challenger, we can easily make k large enough to make the probability of a composite n sneaking through the entire gauntlet vanishingly small. Once this probability is lower than, say, the probability of the computing device performing these calculations being destroyed by a meteorite during the computation that assumes the primality of n should surely satisfy even the most paranoid and ruthless shot caller!

However, as explained both on the Wikipedia page and in the Numberphile video linked above, ditching the randomness altogether by simply using the first twelve prime numbers 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31 and 37 as the fixed set of challengers is guaranteed to reveal the true nature of any composite number that fits inside the top bunk of the `long` integer type in Java. Only God may judge a true prime from the domain of `long`, but every composite is guaranteed to be soon carried by six after having been judged by these twelve. The method

```
public static boolean isPrime(long n)
```

should therefore send precisely these twelve angry primes to challenge the parameter n . If, and only if, these twelve challengers all get their asses handed out to them, the solid "rep" of n as a genuine prime number has been conclusively established. Of course, to make this method fast in practice, the method should first try the trivial trial division of n with the first twelve primes, before engaging in the more expensive Miller–Rabin primality test. For vast majority of integers from the civilian world unaccustomed to prison rules, this simple check establishes their composite nature.

Once the above three methods pass the JUnit test, congratulations, your hard time is now considered fully served so that you can get back to The World again. This lab is complete and you can move on with your life. The rest of this problem specification is just additional education for students who are interested in this sort of stuff. You know me, and how I like to ramble.

The randomized Miller–Rabin algorithm for primality testing can be used as a subroutine to generate a random prime number of b bits on the principle of [rejection sampling](#) so that every b -bit integer is equally likely to be chosen. Simply keep generating random b -bit integers from $b - 1$ random coin flips, the lowest bit of each number hardcoded to 1. For each such integer, use the randomized Miller–Rabin primality test enough times to determine whether that particular number hit the jackpot, in which case that number can be returned as the randomly chosen prime. Otherwise, just keep generating more b -bit integers until you hit the pay dirt. Since the prime numbers are far more dense among all integers than most people might intuitively assume, this process is virtually certain to hit some prime number soon enough, even with b in tens of thousands! We are not asking you to implement this method here, since it has already been implemented in the `BigInteger` class as a static method

```
static BigInteger probablePrime(int bitLength, Random rnd)
```

The parameter `bitLength` is the number of bits the desired prime number should consist of, and `rnd` is the random number generator used to make the random choices during this method. (It is unclear to this author what `rnd` exactly stands for in this part of the Java standard library, since `rng` is more standard and clearly means “random number generator”. Since the formal name of the parameter does not affect the caller anyway, this change would not break any existing user code.)

As the final philosophical point about what we have now witnessed in this lab, we tend to think of randomness as an enemy whose effect on us we try to minimize and preferably avoid. But as you will learn in many later courses, randomness can also be a good friend and an ally in your corner, if you just allow it to be! Analogous to how we use fire properly down here in the physical world always as a servant but never a master, you should not let randomness wildly burn all over everything inside your program (or even worse, write your entire program randomly), but strictly limit its effects to rage inside some small, well insulated part of your program while keeping the rest of the program predictably deterministic. This lets you have the best of the both worlds as the unpredictably raging fire of randomness takes apart the things that you need taken apart, no matter how tricky and convoluted your Adversary has tried to set up these things to hinder your progress.

As you may appreciate by watching this method spit out random prime numbers with thousands of digits in less time than it takes you to snap your fingers for the gesture of such sorcery taking place, the sheer effectiveness of this random prime number generation algorithm might well be the closest thing to magic that we encounter during this course.

Lab 52: Linus Sequence

JUnit: [LinusSequenceTest.java](#)

This lab returns to the theme of interesting integer sequences where iterating some simple rule makes the sequence bounce around in a manner that straddles somewhere between order and chaos. The apparent incongruence between the algorithmic simplicity of the rule, versus the almost lifelike complexity of the outcome that emerges from the repeated application of that rule, is sometimes perceived as adorably comical. It is as if “The Little Sequence That Could”, delightfully portrayed in our imaginations by Zooey Deschanel, Anna Kendrick or some other lovely *ingénue* of equal stature, aspired to become something higher than it has any right to be. We tend to instinctively root for lovable underdogs.

In this spirit, the definition of [Linus sequence](#) tries its best to be as unpredictable as possible, and yet this very attempt makes it trivially predictable! This paradox combines elements of both the zen koan of the tip of the tail of the ox inevitably getting caught in the gate, and the sociological phenomenon of how every “hipster” rebel whose trendy cookie-cutter opinions would not raise an eyebrow in the HR department of any Current Year Fortune 500 company tries hard to be all unique and mysterious, and yet ends up being Yet Another Nonconformist in lockstep with all others.

Before embarking on coding, the seemingly paradoxical nature of this sequence provides us with another more general and potentially useful life lesson. The argument used by Linus in the *Peanuts* comic strip behind the link is the same basic train of thought that runs deterministically towards wherever its by now well-worn tracks carry it. As every mentalist knows, most people will unknowingly repeat the same flawed algorithm if asked to generate a “random” sequence of truth values using purely mental means without use of dice or other physical randomization aids. Somebody whose “random” sequence starts with four equal elements is quite a unicorn among us Earthmen, even though there ought to exist nearly one billion people on Earth doing exactly that, if every human currently alive were somehow asked to mentally generate four random bits! Being in this sense very different from Vulcans, humans are in general notoriously [bad at both emitting and detecting randomness](#). Fortunately, algorithmic techniques exist to combine and convert biased sources of random bits, either from a man or a machine, into higher quality sources of randomness.

Hoping that the readers will excuse the slightly old-timey convention of using lowercase Greek letters to denote character strings (at least around your third year you should be taking a computer science course where this convention is still used, so that it is not “all Greek to you”), the **maximum repeated suffix** of the sequence s is its longest possible suffix β that allows the breakdown of sequence s into three component substrings $s = \alpha\beta\beta$ where α is the prefix of s before the duplicated suffix β . The maximum repeated suffix exists and is well-defined for any finite string s , since the empty string always works as the suffix β to force $s = \alpha$. For example, the maximum repeated suffix β of the string "1100110101101" is "01101", where α equals the leftover prefix "110" that could not be used to extend the repeated suffix.

Create the class `LinusSequence` in your labs project, and there the method

```
public static int maximalRepeatedSuffix(boolean[] bits, int n)
```

that computes and returns the length of the maximum repeated suffix of the sequence that consists of the first n elements of the boolean array `bits`. As seen earlier, it is often a good idea to have a method that operates on array arguments to take additional index parameters that make that method restrict its attention to some particular subarray. The following table showcases some n -bit boolean arrays, written below as character sequences of ones and twos instead of truth values for brevity and readability, and their corresponding maximum repeated suffixes. We might as well use this convention since the construction of the Linus sequence consists of two possible elements that, despite being represented in our code as boolean values, seem to be always called 1 and 2.

bits (of length n)	Expected result
2211212	2
1212121212	4
2112121112121	6
22222212122221212	8
212111221112221121	0
1121111211121122111	1

The first element of the Linus sequence is defined to be 1, from which the sequence will be constructed one element at the time. Just to be fancy and exotic, let θ denote the Linus sequence of length k that has been constructed so far. The element $k+1$ of this sequence is always either 1 and 2, the symbol that minimizes the length of the maximum repeated suffix of the corresponding extended sequence $\theta 1$ and $\theta 2$. (There will never be a need for tie breaker, since the maximum repeated suffixes of these two extended sequences will always have different lengths. DUCY?)

```
public static boolean[] linusSequence(int n)
```

Computes the first n elements of the Linus sequence and returns the sequence as an array of truth values so that `false` stands for 1, and `true` stands for 2. The first element of the sequence is in the position zero, set to `true`. (Absolute position indices do not matter in this problem anyway, as the rule only talks about suffixes of the sequence generated so far, and therefore takes the agnostic position in the Gulliverian schism between the zero- and one-based indexing.)

The emergent behaviour of the Linus sequence is largely unknown, and chock full of mysteries far above this humble instructor's pay grade. Appropriately enough, your instructor first learned about this tricky sequence when browsing the [Futility Closet](#), an excellent blog sitting at the intersection of recreational mathematics and historical curiosities. The category "[Science & Math](#)" is probably the most relevant for both the aims and the spirit of this course.

Lab 53: Tchoukaillon

JUnit: [TchoukaillonTest.java](#)

Variations of [Mancala](#) have been played around the world, but are still relatively unknown in the West. The simple structure of such games that gives birth to a rich complexity of emergent situations would make for many good exercises on integer lists and arrays. Generating the legal moves of the game would be implemented as methods that operate on arrays and lists that represent the current position of the game and the position that emerges as the result of some move. The first course on artificial intelligence, usually taken around the third year, then covers the **minimax algorithm** to find the best move that thwarts the opponent's potential responses.

Until we have that algorithmic machinery available, we have to content ourselves with [Tchoukaillon](#) (bless you!), one of the simplest **solitaire** puzzle versions of this game. Conveniently for our coding purposes, especially compared to the more convoluted boards used in many two-player variations of Mancala (even ignoring the symbolic decorations and other chrome), the Tchoukaillon board is a simple one-dimensional row whose pits are numbered with natural numbers the same way as lists and arrays. Each position contains some number of discrete but identical seeds, coins, pebbles, men, or whatever word you wish to use to refer to your game pieces. In all methods written in this lab, the game board is represented as some kind of instance of `List<Integer>` that contains the number of seeds in each position.

Position zero at the beginning of the row is the **safe home** in which you are trying to bring as many of your coins as you can. A legal move in a state consists of choosing a position k that contains exactly k pebbles in it; for example, the position number five that currently contains five men, no more or less. If no positions that fulfill this requirement exist on the board, it's game over, man, game over! Those k pieces of seed are sown to the k preceding positions, exactly one coin per such position. Each move maintains the total number of pebbles but increases the number that are safely home by exactly one, the game for n men reaches a terminal state after at most n moves.

Okay, I will stop that silliness now. From now on, the game pieces will be called "seed". Were you one of the lucky quick few who read the previous paragraphs too fast to notice? Either way, create a new class `Tchoukaillon` in your labs project, and in that class its first method

```
public static boolean move(List<Integer> board, int k)
```

This method grabs the k seed from position k and sows them into the previous k positions, one seed each. Note that this method must update the state of the `board` argument object, shared by both this method and its caller the same way as all object arguments are shared in Java, to correspond to the new reality after this move. When called for `{3,1,2,0,4,5}` and $k=4$, the contents of that same list object would become `{4,2,3,1,0,5}`. Redundant zeros created from sowing the last position should be trimmed from the end of the list before returning.

The return value of this method indicates whether the move from position k was legal and successful; that is, whether the board position k contained exactly k seed at the time of method call. If the

position `k` contains any other number of seeds, this method should not modify the board in any way, but return `false` to inform the caller that the attempted move was not legal.

```
public static boolean undo(List<Integer> board, int k)
```

This method is the **inverse** of the previous `move` method. Given the state of the board after performing the move at position `k` in some unknown previous state of the board, restore the board to this unknown previous state. It is possible for the parameter `k` to lie past the end of the board, which would happen when trying to undo the sowing from the last position of the board.

The return value of this method indicates whether the arguments `board` and `k` represent an undo of some legitimate move within the rules of Tchoukaillon. The move is legal when the position `k` on `board` contains zero seed, and every position lower than `k` contains at least one seed (including the position zero, that for the purposes of the JUnit test, must contain exactly one seed), so the undo can be performed by collecting these `k` seeds back to position `k` from the previous positions. Otherwise, this method should return `false` without modifying the board. For example, trying to undo the move `k=3` at the position `{3,1,0,0,4}` would be such a [no-op](#), since position 2 doesn't have a seed that it could give back for the undo operation.

Your instructor certainly stands high and mighty on his bully pulpit every time he preaches a sermon about the fruits and blessings of immutable types in programming, and how accepting these types and their associated programming style in your heart will make your code ascend closer to its abstract form that resides eternally in the Platonic paradise that is fundamentally out of reach of our mortal minds. To ruin this pastoral idyll, that vile serpent Beelzebub will now slither in and legitimately point out the giant downside of immutable types; iterating through a large number of separate values using an immutable type necessitates the creation of a separate object for every such value. Furthermore, such a **functional programming** approach is simply [too alien and opposite to the usual way of thinking of even seasoned professionals](#).

Using mutable types, the same object or variable can just keep track of the current value, from which the next value is generated in computational means. For example, we can write arbitrarily long for-loops that use a mutable primitive `long` value as their loop counters, and count up to millions of trillions without ever running out of space when the eight bytes of that variable act as our "peephole" to the long (in both senses of that word) sequence of values being iterated over.

Games that allow the methods `move` and `undo` to operate **in place** on the same mutable game state, instead of having to create a new object to represent that game state that results from each move, make searching for solutions in such games much easier in recursive means. If we imagine the Tchoukaillon board as a crudely simulated war zone so that these seeds have sprouted into an army of identical little army men symbolically represented as some kind of tokens under your command, how many men could you bring to the safety of home before running out of moves?

This problem, exactly the same as other search problems that ask you to design a series of moves that solve the given puzzle, can be solved with a **recursive search** that receives the current state of the puzzle as an argument. The base cases of this recursion are the states that allow no legal moves,

and their values are given by the number of men who are safe. Otherwise, the method should loop through the possible moves in the current state, and recursively compute the values for the states that result from each move, in this problem easily generated in place by using `move` and `undo` around the recursive call. Return the maximum value of these possible moves as the value of the current state. This recursion should be written as one method

```
public static int value(List<Integer> board)
```

that computes and returns the maximum number of men that can be brought safely home from the given starting board. Despite being in principle exponentially branching, this particular recursion usually manages to be quite fast, since most states of `board` contain only one or two possible moves, and most choices for these moves will quickly lead to obvious dead ends. The following table displays some expected results for some initial distributions of these men in the front lines.

board	Expected result
[0, 1, 2, 3, 4]	9
[0, 1, 2, 3, 3]	5
[0, 1, 2, 0, 3, 4, 4]	3
[0, 0, 2, 3, 4, 5, 6, 5, 3]	16
[0, 1, 2, 1, 4, 3, 6, 7, 8]	27

Whenever such a `value` method is available as a black box, we could easily use it to make the actual moves by consulting the `value` method for all possible successors of our present board and simply making the move that the black box claims to be the best. We shall not fear being led astray by the marching orders implied by this box, since the individual evaluations of that box will guide our steps all the way up to at least the best result possible, if not the hoped-for best possible result.

When it comes to the puzzle being fully winnable, Tchoukaillon is a rare bird among all such flightless waterfowl in that for every positive integer n , there exists **exactly one** possible way to place n men on the game board so that it is possible to bring all of them safely home using the moves of Tchoukaillon! Any other way of placing these n men on the board will always see the battlefield eventually quiet down with at least one man hopelessly stranded behind the enemy lines, no matter how you finagle the individual moves in the interim.

Since each move will always decrease the number of men outside safety by exactly one, and these men must be arranged in the game board in a way that makes continuing to the solution possible, it follows that these unique solvable states form a linear chain that springs forth from the goal position where all pebbles are safe. From there, [the chain reaches up towards infinity](#) by moving from the unique solvable state for n men to the next higher unique solvable state for $n + 1$ men, easily generated by simply performing an `undo` to the leftmost empty position k , creating a new empty position to the end of the list when no earlier empty positions exist.

```
public static void previousSolvable(List<Integer> board)
```

This method should perform this operation on the given board. In the JUnit tests, this board is guaranteed to be in some solvable state. Furthermore, to canonize the representations of possible boards, the number of men in the safe zero position is kept at zero. The value of any future action can only be affected by men outside safety, never by the **sunk reward** of men who already reached it... and besides, of what import are brief and nameless lives to Galactus anyway?

Lab 54: All Hail The Mighty Xor

JUnit: [SigVectorTest.java](#)

♪ *Clean code, cool tools, and I don't know where I am going to,
Test suite, black box (black box), I don't need to read the do-o-cs,
Functions running as fast as they can
'Cause every core's crazy 'bout a bit wise man!* ♪

— $\mathbb{Z}^2 \top$, “Bit Wise Man”

The **caret** operator $a \wedge b$ has nothing to do with exponentiation in either Python or Java, but denotes the **bitwise exclusive or** between two integers a and b when these integers are thought as **bit patterns** instead of ordinary integers that allow arithmetic and order comparisons. Unlike the familiar operators of **bitwise and** and **bitwise inclusive or** that tend to erase information as they go, this delightfully symmetric and most importantly **reversible** bitwise operator can be thought to take a bit pattern and a **stencil mask** that determines which bits are **flipped** to the opposite values. This intuition of stencils makes it easy to see why applying the same mask twice on the same bit pattern produces the original pattern. Generalizing this idea explains why the successive operations of the same mask cancel each other out even if other masks are used to flip the same bits in the interim. This seemingly magical property of exclusive or to perform tricks that apparently require memory but without actually needing any memory has several applications in computer science.

For example, it is occasionally convenient to compute and keep track of some sort of **checksum** of a large data structure or a data clump to speed up equality comparisons. If the checksums of two large objects are different, their payload data is also certain to be somehow different without further ado. This is especially convenient when this data is **immutable**, so this checksum can be **cached** alongside the actual data in a separate field. For example, the Java `String` class computes the **hash code** of each object **lazily** the first time that it is actually needed for something, and caches this hash code so that it can be looked up in constant time in every future occasion. After paying the cost of each `String` object having another integer field, practically every negative equality comparison in the future will be correctly resolved in constant time by a single comparison of these cached hash codes.

For **mutable** data, updating the checksum on the fly at every mutation could be inconvenient. It seems especially wrongheaded to diligently perform some linear-time checksum recalculation after every little constant-time update! To avoid such silliness, this lab explains and implements [Zobrist hashing](#) that uses the exclusive or operator to a bunch of random integers to generate scambly and chaotic checksums that can be updated with simple constant-time operations at every update of the data! This ingenious technique is usually presented on chess and other exciting game boards made of two-dimensional arrays. For simplicity, we shall restrain ourselves to compute these Zobrist checksums for one-dimensional **bit vectors**. Higher-dimensional data, or data from larger domains than mere binary, can (and always *will*) be encoded into bit vectors anyway.

To kill two birds with one stone, we shall also finally explore the Java language mechanism to allow **cloning** the instances of some class. In fact, the automated fuzz test will be cloning your objects as if

it was the mid-to-late nineties again with the cloning craze all over both news and entertainment media. Create a new class with the following signature and the three data fields

```
public class SigVector implements Cloneable {
    private boolean[] bits;
    private long[] keys;
    private long sig;
```

The objects of this class will serve as bit vectors that maintain a running checksum of type long to describe their contents. The actual bits are stored in the eponymous array field, and the field sig contains the current checksum. Your class should have the constructor

```
public SigVector(int n, long[] keys) {
    this.bits = new boolean[n];
    this.keys = keys;
}
```

to initialize these fields to create a bit vector of length n for which we maintain a checksum.

Zobrist hashing uses a table of random integer keys. Since these random keys are given to this class from the outside, and the correct operation of its methods is never supposed to mutate these keys, the same keys can be shared between all the instances of this class, instead of feeding the expensive habit to create a separate defensive copy for every individual instance jonesing one.

To access the individual bits and the current signature, the class should have the methods

```
public long set(int i, boolean b) throws IllegalArgumentException
public boolean get(int i) throws IllegalArgumentException
public long getSignature()
```

The first two methods are used to set and get the value of the bit i of this bit vector. The set method should return the checksum after setting the bit i. Furthermore, whenever the bit i flips to a new value, this method should update the current checksum with the assignment

```
sig = sig^keys[i];
```

or more briefly with the shorthand notation analogous to += and other such operators

```
sig ^= keys[i];
```

to mix into the checksum the random key that corresponds to the bit i. It is now easy to see that for any sequence of bit flipping operations, flipping any bit an even number of times does not affect the final checksum, regardless of the other bit flipping operations. Especially if all bits are turned on and off in an arbitrary order, the checksum ends up being the same goose egg as it started as.

With the checksum mechanism available, the following methods should now be embarrassingly easy to make both effective and efficient. For the `hashCode` method, you could simply extract any 32 bits of the `sig`, usually the bottom 32 since they tend to change the most erratically. Even better, use bitwise arithmetic to extract the top 32 bits and the bottom 32 bits into an `int`, and return the bitwise exclusive or of these two halves of a 64-bit `long`. The `equals` method should first compare the checksums, and only if they are equal, compare the actual contents of the bit vectors.

```
@Override public int hashCode()  
@Override public boolean equals(Object other)
```

(In fact, the checksum inequality is always sufficient to infer content inequality for all the pseudo-random bit vectors generated during the run of the JUnit test! Despite this, you should not rely on this happening in general. Even stars do occasionally collide with each other despite the vast emptiness of the intergalactic space.)

```
@Override public String toString()
```

As so often in the real world, the `toString` method can return pretty much anything that aids your debugging purposes instead of hindering them. Finally, implement the important method

```
public Object clone()
```

This method should first call `super.clone` to duplicate the actual bitwise identical clone object, since this method has an **implicit contract** of never using the operator `new` for this purpose. Since the superclass version of `clone` inherited from `Object` performs a **shallow copy**, you need to perform the required **deep copy** by yourself and `clone` those freaking `bits` so that these bit vectors can later be safely mutated without silently affecting any other bit vectors that were supposed to be separate entities.

(Those among us who truly are of pure heart must also actually `clone` these `bits` instead of resorting to `Arrays.copyOf` or similar. Such methods will certainly use the operator `new` behind the scenes, causing us to inadvertently violate the sacred implicit contract passed from the subclass to the superclass without saying a word in the class itself.)

Lab 55: Accumulated Wisdom

JUnit: [AccumulationTest.java](#)

Numerical arrays allow constant-time **random access** to their individual elements, regardless of their positions. The situation is not equally serene for **summation queries** performed over arbitrary subarrays, since those require looping through that entire subarray to add up its elements one at the time. However, with the aid of an **accumulation array** constructed from the original array, arbitrary summation queries over contiguous subarrays can be performed in guaranteed constant time, regardless of the width and the position of that subarray!

In fact, since the elements of the original array can be accessed as trivial summation queries for **singleton** subarrays, you can potentially save a bushel of memory by throwing out the original array once you have constructed its accumulation array. The accumulation array contains all the information that was available in the original array, but in a form more suitable for subarray summations. As long as there is no integer overflow, having to use more bits to store the individual array elements is a reasonable price to pay for this convenience. Now we are cooking with gas.

An accumulation array has the same dimensions as the original array, but each element equals the **sum of all elements up to that particular position**. For example, the accumulation array for the original array `a={4, -2, 3, 1}` would be `acc={4, 2, 5, 6}`. To construct this accumulation array in linear time, we simply need to realize that each element `acc[i]` can be computed with the expression `acc[i-1]+a[i]`, utilizing the fact that the sum of the elements up to the position uses the same terms as the accumulation up to the previous position, with only one more original element included. The initial element is set up with `acc[0]=a[0]` as the base case for the formula.

In your labs project, create a new class `Accumulation`, and in there two `static` methods

```
public static int[] accumulate1D(int[] items)
public static int subarraySum(int[] accum, int start, int end)
```

The first method creates and returns the accumulation array for `items`. The second method uses the given accumulation array to quickly add up the elements of the subarray of `items` from `start` to `end`. (Yes, despite the fact that this method never even sees the original `items` array object!) Also note that these parameters follow the Pythonic convention where the `end` position is treated as **exclusive**, whereas the `start` position is **inclusive**. Therefore when both of these parameters are equal, they span an **empty subarray** whose sum is zero.

(After completing the logic for accumulation over one-dimensional sequence of data, the reader might take a breather to philosophize about the potential usefulness of computing a second-order accumulation array whose elements are the accumulated sums of the accumulations of the original elements. Elegant facts await us there.)

Next, we shall generalize this technique to handle **grids**, two-dimensional arrays whose every row has the exact same number of columns, so that no row is **ragged** and leave gaps or protrusions in

the general rectangular shape of that array. Each element `acc[row][col]` of this accumulation array now equals the sum of elements in the rectangular subarray from the **origin** position `[0][0]` up to and including the position `[row][col]`. Note how the previous one-dimensional accumulation is merely a special case of this two-dimensional accumulation done to a two-dimensional array that contains only one row and is therefore effectively one-dimensional. The corresponding methods for you to write are this time

```
public static int[][] accumulate2D(int[][] grid)
public static int subrectangleSum(
    int[][] accum, int row, int col, int h, int w
)
```

In the second method, the parameters `row` and `col` determine the top left corner of the rectangular subarray of the `grid` whose elements are being summed. Parameters `h` and `w` give the height and width of this subarray, measured in rows and columns. For example, if both parameters `h` and `w` were to equal one, this sum would equal the singleton element `grid[row][col]`.

Analogous to the one-dimensional case, the computation of each element `acc[row][col]` should not require brute force looping through the entire subarray. Instead, each `acc[row][col]` should be computed by adding and subtracting a fixed-size handful of previously computed values from the two arrays `acc` and `grid` in a more sophisticated manner that, as a nifty teachable moment for us, also showcases the [inclusion-exclusion principle](#) often seen in discrete math calculations!

(By the way, this is one of those problems where you should **literally draw a picture** to help you maintain all those parameters and their meanings straight in your head. No rush. I will wait here.)

After completing these two methods, instead of mindlessly repeating this approach once more with a feeling for three-dimensional **cuboids**, we shall instead finish this lab in flamboyant style by solving [a classic coding job interview chestnut](#). Like all the best problems, this one is simple enough to state in one sentence: Given a two-dimensional `grid` of boolean truth values, return the side length of the **maximal square** whose every element is `true`.

```
public static int largestTrueSquare(boolean[][] grid)
```

Every “[Shlemiel](#)” would solve this problem by throwing at it some horrendous amount of brute force, in this case **four** levels of nested loops. The outer two loops iterate the positions of the `grid` that could potentially be the top left (for a symmetric approach, the bottom right) corner of the largest `true` square. For each such potential corner, the two inner loops would gauge the size of the `true` square that emanates from that corner. For arrays that have been randomly filled with truth values taken from random coin flips, the average running time of this technique would be pleasantly low. However, the worst case would absolutely kill us. Simply imagine what would happen to the running time if every element of the original array were `true`!

Any attempt to make the method more clever so that it knows to terminate in this special case can always be thwarted by adding enough `false` values to tactically chosen positions to coax out the

quartic worst case behaviour. You have to implement the method first, and only after that, the **adversary** will choose the arguments given to your method. Since this nasty little imp who we already know from **Murphy's law** will *always* make *everything* turn out the worst possible way for your algorithm, the average running time over all possible arguments is rendered meaningless. When you are fighting against nature whose randomness has no aim to help or hurt anyone, you can rely on algorithms that are good on average, but when facing intentional hostility, the average is *finito* and you need to actively defend against the worst case scenario.

We cry out for a more sophisticated approach to defeat the adversary by solving the problem with only two nested loops whose running time is **oblivious** to the exact values of the array elements, and deterministically depends only on the size of the `grid` itself! The adversary can then fill in this `grid` literally any which way he wants, and still has no chance to gum up the works by making your algorithm run back and forth in some maximally inefficient and silly manner.

Even though the following approach technically requires the technique of **dynamic programming**, taught in some course normally taken around the third year in a typical computer science undergraduate program, this author hopes that completing the previous methods has psychologically primed the student for that one crucial flash of insight that is needed to implement the efficient and correct technique in this problem. Start by creating a two-dimensional integer array `int[][] s` whose size is the exact same as the size of the original truth value `grid`. Fill up this integer array so that each element `s[row][col]` gives the size of the largest `true` square whose bottom right corner lies in that position. Once this array has been completely filled so that each element is correct, its maximum value is returned as the final answer.

The outer two loops are exactly the same as they were in Shlemiel's solution. However, now ask yourself this; how could the three neighbouring values `s[row-1][col]`, `s[row][col-1]`, and `s[row-1][col-1]` adjacent to the north, northwest and west from the current position be used to compute the correct value of the current position `s[row][col]` without **any** inner looping, assuming that your two nested for-loops have already correctly filled the values of these three neighbouring elements? Once you see this, bang, that's all the fat lady wrote about this problem.

The following optimization is not required for your code to pass the JUnit test in reasonable time, but observant viewers may still note that to save another big bushel of memory, the array `s` does not actually have to be the same size as the original `grid`. This array can simply act as a **sliding window** to keep track of only two consecutive rows; the **current row** that is presently being filled in the inner loop, and its **previous row** whose values are needed for correctly filling the current row. Once the current row has been filled, its previous row will never again be needed, assuming that you keep track of the overall maximum value that has been seen in all previously generated rows, even those already memory-holed.) The current row becomes the new previous row, from which the next current row (recycling the space of the now redundant previous row) is computed.

One final optimization is lurking inside the previous optimization. There is no need to explicitly copy the elements of the current row to the previous row every time the sliding window is moved one notch downward. Once we realize that those things that walk, look and quack like two-dimensional Java arrays are under the hood one-dimensional arrays whose elements themselves are one-

dimensional arrays, swapping the references to these two rows, which can be done in constant time regardless of the length of these rows, achieves the desired effect.

One final thing that you might want to notice is that this very technique can be used to solve any well-formed recursive equation whose subproblem parameter domains are contiguous subsets of integers. You can even fill in the subproblem solutions in literally any order that guarantees that when at some particular position, the positions that store the recursive subproblem solutions needed to fill in the current position have already been computed. Recursive calls to solve these subproblems can be replaced by trivial array lookups!

A ravishing beauty of a technique, **dynamic programming** is truly [a skeleton key to unlock a thousand doors](#) that Hercules himself could not pry open with sheer brute force. The hardest part of all such problems is coming up with the recursive equation to describe the system. Once you have written down that equation, you should draw the picture of the subproblems and their recursive dependencies as a grid with arrows pointing from cells to other cells (even if only as a mental image), to make the actual coding stage basically a kabuki act.

Lab 56: find() Me A Find, catch Me A Catch

JUnit: [MatchmakerTest.java](#)

In the famous [stable marriage problem](#) in **combinatorics**, a group of n boys is to be paired one-to-one with a group of n girls. Each boy and each girl has privately **ranked** all the members of the opposite sex in some total preference ordering. Everyone is honest about their true preferences without shame, not trying to go after some hot babe just because that is what the mass media told them to do while every other guy also tries to follow suit, or in this case, a skirt.

An arrangement that pairs each boy to exactly one girl is called a **matching**. When both boys and girls are numbered from 0 to $n - 1$, such a matching can be expressed as an n -element array that describes which girl each particular boy has been paired with. Since this array must be a **permutation** of n elements, its inverse permutation describes which boy each girl has been paired with. This problem is completely symmetric between boys and girls, but is conventionally phrased in line with traditional gender roles so that the boys take turns to individually approach girls.

Suppose that the matching contains two pairings Chad–Becky and Norbert–Stacy so that **both** Chad and Stacy would prefer to break off their current pairings and elope with each other, leaving Norbert and Becky in the dust to settle for each other. The existence of even one such unstable pair makes that entire matching **unstable**. Note that merely having Chad desire Stacy (or just Stacy desire Chad) over his current partner does not make that pairing unstable, unless Stacy (Chad) is equally willing to break away from his current pairing.

Regardless of how each boy and girl privately ranks the members of the opposite sex, at least one **stable matching** with no unstable pairings will always exist. In fact, depending on how these preferences mesh and conflict with each other, even exponentially many stable matchings may well exist! For example, consider the situation for $n = 2$ with two boys Chad and Norbert and two girls Stacy and Becky. Chad prefers Stacy over Becky, whereas Norbert prefers Becky over Stacy. So far, no conflict. Unfortunately, the girls have opposite preferences so that Becky prefers Chad and Stacy prefers Norbert, setting up the scene for a raucous nineties teen comedy. Both possible matchings in this situation are stable, as the reader can verify. However, these two matchings differ drastically in their **fairness** for whose preferences are satisfied. One of these matchings gives each boy his preferred girl and leaves both girls to settle for her lower-ranked boy, whereas the other matching gives each girl her preferred boy by having the boys settle for their respective lower-ranked girls.

The famous [Gale–Shapley algorithm](#) can rapidly construct the stable matching that is optimal for boys and pessimal for girls. In the guaranteed result, each boy skips to the loo with his highest ranking “best gal” that he has any chance to pair with at all in any possible stable matching. Conversely, each girl will be paired with her lowest-ranking boy that she could pair with without making the matching to be unstable. Since the roles of boys and girls are perfectly symmetric, this algorithm could be run as its mirror image version to favour girls instead of boys. In our more enlightened times since the swinging sixties back when this algorithm was first developed, it also serves as an example of **algorithmic bias** that systematically oppresses a particular subgroup of participants. (As in so many other walks of life, even those who instinctively rightfully sense that the entire sys-

tem is somehow rigged against them still perceive only the very surface level from how deep that entire rigging *really* goes.)

The Gale–Shapley algorithm makes for an excellent exercise of handling two-dimensional integer arrays for our Java programming course. To visualize this algorithm, imagine these boys and girls coyly standing near the opposite walls of a high school gym, everybody dressed and made up as if they had been cast in summer stock revival of *Grease*. After setting up the necessary arrays to keep track of the current state of this high school dance, the algorithm itself is a single while-loop that keeps running until every boy have been matched to some girl, which must eventually happen when there are at least as many girls as there are boys.

On his turn, each boy approaches his highest-ranked girl that he has not approached before during this process. Since the Bro Code is still decades away from being codified into law, the boy must approach that particular girl even if she has already been provisionally paired with someone else. If Stacy is still unpaired at the time when Norbert approaches her, they are provisionally paired with each other for the time being. However, if Stacy has already been paired with someone else, say, Melvin, the one of the two boys currently vying for Stacy’s hand must slink back to the opposite wall with his shoulders drooped while the other boy can stay to twist his limbs in a victory pose will be determined by Stacy’s preference between her two suitors.

Whichever boy was sent back will get his turn again in the future, but may not approach Stacy again (as any repeated approach would be futile anyway and only deepen his potential future incel rage), so on his next turn, he will be approaching the girl next down the line in his preference rankings. Since the preference ranking of each boy contains exactly n girls, and each boy gets to approach each girl at most once, every boy will necessarily become paired with some girl after at most n^2 total approaches, usually substantially fewer.

In your labs project, create a class `Matchmaker`, and there the method

```
public static int[] galeShapley(int[][] boysPref, int[][] girlsPref)
```

that receives the preference rankings of both boys and girls as n -by- n integer arrays so that the i :th row of each array contains the preference ranking list of the i :th boy or girl, respectively. This method should create and return a one-dimensional integer array `boysPair` whose each element `boysPair[boy]` shows which girl that particular boy has been paired with. During the execution, you will also need two other one-dimensional integer arrays `girlsPair` and `boysPos`. The former keeps track of which boy each girl has been paired up with, which is necessary for deciding whether an existing pair should be split up. The second array keeps track of how far down the preference ranking each boy has descended during the algorithm execution.

Curiously enough, the unique end result of this algorithm does not depend whatsoever on the order in which the individual boys get to approach these girls. Confident of the final outcome that is set in stone by the iron hand of mathematical necessity from the preference rankings of these individual boys and girls, Chad can be a suave gentleman and nonchalantly comb his hair in his best imitation of Kookie Byrnes while he lets Norbert to first approach Stacy, coyly waiting near the other wall

presumably wearing some kind of pencil skirt and bobby socks with starry-eyed visions of of hunky Chad coming to sweep her away from chunky Norbert.

The bottleneck for the execution time of this algorithm is quickly finding some unpaired boy, again literally any such boy. Details of how to best implement such **nondeterministic** just-pick-any-one-I-don't-care-which-one selection to run efficiently are left for the reader; several good ways exist to do this. Furthermore, since this algorithm needs to be able to quickly determine whether Stacy prefers Norbert or Melvin; some preprocessing of the `girlsPref` array might help Stacy make up her mind faster during the algorithm's main loop. The girl that the boy will approach in his current turn is then given by the expression

```
boysPref[boy][boysPos[boy]++]
```

that **post-increments** the `boysPos` counter for that particular boy so that the next time when his turn to approach a girl comes, he will approach the next girl down the line along his preference ordering. Since the pre- and post-increment operators are a highly convenient feature of Java that Python doesn't have at all, we might as well smoke 'em since we got 'em, in this rare occurrence of the Java syntax being superior to that of Python for the everyday programmer.

As discussed earlier, the result produced by the Gale–Shapley algorithm is optimal for the boys and pessimal for the girls. Advanced algorithms for the stable marriages and other matching problems attempt to make such unfairness a bit less lopsided.

When n boys are to be paired off with m girls so that n and m are not equal, we can always invite Rosie and her sisters to balance the ranks. Even so, the terms “boys” and “girls” are mere placeholder names that can stand for anything in the **bipartite graph** whose nodes they comprise; for example, **hospitals and residents** in the most famous practical application of the stable marriages problem. This problem setting can be further generalized so that instead of being gathered all together in the gym and everybody seeing everybody else, each boy knows only some subset of girls, and each girl knows only some subset of boys. [Hall's Marriage Theorem](#), a famous result of graph theory, shows that some matching exists in the first place if and only if every subset of boys together knows at least as many girls as there are boys in that subset, and vice versa. (As usual, one of those ifs is trivial while the other one is world-changing.)

Any societal implications of this phenomenon in the real world are left as an exercise for the reader. In spirit of the fluid modernity of the Current Year, this problem can be further generalized to the euphemistically named [stable roommate problem](#) where the participants are not divided into binary sexes but anybody can be paired with anybody. Unfortunately, no algorithm can now possibly guarantee finding some stable matching in all situations, for the simple reason that some preference ranking combinations just do not allow *any* stable matching. The simplest example of such a situation is when Andy prefers Bob who prefers Carl who prefers Andy to create a **non-transitive preference cycle**, and all three men equally agree on their dislike of Dave whose preferences don't really matter here. (Note the similarity to the famous [Arrow's Impossibility Theorem](#) for incoherent group preferences, even when the preferences of each individual are perfectly coherent.)

Lab 57: Flood Fill

JUnit: [FloodFillTest.java](#)

Given a pixel image that consists of a solid background colour (in the example on the left below, white) that has been divided into disjoint areas using lines, curves and other shapes with some other colours, [flood fill](#) is a classic technique to paint over all pixels inside the same contiguous area. The flood fill algorithm carves its way through precisely the pixels that belong to the same contiguous area, regardless of the shape of that area and the curvature of its boundaries. The area to be painted is defined by any given **seed** pixel (x, y) known to be inside that area. See the screenshot from the `main` method of the automated test for an illustration for the original image of random arcs on the left to create a maze, and a whole bunch of flood fill operations separately applied to its white areas, using a randomly chosen colour for each such area.



The basic flood fill algorithm is explained well enough in pseudocode in the linked Wikipedia page, from where your task is to translate this given algorithm to Java. For once, you don't need to come up with the algorithm yourself from the specification for what it is supposed to achieve! However, your flood fill method certainly should not be recursive, since the recursive flood fill that snakes through the white area in the maximally convoluted fashion will almost certainly require a large recursion depth and run out of the allocated stack space for recursion bookkeeping. Instead of once again heading over to [Stack Overflow](#) with your hat in hand to ask how to increase the recursion limit inside your Java Virtual Machine, your method should be completely non-recursive from the get-go, and instead use some kind of **queue** of pixels waiting to be processed.

In your project folder, create a new class `FloodFill`, and there the method

```
public static void floodFill(BufferedImage img, int x, int y, Color c)
```

Given an instance of `BufferedImage` and the pixel coordinates `x` and `y` of the pixel inside the current area, this method should paint over that entire area, and not one pixel more and no less, with the new colour `c`. Instead of the more general `Image`, this method expects its more specific subtype `BufferedImage` that allows additional methods `getRGB` and `setRGB` to read and write the colours of its individual pixels. Note that these two methods expect the colours to be represented as primitive `int` values, whose four bytes encode the ARGB components of the actual colour, one byte per colour component. However, this representation should really not be a problem, since the Java `Color` objects come with the method `getRGB` to extract this compact `int` representation from that object. You don't need to perform any bitwise arithmetic to convert the colours to bytes and vice versa.

The original colour of the seed pixel (x, y) is treated as the background colour that the flood fill should paint over, whereas every other colour found in the image is considered to be a wall that prevents the current fill colour from spreading over to the rest of the image. (If the original pixel colour equals the colour `c`, the fill with therefore terminate immediately.) Note also that the return type of this method is `void`, so that this method should paint into the contents of the original `img` object shared by the caller and this method, instead of creating a brand new image to represent the result while keeping the original `img` intact.

After initializing your necessary variables and the queue data structure, the flood fill algorithm is yet another application of the classic **breadth-first search**, already seen a couple of times before in this problem collection. Initially, the **frontier queue** contains only the seed pixel (x, y) . The algorithm main loop keeps going as long as the frontier queue is nonempty. In each round of this main loop, pop out the next pixel to be processed from the front of the queue. If this pixel at the current moment has some other colour than the background colour, just ignore it and go to the next round. Otherwise, colour that pixel with the new colour `c`, and push its **four** neighbours in the main compass directions that are inside the image into the queue. The frontier queue will become empty precisely when all the background pixels reachable from the seed pixel have been processed and painted with the new colour.

To simplify and streamline the translation of the algorithm from the Wikipedia pseudocode presentation to actual executable Java, it might be convenient to first define the utility method

```
public static boolean inside(BufferedImage img, int x, int y,
                             int bgCol)
```

that checks whether the pixel (x, y) lies inside the dimensions of `img` and that its colour is still equal to the given background colour `bgCol`. Note how the methods `getWidth` and `getHeight` of the `BufferedImage` class will provide you the exact pixel dimensions of that image.

The `main` method of the automated test class tries out your flood fill method on all of the areas of the pseudorandom image of black curves and lines seen in the above screenshot. Since your method should produce that exact same result when applied to the pseudorandom test image, you can use this `main` method to verify by eyeballing that your flood fill algorithm works as intended. The actual `testFloodFill` method does not display the thirty pseudorandom images of various dimensions that it creates and paints over using your flood fill method. Instead, the test method computes the checksum from all the pixels of each of these pseudorandom test images, to ensure that your flood fill algorithm fills in **precisely** the same pixels with the same colours into every such test image as this author's private model solution would fill.

Once they get the basic flood fill method working, interested students can also try out implementing the more advanced and memory-efficient **scanline fill** techniques also described in the Wikipedia page. Even though we are most familiar with the result of flood fill from the now mostly just meme-worthy MS-PAINT images, this algorithm allows many optimizations for both time and space that, back in the day, allowed it to run even in an eighties 8-bit home computer such as Commodore 64 and Sinclair ZX Spectrum, both of which this author still has both fond and not so fond memories of.

Despite being adorably primitive from our point of view up here in the cyberpunk future, even the Basic programming language of the ZX Spectrum had the flood fill operation available as a keyword in the language, along with the more elementary graphical primitives such as drawing a line segment or a circle. However, the mighty one megahertz processing power of the Z80 processor could only execute this operation so slowly over the 256-by-192 screen resolution (and even that split into 8-by-8 regions with only two colours per region) that you could watch the algorithm paint the pixels, almost as if the whole thing were some animated YouTube demo of the algorithm.

(To see how far humanity has gradually advanced in this respect, and [what the exponential growth of processing power means in practice](#), you can try out this author's hand-rolled and non-optimized [Mandelbrot.java](#) demo that not only flood fills the Mandelbrot fractal image with twenty times as many pixels, but determines the positions of the walls and the colour of each pixel on the fly from a complicated iterative process of high-precision complex number arithmetic!)

Lab 58: Block Cellular Automata

JUnit: [BlockCellularAutomatonTest.java](#)

Various [cellular automata](#), the most famous of which surely are the variations of [Conway's Game Of Life](#), produce a surprisingly chaotic emergent complexity of interesting patterns from the local interactions of applying simple rules uniformly over a grid of truth-valued cells. We have already seen Conway's Game of Life as one of the example programs [GameOfLife.java](#) in this course material, and the [Ulam-Warburton crystals](#) as an earlier lab problem in this very document.

[Block cellular automata](#) are an interesting generalization of the basic idea of cellular automata. In a block cellular automaton, the local rules are not applied directly to the individual cells, but to some larger neighbourhood, in the simplest non trivial case a 2-by-2 square that the grid has been subdivided into. For brevity, each such 2-by-2 square is called a **Margolus square**. Since the width and the height of the entire grid are assumed to be even, this subdivision neatly places every cell into exactly one Margolus square. Since each `boolean` cell inside the same Margolus square can be in two possible states (as usual, these states are again colourfully named "dead" for `false` and "alive" for `true`), that Margolus square can be in one of $2^4 = 16$ possible configurations.

Same as in all other cellular automata that we have seen so far, execution of a block cellular automaton proceeds forward in discrete time steps. At each time step, a hardcoded rule is used to separately convert each Margolus square to its next configuration. So that these squares could interact with each other instead of being small isolated islands, the origin location from where the grid is subdivided into these squares is placed at position (0, 0) at the even time steps, and at the position (1, 1) at the odd time steps. So that the cells on the edges of the grid will also get properly treated in both odd and even time steps, this grid is treated as **toroidal**, meaning that its east and west edge are treated as to have been logically wrapped together. The wrapping of its north and south edges is analogous. For a cell located at the right edge of the grid, its immediate right neighbour lies on that same row at the left edge, and vice versa.

The four truth values inside the Margolus square are treated as four bits of the binary representation of an integer, so that the northwest cell adds 1, the northeast cell adds 2, the southwest cell adds 4, and the southeast cell adds 8 to the value of the square. For example, if the northwest and southeast cells of some Margolus square are currently alive, and the other two cells currently dead, that square as a whole will have the numerical value of $1 + 8 = 9$. Another Margolus square whose all cells are dead would have the numerical value of zero, whereas a third square whose all cells are alive would have its maximum numerical value of $1 + 2 + 4 + 8 = 15$.

Since the chosen rule for the block cellular automaton turns the configuration of each Margolus square into another configuration of that same square, the entire rule can be compactly expressed as an array of 16 integers. Once you have computed the numerical value of some Margolus square, the numerical value of that square will simply equal `rule[value]` in the next time step. You can then convert this numerical value back into the individual dead and live cells inside that Margolus square by expressing that value in binary and making precisely those cells be alive in the next time step for which the bits for the corresponding powers 1, 2, 4 and 8 are turned on in that number.

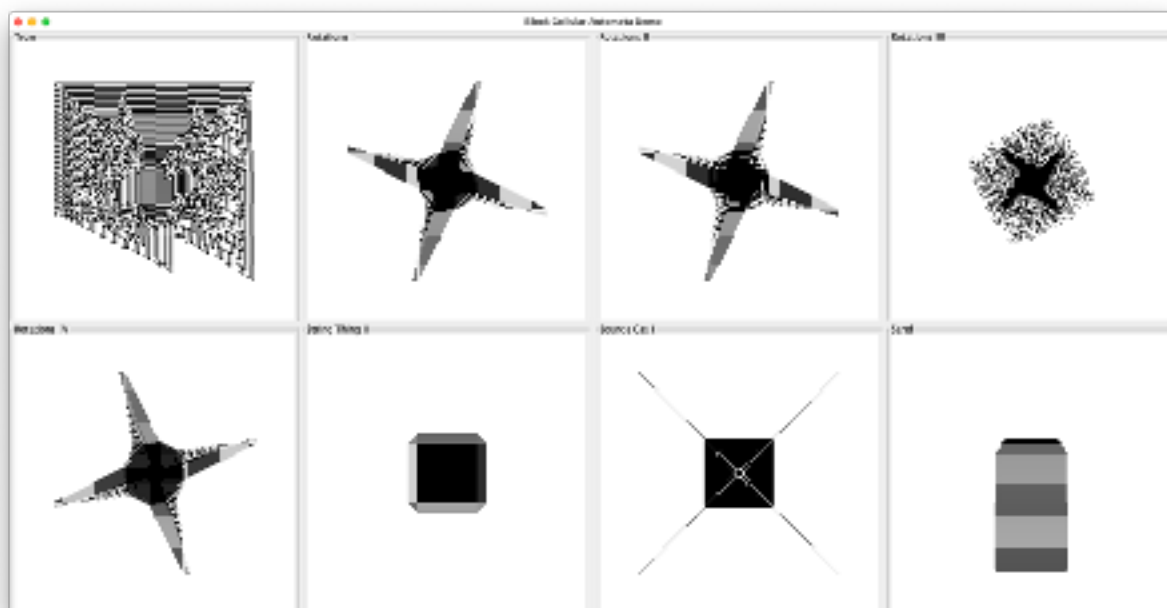
In your labs project, create a new class `BlockCellularAutomata`, and there the method

```
public static void margolusNextState(boolean[][] current, boolean[][] next, int[] rule, int off)
```

that is given the `current` state of the grid of cells, from which this method computes the next state of that grid using the given `rule`. The parameter `off` indicates whether the current time step is odd or even, so that this parameter equals 0 for even time steps, and 1 for the odd ones. (The method doesn't need to know the current time to work, only the parity of the current time.)

This method does not create and return the new `next` array object, but the caller provides this array object `next` to be filled in as a method parameter. In the automated JUnit test, this `next` array is guaranteed to have the exact same dimensions as the `current` array from which the next state of the block cellular automaton is computed. This handy scheme eases the burden of the Java garbage collector when the execution thread of the JUnit tester (provided by the instructor here; what are you so worried about?) does not constantly have to be creating new grid arrays only to then throw these perfectly good brand new array objects into garbage at the very next time step! Instead, this animation logic can simply keep recycling the same old array objects back and forth to keep track of the current state and the next state of the automaton, interchanging the roles of the two array objects at every time in this dance. (However, unlike with freshly created arrays, the recycled array is not guaranteed to be filled with false values at the call, so your method needs to compute and fill in every element.)

The JUnit test class for this lab also contains a `main` method that launches eight separate instances of a block cellular automaton, each instance using eight aesthetically behaving rules copied from the page "[Reversible Cellular Automata](#)". When run with your method, the initial state of a solid block of live cells as the center of each grid should evolve towards the screenshot below.



Students interested in GUI programming with Swing should again note in the `main` method that out of the box, the `JFrame` instances that represent the underlying GUI windows inside Java and Swing do not obey the close button and go gentle into that good night. Instead, they will rage, rage against the machine by ignoring the order to ride into the valley of unreachable objects whose role is only to do or die without making a reply. Normally we use the method `setDefaultCloseOperation` to tell a `JFrame` instance to become more respectful of the chain of command. However, we cannot do so in this example, since launching a new `Timer` instance creates a new execution thread that we must ensure will also cease to run when the user closes the window. Otherwise the Java Virtual Machine process would not be able to terminate, since it has to keep the timer ticking like a forgotten metronome that nobody is listening to any longer!

As an interesting aside, all instances of the Swing `Timer` are managed by the same execution thread that internally keeps all active timers in a **priority queue** data structure (presumably implemented as a **binary heap**) to quickly find the active timer that currently has the least amount of time left until its next moment to step in the limelight. This common thread that manages all timers will repeatedly `sleep` until that time to shine comes up for the first timer waiting in this priority queue. It then executes the action associated with that timer, and pushes that timer back to the priority queue where the binary heap logic quickly moves it to its correct place in the priority queue. Unlike actual execution threads whose creation is expensive, you could easily launch thousands of timers to simultaneously tick in background at arbitrary speeds! How wonderful it is that things that `sleep` quietly so that the processor does not need to be constantly checking up on them do not consume even one processor cycle by their mere existence in the memory!

In addition to the eight example rules illustrated in the test class, this mechanism allows so many possible rules even for 2-by-2 squares and `boolean` cells that humanity will surely never explore the possibilities of but a tiny portion of these $16^{16} = 2^{64}$ possible rules. Even if the `rule` array is restricted to be a permutation of integers 0 to 15, this still leaves us $16! = 20,922,789,888,000$ possible rules to play around with. All such **bijective** permutation rules are guaranteed to be **reversible**; any such automaton can always be run backwards in time from any arbitrary future state all the way back to the starting state and even beyond that merely by inverting that permutation to use as the new rule! (Such an automaton therefore cannot have any [Gardens of Eden](#).)

Further restricting these rules to be **symmetric** to all four compass directions will also drastically lower this count. Furthermore, the configuration where all cells are dead is usually restricted to stay dead (that is, a legal rule must map the Margolus squares of value zero to the same squares of value zero), as is the case with all of the rules used here. All this still leaves us a whole bunch of possible rules to explore. In absence of this natural restriction of something ever coming out of nothing, the entire automaton grid would immediately fill with nonzero stuff to create an annoying flashing that can potentially harm students with epilepsy, as is the case with the famous [Critters](#) rule.

The block cellular automaton mechanism also trivially generalizes to larger k -neighbourhoods, of which Margolus is a trivial special case of $k = 2$ (the neighbourhood shift offsets must then also use the same period of k), and a larger number of possible states for the individual cells (usually then called the **colour** of that cell), with the number of possible rules blowing up exponentially into unimaginable numbers and emergent complexity. Such automata can be used for generative and computational art; see the web page "[6-Colour Block Cellular Automata](#)" for illustrative examples.

Lab 59: Hitomezashi Golygons

For testing: [HitomezashiMain.java](#)

This problem was inspired by the [Numberphile](#) video “[Hitomezashi Stitch Patterns](#)” that the reader might want to first check out to get a quick idea what these patterns are all about. (You should also note the clever mental technique used in the same video to create a random bit pattern!) The blog post “[MathArtChallenge Day 14: Hitomezashi stitching](#)” at [Arbitrarily Close](#) also succinctly explains the mathematics behind these patterns. Such patterns are used in [traditional Japanese stitching](#) to create regular designs that are both useful and pleasant to look at, for the best of both worlds. However, sprinkling just a pinch of randomness on this surprisingly simple process reveals an emergent complexity that was waiting underneath the surface this whole time for its chance to blossom.

A **hitomezashi** pattern is drawn on a rectangular area that has been subdivided into a grid of squares. In the space that separates each row of squares, a horizontal stitching line is drawn so that a line segment is rendered visible only under every other square, as if creating a stitch on a fabric with a needle and a thread. Each row is initially assigned a truth value to decide whether the squares in the odd positions (if `true`) or in the even positions (if `false`) get a visible line segment under them. This same process is repeated for the vertical columns to complete the pattern.

If the rows and columns are assigned truth values in some simple repeated patterns, the entire two-dimensional pattern will end up being uniformly repeated with the same period. However, a random assignment of truth values to each row and column produces a chaotic fractal pattern of [unit golygon](#) shapes whose every edge has length of exactly one. As long as there is room inside a shape, that shape will properly contain smaller such fractal shapes tucked inside it, with none of these nested inner shapes ever abutting any of their brothers or the larger shapes that immediately surround them.

This lab is another exercise of how to create and draw on `BufferedImage` objects to create two-dimensional pixel raster images. Create a new class named `Hitomezashi` in your labs project folder, and in that class the following `static` method

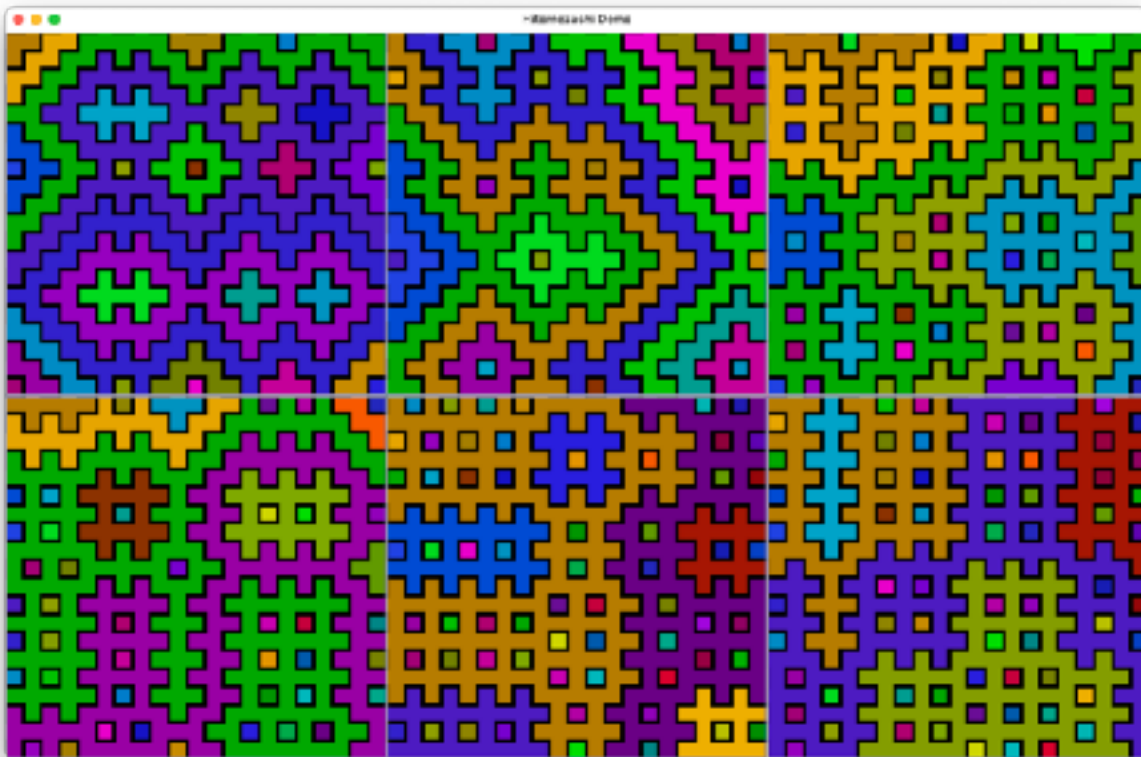
```
public static BufferedImage createPattern(  
    int width, int height, int square,  
    boolean[] horizontal, boolean[] vertical  
)
```

The first two parameters `width` and `height` define the dimensions of the pattern, as measured in squares, and the parameter `square` gives the size of each individual square in pixels. For example, if both `width` and `height` equal 25, and `square` equals 20, the pattern created by this method will be $25 \times 20 = 500$ pixels wide and tall. The bit vectors `horizontal` and `vertical` determine the parity of squares that get a visible line segment, given separately for rows and columns.

From these parameters, this method should create a new `BufferedImage` instance producing that pattern as the line segments rendered in black against a white background. The instructor provides

a main class [HitomezashiMain.java](#) that calls this method with six different random horizontal and vertical bit vectors, and displays the gallery of returned images inside a Swing JFrame window. The bit vectors that produce the six patterns of the screenshot below are created by making each bit to be equal to the previous bit with a probability that ranges from 30% to 80%, with a ten point increment between the successive patterns. To ensure that your code treats both width and height parameters as it should, each of these six shapes is 21 squares wide and 20 squares tall, to catch any accidental misuse of width as height, or vice versa.

Colouring the golygons is not required to complete this lab. You can just render only the black line segments that form the outline of that shape. Even so, make sure that your golygons end up having the exact same outline as those given in the example screenshot from the main class shown below. However, if you have already completed the Flood Fill exercise specified earlier in this document, you might as well use its `floodFill` method to paint the insides of these contiguous golygons with random colours to make them pop out powerfully from their statistically similar surroundings.



At least to the untrained eye of this author, these patterns evoke nostalgic visuals of Native American artwork, at least if the colour knob used to fill in these golygons were adjusted towards more earthy and muted tones. Readers who have completed the Flood Fill project can experiment further with such tones. Furthermore, as explained in the above Numberphile video, this technique can be applied to other grids such as **triangular** grids, not just to the familiar grid of integers.

The same technique can be further generalized by using some more complex pattern to fill in the line segments between the rows and columns, instead of the regular on-off metronome stitch. If such stitches are rendered randomly using line segments aligned to the four intercardinal compass directions, the result is a **maze generator** made famous in the classic Commodore 64 Basic one-liner [10 Print](#), arguably the spiritual great granddaddy of all such later generative art.

Lab 60: Bit Deque

JUnit: [BitDequeTest.java](#)

Same as other imperative programming languages, Java guarantees that the elements in an array are stored in consecutive memory locations. Assuming that these arrays are **homogeneous** in that each element takes the same number of bytes to store, such arrays enjoy the advantage of being **random access data structures**. This conceptual misfire of a terminological chimera has nothing to do with randomness, and would be much more truthfully called an *arbitrary* access data structure, but this linguistic hill has long since been turned into condos. In a random access structure, as opposed to a sequential structure that must be accessed in order from beginning to end, the element in any position can be accessed for both reading and writing in the same constant time, regardless of the earlier operations performed on that array.

(Sure, *ackshually* the above isn't technically true for truly huge arrays, due to **memory paging** and **processor caching** effects. However, this is about as relevant for the discussion here as planning your next supermarket trip using the mental model of travelling on a spherical surface: globally true, yet locally utterly irrelevant.)

The downside of such rigid memory order is that every array object must maintain the same length during its entire lifetime. This makes arrays inconvenient for storing dynamic data structures where the number of elements can increase and decrease during execution. At this point, old books and other such material on data structures used to recommend the use of **linked lists** to represent such dynamic structures. Each element stored in a linked list is kept in a separate **node** object allocated from the heap, requiring all the associated bookkeeping needed for the heap management and garbage collection to take place behind the scenes.

Furthermore, each node object still has to use an additional 8 or 16 bytes to keep track the memory address of its successor and predecessor. Spending all this extra memory in addition to the **payload** of the data elements allows adding and removing an element from a known position in guaranteed constant time, the upside of the flexible chainlike structure of linked lists whose nodes and chains of nodes can be extracted and inserted in arbitrary positions in constant time. Arrays can contain only the payload data bytes, and don't need anything else to keep them in order, since the memory hardware already does that job for them for free.

However, sometimes we know for a fact that elements will always be added and removed at the end of the current structure. Consider the [StringBuilder](#) class of the Java standard library, where new characters and strings are always appended to the end of the string that is currently being built piecemeal. A linked list would be overpaying for overkill in such situations. A superior technique uses an ordinary array to store the payload elements, with a separate integer data field (or two) on the side to keep track how much of this array is actually being used to store the current payload cargo. The positions that do not currently contain any payload elements are **slack** available for future additions to the structure. The element values in the slack area of the array can be anything, since they will never be read out as actual elements.

The standard terminology is to define the **capacity** of the structure to equal the actual length of the underlying array, and define the **size** of the structure to equal the number of actual payload elements stored inside the array. This terminology can be seen in the public interface of the good old Collection workhorse [java.util.ArrayList<E>](#), for example. The difference between these two quantities gives the slack space currently available. As long as some slack space exists, adding new elements to the structure is a simple constant time operation, regardless of the size and the capacity of the structure. Once the slack runs out when a new element is being added, a new array twice the size the previously used array is created to copy the existing elements into. The structure then says *sayonara* to the old array as thanks for its completed service, and continues to use the new bigger array from that point onwards, as if that new array had been the original array all along, now just with plenty of slack space available for future additions. (Convenience methods `copyOf` and `copyOfRange` of the [java.util.Arrays](#) utility class make this simulated resizing practically a one-liner.)

Primitive arrays are often used to implement various forms of **queues**, a generic name for a certain type of data structure of which the “queue” in the everyday sense of this word is but one possible special case. In the terminology of data structures and algorithms, a queue is something that you can at any time **push** elements into one by one, and can any time **pop** to ask for one element to come out. The actual subtype of the queue determines the element that comes out at the pop operation. A **first-in-first-out queue (FIFO)** behaves like a queue in the everyday sense of this word, popping out the element that has waited in line inside the queue the longest. A **last-in-first-out queue (LIFO)** behaves as a **stack** so that the element popped out is always the one who has waited the least time inside the queue. Both FIFO and LIFO queues have useful applications in many algorithms that we have already seen. For example, the **breadth-first search** algorithm used to solve the “Big Ten-Four” and “Flood Fill” exercises earlier in this document uses a FIFO queue to store the frontier of elements that are waiting to be processed.

Of course, the standard library of not just Java but any programming language that we could ever seriously consider for real work in this millennium offers both these queue structures, and more. Nobody should ever have to implement either one from scratch any more than, say, they would implement their sorting algorithm. (What separates computer science from most other fields is that no problem ever *needs* to be solved twice.) However, in the same spirit as the [NatSet](#) example class used to store a monotonic set of natural numbers with one bit per element, we realize that these generic queue classes that are able to store any type of elements will be horrendously inefficient with respect to the memory use when used to store primitive data types. You don’t have to be Uncle Scrooge to viscerally shudder at the idea of wrapping each payload item, itself only a couple of bytes, inside some object that, even in the best scenario, takes at least 24 bytes in the object heap!

In this lab, we implement an array-based **double-ended queue** data structure to store boolean values, for as tight use of memory as this can possibly get. A double-ended queue, or just **deque** from now on, is a linear queue that allows elements to be pushed and popped from either end at any time, for four different operations `pushFront`, `pushBack`, `popFront` and `popBack`. If you have a deque available, you can always voluntarily restrain yourself to use only the operations of the LIFO stack and FIFO queues, making these structures special cases of a deque. The added flexibility of the deque can come handy in special situations... provided that we first figure out a way to implement all these deque operations to work in constant time, at least when **amortized** over a long series of

such operations. If each deque operation needs a linear time internal reorganization after every operation, its user code might as well just keep all the data in its own arrays anyway.

In your project folder, create a new class named `BitDeque`. This class will have the following six public methods for you to fill in once we get the preliminaries out of the way. The first four methods implement the deque operations, and the last two allow the user code the query for the current size and the current capacity of the queue.

```
public void pushFront(boolean v)
public void pushBack(boolean v)
public boolean popFront()
public boolean popBack()
public int getSize()
public int getCapacity()
```

To store the elements currently inside the deque, this class should first define the data fields

```
private boolean data[];
private int startIdx;
private int endIdx;
```

The elements are kept in the bit vector, here named `data`, that will have slack in both ends to allow pushing and popping elements from both ends in constant time. The position indices `startIdx` and `endIdx` delimit the payload area of stored elements in the middle. Conventionally, the position `startIdx` is treated as being **inclusive**, whereas the position `endIdx` is treated as being **exclusive**. For example, let `data.length==20` so that the array can store twenty elements. Supposing that `startIdx` equals 5 and `endIdx` equals 13, the deque currently contains exactly eight payload elements, as given by the elegant formula `endIdx-startIdx`. (If the end index were also treated as inclusive “because *muh symmetry*” or some other sentimental *ad hoc* rationalization, you would always have to remember to add one to the result given by this formula. Other benefits of using an exclusive end index will follow until the morale improves.)

Once you understand how this structure works by packing everything inside a simple bit vector, all four deque operations can be implemented with just a couple of lines of code, with delightful symmetry to boot! For example, the `pushFront` method will first decrement `startIdx`, and place the new element into the new `startIdx` position. Because the end index is treated as exclusive, the `pushBack` method will first place the new element in position `endIdx` that is known to be the first position of the slack after the payload, and increment `endIdx` only after adding the element. (Knowing the distinction between the prefix and postfix forms of the operators `++` and `--` will now pay its dividend.) The methods `popFront` and `popBack` operate symmetrically to undo the effect of the corresponding `push` and, analogous to the two pushing methods, differ by performing the element insertion and updating the index in the opposite order.

The class should also define the following constructor to initialize the deque. Since there is no point being miserly with individual bits, the capacity of the `data` array will be kept at a minimum of 512

bits during the lifetime of the deque, regardless of how many payload elements actually happen to be stored in the deque. The payload area in the middle is initially empty, but thanks to the decision to use the `endIdx` as exclusive, recognizing this situation and handling it correctly is no problem, since the deque is empty if and only if both indices `startIdx` and `endIdx` are equal.

```
private static final int MIN_CAPACITY = 512;
public BitDeque() {
    data = new boolean[MIN_CAPACITY];
    startIdx = endIdx = MIN_CAPACITY / 2;
}
```

As long as there is slack available in the direction of the push, everything is straightforward. But what happens when this slack inevitably runs out after enough additions? To get out of this predicament, your class should define a private utility method

```
private void expand()
```

that both methods `pushFront` and `pushBack` call when there is no slack available in the desired direction. This method creates a new bit vector **twice the size** of the current data array. After the existing payload elements have been copied to the center of this new vector, the previous data array is discarded in the loving arms of garbage collection. The `startIdx` and `endIdx` indices are updated to reflect the correct location of the payload elements in the new larger array, after which there will be plenty of slack available in both ends for future deque operations.

Note that the automated JUnit test class will enforce the fact that the threshold and logic of calling `expand` is symmetric in both directions, and your code will not pass this test unless it does this. (Sorry guys, I didn't make the rules; I am but a humble messenger of far higher forces responsible.)

For the common use case where the deque is used as a FIFO queue where the payload area is always moving in one direction, there will be plenty of slack available in the other end of the array, all seemingly sitting there doing nothing but twiddling its virtual thumbs. Instead of wasting perfectly good memory by allocating a new bigger array and then continuing to essentially use only one quarter of its space, you can implement a private utility methods that moves all payload elements so that they are again centered in the array. you should use the method `System.arraycopy` to perform this move efficiently in bulk, instead of blithely using a for-loop through the relevant positions and copying their elements inefficiently one at the time. This method has been implemented natively inside the Java Virtual Machine, and will therefore be significantly faster than any for-loop that we could ever write in the higher level of abstraction of Java. Remember to also update the `startIdx` and `endIdx` indices yourself, though, since no method can magically update your fields and local variables for you.

(You may also note how the method `System.arraycopy` shows its age all the way back to the internal development versions of Java in how its name does not follow the convention by being named `arrayCopy`. This method has been grandfathered in the language the same way as `Color.Black`

and other named constants for the common primary colours that should have originally been named `Color.BLACK` in accordance to the all-uppercase naming convention of named constants.)

An alternative solution to using the previous two methods, although somewhat more complicated to code, is to treat the `data` array as being **circular**, so that either index that goes past the edge of the array will appear at the other edge. This scheme is more efficient in its memory use in that the space available on one side is available for pushing elements from the other side. Data array expansion would be triggered by a push operation that would make both `startIdx` and `endIdx` become equal, a situation that was supposed to mean that the entire deque is empty. The expansion code will also become somewhat more complicated. Students who implement their deque this way need to be mindful of (in this case quite literal) edge cases where these indices are pointing to the first and the last position of the array, which can happen either way during the use of the deque. Even outside this edge case, the contents of the deque are generally split into two contiguous subarrays, which the expansion must copy correctly into the newly allocated larger array.

The last situation that you unfortunately have to consider in this lab is when a large number of elements is first pushed in the deque, after which almost all these elements are popped out. It seems very inefficient to end up using a giant array to store a handful of elements, most of this giant array ending up as slack that may never be filled in again. Even if we knew for certain that it will be filled in later, all those megabytes might as well be put in some more productive use before that happens! (A megabyte here and a megabyte there, and pretty soon you are talking about real memory.) An immediate idea would be to define a utility method for contracting the deque as

```
private void contract()
```

Your `popFront` and `popBack` methods must call this method whenever the size of the payload block becomes too small compared to the overall capacity of the `data` array. The automated tester will enforce that unless that capacity of the deque is at its smallest possible value of 512 bits, the capacity of the deque **may not be more than four times the current number of elements** in that deque. Again, be mindful of edge cases when implementing the `contract` method. Symmetric to the previous `expand` method to be able to undo its effect, `contract` should allocate a new array half the size of the original `data`, copy the existing elements around the center of this smaller array, and discard the previous `data` array hoping that the bits and bytes that comprise it will hopefully be put in better uses. However, realize that intuition of when to trigger the contraction can now easily lead you astray, especially if you again vigorously hand wave something about symmetry!

The threshold for triggering a contraction operation absolutely positively **cannot** be that the `data` array is only half full, even when the automated tester does not enforce this. Following the general principle of [hysteresis](#), this threshold must be that the array is **one quarter full**. Otherwise, a clever adversary could engineer a situation where immediately after an expansion has taken place, precisely two elements are popped out to trigger a contraction, which is then followed by two elements being pushed in to trigger an expansion. The malicious adversary could repeat such a back-and-forth cycle any number of times, and the push and pop operations would no longer be guaranteed to work in amortized constant time, making the surrounding system that relies on the efficient operation of this deque to slow down to a glacial crawl.

Lab 61: Save The Date

JUnit: [CalendarQueueTest.java](#)

Continuing on the general theme of queues from the previous problem, this problem looks at **priority queues** where the universe of elements that can potentially get pushed in the queue must be **comparable** with some total order that, for any given two elements, can unambiguously determine which one of these elements is smaller. This ordering determines a **priority** for these elements, so that the pop operation always brings out the element with the lowest priority among those that are currently in the queue.

The class [PriorityQueue<E>](#) in the Java standard library uses the classic **binary heap** to maintain the priority ordering behind the scenes. This class normally compares the elements by their built-in comparison `compareTo`, but it can also be given some [Comparator<E>](#) strategy object to perform the order comparisons according to any desired criteria to settle the relative priorities of these objects. However, akin to how bees make honey and beavers build dams all by instinct, as a species humanity has come up with [a multitude of data structures](#) that implement a priority queue with efficient operations even for millions of elements. These data structures allow the basic push and pop operations to run significantly faster compared to keeping all elements in a messy unsorted array, and always looking up the smallest element with a linear loop every time the next element is popped out. (Some of these advanced priority queue data structures allow additional operations such as **melding** two priority queues together into a single priority queue, or **updating the priority** of an element currently somewhere in the queue, to be done significantly faster than they are when keeping the elements in an unsorted array.)

This lab looks at a classic but now mostly forgotten data structure of [calendar queue](#), another has-been old wino from a simpler age that might still have some vinegar in its veins to save the day in situations that makes the more general structures grind down to a crawl. Consider a typical algorithmic application of any priority queue in an **event simulation**, where each **event** consists of a time when that event will take place in the future, plus some payload information associated with that event to describe how that event affects the simulation. The event simulator algorithm itself is a simple while-loop that repeatedly pops out the earliest event from the priority queue, and executes the effect of that event in the world being simulated. Each event can and will produce new events to take place in the future. These new events are added in the priority queue during the execution of that event that begat them. In this technique, the event simulation algorithm can always instantly jump to the next interesting event to happen in the future, instead of patiently waiting for something interesting to happen while advancing the simulation only one time unit at the time.

If the world being simulated follows the usual laws of **causality** in that events taking place at the present time cannot affect the past by creating new events that do take place in the past (thus potentially creating ill-defined situations inside the entire simulation, such as whatever would be the equivalent of the famous [grandfather paradox](#) in the world being simulated), the event popped from the head of the priority queue can only cause later events to be pushed into the queue. Furthermore, in most simulations, most events will cause new events to take place in the very near future. Under these two reasonable assumptions about how events beget other events to follow them, a calendar queue can perform its basic push and pop operations of a priority queue in expected constant-time

that is almost a cinch. The associated algorithms are also among the simplest to be found among the of the priority queue data structures linked above.

Analogous to an ordinary desk calendar, the calendar queue assumes that time proceeds in discrete **years** that have been divided into **days**, the smallest unit of granularity available for scheduling these events. To get started with this lab, add the instructor's class [CalendarEvent](#) in your labs project. **Do not modify the contents of this class in any way while completing this lab.** Instead, just read through this simple **data class** to see how its objects represent events that take place in a given year and day, with the payload information of each event given as a message string, and the year and day given as integers. Unlike us glorified apes in trousers who have trouble remembering anything symbolic better than a toaster could, the computer does not need the convenience division of years into months, but each year will consist of exactly `DAYS_IN_YEAR` days, with no leap years existing. Days are numbered starting from zero, and count up to `DAYS_IN_YEAR-1`, after which the year increases by one as the day rolls back to zero to begin a new year.

For the purposes of automated testing of this lab, this named constant `DAYS_IN_YEAR` has been defined to equal ten inside the `CalendarEvent` class. (We can pretend that this exercise takes place on those nanoplanets in *The Little Prince* populated by various melancholic yet tragically colourful characters.) However, changing this named constant to 365, or any other positive integer for that matter depending on which planet we live on, should not change anything in the logic of your solution code. Never hardcode any **magic numbers** directly in your code, but always refer to such quantities using the named constants!

Internally, a calendar queue consists of an array that has a slot for each of the `DAYS_IN_YEAR` possible days that exist in our time keeping. From each such slot hangs a **singly-linked list** of events that fall on that particular day. The events of each list of all such same-day events are kept sorted by the year that those events will take place. For example, if the queue contains three events that all take place on day 7, but in three different years 2525, 3535 and 6565 as in the famous song of yore, these three events are guaranteed to be stored inside the list so that the event of the year 2525 is first, the event of taking some pills in the year 3535 comes right after that, and the event of picking your son and daughter from a long glass tube in the year 6565 will be the last. These events are distributed over these linked lists in this manner, finding the earliest event that takes place on any given particular day is now quick and easy, regardless of the total number of events stored in the calendar queue. (Readers who are familiar with the internal operation of **chained hash tables** may recognize this same basic technique being productively applied in a different context.)

As for representing these chains, the Java Collection Framework offers the [LinkedList<E>](#) data structure for linked list operations. However, because of the utterly idiotic design decision done in the pre-generics Java [of making arrays covariant](#) that requires every Java array object to know its full element type at the runtime, the implementation of Java generics under the hood with **type erasure** that throws out all this runtime information after compilation will by definition make it impossible to have arrays whose element type is any generic class, even if the type of that generic class is fully instantiated! However, this lab will be an exercise in implementing the basic linked list operations from scratch, as has historically usually been done in second programming courses such as this one. We all have to eat a little bit of dirt before we die, even if we otherwise get to reach towards the higher Platonic planes in our coding.

In your labs project, create a new class `CalendarQueue`. Inside this class, copy-paste the following nested utility class whose objects represent individual **nodes** of a **singly linked list**. Each such node contains the payload of the event that is stored in that node, and a reference `next` that, just like it says on the tin, points to the next node in the linked chain.

```
private static class CalendarNode {
    public CalendarEvent event;
    public CalendarNode next;
    public CalendarNode(CalendarEvent event, CalendarNode next) {
        this.event = event;
        this.next = next;
    }
}
```

In the last node object currently stored in a chain, the reference `next` equals `null`, to indicate in a controlled fashion the absence of a successor node object. To implement all the operations of a calendar queue, we only need the operations of a singly linked list, so these node objects don't need to waste memory for another reference to point to the predecessors in that chain.

Note also that this nested class is not an **inner class**, seeing that it has been defined to be `static`. Unlike the non-static inner classes whose objects cannot exist on their own but must always exist in the context of some mothership of an outer class object (this allows the inner class objects to access all public or private members of this mothership object directly exactly as if these members were their own, without even any special syntax), objects from static nested classes don't need anything from the outer class and may therefore exist on their own, completely independent of whatever outer class objects have come into existence so far. The advantage of using static nested classes whenever feasible is saving eight bytes of heap storage per object compared to inner classes, due to the absence of the hidden reference to the mothership object that must be stored separately inside every inner class object, as the inner class objects can have different motherships.

(Students who have been paying attention during this instructor's lectures might at this point object that this nested class violates the proper principles of **encapsulation** by defining its two members `event` and `next` to be `public`! Hold your horses; no violation of any noble engineering principles is actually taking place here, since this nested class has itself been defined `private` inside the outer class. No information about implementation details leaks out to the outside world any more than would leak from using local variables inside some method.)

After this nested class, define the following fields inside the class `CalendarQueue`:

```
private CalendarNode[] head;
private int size;
private int currentYear, currentDay;
```

The `head` array contains a total of `DAYS_IN_YEAR` elements, each element the reference to the first node of the linked list hanging from that slot. For days that have no events scheduled to happen during that day, this slot is initially `null`. The field `size` contains the current number of events stored in this calendar queue, updated inside every `push` and `pop` for quick constant time lookup in the `getSize` method. The calendar queue object must also keep track of the current year and the current day of the simulation, here stored in two fields `currentYear` and `currentDay`. As the last piece of code that this specification will give you before unleashing you to complete the rest yourself, here is the constructor of `CalendarQueue` to properly initialize these fields.

```
public CalendarQueue(int currentYear, int currentDay) {
    head = new CalendarNode[CalendarEvent.DAYS_IN_YEAR];
    this.currentYear = currentYear;
    this.currentDay = currentDay;
}
```

The field `size` is already zero to begin with, and thus needs no initialization. (Of course, it never hurts to be explicit.) The three public methods left for you to implement are

```
public int getSize()
public void push(CalendarEvent event)
public CalendarEvent pop()
```

The automated JUnit test for this problem guarantees that it will never push in any events that take place earlier than the current year and day of the calendar queue, and will never try to pop out an event from a queue that is empty. Nobody can be reasonably expected to achieve anything that is logically impossible, after all.

The method `push` should first access the day of its argument `event` to decide which slot the new `CalendarNode` object that contains that event as a payload will be linked into, and from that chain, find the position where this new node should be linked to. There are essentially three possible cases to get right. First, if the linked list hanging from that slot is empty, or if the event stored in the first node of the chain takes place later than the event currently being pushed in, the node that stores the new event becomes the first node in the chain. (Your code must correctly break the ties between two events with the same year and day by the message component. This happens automatically if you use the `compare` method of `CalendarEvent` to perform this comparison.) This is easily achieved by assigning the reference in the proper head slot to point to this new node, whose `next` reference in turn is made to point to the node that was previously the first node hanging in that chain. That previous first node will therefore now become the second banana, properly positioned to become the successor node of the new node.

(Note how the same unconditional logic for assigning the `next` reference works for both cases. It is not okay to try to follow a `null` reference to get to an object, whereas merely assigning a `null` value to some reference is hunky dunky.)

Otherwise, if the chain hanging from that day already contains at least one node, loop through the nodes of this chain by repeatedly following their `next` references to step to the next node in the chain. Keep going until you find the **predecessor node** whose successor the new node should become to maintain the sorted order of the events stored in that chain. Reassign the `next` reference of this found predecessor node to point to the new node, and assign the `next` reference of the new node to point to the earlier successor of that predecessor node. (Again, the same unconditional logic works even when the predecessor is the last node of the chain.)

The method `pop` should first look at the slot of the `currentDay`. If the year of the event stored in the first node of that chain equals `currentYear`, remove that node from the chain using a symmetric approach to the one used to insert a node, and return the event stored in that node as the answer. Otherwise, keep incrementing `currentDay`, rolling back to zero and incrementing the `currentYear` every time you go past the end of the array, until you eventually must come to a slot that satisfies the previous condition for its first node to allow you to terminate. Assuming a reasonable distribution of event times in the near future, this approach will usually return the next event faster than you could say boo to a goose, regardless of the number of events currently stored inside that priority queue.

To admire the simple elegance of the calendar queue approach, just realize that this search algorithm will only look at the first node of any slot! Even if a billion other future events that take place that same day in some far future year (perhaps something funny has already been destined to take place at [The Restaurant at the End of the Universe](#)) are hanging from every chain, these unseen nodes do not affect the time it takes to find the event to take place next. More remarkably, these unseen nodes will also not affect the speed of updating the priority queue data structure after extracting that first event! This is something truly unusual among not just with priority queues, but with all data structures that this author has ever seen. Typically, even if that first element can be found and extracted in constant time, the time needed to update the rest of the structure to reflect the new state is usually at least **logarithmic** with respect to the overall size of the data structure. But not today, Satan. Today we will feast and rejoice for the gift that we have received in this.

Lab 62: Worley Noise

For testing: [WorleyNoiseMain.java](#)

The advantage of using pseudorandom number generators instead of true randomness is that any algorithm that uses the same RNG initialized with the same seed value is guaranteed to produce the same results every time. Since the golden days of [Elite](#), many computer games have utilized this approach by using [procedural generation](#) to build the levels of the game. Even confined to use 64 kilobytes of memory for literally everything in that game, such games can potentially pack truly enormous game worlds into such tiny memory space. Only the level generation algorithm needs to be stored, and its RNG is each time seeded with, say, the level number or the coordinates of the level that is being generated for the player to enter. Of course, such levels can't remember any changes that the player caused in them, but are generated afresh each time the player returns to them.

In general, using raw randomness to decide what goes where in each level is usually not the best for an aesthetically optimal outcome. Imagine playing a game of *Civilization* where the type of each tile has been randomly chosen among all possible types of tiles in that game, independent of the tiles surrounding it! As amusing as playing in this kind of *Secret Wars* patchwork battleworld might be, even if only as a thought experiment, we normally want to place our thumb on the scale to produce structures whose complexity falls somewhere around the sweet spot in the continuum between the two end points of the rigorous uniform determinism that allows no variety, versus the boiling pot of random noise that doesn't allow permanent meanings to coalesce.

If the randomly generated structure is supposed to resemble something familiar to us that has an organic existence in our physical reality, such as an island, patterns on granite or other types of stone, or biological cells, finding the right balance between chaos and order by computational means might at first seem an impossible task. Fortunately, many techniques to create procedural noise have been discovered, so tweaking the parameters of the noise generation and the palette user to render the result can produce surprisingly natural looking patterns. Even better, the algorithm can produce the result image in any desired resolution, without any pixellation! In this lab we implement one famous procedural noise generation algorithm known as the [Worley noise](#).

To understand how Worley noise works, we should first look at a classic computational geometry mainstay of a [Voronoi diagram](#). For the given set of points P that can be arbitrary (but have typically been placed pseudorandomly on the image to be generated), the Voronoi diagram divides the image into disjoint convex polygons so that each polygon contains precisely the points that share the same nearest point in P . Once the Voronoi diagram for the given set of points has been constructed, many other important computational geometry properties can be conveniently computed from it. However, colouring all the pixels of the same polygon using the same flat uniform colour does not create a very aesthetically pleasing image, which is our goal and purpose in this lab.

Worley noise takes the same approach and runs with it by considering not just the point in P that is closest to the pixel being coloured, but potentially all points in P . The colour of each pixel is individually computed from the weighted sum of its sorted distances to all points in P , although in practice, most of these weights are set to zero. For example, the colour of a pixel might be determined by its

distances to its second, fifth and seventh nearest points in P , and the rest of the points do not matter for colourization of that particular pixel.

Since weighted distances can become arbitrarily large and we want to normalize the colours to be between zero and one, we first need some function to map any nonnegative real number distance into the real line interval $[0,1]$, preferably in some efficiently computable but generously aesthetic fashion. Even at the slight risk of giving the readers painful flashbacks to their high school trigonometry classes, we recall that the trigonometric functions *sin* and *cos* not only oscillate between values -1 and 1 , but are smoother than a baby's bottom in all their differentials. A slight adjustment to also consider the desired frequency gives us the utility function

```
private static double wave(double x, double a, double f) {  
    return (1 + Math.sin(Math.pow(Math.abs(x), 0.95) / f)) / 2 * a;  
}
```

where the parameter a is the **amplitude** and the parameter f is the **frequency** of the wave that maps the given real number x into the interval $[0, a]$. To make this transformation a bit less boring, the parameter x is first raised to some fractional power before the actual transformation, to make the frequency changes slow down as x increases. (After completing this lab, the student may wish to change the exponent from its present value of 0.95 to something larger or smaller to observe the effect that this exponent has on the shape of the noise.)

We next need to refresh our high school algebra for the formulas of computing the distance between two points (x_0, y_0) and (x_1, y_1) on the two-dimensional plane. The **Euclidean distance** is derived from the Pythagorean theorem, but **Manhattan distance** and **chessboard distance** could just as well be used as alternative (but mathematically perfectly sound) distance metrics, from the set of infinitely many other functions that could serve in such a role.

```
private static double euclidean(  
    double x0, double y0, double x1, double y1  
) {  
    return Math.sqrt((x1-x0)*(x1-x0) + (y1-y0)*(y1-y0));  
}
```

```
private static double manhattan(  
    double x0, double y0, double x1, double y1  
) {  
    return Math.abs(x1-x0) + Math.abs(y1-y0);  
}
```

```
private static double chessboard(  
    double x0, double y0, double x1, double y1  
) {  
    return Math.max(Math.abs(x1-x0), Math.abs(y1-y0));  
}
```

These preliminaries out of the way, we now have everything that is needed to generate some Worley noise. In your project folder, create a new class `WorleyNoise`, and in there the following method that probably takes the largest number of parameters among all the methods seen somewhere in this specification.

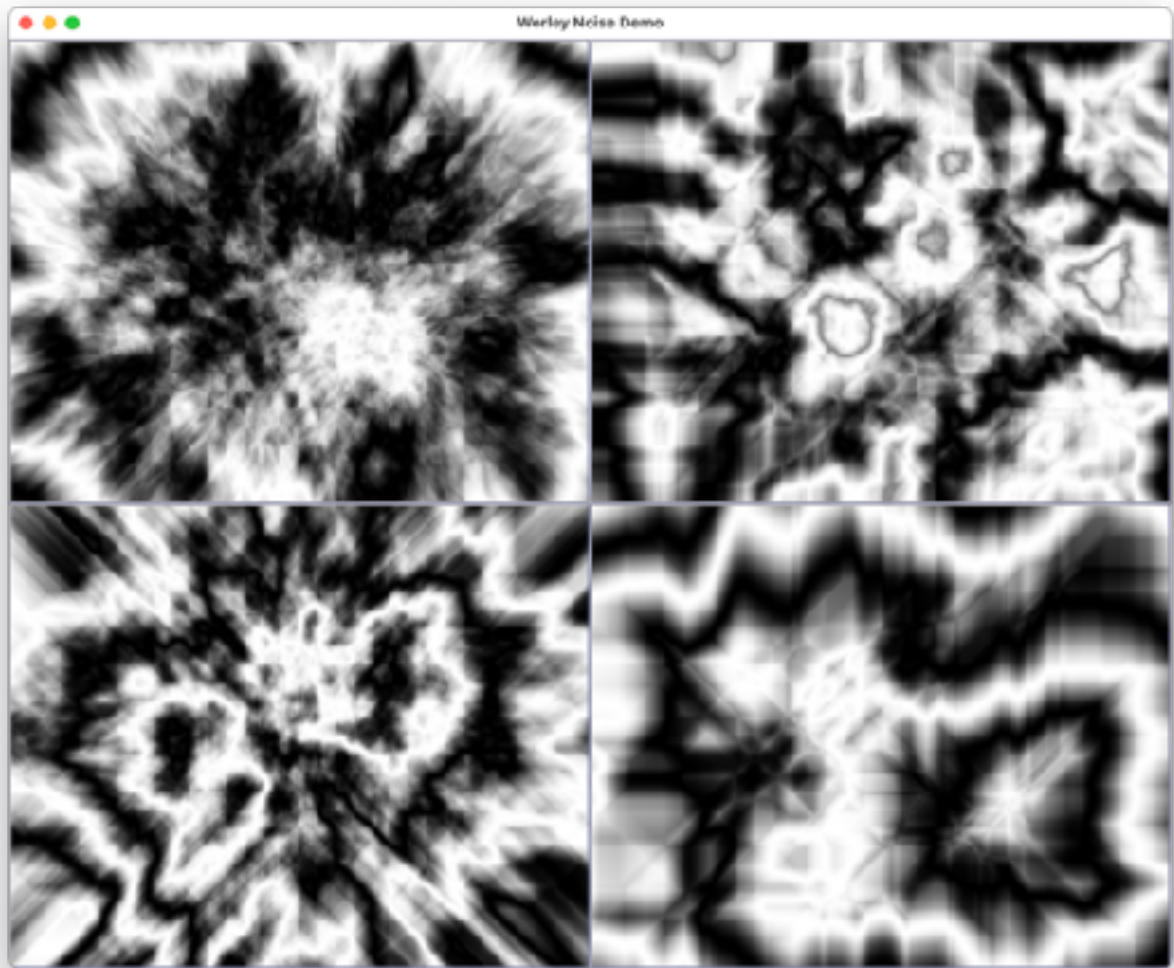
```
public static void compute(  
    double[][] out, double[] xs, double[] ys, double[] weight,  
    double a, double f, String distance  
)
```

Note again that for efficiency, this method does not create a new array to be returned as the answer each time, but fills in the computed result to the given array object argument `out`. The raw numerical values computed into this `out` array will later be turned into a `BufferedImage` whose height equals `out.length` and width equals `out[0].length`. As always, the caller is fully responsible for ensuring that the dimensions of the array object passed in as the `out` argument are large enough to contain the result. As the fundamental principles of the contract law dictate, your method can fail any which way it wants, should the caller not hold on to their end of the contract. This is not the time to be defensive.

The arrays `xs` and `ys` contain the x - and y -coordinates of the points P from which the value of each element of `out` is calculated, both arrays containing same nonzero number of elements. Your method should loop through all the positions of the `out` array and fill in the value of each `out[y][x]` that corresponds to the point (x, y) on the plane in the following manner. (The swap of coordinate roles is done here because Java arrays are stored in [row-major order](#).) First, calculate the array of distances from the point (x, y) to all points of P . The value of the `distance` parameter determines which distance metric you should use for this calculation, for the three possible **option constant** values of "euclidean", "manhattan" and "chessboard" that the automated tester will ever give it. (A more modern implementation of this problem would use an **enumerated type** here, instead of using `String` as a fake enumerated type as if it were the early nineties again.)

Next, sort this array of distances in ascending order, without caring which distance corresponds to which point in P , since we need only these distances themselves to compute the value of the element `out[y][x]` currently being filled in. Then, calculate the weighted sum of these sorted distances so that the i :th distance in this sorted order is weighted by `weight[i]`. Finally, transform that weighted sum of distances into a value in the interval $[0, a]$ using the previous wave function with the two remaining parameters `a` and `f` finally coming in to play their part at this point.

Note that the `weight` array can be shorter than the number of points in P given in the arrays `xs` and `ys`, in which case the weights past the end of the `weight` array should simply be considered to be zeros. For example, should the `weight` array happen to equal `{1, 0, -0.5, 0.5}`, this would mean that the distance of the closest point is given the weight 1, the distance to the third closest point is given the negative weight of -0.5 , and the distance to the fourth closest point is given weight 0.5. Other points in this ranked order are ignored in this calculation.



This is it for this lab: the `main` method of the [WorleyNoiseMain.java](#) test class takes care of converting these noise values into greyscale pixel values to be proudly displayed as images inside a new `JFrame` instance. When run, your code should produce the result visually identical to the screenshot below. The top left image uses Euclidean distance applied to hundred points and fifty randomly chosen weights for a chaotic explosive result. The top right image uses Manhattan distance and fewer points. The bottom left image uses chessboard distance with the same number of points, and the bottom right image goes back to Euclidean distance but with only twenty points.

In practice, the points of P are not thrown on the image the way that a truly blind man throws darts, but the image is first subdivided into some smaller grid, and each cell of this grid will receive the same number of points, typically just one point per cell. Such a horrendous zero-variance deviation from the way that true randomness would be, with probability almost close to one, guaranteed to create clusters and clumps somewhere, ensures that the pixels near the edges of the image don't all essentially behave the same way as the pixels in their local neighbourhood, but the noise remains pleasantly detailed and varied even near the edges. (As discussed in the earlier Linus Sequence problem, the easiest way to detect human-generated fake randomness is in the notable lack of such clustering. If the image were divided five-by-five into 25 grid cells, not only would most people au-

tomatically place one point in each cell when asked to simulate the results of throwing darts randomly, but even fancy themselves especially devious in doing so!)

To create even more natural results, multiple images generated with either Worley noise or other suitable types of noise, for example [Perlin noise](#), can be combined in layers to produce **fractal noise** to further smooth out the result to make it seem more natural without merely just making it blurry by losing all the interesting little detail. To achieve this outcome, these layers must be computed with different amplitudes and frequencies so that the higher the amplitude, the lower the frequency and the number of points in P used to generate that particular layer. The high-amplitude layer therefore defines the main features of the landscape, on which the lower-amplitude layers then build on detail of smaller amplitude but higher frequency without distorting the general features too much.

If the amplitude and frequency of each layer are chosen accurately, zooming in this landscape brings forth the fractal nature of the result in that the result looks similar in every scale, the way that all fractals are by convention expected to do. Of course, same as in the real world, zooming in deep enough will eventually reach the final layer, after which the microstructure will be very different from the way that the whole thing looks like at the macroscopic level. Once again, as above, so below; repeatedly [subdividing the coastline of England](#) can make that coastline to be as long as you want it to be, but eventually you will reach the quantum subatomic level and the whole approach breaks down.

Up here at the practical level of things, suitable choices of noise parameters and the associated colour palette can make this fractal noise pass the Pepsi Challenge when paired with actual photos of real mountain landscapes or other natural and organic entities. However, the parameter space has enough degrees of freedom to allow simulation of not just alien landscapes, but realistic biological textures and structures, be they either animal, vegetable or mineral.

Lab 63: Recursive Subdivision Of Implicit Shapes

For testing: [ImplicitShapeRendererMain.java](#)

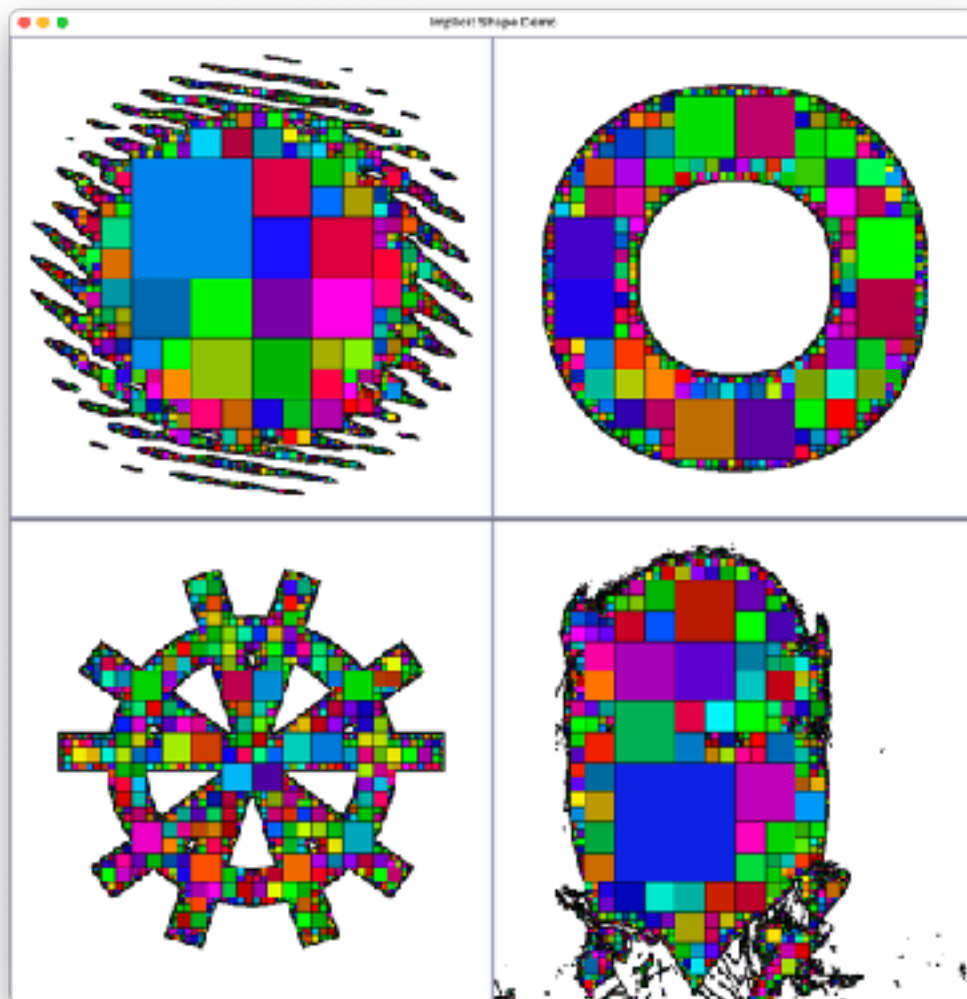
The earlier lab about Lissajous curves demonstrated the important technique of rendering one-dimensional **parametric curves** on the plane. A parametric curve is defined by two functions $x(t)$ and $y(t)$ of one parameter variable t so that the curve consists of all points $(x(t), y(t))$ produced by taking the parameter t for a walk through some range of values, typically from 0 to 1. To render such a continuous curve into a discrete pixel raster, the curve is subdivided into smaller pieces to be rendered in a piecemeal fashion typically as quadratic or cubic segments. Each piece should smoothly connect with the previous and next piece around it without any visible discontinuities in either the curve or its tangents. Viewed from a distance, the pixel representation of that continuous curve will be perceived as actually being that continuous and smooth curve, of course within the limitations of the granularity of the pixel raster itself.

In this same spirit, [implicit functions](#) define two-dimensional **implicit shapes** on the plane. One function f receives as its arguments both the x - and y -coordinates of an arbitrary point (x, y) , and returns a numerical result $f(x, y)$ whose sign indicates whether that point lies inside the shape (negative), on the shape boundary (zero), or outside the shape (positive). For example, the implicit function of the unit disk centered at the origin is $x^2 + y^2 - 1$, which the reader can verify is negative for all points (x, y) inside the unit disk, zero for all points on the circle wrapped around that disk, and positive for all points outside the disk. (The implicit formula for the general case of a disk centered at the point (x_c, y_c) with the radius of r would be $(x - x_c)^2 + (y - y_c)^2 - r^2$, of course.)

To render a continuous two-dimensional shape into a discrete pixel raster, some subdivision scheme is again used to sample individual points on the plane, and the information gleaned from those points is used to render something reasonable to depict the infinite continuum of points that did not get sampled. The technique of [marching squares](#) subdivides the plane into a regular grid of boxes of uniform size. The contents of each box are filled with triangular shapes of foreground colours depending on which one of the four corner points of that box lie inside and which lie outside the implicit shape. Ambiguous ties that occur around **saddle points** are resolved by letting the center point of the box determine which direction is more dominant.

At most five points sampled from the infinite continuum inside the box hopefully reveal the entire continuum with reasonable accuracy. Same as with any numerical method based on finite sampling, the implicit function given as a black box is assumed to be sufficiently well-behaved so that the global and local structure of the shape can be reasonably estimated from such point samples. Of course, a crafty adversary who knows our sampling scheme could always design this black-box to be a fractal made of holes, islands and tendrils to make the result of any such sampling scheme to be arbitrarily far off from reality. No algorithm based on finite sampling of a black box can possibly hope to fight back against this... at least unless we again resort to randomness to prevent the adversary from predicting which points we are going to be sampling!

This lab concerns another rendering technique that also subdivides the implicit shape into rectangles. However, unlike for the marching squares, our subdivision will be performed recursively so that the depth of this subdivision is not uniform across the plane, but automagically becomes more



fine-tuned near the edges of the shape where such precision is needed. In your project folder, create a new class named `ImplicitShapeRenderer`, and there the recursive method

```
public static int render(
    DoubleBinaryOperator f, double xStart, double xEnd,
    double yStart, double yEnd, Graphics2D g
)
```

The first parameter is the implicit function f that determines the shape to be rendered with recursive subdivision. Since functions are not **first-class objects** in Java so that they could be passed back and forth in method calls, the implicit function is wrapped inside a **strategy object** of type [DoubleBinaryOperator](#) defined in the standard library package [java.util.function](#), and this strategy object is then passed as an argument to this method.

The next four parameters define the rectangular box whose contents are rendered in this method call into the `Graphics2D` object `g` given as the last parameter. This box is an axis-aligned rectangle

that covers all points whose x -coordinate lies anywhere between the values `xStart` and `xEnd`, inclusive, and whose y -coordinate lies anywhere between the values `yStart` and `yEnd`. The width and the height of this box are given by formulas `xEnd-xStart` and `yEnd-yStart`.

This method needs to choose random colour for each box that it renders to create the effect seen above in the style of the psychedelic sixties. To create a random colour, it is bad form to just choose a random red, green and blue component for the RGB colour, as the palette created this way will usually end up looking like something that you would expect to find hanging from the rack inside a Goodwill store. A better technique, as illustrated inside the example program [Sliders](#), first chooses a random **hue** from the visible spectrum, and then adjusts it with the random **saturation** and **brightness** between 0.5 and 1.0 each. Fortunately, the [Color](#) class in the AWT package knows how to perform the necessary conversions between the RGB and HSB colour spaces.

As you can see from the trippy screenshot, we aim to subdivide the implicit shape by making each box as large a rectangle as we can, and have the subdivision recurse deeper to produce smaller boxes only near the edges of the shape. This approach very much resembles the idea that we have already seen in the earlier lab 31 where a binary grid of pure black and white cells was subdivided into a **region quadtree** whose branches were on different depths depending on the uniformity of the shape represented by that branch.

At the top level call of each of the four example images in the screenshot, the box has dimensions of 500 by 500 pixels, with the origin set at the top left corner. The shape depicted on top left is a circle where the x - and y -coordinates have been perturbed with some trigonometry before passing them to the circle formula. This example illustrates one big advantage of implicit shapes in that they can be easily **geometrically transformed** by *any* invertible transformation (not merely by the affine transformations of **translation**, **rotation** and **scaling**, for when the shape must consist of a **triangle mesh**) by first applying the inverse transformation to the point (x, y) whose membership inside the implicit shape is to be determined.

The shape depicted in the top right image is an **annulus** created by subtracting a small [superellipse](#) from inside a larger one, both superellipses using the exponent of 2.3. This example showcases another nifty advantage of implicit shapes in that their **constructive geometry** operations of **union**, **intersection** and **difference** (either symmetric or not) become trivial by combining the results of two implicit functions by appropriate logical operators. The resulting shape will appear seamless when viewed by the subdivision algorithm that treats the combined shape as a black box.

The bottom left shape illustrates the use of the [Area](#) decorator that can be applied to arbitrary instances of [Shape](#) to give them superpowers of not just constructive geometry, but arbitrary affine transformations. The function to determine whether the point (x, y) lies inside the shape then translates into a one-liner that asks the `Area` object the very same thing, and converting the received boolean answer into a signed int. The bottom right shape uses an implicit function that determines the membership of point (x, y) in the area from the colour of that pixel in the underlying image, here again the example image [ilkka.jpg](#), this colourful subdivision transforming your instructor's ugly mug into something more enjoyable.

To achieve this fabulous effect, the integer return value of the `render` method should always be one of the three possibilities `-1`, `0` or `+1`. The return value of `-1` means that the entire box lies inside the implicit shape without any holes anywhere inside it, at least to the extent that our finite sampling of the infinite continuum would be able to detect. (The Adversary is assumed to be a gentleman who plays ball with this assumption.) Symmetrically, the return value of `+1` means that the entire box lies outside the implicit shape. The return value of `0` conveys the message that some parts of the box are inside the implicit shape and some are outside it.

The base cases of this recursion are when either the width or the height of the box is less than `1.0`, the size of a single pixel. In this base case, this method should pass the coordinates of the center point inside this box to the function `f`, and return `-1` if that center point lies inside the shape, versus returning `+1` if it does not. (In this finite sampling of the continuum, we treat the vanishingly small subset of points that lie precisely on the edge of the shape as if they were inside the shape.) Note that nothing gets rendered on `g` at these base cases, not even if the center point is inside the shape! After all, we cannot know whether this small base case box will eventually become a part of some larger box whose points also all lie inside the implicit shape.

This method should subdivide boxes that are too big to qualify as a base case into four equal-sized quadrants, and recursively call itself for each of these subdivided quadrants. The four answers received from the NW, NE, SE and SW quadrants determine whether the method will perform any rendering at this level of recursion, and what answer it will return to the caller at the previous level. If all four quadrants returned the same answer that is either `-1` or `+1`, this method should also return that same answer without rendering anything. This is exactly what happened in the base case, since the entire box lies fully inside (`-1`) or fully outside (`+1`) the shape.

However, if any of the four recursive calls returns a zero, or if any two of the four quadrants return a different answer, this method should itself return zero, and before that render a filled rectangle over precisely those quadrants whose recursive call returned the answer `-1`. First fill the rectangle with a randomly chosen colour as explained earlier, then draw a thin black outline around it. Pay attention to calculating the arguments given to the constructor of the [Rectangle2D.Double](#) object, seeing that this constructor expects to be given the coordinates of the top left corner along with the width and the height of the rectangle, whereas the parameters of the method that you are writing here essentially define the coordinates of the top left and the bottom right corners of the box. Some care is needed in calculating the arguments for the rectangle constructor from the coordinates of the box corners.

Nothing is rendered for the quadrants whose recursive call returned the answer `+1`, since those quadrants told us that their contents lie fully outside the shape currently being rendered. Most importantly, nothing is also rendered for the quadrants whose recursive calls returned the answer zero, since that answer indicates that that recursive call already properly rendered the contents of such quadrants! Anything that we could possibly render on top of the work done by that recursive call would at best superfluous, and at worst mess up the perfectly good rendering that was painstakingly achieved by those recursive calls.

Lab 64: Learning The Ropes I: Composition

JUnit: [RopeTestOne.java](#)

The `String` data type in Java is both powerful and useful, especially due to its guarantee of **immutability**. Since the `String` type is basically a Potemkin village draped over an ordinary character array stored privately inside that `String` object, accessing the individual characters with the `charAt` method based on their position can happen in guaranteed constant time. However, the operations of string **concatenation** and **substring slicing** will always take linear time with respect to the length of the strings. Both operations require creating an entirely new `String` object, since the rigidity of the underlying character array necessitates the creation of a second brand new character array to store the individual characters of the result.

However, the raw character strings are not the only gunslinger in this dusty backwater town to represent immutable sequences of characters that we would like to concatenate and slice pieces out of as the need arises. This lab and its followup introduce a more flexible representation of character sequences with a data structure known as a [rope](#). Just like the ordinary `String` type, each instance of `Rope` represents a linear sequence of Unicode characters. For example, we could start with two ropes that represent the character sequences "hello" and "world". The truly astonishing property of the rope data structure is that any two instances of `Rope` can be concatenated into a new rope "helloworld" so that both time and memory use of this operation are guaranteed to be constant, regardless of the length of the two ropes being concatenated! Even more amazingly, the exact same guarantee of constant time and memory use holds for the operation of slicing out a sub-rope from any existing rope.

Since there is no free lunch, there must be a hidden catch somewhere. Otherwise, why should the ordinary `String` data type even exist in Java, if these ropes could just waltz in stage right at any time and belt out how “♪ Anything strings can do, we can do better ♪” to lasso not just our minds but also our hearts? The hidden disadvantage of ropes is that accessing their individual characters by their position with the `charAt` method can no longer take place in a guaranteed constant time, but the time required will be **logarithmic** with respect to the `length` of the rope, and even that guarantee assumes an industrial strength implementation of this data structure that can perform **rebalancing** operations behind the scenes whenever the structure starts getting too lopsided.

In this lab, we implement a slightly simplified version of ropes that do not perform any complicated rebalancing, since the student will surely agree that even the toy version presented here is already amazing not just in its algorithmic elegance, but as an example of good object oriented design that we are supposed to be learning in this very course!

Unlike a string object "helloworld", a rope structure is not necessarily a single object, but a tree structure made of several rope **nodes**. All these nodes are instances of subtypes of the abstract superclass `Rope` that defines the public functionality common to all ropes.

Start by adding the following implementation of that superclass in your labs project.

```

abstract public class Rope {
    abstract public char charAt(int index);
    private int len = -1;
    public final int length() {
        if(len == -1) {
            // Call the template method to initialize len.
            len = computeLength();
        }
        // Either way, we can now return the cached value.
        return len;
    }

    // Subclasses must override the actual length computation.
    abstract protected int computeLength();
}

```

Each concrete subclass of `Rope` must implement the public method `charAt` to return the character at the given position `index`, done in some appropriate fashion that depends on how the necessary data is stored inside that subclass. If the given `index` is out of bounds for the sequence of characters represented by that rope node, this method must throw an `IndexOutOfBoundsException` instead of returning anything. Note that since this exception type is **unchecked**, it does not show in the signature of the `charAt` method.

Each instance of any subtype of `Rope` will also be able to tell you its length. However, since all instances of `Rope` are designed to be **immutable**, their length cannot possibly change after object creation, so it would be redundant to compute this same length over and over again each time the user code requests to know it. Therefore, the method `length` has been implemented as a `final` method in the `Rope` superclass to perform **caching** of the computed length into the private data field `len`, in a suave application of **trading space for time**. Since no rope can possibly have a negative length, this field is initially set to `-1` to indicate that its true value has not been computed yet.

The first time someone requests the length of this rope, the method `length` calls the protected **template method** named `computeLength` that each concrete subclass must implement to correctly compute the length of that particular rope instance. The template method design pattern allows both the `Rope` superclass and all the user code to be blissfully ignorant of the gory details of how the sausage of computing the length of each instance is actually made in the smoke-filled rooms hidden behind the encapsulation interface.

The first concrete subclass of `Rope` represents a simple rope that has been constructed from the given instance of `String`. Create the class `StringRope` in your lab project, with the signature

```

public class StringRope extends Rope

```


This class should have a public constructor that receives a `String` parameter that it simply stores into some private data field. Both methods `charAt` and `length` can then simply call these very same methods of that privately stored string, and return whatever those calls returned.

Same as with the other subclasses that we extend from the abstract superclass `Rope` in this lab, the subclass `StringRope` must also override the method `toString` that allows converting the contents of that rope instance back into a Java string used by the rest of the user code program. Inside this class, this method can simply return that privately stored string.

Instances of the second concrete subclass written in this lab represent ropes that have been constructed by concatenating two existing ropes. In your lab project, create the class

```
public class ConcatRope extends Rope
```

whose polymorphic constructor receives two parameters `left` and `right`, both instances of the type `Rope`. Same as what `StringRope` did for the parameter `String` that it was given, the constructor of `ConcatRope` should store these parameters into data fields with the same names.

The method `computeLength` should call the method `length` of its `left` and `right` pieces and return the sum of these lengths. (Note that you need to specifically call the method `length`, not the method `computeLength`, to enjoy the benefits of caching the length for these pieces!) The method `charAt` should decide which piece it recursively consults for the character in the given position `index`, based on whether that `index` falls in the `left` piece or in the `right` piece. This decision is easily made by comparing the `index` to the `length()` of the `left` piece. Continue from the `left` piece if the `index` falls in there. Otherwise continue from the `right` piece, but don't forget to first subtract the length of the `left` piece from the `index` that you pass on!

Instances of the third concrete subclass in this lab represent ropes that have been constructed by slicing off a subrope from an existing rope. In your lab project, create the class

```
public class SubRope extends Rope
```

The constructor of this class should expect three parameters, the original rope being sliced, and the `start` and `end` indices that define the `start` and `end` position of the subrope inside the original. To be consistent with the behaviour of the `substring` method in Java `String` class that this operation mimics, the `start` index is defined to be inclusive, but the `end` index is guaranteed to be exclusive. Also, again in spirit of the `substring` method in `String`, the constructor of this method must immediately throw an `IndexOutOfBoundsException` if the `start` index is negative or greater than the `end` index (these indices can also be equal, to denote slicing off the empty substring), or where the `end` index is greater than the length of the original rope. (The corner case where both `start` and `end` equal the length of the string returns an empty rope.)

The method `charAt` can now consult the same method of the original rope, but must remember to shift the `index` to the right by adding the `start` value to it. The `length` method simply returns

the value of `end-start`. Without access to the actual type of the original (which the method should not care one whit about anyway, following to the bumper sticker version of the **Liskov substitution principle**), the `toString` method has no choice but create a new `StringBuilder` and append in there all its individual characters extracted with the `charAt` method.

In `SubRope` construction, it is possible to flatten out the resulting tree structure by noticing that if the original rope is a `ConcatRope` so that the subrope defined by the parameter indices `start` and `end` falls entirely within either its `left` or `right` part, the newly created `SubRope` can bypass this redundant middleman and point directly to its `left` or `right` part, whichever is appropriate. Similarly, if the original rope is itself a `SubRope`, the newly created `SubRope` can always unconditionally bypass this redundant middleman. These optimizations can then descend down the tree structure as deep as possible. When these ropes are used to construct complex strings by chopping and gluing together existing ropes, making the resulting tree structure be as flat as possible also makes the `charAt` operations performed on those ropes significantly faster.

The JUnit test class [RopeTestOne.java](#) uses these three subtypes to build up and slice out ropes of various lengths and contents, including one giant instance of rope that consists of over one billion characters, far too many to be represented as an ordinary `String`.

One major criticism of this design still lingers in that the user code must be aware of all of the subtypes `StringRope`, `ConcatRope` and `SubRope`, since the user code operations on ropes involve explicit creation of instances of these subtypes with the operator `new`. However, this is easily remedied by defining appropriate methods in the superclass `Rope` to achieve these desired effects. For example, concatenating two existing ropes could be defined as a one-liner **factory method**

```
public final Rope concatenate(Rope other) {  
    return new ConcatRope(this, other);  
}
```

to allow the user code then to simply say

```
Rope result = r1.concatenate(r2);
```

to concatenate their two existing `Rope` instances `r1` and `r2`, without knowing or caring about the type of the subclass that represents this concatenation. A superior alternative to using a constructor and the operator `new`, these factory methods can also potentially involve other optimizations. For example, when extracting an empty subrope where `start == end`, the subrope extraction utility method could always return a reference to one and the same `Rope.EMPTY` object, instead of redundantly creating a new object for the empty rope. When new objects are created with the operator `new` instead of using a factory method, a brand new object will always be created, and the type of that new object is set in stone at the compile time. However, a factory method is not under any such constraints, but can return an existing object when one already fits. Since the `Rope` type was designed to be immutable, no harm can follow from sharing the same `Rope` objects between multiple parts of the user code.

Since the rope data structure is designed to be immutable, the operation of **deleting** a subrope from an existing rope would not actually modify that existing rope, but again create a new instance of `Rope` to represent the result. We don't even need to define a new subclass of `Rope` to represent instances created in this fashion, since even this operation can be made to work in guaranteed constant time and memory once we realize that it is merely **syntactic sugar** for concatenating two subropes tactically sliced out of the original rope, implemented in `Rope` as something like

```
public final Rope delete(int start, int end) {
    Rope left = new SubRope(this, 0, start);
    Rope right = new SubRope(this, end, original.length());
    return new ConcatRope(left, right);
}
```

One more reasonably common and useful operation on text string would be reversal. The `String` class in Java doesn't have a `reverse()` method at all, so the reversal is canonically performed with the aid of a `StringBuilder`. For ropes, we can define a class `ReverseRope` that contains a reference to the original rope that is reversed. The method `charAt` in this class would simply mirror the given position index relative to the `length` of that rope, and consult the original rope about the character in that mirrored position. If the construction of the `ReverseRope` is done via the factory method `reverse` defined in the `Rope` superclass, this method can optimize the situation where the rope being reversed is itself a reversal of some rope, and just return that original unreversed rope without creating any new objects at all.

This lab is followed by the Part II right around the corner, but before that, you could already take a little side tour to put your ropes in good use in the bonus lab 104, "Beklemishev's Worms".

Lab 65: Learning The Ropes II: Comparisons

JUnit: [RopeTestTwo.java](#)

After completing the basic rope operations for character sequence in the previous lab, the time has come to implement the core methods that are necessary and sufficient for the `Rope` data type to stand proud and tall as a first class citizen among all the other classes in the Java language and its standard library. That is, these ropes have a meaningful method `equals` for equality comparisons, the method `hashCode` for calculating an integer index needed to store arbitrary ropes inside a `HashSet<Rope>` data structure for fast lookup and retrieval, and the method `compareTo` to compare two `Rope` objects to each other in lexicographic order.

We shall start with hash codes, since they can later be used to massively speed up the equality comparison. Since the instances of `Rope` are designed to be immutable, their hash codes can never change. We can therefore cache the hash codes the exact same way as was done with the lengths. Inside the abstract superclass `Rope` that was given to you in the last lab, define a new data field

```
private int hash = -1;
```

that will cache the computed value of the hash code the first time that someone requests to see it, available in constant time for all future hash code queries after that. The special value `-1` again denotes that the actual hash code has not yet been computed. Then, write the method

```
@Override public int hashCode()
```

that computes and caches the hash code whenever `hash == -1`, and returns the value of the cached hash. The simplest way to turn a character sequence into a reasonably chaotic hash code so that every character will affect the computed hash code is to use a for-loop to iterate through the characters of the sequence from beginning to end. For each character `c` encountered along the way, update the hash with the assignment

```
hash = 33 * hash + (int)c;
```

After the loop has terminated, return the accumulated hash, although you also need to ensure that in the rare case where the hash ends up being equal to the special value `-1`, the method returns some other fixed value instead. (Pick any value, for example 17 or 42 or 1234567, since any fixed value is as good as any other value.) Alternatively, if you dislike negative numbers and want the hash code always be nonnegative, if only to make the job a bit easier for those guys who thoughtlessly implement the data structure to use these hash codes, the fast and correct way to do so is to use **bitwise arithmetic** to chop off all but the lowest 31 bits of the signed integer with the expression

```
hash = hash & 0x7FFFFFFF;
```

After implementing `hashCode`, the next job in our TODO list is to allow lexicographic ordering comparisons between arbitrary ropes. Modify the signature line of the entire class to say

```
public class Rope implements Comparable<Rope>
```

to promise that the order comparison is a meaningful operation between any two instances of this type. Implementing the interface `Comparable<E>` then requires you to implement one method

```
@Override public int compareTo(Rope other)
```

whose return value should be `-1` if `this` rope is lexicographically smaller than the `other` rope, `+1` if `this` rope is lexicographically greater than the `other` rope, and `0` when both ropes are equal. To perform this order comparison, loop through the positions of `this` and `other` rope in lockstep from beginning to end, comparing the two corresponding characters in each position. As soon as those two characters are in any way different, or if this loop runs past the end of either rope, you have your answer and don't have to even glance at the rest of the characters in either rope. Only if you run past the end of both ropes simultaneously, you may conclude that both ropes are equal, and the method can return `0` in the end.

We intentionally left the equality comparison method to be the last method to implement, since its implementation greatly benefits from the existence of the `hashCode` and `compareTo` methods. However, remember that the equality comparison method

```
@Override public boolean equals(Object other)
```

must accept any `Object` whatsoever as its argument, since in the Java type system, equality comparison is a perfectly meaningful concept between any two objects. (Even if the answer to a question happens to be some laconic variation of "No way, José", this by itself doesn't render the question itself in any way meaningless or malformed!) After using `instanceof` to verify that the other instance is also some kind of a `Rope`, create a new properly typed reference to the argument object with a **downcast**, to allow you to access the `Rope` functionality of that `other` object.

Since the inequality of the hash codes entails the inequality of the objects themselves, this method should start by first checking if both `this` and `other` ropes have their cached hash values set to something other than `-1`. If that is indeed the case, and those cached hash values are different, you can smoothly return `false` without further ado. Otherwise, call the previous `compareTo` method to find out how `this` and `other` ropes relate to each other in lexicographic order, and return `true` if and only if these ropes are equal in that sense, as this can only happen when the ropes themselves represent equal sequences of characters. It is theoretically possible for the two ropes to have the same hash code despite representing distinct character sequences. As unlikely as this possibility may be, it is actually far more likely to happen over time than most people would intuit, so your code needs to be prepared for that possibility.

Lab 66: Slater-Vélez Sequence

JUnit: [SlaterVelezTest.java](#)

The lab problem 17, “Diamond Sequence”, as well as the two more recent sequences implemented in the problem 0(L), “I Can, Therefore I Must”, were based on the idea of always continuing the given positive integer sequence with the smallest previously unseen positive integer that does not create a violation of the sequence-defining rule with the earlier values in the sequence generated so far. This lab continues the same grand theme by generating the elements of the remarkable [Slater-Vélez sequence](#), this author becoming aware of this lively sequence only from the Code Golf problem that the link leads to.

Each element of the Slater-Vélez sequence is the smallest positive integer n that has not appeared so far in the sequence, under the constraint that the absolute difference of n and its immediate predecessor in the previous position of the sequence also has not previously appeared as the absolute difference of any two consecutive elements in the sequence. This definition makes the sequence unique, and the reader can verify that the Slater-Vélez sequence starts as

1, 2, 4, 7, 3, 8, 14, 5, 12, 20, 6, 16, 27, 9, 21, 34, 10, 25, ...

with the corresponding sequence of absolute differences between consecutive elements being

1, 2, 3, 4, 5, 6, 9, 7, 8, 14, 10, 11, 18, 12, 13, 24, 15, ...

It follows trivially from the rule used in the construction of this sequence that no element can ever repeat. However, the same way as is guaranteed to happen in the diamond sequence, this sequence doesn't leave any holes behind in its journey towards the omega, but will eventually visit every positive integer, as proven by those two jolly *caballeros* Slater and Vélez back in 1979. This sequence is therefore an infinite permutation of the positive integers, although its cycles are not quite as tight as those in the diamond sequence where every cycle had the exact same length of two.

As was done in the diamond sequence lab, we implement this sequence as a **virtual iterator** that computes the next element of the sequence lazily on command. Unlike Python, Java doesn't have lazy sequences and generators built in the language itself. The **iterator framework** is then the next best way to achieve the zen bliss of enlightened laziness. In your lab project, create the class

```
public class SlaterVelez implements Iterator<Long>
```

Since the sequence is infinite, the method `hasNext` should always unconditionally return `true`. We will tacitly assume that the sequence won't be generated far enough for our values to get their heads chopped off by spilling outside the Procrustean bed of 64-bit `long` integers.

Since the sequence generator needs to keep track of the numbers and their absolute differences that it has already seen, we can again follow the lead of the diamond sequence problem and use two instances of the [NatSet](#) utility class to achieve this. Inside your class, define three data fields

```
private NatSet seen = new NatSet();
private NatSet diffs = new NatSet();
private long previous = 0;
```

The two instances of the [NatSet](#) utility class, named above `seen` and `diffs`, keep an eye on the positive integers and their absolute differences that have already been `seen` in the sequence generated so far. Furthermore, the class needs to remember the `previous` number in the sequence, which we might as well initially set to zero to indicate that nothing has been generated. To complete the obligations of implementing the interface `Iterator<Long>`, write the method

```
public Long next()
```

to virtually extend the sequence by one more element that becomes the new `previous` number and gets returned from this method. This method should also update the `seen` and `diffs` sets to correspond to the new sequence that is now one element longer than it was before this method call.

Since the `NatSet` instances keep track of natural numbers that include a zero, unlike the positive integers that start from one, make sure to add the zero as a member to both `seen` and `diffs` sets in the constructor of this class, before you start generating the actual values. Furthermore, to speed up the generation of the `next` element of the sequence, you should note that the `NatSet` utility class comes with the convenience method

```
public long allTrueUpTo()
```

that returns the smallest natural number that is not a member of this set. By getting this invaluable information about the numbers that you have already `seen` as part of this sequence, you know right away where to start looking for the next number in the sequence, instead of always starting to look for that number by counting up all the way from one. By cleverly using also the information about the `previous` value and the smallest difference that is not yet a member of the `diffs` set, you can sometimes jump even higher for your starting point to look for the next element, without any risk of jumping over the smallest positive integer that would have satisfied the rule of this sequence. With these optimizations, your code should be able to produce the first half a million elements of the Slater-Vélez sequence in a couple of seconds.

Lab 67: Lemire Dynamic Frequency Sampling

JUnit: [LemireFrequencySamplingTest.java](#)

To sample a random element fairly from the given array `arr` so that every element has an equal chance of being chosen, the expression `rng.nextInt(arr.length)` can be used to get a random position to fetch the actual element from, where `rng` is your random number generator. However, sometimes we don't want all elements to get chosen with the same uniform probability, but we want these random choices to be performed according to the given **frequency array** of the exact same length as the array `arr`.

Each element of the frequency array gives the probabilistic **frequency** for the corresponding original element to be chosen in the long run. For example, if the original array `arr` has five elements, the frequency array `{3, 4, 0, 1, 4}` would indicate that the element in the position 0 with frequency 3 would get sampled roughly three times as often as the element in the position 3 with frequency 1, and roughly 3/4 as often as the element in the position 1 or in the position 4. As you see here, some frequencies can also be zero, so that those elements should never be sampled.

To sample the next element from the given frequencies, each element must have the probability f / s to get sampled, where f is the frequency of that particular element and s is the sum of all frequencies in the frequency table. For example, for the above example frequencies, the probability for the element in the position zero to get sampled would be $3 / (3 + 4 + 0 + 1 + 4) = 3/12 = 1/4$.

Note that the choice of each random element must be done independently of the history of the past selections, with no attempt to "correct" the frequency of some element that so far has not appeared as much as its desired frequency would predict. In the short run, all random walks exhibit variance and clustering, so that the frequencies of the sampled elements will basically never be exactly equal to what the frequency table says that they would ideally be. In the long run, the random sampling algorithm should see the proportion of each sampled element converge to its given frequency divided by the sample size s , in line with the [frequentist interpretation of probabilities](#).

The simple algorithm that samples a random element according to the given frequencies starts by creating a random number r between 0 and $s - 1$. It then adds up element frequencies from the beginning until r becomes strictly smaller than the sum of the frequencies up to that position. The element in the position where this happens is then returned as the answer. Written in Java, the method might look something like the following. (Note the zero-based counting used in both quantities `total` and `pos`.)

```
public int frequencyChoice(int[] arr, int[] freq, Random rng) {
    int s = 0, total = 0, pos = 0;
    for(int f: freq) { s += f; }
    int r = rng.nextInt(s);
    while(total <= r) { total += freq[pos++]; }
    return arr[pos - 1];
}
```


This algorithm is fine for sampling a single element from a handful of elements. However, if a large number of elements are to be randomly sampled one at the time, the first obvious optimization is to compute the sum of frequencies `s` only once, and pass this quantity to this method as an additional parameter instead of redundantly recomputing it every time in the first for-loop. If the array `arr` is large, the next opportunity to optimize is to precompute the **accumulation array** for these frequencies, as seen back in the Lab 55, “Accumulated wisdom”. From this accumulation array, the **binary search** algorithm finds the position `pos` where the sum of elements first becomes larger than `r` quickly in logarithmic time, instead of using a for-loop to find that position in linear time.

The previous discussion assumed that we are performing **sampling with replacement**, so that sampling a random element does not affect the frequencies used in future sampling. However, this problem becomes more interesting when adapted to **sampling without replacement**, so that each time the element in the position `i` gets randomly sampled, its associated frequency `freq[i]` should decrease by one! It is not immediately as obvious how to perform such sampling efficiently. However, in this lab we shall implement a slight variation of the ingenious technique to achieve this goal, as explained by [Daniel Lemire](#) in his blog post “[Sampling efficiently from groups](#)”.

In your labs project, create a new class named `LemireFrequencySampling`, in which you will be writing the following two static methods.

```
public static int[][] buildSamplingTable(int[] freq)
public static int updateSamplingTable(int[][] table, int r)
```

Given an arbitrary frequency array `freq`, the first method will build and return a two-dimensional sampling array. The topmost (zeroth) row of this sampling array has the same elements as the original frequency array `freq`. After this zeroth row, the length of each following row is always exactly one half of the length of the row immediately above it. The i :th element of each row equals the sum of the two elements in the positions $2i$ and $2i + 1$ in the row immediately above it. For simplicity, this method may assume that the length of frequency array is an exact power of two. (Any frequency array of some other length can always be **padded** with zeroes until its length becomes some exact power of two, without affecting the probability of any element being sampled.)

For example, given the frequency array `{4, 0, 1, 5, 2, 2, 3, 4}`, the two-dimensional sampling array would have four rows. Row 0 would have eight elements, equal that same frequency array. The length of the next row 1 would be 4, and its elements would equal `{4, 6, 4, 7}`, the sums of pairs of elements in the previous row. The length of the next row 2 would be 2, and its elements would be `{10, 11}`, since $4+6 = 10$ and $4 + 7 = 11$. The bottom row 3 would have just one element `{21}`, always equal the sum of the original frequencies.

The technique introduced in the good professor Lemire’s blog post cleverly performs these accumulations **in place** in the original frequency array. However, the resulting compact representation of data also makes the algorithm that performs the actual random sampling without replacement slightly more tricky. To nimbly sidestep such pitfalls, we choose to **trade space for clarity**, a tacit

tradeoff similar to its more famous cousin of **trading space for time** but surely more common in practice.

The second method is given as argument the very sampling table constructed in the previous method, and a nonnegative integer `r` that serves the same purpose as the local variable `r` in the earlier linear time algorithm for frequency sampling. This method should find and return the position in the current frequency array (that is, the current zeroth row of the given sampling table) where the sum of frequencies up to that position becomes strictly greater than `r`. Before returning that position to the caller, the frequency of that position should also decrease by one, since the randomly sampled element is not returned back in the population.

Since our automated JUnit tester will repeatedly try out this method with frequency tables of over a hundred thousand elements, updating the entire sampling table to correspond to the new situation must be blazingly fast for such tests to finish in an acceptable total time. Fortunately, the ingenious construction of the sampling table allows us to achieve this feat in **guaranteed logarithmic time** with respect to the length of the frequency table for each individual sample. The sampling algorithm is simplest to implement as a while-loop that starts at the bottom row of the sampling table, and keeps climbing up until it has reached the zeroth row. During this climb, the body of the loop will update exactly one element of the current row, and look at exactly one element in the previous row to decide which direction to continue. The total work done by this loop is therefore guaranteed to be linear with respect to the number of rows in the sampling table. This quantity is in turn logarithmic with respect to the number of elements in the original array and its frequency table.

The while-loop of this method should keep track of the `row` and `col` position that its traversal up the sampling table is currently at, starting at the lone element at the bottom row. The body of this loop should first decrement the element at the current `row` and `col`, and then look at the element `e` at the position `[row-1][2*col]` to decide which direction to continue. If `r < e`, continue left to the column `2*col` in the `row-1`, keeping the value of `r` as it were. Otherwise, continue right to the column `2*col+1` in the `row-1`, but also decrease `r` by `e`. Once you reach the zeroth row of the sampling table, decrement the frequency stored in the column that you ended up at, and return the index of that column (not the value in that column!) to the caller.

Lab 68: Interval Sets I: Dynamic Set Operations

JUnit: [IntervalSetTest.java](#)

Bit vectors can be used to store a set of natural numbers in a compact fashion, needing only one bit of memory for every element of the appropriate type and range that could potentially become a made member of this set. However, this representation might still use space very inefficiently for **sparse** sets where most of the bits end up being `false`, so that only some dust of `true` bits is scattered throughout this vast astronomical emptiness. In situations where both the members and the nonmembers of the set tend to form contiguous **clusters**, a vastly more efficient representation for such sets can be achieved with ideas adapted from the technique of **run-length encoding** by storing not merely the individual elements, but entire consecutive **intervals** of contiguous elements that are members of the set.

This lab has you implement a rudimentary version of this kind of approach to represent such clustered sets of natural numbers in a highly compact fashion so that their basic **dynamic set** operations `add`, `contains` and `remove` end up being as efficient as the student will hopefully end up becoming effervescent while writing this code. This lab also serves an exercise of using the [LinkedList<E>](#) subtype from the Java Collection Framework along with its **list iterators** that not only allow the **bidirectional traversal** back and forth through the elements of the list one step at the time, but also constant time modifications with the operations `add` and `remove` performed at the current location of the iterator, as opposed to the eponymous list methods that operate on that list as a whole.

Compared to the more commonly used `ArrayList<E>` collections, linked lists are appropriate for situations where we require constant time **random access** to the elements in arbitrary positions, but always traverse the elements sequentially from left to right, with the option to occasionally update the contents of the list in constant time at the current position of the iterator traversal. For the `ArrayList<E>` collections, the rigidity of the underlying array makes such local modifications inefficient, since the elements that follow the current the position all have to be moved one step forward or backward after the modification to either make room for the new element that was added, or fill in the gap left behind by the element that was removed.

The best way to understand the behaviour of a [ListIterator<E>](#) instance acquired from the underlying [LinkedList<E>](#) object is to think of it as a **cursor** that points not at an element directly, but at the invisible space that separates each two consecutive elements of the list. For example, the underlying linked list of integers might contain the following elements, with the current location of the iterator cursor shown as a vertical bar between the elements 42 and 333:

17 99 42 | 333 81 4 87

Calling the method `next ( )` for the iterator makes it skip over the element that immediately follows it, and return the value of the element that was skipped over. The method `hasNext ( )` should generally be called before this move to determine whether the cursor has traversed past the end of the

list, to avoid throwing the `NoSuchElementException` for trying to skip over the nonexistent element past the end of the list.

For example, calling `next()` in the situation depicted above would return 333, with the new cursor position inside the list becoming

17 99 42 333 | 81 4 87

Since each **node** of the underlying linked list contains references to both its successor and predecessor nodes, the `ListIterator<E>` can support bidirectional traversal. The methods `previous` and `hasPrevious` operate symmetrically to the methods `next` and `hasNext`.

The method `remove` of the `ListIterator<E>` cursor removes the element that the cursor stepped over during its most recent `next` or `previous` operator. The iterator cursor remembers only the most recent element that was stepped over. Therefore, calling the `remove` method twice without calling either of the methods `next` or `previous` in between these two calls, the second call to `remove` will fail and throw an `IllegalStateException` to announce that the iterator object was not in a proper state to perform the requested operation.

For example, calling `remove` after stepping over the element 333 with the previous call to the method `next` would create the following situation, with the cursor position unchanged:

17 99 42 | 81 4 87

The method `add` of the iterator cursor adds a new element before the cursor position. Unlike the method `remove`, this method can be called any number of times regardless of whatever calls to `next` and `previous` have taken place in the interim. For example, calling the method `add(38)` for the iterator cursor would produce the new situation

17 99 42 38 | 81 4 87

Make sure to appreciate what a different thing it is to call the method `add` for the iterator to add the new element to the current position of the iterator cursor, as opposed to calling the method `add` in the list object as a whole to add the new element at the end of the list, regardless of any iterator cursors that might exist at that moment.

These preliminaries give us enough to begin implementing the set of intervals. In your labs project, create a new class `IntervalSet` whose instances represent sets of natural numbers stored as **intervals** of consecutive integers that are members of that set. For example, an `IntervalSet` instance that will print out as "[4-10, 15, 20-27]" would contain all integers from 4 to 10, the integer 15, all integers from 20 to 27, and no other integers. Each interval is considered to be inclusive at both ends. **Singleton** intervals whose `start` and `end` values are equal are printed as the starting value only, without the separator dash and the end value.

You should start by first defining the following **nested class** inside the `IntervalSet` class to represent an individual interval:

```
private static class Interval {
```

This class should have two public integer data fields `start` and `end`, and a constructor to initialize these fields from its parameters. Note that we are not designing this nested private class to be immutable, but allow the methods of the surrounding `IntervalSet` to modify the `start` and `end` values of any given `Interval`. You should also override the `toString` method of the `Interval` class to return the string of the form "`start-end`" whenever `end > start`, and just "`start`" when these two values are equal.

Each instance of `Interval` represents the fact that the set contains all elements from `start` to `end`, inclusive. Each `IntervalSet` is therefore a collection of disjoint intervals stored in a linked list. Each `IntervalSet` instance should therefore contain the field

```
private LinkedList<Interval> intervals = new LinkedList<>();
```

in which the intervals that form the interval set are kept in strictly increasing order. Furthermore, we will ensure during the execution that all the intervals are **maximal** so that no two of them could be combined into one larger interval that contains both of the original intervals. For example, instead of storing two intervals `10-20` and `21-30`, the list should only contain one interval `10-30` with the same information in a more compact form.

To help debugging the dynamic set operations that you will be writing in this class, you should first override the `toString` method for the `IntervalSet` class to return a string that contains square brackets as its first and last characters, followed by the individual intervals as produced by the `toString` method of the `Interval` class, consecutive intervals separated by exactly one comma and one whitespace character. Make sure to also not have any silly trailing commas or whitespace in the end. Once you have completed this method, you can start filling in the bodies of the required dynamic set methods

```
public void add(int start, int end)
public void add(start)
public boolean contains(int start, int end)
public boolean contains(int start)
public void remove(int start, int end)
public void remove(int start)
```

Since the operations on **singleton** intervals whose `start` and `end` are equal would presumably be common in the user code of this data structure, we will provide the convenience versions of the dynamic set methods that take only one parameter each. All three convenience methods should be simple one-liners that call the more general two-parameter version of that same method by duplicating that parameter. The student should read through the explicit test cases of our JUnit test class

to get an idea how these methods are used, and what their expected results are expected to be in the provided test situations.

The method `add` should first create a new `Interval` object for its given `start` and `end`, and ask the list of `intervals` for a list iterator to traverse through the existing `intervals` in the list. For each existing interval, check whether it could be **absorbed** into the new interval that is being added to the set. To absorb an existing interval, remove it from the list, and update the `start` and `end` values of the new interval accordingly to make the new interval fully contain the absorbed one. For example, the interval 10–20 currently being added to the set could absorb an existing interval 5–15 and become the interval 5–20, so the now redundant interval 5–15 can be removed from the list. Note that an interval can also absorb another interval if either one of these intervals starts at the value that is one larger than the end value of the other interval. For example, the new interval 4–9 can absorb an existing interval 10–15, and become a single interval 4–15.

As the traversal through the existing `intervals` removes some of the existing intervals that get absorbed into the new interval that is currently being added to the set, one of the two things must eventually happen. Once some existing interval reached in the traversal starts after the end of the new interval being added, the new interval is simply added at that position, and the method can terminate. Should the traversal of the list reach the end of the list before such an existing interval is found, the new interval is tacked on at the end of the list.

Thanks to our design decision to store the intervals in the maximal fashion without redundancy, the logic of the `contains` method that determines whether all numbers from `start` to `end` are currently members of the set is pretty straightforward. This method only needs to iterate through the list of existing intervals to find an interval that fully contains the entire interval being searched for. If no such interval exists, the method can return `false`, safe in the knowledge that at least one of the numbers from `start` to `end` is not a member of this set.

As usual with dynamic sets, the logic of the `remove` method is the most complicated of the three dynamic set operations. (Removing an element from any data structure decreases the **entropy** of that data structure, so the operation must perform work against the grain of the natural flow of the universe.) The `remove` method should iterate through the existing `intervals` in the list, and update the list and the intervals stored therein depending on the way that the existing interval overlaps the interval of the given values being removed. To illustrate the different possible cases and the treatment that they require, let us assume for simplicity of numbers that we are currently removing the particular interval 10–20, to make it easier to visualize what ought to be done for the given existing interval in the list.

- If the existing interval is fully inside the interval being removed, such as 10–14, 15–18 or 13–20, simply remove that existing interval from the list.
- If the existing interval contains the interval being removed so that their either endpoint coincides, such as 10–30 or 5–20, update the opposite endpoint of the existing interval, and terminate.
- If the existing interval properly contains the interval being removed so that the endpoints do not coincide, such as 7–30, modify that existing interval to end one step before the removed

interval begins, and add a new interval for the values that follow the interval being removed. For example, removing 10–20 would split the existing interval 7–30 into two separate intervals 7–9 and 21–30. Again, you can terminate the method after this operation.

- If the interval being removed partially overlaps the existing interval from the left, such as with the existing interval 15–40, update the `start` value of that existing interval to one past the end of the interval being removed. For example, removing 10–20 from the existing interval 15–40 makes that interval to be 21–40.
- Symmetrically to the previous item, if the interval being removed partially overlaps the existing interval from the right, such as with the existing interval 0–14, update the end value of that existing interval to one less the `start` of the interval being removed. For example, removing 10–20 from the existing interval 0–14 makes that interval to be 0–9.

The mass test of these three operations creates a large number of pseudorandom intervals to be added to and removed from the current `IntervalSet`. You can uncomment the lines that print out the current contents of the interval set in the mass test method of the `JUnit` test class to watch the results that your methods produce for all these pseudorandom additions and removals.

Lab 69: Interval Sets II: Iteration

JUnit: [IntervalIteratorTest.java](#)

The three dynamic set operations `add`, `contains` and `remove` by themselves are not enough for the common use case of **iterating** through the elements of that dynamic set so that each element is visited exactly once through this traversal. To follow the lead of the classes in Java Collection Framework, we shall adapt the `IntervalSet` class from the previous lab to also allow this. Since it must be possible for multiple iterators to simultaneously exist and point to different locations inside the same collection, the collection object itself cannot act as its own iterator. The iterator objects have been constructed from a separate class that has sufficient access to the `private` internal details of the collection that it iterates through. An **inner class** is just what the doctor ordered for this purpose. Therefore, inside your `IntervalSet` class, define the inner class

```
private class IntervalIterator implements Iterator<Integer>
```

whose instances act as independent iterators on the particular `IntervalSet` instance from which they were originally acquired. Since we define `IntervalIterator` to be an inner class, instead of a `static` nested class, its instances cannot exist on their own independently of whatever other objects may exist at the moment, but these inner class objects can only exist in the context of some mothership object of the outer class `IntervalSet`. Making the class `IntervalIterator` to be specifically an inner class allows its methods direct access to all members of the outer class, even those that are `private`. This doesn't even require any special syntax, but this access happens exactly the same way as these methods access their own actual members.

Your `IntervalIterator` objects should themselves contain a data field that is a reference of type `ListIterator<Interval>`, to keep track of which `Interval` in the list of intervals in the outer class object their traversal is currently at. Another `int` data field named `nextValue` will then keep track of the particular integer value inside the current `Interval` the iterator's traversal is currently at. After initializing these fields in the default constructor of `IntervalIterator`, you have everything you need to implement the two required methods of the `Iterator` interface

```
@Override public boolean hasNext()  
@Override public Integer next()
```

Even though we could make the iteration more general by defining it to be a `ListIterator` with `hasPrev` and `prev` methods symmetric to the above two methods for forward iteration, we will keep this lab simple by allowing iteration only to the forward direction. The method `hasNext` should return `true` if the previous `ListIterator<Interval>` reference is non-null, which means that at least one value remains to be visited during the current iteration, knowing that we will never store empty intervals in the `intervals` list. The method `next` will return the current value of `nextValue` inside that current `Interval` and meanwhile also increment `nextValue`, telling the list iterator to step to the next `Interval` in the outer `intervals` list whenever the value of `nextValue` exceeds the end value of the current `Interval`.

To keep things simple, we also choose not to allow mutation of the `IntervalSet` directly through the iterator methods `add` and `remove`, so these two methods don't have to be implemented in the inner class. Once you have completed this inner class, you can add the following instance method to your `IntervalSet` class

```
public Iterator<Integer> iterator() { return new IntervalIterator(); }
```

to allow the outside user code to acquire iterator instances to the particular `IntervalSet`, and finally change the signature of your `IntervalSet` class to the form

```
public class IntervalSet implements Iterable<Integer> {
```

to proudly come out of the closet and announce to the whole world the good news that it has now become possible to iterate through the instances of your `IntervalSet`! Even the **for-each loop** will now work for the `IntervalSet` instances out of the box, just like it does for all those snobby Boston brahmin array and collection types who landed on Plymouth Rock back then the Java language and its standard library were first founded.

The iterators in the Java Collection Framework and elsewhere tend to display great variation on how tolerant they are towards their underlying collection being mutated during the iteration. In one extreme, some collection subtypes have been hardened to tolerate even **concurrent** access and mutation from multiple execution threads, without interfering with the iteration over the contents of that set at the time the iterator was created. However, in this lab we will make the `IntervalSet` iterators to be of the other extreme by being **fail-fast**; any mutation of the collection while it is being iterated makes any active iterators immediately fail the next time that their methods `hasNext` or `next` get called.

This design decision not only makes our lives simpler, but also allows us to demonstrate the useful programming technique of **timestamping** that allows our iterators to immediately detect any mutations done on the underlying collection, without these iterators actually having to actually look at the elements of the underlying collection to determine whether some stealthy mutation has occurred since that iterator was last accessed. In the outer class `IntervalSet`, define a field

```
private long timestamp = Long.MIN_VALUE;
```

and modify the methods `add` and `remove` of the `IntervalSet` class to increment this timestamp every time that they are called. The inner class `IntervalSetIterator` should then also have a new data field

```
private long initialTimestamp;
```

initialized in the constructor of `IntervalSetIterator` to the current value of the outer class `timestamp` field. The iterator's two methods `hasNext` and `next` should then first check that the initial timestamp stored inside that iterator object still equals the timestamp of the outer class, and

if this is not the case, throw a `ConcurrentModificationException` instead of performing the requested operation.

Since the timestamp needs to be stored only once per each `IntervalSet` instance and once per iterator object, the use case of millions of iterators being vanishingly unlikely in practice, we might as well use the eight-byte `long` integer type to keep track of the timestamp values to ensure that they can't realistically overflow during the use of this data type. ("Four billion mutations on a set should be enough for anybody" just somehow sound like famous last words even if given in the voice of Carl Sagan, unlike the same sentence starting with the more impressive "Sixteen billion billion mutations...")

Lab 70: Cup Tournament Survival

JUnit: [CupSurvivalTest.java](#)

After these commercial interruptions, we again join live the national sportsball championship tournament between n teams numbered from 0 to $n - 1$, where n is guaranteed to be some power of two. Each round, pairs of teams play a match that determines which team advances to the next round. The number of teams is halved in each round, until the final match determines the winner of the entire tournament. For example, if $n = 8$ so that the teams are numbered from 0 to 7, the first round sees the matches between teams 0–1, 2–3, 4–5 and 6–7. The method that you write in this lab will compute the exact probability for each team to win the tournament, using an iterative process described below to build up these probabilities bottom up, using the [Fraction](#) type to exactly represent the intermediate and final probabilities.

In your lab project, create a new class named `CupSurvival`, and in there the static method

```
public static Fraction[] computeSurvival(Fraction[][] winProb)
```

that returns an array of n fractions that together add up to one, giving the winning probabilities for each individual team. The parameter `winProb` is an n -by- n array of fractions so that each element `winProb[i][j]` is the exact probability that the team i wins the game playing against the team j . (Perhaps these probabilities were computed by the famous sportsball handicapper “Ilk the Finn”.) Since there is no crying or ties in sportsball, each match is guaranteed to see exactly one of the participants advancing to the next round. The two symmetric probabilities `winProb[i][j]` and `winProb[j][i]` always add up to exactly one.

Instead of estimating the winning probability for each team in **frequentist** spirit by simulating the entire tournament a large number of times and estimating the winning chances of each team as the percentage of these simulated tournaments that the team ended up being the top dog, we will rather compute these probabilities as exact fractions from the bottom up without any randomness. The summations in the **marginalization** formulas for these probability calculations can spread out to be pretty hairy, but that is why we have bred our machines to perform these calculations, instead of expecting us slow hairless apes to grind through these formulas by hand.

For the bookkeeping needed to compute the winning probabilities from the bottom up, the method should define as a local variable a two-dimensional array

```
Fraction[][] survival = new Fraction[k][n];
```

where k is the number of levels in the tournament cup tree. For example, $k = 3$ when $n = 8$. Each quantity `survival[r][i]` will give the probability that the team i will win the match against its opponent in round r . Once the method has filled in this table, the method can simply return the entire row $k - 1$ of this array as the final answer, since the probabilities of that row are precisely the probabilities for each team to become the winner of the entire tournament.

This method should start by filling all elements of the bottom row `survival[0]` with values equal to one. Before you start coding this method, you might want to define two named constants

```
private static final Fraction ZERO = new Fraction(0, 1);
private static final Fraction ONE = new Fraction(1, 1);
```

inside your class to make your method code easier to read.

After initializing the bottom row with ones, each probability `survival[r][i]` in the higher rows can be computed with a recursive formula that involves the probabilities of the previous row $r-1$ and the winning probabilities for the individual matches between the given two teams given as a parameter to this method. However, to prevent this recursive computation from spending an exponential time by redundantly recomputing the same entries in the lower rows over and over again, your method should fill in the `survival` array in **dynamic programming** style using three nested for-loops, with no actual recursion used anywhere! The outer loop will count through the rows from 1 up to $k-1$. For each row, the inner loop will count from 0 up to $n-1$ to fill in the survival probabilities for the teams during that round. This computation will require another loop inside its body, which makes the entire algorithm consist of three levels of nested loops. (Hey, it's action, isn't it?)

The probability for the team i to emerge as the winner in its match in the round r is the product of two separate probabilities, representing the **parlay** of two separate things that both must happen for the team i to win its match in the round r . First, the team i must reach the round r to get to play there for a chance to advance; that is, the team i must emerge as the winner of its match in the previous round $r-1$. Conveniently for us, this very value was already filled in the `survival` table during the previous round of the outer loop! Having crossed the tall hurdle of showing up, the team i must win its match against the team that becomes its opponent in the round r . Marginalization of the combined probability into the sum of its disjoint elementary possibilities expresses the desired probability as the sum of simple terms of the form `survival[r-1][j]*winProb[i][j]`, where the innermost loop variable j counts through all the teams that could end up becoming the opponent for team i during the round r .

The only remaining problem is to determine which teams this innermost loop must go through to cover precisely the teams that can meet the team i in round r . We start by noting that if the team i starts at the position i in the cup tournament tree and wins its first match, it will be in position $i/2$ in the next round, where the division operator $/$ is the **truncating integer division**. Continuing this reasoning, team i will reach the position $i/2^k$ of that level in the cup tournament tree after winning k matches in a row. Even though Java does not have integer exponentiation function in the language, powers of two can be handily computed with **bit shifts**, since the expression `(1 << k)` always gives the desired power 2^k .

To determine the potential opponents of team i in the round r , first compute the position p where the team i would end up in the round $r-1$, were that team to win all its matches to reach that far. If that position p is odd, the opponent would play in position $p-1$ in that round; and if the position p is even, the opponent would play in position $p+1$. To traverse down the cup tournament tree back to

the original teams that could end up in that opponent's position in the round $r-1$, simply multiply the opponent's position p by the factor 2^{r-1} to find the first team in the bottom row that could become the opponent for the team i in that round. For round r , there are exactly 2^{r-1} such teams, and since they are all consecutive in the original bottom row, it is easy to loop over them. For example, the potential opponents of whichever of the teams 0 and 1 wins the first match are the teams 2 and 3 in round two, and in round three, the teams 4, 5, 6 and 7.

(Readers who will next take the course on data structures and algorithms after this course may there note the similarity between the previous formulas of moving up and down the cup tournament tree, and the formulas used to move up and down the positions of a [binary heap](#) data structure, to implement an efficient **priority queue**.)

Even when the original winning probabilities are simple fractions such as one half or one third, the summation formulas over all possible opponents for round r will quickly produce complicated fractions that have dozens of digits in their numerators and denominators. To make these results exactly reproducible over all computer systems of past, present and future so that we can write a meaningful JUnit test for them, we cannot compute these results in floating point, but `Fraction` objects must be used. To paraphrase the famous slogans of yesteryear, winners don't use floating point. Floating point — Just Say No! Friends don't let friends use floating point. Dare to keep kids away from floating point. It's float o'clock — do you know where your kids are? A friend with floating point is no friend indeed. For most programs in this computer, gone are the days of there being a floating point in every register. Are you better off now than you were four floating point operations ago? In your heart, you know `int`'s right!

You starting to get the floating point?

Lab 71: The Gamma And The Omega

JUnit: [EliasCodingTest.java](#)

Positive integers are simple enough to encode to be processed by mindless machines as **binary numbers** using sequences of zeros and ones. However, encoding not merely a single integer but an arbitrary sequence of integers requires some way to recognize where the binary representation of each individual integer ends and the next one begins. For example, would some given sequence made of 32 bits encode a single 32-bit integer, two 16-bit integers, one 23-bit integer followed by a 9-bit integer, or some other possible way to partition those bits into separate pieces? This question has to be resolved somehow without any hand waving such as using additional separator symbols between the binary numbers to make the boundaries between the payload bits stand out.

Of course, we can always agree that each number is stored in some fixed number of bits, typically 16 or 32, or these days actually 64. (As an aside, whenever you need to make up a **magic number** to limit something in computing, make that limit some power of two, or some power of two minus one, and suddenly nobody will question even the existence of your made-up limit, but will imagine the implementation artifact that created this limit to be some kind of immutable law of the universe.) However, not only would such policy set a hard upper bound to the magnitude of the integers that could be encoded in this system, but also end up transporting air by stretching all those itty bitty numbers to fill up this Procrustean bed by padding them from the front with redundant zero bits that convey no additional information.

We already saw one way to solve this problem back in lab 15 with the [Zeckendorf encoding](#). Break down the given positive integer as the unique sum of nonconsecutive Fibonacci numbers, and then use the bits to indicate which Fibonacci numbers appear in this sum. The impossible bit sequence 11 will then act as a separator between the encodings of two adjacent integers in the sequence. In the Zeckendorf encoding, the separator pattern 11 is placed at the end of the encoding of each integer, akin to the period that ends each sentence in English. In this lab, we take a look at two alternative ways to solve this same problem by prefixing each integer with an unambiguous indicator of its bit length. We start with the simpler but less efficient technique of [Elias gamma coding](#), and follow with its more advanced and tight cousin of [Elias omega coding](#).

To encode some positive integer x whose binary representation consists of $N + 1$ bits (that is, N is the highest power of two that fits inside x), the Elias gamma coding prefixes the length N in **unary** by using exactly N zeroes, followed by the $N + 1$ bits of the integer x itself from highest to lowest. For example, consider the integer 13, in binary $8 + 4 + 1 = 1101_2$, which makes $N + 1 = 4$. This number would be gamma encoded as the bit sequence 0001101. Note that since the highest bit of any positive integer is known to be 1, the 1 in the middle cleverly plays a dual role as both the terminator of the length prefix of unary zeros, and the highest order bit of the number x itself.

In your labs project, create a new class named `EliasCoding`, and there the following two methods to encode and decode a list of `BigInteger` values into a seamless sequence of zeros and ones using the Elias gamma coding. Of course, an adult realises implementation of these method would emit actual bits and bytes instead of Unicode characters, for maximal compression. However, this

lab exercise should be considered to be more of a **proof of concept** meant for learning and demonstration purposes, so the resulting sequences of bits are expressed as strings that can contain only the characters 0 and 1 to simulate the bit sequences that would actually be emitted.

```
public static String encodeEliasGamma(List<BigInteger> items)
public static List<BigInteger> decodeEliasGamma(String bits)
```

In the spirit of “**design by contract**”, the encoding method can assume that all `items` are positive, and the decoding method may assume that `bits` is a complete and legal Elias gamma coding of some list of `BigInteger` values. Neither method has to be able to handle and recover from errors in situations where this is not the case. Being the party that is in complete control of the material facts that affect the preconditions, the caller always bears the fault of the consequences for passing the method argument values that violate the preconditions of that method.

Instead of performing all the bitwise computations by yourself, it pays to remember that the people who created the Java standard library did not choose the classes and methods included in this club as some sort of dadaist abstract performance art event, but to serve programmers by giving them operations that were not important enough to gain entry to the inner circle of the language itself, but are still common enough to show up in all problem domains that it would just be wrong to expect every programmer to roll their own versions of those operations.

Therefore, before starting to hammer in the above two methods, you should take a moment to appreciate how the constructor of the [BigInteger](#) class has been overloaded to accept the **radix** as its second parameter, so you can have the first parameter string to be in binary instead of base ten just by using the radix of two. Similarly, the method `toString` has been overloaded to accept the desired radix of its result as the second parameter. Finally, the method `bitLength` returns the length of the binary encoding of the positive `BigInteger`, without actually constructing that string representation itself. Subtracting one from the result given by the `bitLength` method therefore tells you the highest power of two that is less than or equal to that `BigInteger`.

The Elias gamma coding is simple enough to understand and implement. However, a moment’s meditation reveals its inefficiency in that it almost doubles the number of bits needed to encode each individual integer! We can do *far* better than this. Skipping over the intermediate scheme of the [Elias delta coding](#) to get straight to the more advanced **Elias omega coding**, the bit length of each integer is not prepended to its binary representation in such a wasteful splurge, but is encoded recursively using the very same Elias omega coding!

This clever idea has an obvious chicken-and-egg problem of how we can possibly know whether the integer that was currently decoded from the sequence of bits is actually a member of the integer sequence being encoded, or denotes the length of the integer whose encoding follows it in the bit sequence. Since the highest order bit of any integer is known to be 1, this highest order bit can again play a dual role as both the highest order bit of that number, and the **marker** that indicates whether you need to look at the continuation of the bit sequence after decoding the current integer. Once this highest bit is zero, the decoding of the preceding integer is complete, and you can start decoding the next integer in the sequence after that zero bit.

```
public static String encodeEliasOmega(List<BigInteger> items)
public static List<BigInteger> decodeEliasOmega(String bits)
```

The computational steps needed to encode and decode any positive integer in the Elias omega coding are given on the Wikipedia page, and we shall leave the task of translating them to the higher-level syntactic structures of Java as an exercise for the reader. We shall still note that instead of using recursion, this method is more stylishly written as a while-loop where the integer N decreases in each round of prepending the length of the current number to the encoding. The loop terminates when $N = 1$, meaning that the length of the next block of bits plus one can be expressed in two bits.

Anyway, it's not like this recursion could realistically end up being very deep. As the bit length of an integer grows only **logarithmically** relative to the absolute magnitude of that integer, the extra zeros and ones needed to encode the length of an integer will be but a rounding error of a rounding error compared to that integer itself. The Elias omega coding becomes even more efficient as the numbers in the sequence grow astronomically large. For example, to quote the example given on the Wikipedia page, the Elias omega coding of 10^{100} , one **googol**, requires only 15 additional bits for the length header before the 333 bits that encode the actual one googol in binary, plus that extra zero bit tacked to the end to indicate that one googol was the integer that was actually encoded, instead of giving the decoder the bit length of the actual integer that follows it. (Such an unimaginably huge integer would not even fit inside our physical universe, even if we could somehow carve each one of its individual bits into one elementary particle!)

As the encoded integers grow even more gigantic, the extra bits for the recursive length header vanish into utter insignificance. For example, according to the same Wikipedia page, encoding the behemoth 10^{10000} requires only 22 additional header bits for its length (and the zero marker bit in the end) in addition to the 33,243 bits needed to encode the hundredth power of the already enormous one googol. We will gladly pay this chump change in exchange for the ability to reliably determine without any ambiguity or error where the encoding of one integer ends and the next one in the sequence begins. Boundaries and their maintenance are important in all walks of life, since any ambiguity usually only benefits the dishonest man.

Lab 72: Lazy Generation Of Permutations

JUnit: [PermutationGenerationTest.java](#)

Properties of individual permutations were already thoroughly examined back in labs 45 to 47 where we implemented methods for the algebraic operations, cyclic forms and the Lehmer coding of permutations. This lab examines the extremely important related combinatorial problem of systematically **generating** all permutations of the given length n , one permutation at the time, so that each one of the $n!$ possible permutations is visited exactly once during this process.

Algorithms to systematically generate permutations are well known classics in computer science and combinatorics. However, most of the online material that discusses these algorithms such as the [Steinhaus-Johnson-Trotter plain changes algorithm](#) and [Heap's algorithm](#) tends to present these algorithms in a recursive form, for simplicity of understanding. However, this recursive form is not practical for generating permutations or other similar combinatorial sequences that consist of an exponential number of elements. Instead of wasting memory needlessly by first collecting all permutations into a list for the user code to then iterate over, we would like the permutation generator to produce these permutations in a **lazy** fashion, one permutation at the time.

The permutation generator should keep track of only the current permutation along with possibly some **auxiliary data** whose exact nature depends on that permutation generator algorithm. This information should be enough for the generator to swap the correct elements inside the current permutation to produce the next permutation that follows the current permutation in the sequence of permutations being generated, making only local decisions based on this information without using any recursion. This way the outside user code can at any time tell the generator when it is done with the current permutation and needs to see the next one, that is then filled in the same permutation array that previously held the current permutation. Instead of having the first construct the entire sequence of permutations into a list, which would soon become prohibitive when iterating through billions of permutations, the generator needs memory only for the current permutation and the auxiliary data that is necessary to take the next step in the sequence.

To allow different algorithms for permutation generation to exist within the same class hierarchy, we make each of these algorithms a subclass of an abstract superclass [PermutationGenerator](#), as provided by your instructor. You should not modify this class in any way whatsoever, but implement the following two algorithms as concrete subclasses of this abstract superclass. Despite being abstract, the superclass defines a constructor for the user code to give it an integer array that both the user code and the permutation generator share to represent the current permutation of the sequence. The user code is responsible for not mutating the contents of this shared array object between computing the next permutation from the current one.

Most importantly, this abstract superclass defines the **template method**

```
protected abstract boolean next(int pos, int[] perm)
```

that the `public` method `next` in the abstract superclass calls to ask the algorithm implemented in the concrete subclass to rearrange the elements of the current permutation to turn that array into

the next permutation in the sequence. This method also receives the position of the current permutation in the sequence as the parameter `pos`, since the position information is needed in some permutation generation algorithms to decide what to do next. This method should return `true` if a new permutation was produced, and return `false` when the current permutation is the last one in the generated sequence so that no further permutations exist in the generated sequence.

We start with the classic algorithm that produces the permutations in lexicographic order, by each time first swapping two tactically chosen elements and then reversing the suffix of the array after the second swapped position. These tactical steps are explained in the section “[Generation in lexicographic order](#)” of the Wikipedia page “[Permutation](#)”, but we can illustrate them here with an example. Let us assume that the current permutation is {1, 5, 0, 3, 4, 2}. whose successor permutation in the sequence we are trying to determine.

Counting down from the second last position before the end of the array, find the first position `k` in `perm` that satisfies `perm[k] < perm[k+1]`. If no such position exists, this permutation is the lexicographically largest permutation that ends the sequence, and the method `next` should return `false`. In this example permutation, this is the position marked with a green background. Then, again counting down from the last position of the array, find the first position `l > k` that satisfies `perm[k] < perm[l]`. In this example permutation, this is the position marked with an orange background. Swap the two elements in positions `k` and `l`, after which you reverse the subarray from position `k+1` all the way to the end. In our example permutation, swapping the elements 3 and 4 changes the array into {1, 5, 0, 4, 3, 2}, and reversing the subarray from position `k+1` to the end turns this into {1, 5, 0, 4, 2, 3}, the next higher permutation that follows the previous permutation in lexicographic order.

To implement this algorithm, create a class named `LexicographicPermutation` in your lab project, and have this class extend the earlier `PermutationGenerator`. To allow creation of objects of this concrete subclass, it also needs to have the constructor

```
public LexicographicPermutation(int[] perm) { super(perm); }
```

that passes the argument array `perm` on to the superclass constructor.

Zaks’ algorithm of **prefix reversals**, as discovered back in 1984 by Shmuel Zaks, is not as well known as its more famous cousins such as the **plain changes algorithm** or **Heap’s algorithm**. However, Zaks’ algorithm is interesting in how it always performs exactly one prefix reversal to the current permutation to produce the next permutation to follow it in this sequence. A precomputed scheme is used to quickly calculate the length of the prefix that needs to be reversed to turn the current permutation into its successor. Repeated iteration of this scheme will then generate the entire sequence of $n!$ different permutations, each permutation visited exactly once as if by magic.

Create another subclass of `PermutationGenerator` named `ZaksPermutation`, with a similar constructor as the one seen in the class `LexicographicPermutation`. Before implementing the template method `next` in the `ZaksPermutation` subclass, you need to first write the following

static utility function used to determine the length of the prefix to be reversed in the current permutation, based on the position of that permutation in the generated sequence:

```
public static int zaksSequenceElement(int n, int pos)
```

This utility method is given the length of the permutation n , and the position pos of the current permutation in the permutation sequence, the position counting starting from zero. This method should compute and return the value in the position pos of the prefix length sequence S_n . Each sequence S_n is defined to contain exactly $n! - 1$ positive integers. As the base case of this recursive definition, the sequence S_2 contains just one element whose value is 2. Reversing the prefix of length two is required to turn the first permutation $\{0, 1\}$ of length two into its successor permutation $\{1, 0\}$ that ends that sequence.

After this base case, each sequence S_n is a concatenation of exactly n copies of the sequence S_{n-1} , with the number n tacked in as a separator between these concatenated subsequences. For example, the sequence S_3 consists of three copies of the singleton sequence S_2 with the new element 3 between each S_2 , giving us the five-element sequence 2, 3, 2, 3, 2. The sequence S_4 consists of four copies of the sequence S_3 with the new element 4 inserted as a separator between them, which makes this sequence to be 2, 3, 2, 3, 2, 4, 2, 3, 2, 3, 2, 4, 2, 3, 2, 3, 2, 4, 2, 3, 2, 3, 2. Note how the length of S_3 equals $3! - 1 = 5$, and the length of S_4 equals $4! - 1 = 23$.

To make the method run efficiently even when n gets larger, it definitely should not attempt to construct the sequence S_n explicitly. The element in the requested position should be computed using only integer arithmetic whose details are left for the reader to discover. It would again be useful to precompute the factorials of integers from 2 to 13 to some static array for a quick lookup, seeing that these factorials are needed in every turn during this generation. Once this utility method is working as confirmed by the JUnit tests, you can finally implement the method

```
protected abstract boolean next(int pos, int[] perm)
```

in this subclass to compute `zaksSequenceElement(perm.length, pos)`, and use that result to decide the length of the prefix of `perm` that needs to be reversed to take us to the next permutation. Before doing this, this method should first determine whether `pos` plus one is greater or equal to the factorial of `perm.length`, which happens when the entire sequence of permutations has been generated and the method should just return `false`.

Zaks' permutation generator will work just as well if converted to reverse the suffix of the array each time, instead of reversing the prefix. This sequence will then go through the permutations in a different order, but finish with the same lexicographically largest permutation as the prefix version.

Readers who have come this far can check out [Combinatorial Object Server](#), an interactive website to produce colourful visualizations for various algorithms to generate permutations and other combinatorial objects. The article "[A Unified Framework to Discover Permutation Generation Algorithms](#)" introduces a general template of a permutation generator algorithm that gives all these classic permutation generators as special cases.

Lab 73: Newton–Raphson Fractals

For testing: [NewtonRaphsonMain.java](#)

This problem was inspired by the video “[Newton’s Fractal](#)” by the popular mathematics channel [3Blue1Brown](#), and is for many reasons suitable for the subset of computer science students who are interested in properties of both the [complex numbers](#) and their functions. Despite that fact that neither man could have possibly anticipated them, these so-called [Newton fractals](#) naturally emerge out of the classic [Newton–Raphson method](#), a famous iterative numerical approximation method to find a root for any real-valued function that is differentiable everywhere and is in certain sense sufficiently well-behaved, the tacit assumption that underlies all numerical methods. The choice of the initial guess to start this iteration at determines not just how quickly this method converges to a root, but also which root the method converges to, whenever the function has more than one root.

As explained in the video, Newton’s method works just as well for functions of complex variables, needing nothing more but a simple substitution of a complex variable z in place of the real variable x . However, when the iteration of some function is not confined to the narrow straitjacket of the real line, but takes place on the two-dimensional complex plane that allows things go around other things in infinitely many squiggly ways without bumping heads with each other all the time, the complex plane gets partitioned into disjoint areas depending on which root the iteration converges to when started with a guess from that area.

Even for simple polynomials, these areas are not some boring [Voronoi diagrams](#), but fractals with infinitely deep complex details on the boundaries between these areas. Once again illustrating the principle of how the number three is fundamentally different from the number two in computer science, a moment’s thought reveals that between any two points in two different areas, there must exist a point between them where the invisible forces pulling towards those two roots are in perfect balance so that the iteration has no choice but to lead into some third root. Since the same reasoning can be repeated using the third point as one of these two endpoints, this detail will keep on keeping on like the Energizer bunny, no matter how deep you zoom into it!

Since Java does not have a `Complex` number type in the language, our first task for this lab is to define this data type. In your labs project, create a new class (or these days in a more modern Java 21+ fashion, a **record**) named `Complex`, whose instances represent immutable complex numbers made up of `real` and `imaginary` parts, each of which is, unusually for these labs and the overall worldview of this author, a `double` precision floating point number. We will not be zooming in deep enough into the resulting image for the limited precision of the `double` type to become an issue, and besides, no matter what finite precision you use, every image of a fractal of this nature is doomed to inexact anyway due to the inaccuracies of floating point iteration. (Curiously enough, any inaccurate rendering is still the correct rendering of *some* fractal, we just can’t control which one. Furthermore, rendering fractals with some woefully insufficient floating point precision can create intentionally dramatic displays of the [glitch art](#) aesthetic.)

This class should have a constructor to initialize these fields from its arguments, the **accessor** methods `getRe` and `getIm` to read their values, and a `toString` method to produce a human-

readable description of that complex number to help you in debugging the following methods for arithmetic and the **absolute magnitude** of the complex number, analogous to the four methods for the arithmetic operations that were seen in our lecture example class [Fraction](#):

```
public Complex add(Complex other)
public Complex subtract(Complex other)
public Complex multiply(Complex other)
public Complex divide(Complex other)
public double abs()
```

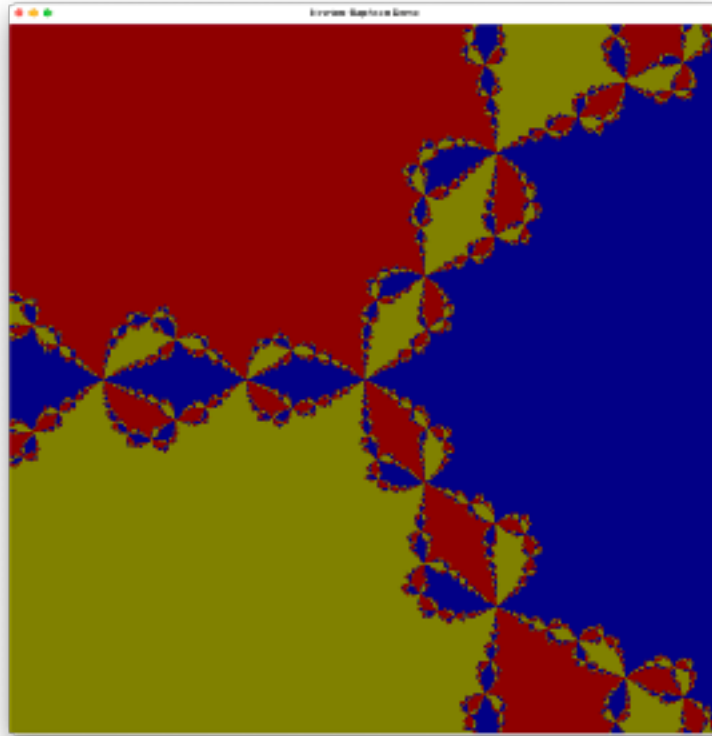
Consult any introductory online or printed material on complex numbers for the formulas to perform arithmetic on complex numbers (especially the division), and how to calculate the absolute value of a complex number. (Instead of computing the absolute value itself, it might actually be better to define the `abs` function to return its square, to avoid the expensive square root calculation that usually isn't even needed in many practical use cases of this absolute value.)

Once you have defined the class for `Complex` numbers, create a class `NewtonRaphson` in your labs project. This class needs to have just one method that computes the fractal. It receives seven parameters; those that define the finite square **peephole** into the infinitely large complex plane; the function `f` whose root we are looking for; the first derivative `fp` of the said function; and finally a `hueFunction` that maps `Complex` points to the hues used to colour that partition with. (The last function allows us to render this fractal without having to first solve all those roots beforehand.)

```
public static BufferedImage createNewtonRaphson(
    Complex top, int pixelN, double pixelSize,
    Function<Complex, Complex> f,
    Function<Complex, Complex> fp,
    Function<Complex, Float> hueFunction
)
```

The first parameter `top` gives the top left corner of the peephole that is being rendered into a new `BufferedImage` instance created by this method. The width and the height of this square image are `pixelN` pixels. The parameter `pixelSize` gives the size of the square on the complex plane represented by each individual pixel. The `Function` object `f` is a black box that contains the function whose root is being numerically iterated, and the similar black box `fp` gives its first derivative. Once the iteration from the given starting point z_0 has converged numerically into some z_n , the pixel that corresponds to the point z_0 is coloured using the **hue** acquired by applying the `hueFunction` to the point of convergence z_n .

This method should start by creating the new `BufferedImage` instance that will be returned from this method in the end. Then loop through the pixels (x, y) of that peephole (with modern Java, this **embarrassingly parallel** part could even be parallelized to massively increase the rendering speed), and compute for each pixel which complex number z_0 on the underlying infinite complex plane it corresponds to. Iterate the Newton-Raphson formula $z_{n+1} = z_n - f(z_n)/f'(z_n)$ until convergence, using the methods of your `Complex` class to perform the arithmetic operations in absence of



operator overloading that for some mysterious reason, the Java language still does not have, at least as of this writing. After convergence, colour that pixel (x, y) with a colour whose **hue** comes from applying the given `hueFunction` to whatever point this iteration ended up, whose **saturation** is `1.0f`, and whose **brightness** is computed from the number of iterations needed to reach convergence within some reasonably small `EPSILON` that you have defined as a named constant in your class. Note the method `HSBtoRGB` in the [Color](#) class that makes this colour computation a breeze.

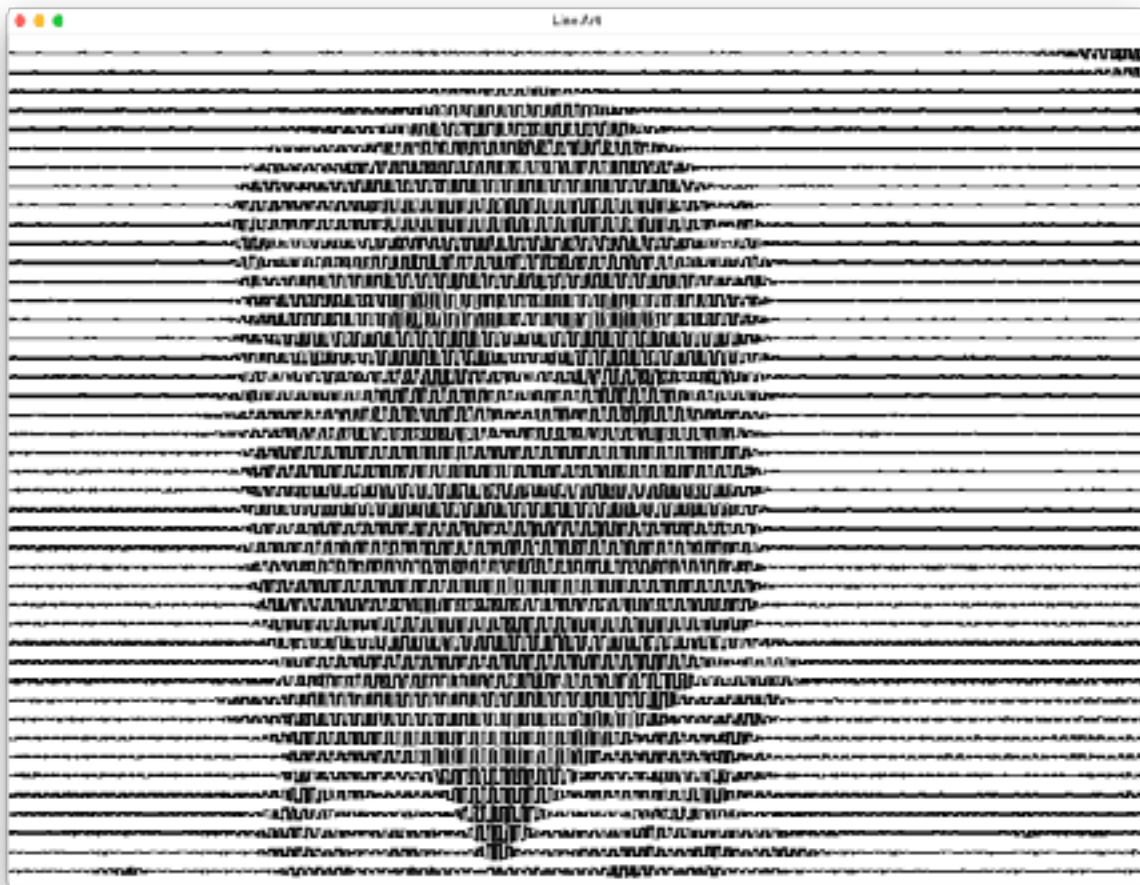
The class `NewtonRaphsonMain` uses your method to plot the Newton-Raphson fractal for the simple example polynomial $z^3 - 1$. According to the iron hand of the [Fundamental Theorem of Algebra](#), this polynomial has exactly three roots, one of which is real and the other two are each other's [complex conjugates](#) with nonzero imaginary parts. Running the `main` method of this class should therefore soon enough pop up the result window that looks like the above screenshot. If the colours of your result image end up being mirrored or rotated, remember that the y -coordinates are mirrored inside a `BufferedImage` so that they increase downwards, as is conventional in computer graphics. Therefore, make sure to use the formula with a correct sign to calculate the starting point z_0 for each pixel (x, y) from the given top left corner point `top` that should be anchored to map to the top left corner pixel $(0, 0)$.

The Wikipedia page and countless other places on the web showcase pictures of Newton-Raphson fractals computed from the roots of more complex (heh) functions than the mere three **roots of unity**. As also explained on that same Wikipedia page, the iterative formula itself can also be generalized to produce wackier results that no longer correspond to actual root finding, but nonetheless prove their worth by being pretty to look at, which is always a good plan B in life in general.

Lab 74: Between The Lines

For testing: [LineArtMain.java](#)

[Op art](#) exploits the inborn tendency of the human visual system to extrapolate patterns from simple but suggestive cues. Complex three-dimensional shapes emerge from the interplay of some simple two-dimensional shapes that are usually rendered in black and white to make this emergence even more dramatic in effect. As seen in the example screenshot below, the method written in this lab can convert the given colour pixel image into a bunch of jaggy horizontal curves drawn with a black pen on a white background, so that squinting at this black and white op art picture from a sufficient distance will make the visual sensation of the original image pop out in almost three dimensions.



In your labs project, create a new class named `LineArt` where you will first be writing two brief utility methods followed by the actual method to perform the above conversion to the pixel image. The first one of these utility methods is used to calculate the **luminance** of the given RGB colour; that is, how bright this colour will appear to the human eye compared to other RGB colours.

```
private static double getLuminance(int rgb)
```

With the RGB colour given as a single int value `rgb` whose bytes from low to high encode the blue, green, red and alpha components of that colour, respectively, this method should return a double between 0 and 1 that estimates its luminance. You should use **bitwise arithmetic** to extract the individual colour components from the `rgb` integer (you can consult either one of our two example programs [ImageOpsDemo.java](#) and [FloydSteinberg.java](#) for the formulas to do this), and from these use one of the possible luminance formulas given in the Stack Overflow page “[Formula to determine perceived brightness of RGB color](#)”, or some other equivalent online resource. Then, you should write the second required utility method

```
private static double averageLuminance(  
    BufferedImage source, int x, int y, int r  
)
```

that calculates the average luminance of the pixels inside the square box in the `source` image whose `x`-coordinates start and end at `x-r` and `x+r`, and whose `y`-coordinates start and end at `y-r` and `y+r`. Be careful at the edges of the image where the boundaries of this box would reach outside the boundaries of the `source` image. With this utility function available, the actual conversion of the pixel image into the black and white op art image will be done by the method

```
public static BufferedImage lineArt(  
    BufferedImage source, double density,  
    double frequency, double amplitude  
)
```

This method should start by creating the `result` image as a `BufferedImage` with the same width and height as the original `source`. Acquire the `Graphics2D` object needed to render things into the image, and fill this image with a white background to act as your canvas. Then set the stroke to be a solid black pen whose width equals `0.15*density`, converted to a float.

Each curve starts at the left edge of the image, and corresponds to some `y`-coordinate baseline. The `density` parameter defines how many pixels apart the `y`-coordinate baselines of these individual curves are spaced in the vertical direction. Each individual curve should be rendered with a loop that starts with `x` equal to zero, and terminates once `x` reaches the right edge of the image. Every other round of this loop, the value of `x` should increase by `frequency` for the horizontal curve segments, whereas every other round `x` should stay the same, for the vertical curve segments.

The loop should also maintain another local variable `int sign`, initially set to equal to `+1`. This `sign` should alternate between the values `+1` and `-1` every two rounds, flipping its sign in precisely those rounds when the `x`-coordinate stays put. Each round of the loop, draw a line segment (for nicer results, use `Line2D.Double` and don't forget to turn on anti-aliasing before drawing) from the previous point to the new point whose `x`-coordinate is given by your variable `x`, and whose `y`-coordinate is computed with the formula

```
y + sign * amplitude * (1-luminance)
```

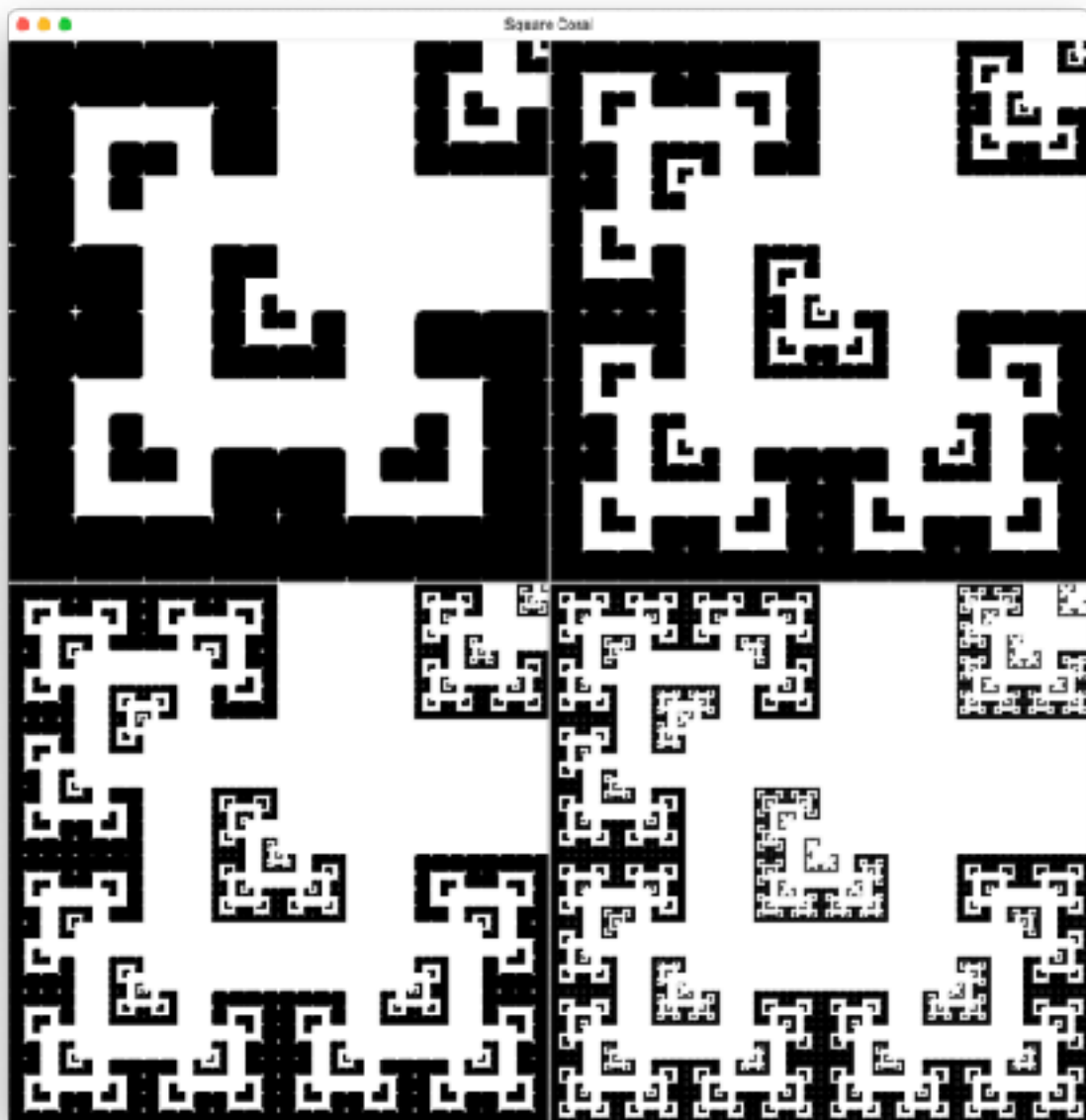

where y is the baseline coordinate of the curve, and `luminance` is the average luminance computed over the box centered at the pixel (x, y) with r equal half of `density`. Once you have drawn this line segment, the new point becomes the previous point for the next round of this loop. Finally, to give your result some kind of artisanal hand-drawn feel of the quaint past when everything was analog and had a constant component of noise, you should also perturb each individual corner point of your jaggy curves by some small random amount. Use the method `nextGaussian` of your random number generator multiplied by suitable amount to get a good distribution of perturbations.

Expanding this same idea, the lines that cover the image plane don't need to be horizontal, but could, for example, be some variation of **spiral** emanating from the center of the image, or even some [space-filling curve](#) such as Peano curve or Hilbert's curve. No matter which way your chosen curve meanders and squiggles, the displacement at each point is always done to the direction of the **normal vector** of the curve at that particular point.

Lab 75: Square Coral

For testing: [SquareCoralMain.java](#)

The page “[Fractals, Chaos and Self-Similarity](#)” by [Paul Bourke](#) features a plethora of formulas and visualizations of exotic fractals for us to explore after all those Sierpinski gaskets and Dragon curves eventually start getting a bit stale. Most of the fractals seen there emerge from iterating all kinds of inventive formulas on a complex plane around some simple poles, same way as in the earlier lab on Newton–Raphson fractals. However, “[Square Coral](#)” is a rare example in that collection of a **subdivision fractal** where some simple geometric shape is recursively cut into smaller pieces whose areas together comprise the original shape, and some smaller pieces are systematically discarded during this process so that the resulting shape will meander around fractal-shaped holes.



The earlier lab problem of imitating the famous **Mondrian art** also concerned subdividing a rectangle into smaller rectangles. However, in that lab, the cutting points inside the rectangle were chosen randomly and none of the smaller rectangles were thrown away, so the distinction between these pieces was made with black outlines filled in with simple primary colours. Construction of the Square Coral fractal begins with a square that is subdivided into four square quadrants, the side length of each quadrant thus exactly one half of the original square. However, depending on one of the four possible **orientations** of the square currently being subdivided, one of these four quadrants is shrunk to half size, so that its side length is only one quarter of the original, to reveal some background colour from behind the shape.

Each of the four quadrants is also given an orientation that depends on the compass position of that quadrant inside the current square, and the given orientation of the current square. This orientation will affect the recursive subdivision of that particular quadrant. By choosing the orientations of the quadrants according to the rule illustrated with the example on the page linked above, the resulting fractal ends up having the overall coral shape seen in the example screenshot.

In your lab project, create a new class named `SquareCoral`, and there the recursive method

```
public static void render(  
    Graphics2D g2, int depth, int x, int y, int width, int orientation  
)
```

This method is given the x - and y -coordinates of the top left corner of the current square, along with the desired width of the result image measured in pixels. The parameter `depth` gives the depth of the resulting fractal image. The base case of this recursion is when `depth` equals zero, indicating that that shape is small enough and should not be subdivided any further, but should be rendered as an instance of [RoundRectangle.Double](#). (For a more aesthetic appearance, why don't we give its corners both horizontal and vertical roundness of `width/5.0`.) Otherwise, depending on the `orientation` of the current square, this method should recursively render the four smaller squares to their rightful positions with `depth-1`. (To simplify these formulas and make them more readable, you might want to first define local variables `h` and `q` to denote the one half and the one quarter of the given `width`, respectively.)

The above screenshot shows the expected visual outcome of performing the Square Coral subdivision up to depths from three to six. To ensure that the result of our [SquareCoralMain.java](#) test bed will match this screenshot when executed with your implementation of the recursive `render` method, the `orientation` of 0 should mean that the northeast quadrant contains the quarter-sized square; 1 means southeast; 2 means southwest; and 3 means northwest. Again, make sure to pass on the correct orientation to every one of the four recursive calls for the subdivided quarters. To simplify your coding and debugging, you should note that for each of the four orientations, exactly two of the subdivided quadrants will have that same orientation, and the quadrant that was shrunk to half the size should be one of these two. When the shape has an underlying symmetry, so will the rules that define that shape, even if that symmetry happens to be hiding behind the veil of some abstraction layers.

Lab 76: Sultan's Daughter

JUnit: [SultansDaughterTest.java](#)

Students whose first programming course was given in Python while this second course is given in Java have already realized how very different these two major programming languages are, not merely on the syntactic level but in their deeper foundations. Even so, these two languages are still relatively close to each other in the space of all possible [Turing-complete](#) programming languages. As a crude analogy of what I am trying to say here, consider how our Old Blazey is way too distant from its nearest neighbour Proxima Centauri in terms of practical space travel for humans and all other species who must do what all that eat and drink must do, and yet even this immensely vast distance becomes meaningless in terms of galactic superclusters truly out there.

Along most of the dimensions along which programming languages can possibly differ, Java and Python turn out to be practically identical. We just tend to have difficulty seeing this, since most of those dimensions seem virtually nonexistent from our everyday point of view as if their width were zero, so that some kind of a law of nature dictates that all programming languages must behave in a certain manner along that particular dimension. However, even a casual perusal of [esoteric programming languages](#) ought to cure any sweet summer child from this misconception.

Esoteric programming languages are not intended as serious tools for working programmers to solve actual problems. They are meant to be works of **abstract performance art** to be appreciated in the sense of opening your mind to esoteric possibilities that otherwise remain hidden from the masses. (You know, all those sheeple around you who don't have a clue what really is going on and what will be coming to them.) The very existence of a good esoteric programming language is intended to be a comically hubristic affront towards and a glove slap in the face of not just what is good and ordered in humanity itself, but also the Janus faces of Nature and Nature's God. Don't worry, they won't get angry for this insult. By its sheer existence, an esoteric programming language makes some poignant point about how many assumptions and ideas about design of programming languages turn out to be far less solid as originally thought.

To play around with this concept, this lab has you implement an **evaluator** for expressions written in [Sultan's daughter](#), an esoteric programming language dreamed together into form by no less of a tinkerer in the kingdom of the abstract than [Raymond Smullyan](#) himself. Better known for his logic puzzles set on islands populated by deeply obsessive-compulsive characters such as knights who always tell the truth and knaves who always lie, some of Smullyan's other whimsical creations also illustrate the surprising powers of various formal logics. This little language mixes ideas of [combinatory logic](#) and the [string rewriting paradigm](#), both in fact ancestors of actual programming languages more suited for elegant salons and tea rooms. *Ad fontes!* Come, come, the game is afoot.

Expressions written in Sultan's daughter are strings that consist of digit characters 0 to 9 only, with no other characters allowed. (If this seems somehow restrictive, you should remember that inside your computer, all your Java programs, just like this PDF document and everything else that exists inside your computer, consist of sequences of nothing but virtual zeros and ones somehow encoded as physical atoms!) To evaluate an expression of the form d^x , where d is some leading digit charac-

ter and X some string of digit characters (possibly empty), the first digit d determines which operation is applied to the recursively computed value of the rest of the expression X.

The evaluation rules for the possible values of d are listed on [the Wiki page for Sultan's daughter](http://esolangs.org/wiki/Sultan's_daughter) at esolangs.org for the reader to consult. For example, since the expression 19872 evaluates to 987, the expression 319872 evaluates to 987987, and the expression 4319872 evaluates to 789789.

In your labs project, create a new class named `SultansDaughter`, and there the first method

```
public static String sultansDaughter(String expression)
```

that evaluates the `expression` according to the rules of Sultan's daughter. Our automated tester is guaranteed to only give this method expressions whose evaluation terminates within reasonable bounds of both time and space. If the value of the `expression` is not defined according to the rules of Sultan's daughter, such as for the expressions 143 and 81932, this method should return `null`.

Once you get the previous method working so that it passes the instructor's JUnit tests, you can use it as a subroutine in the second method

```
public static int iterateSultansDaughter(String expression)
```

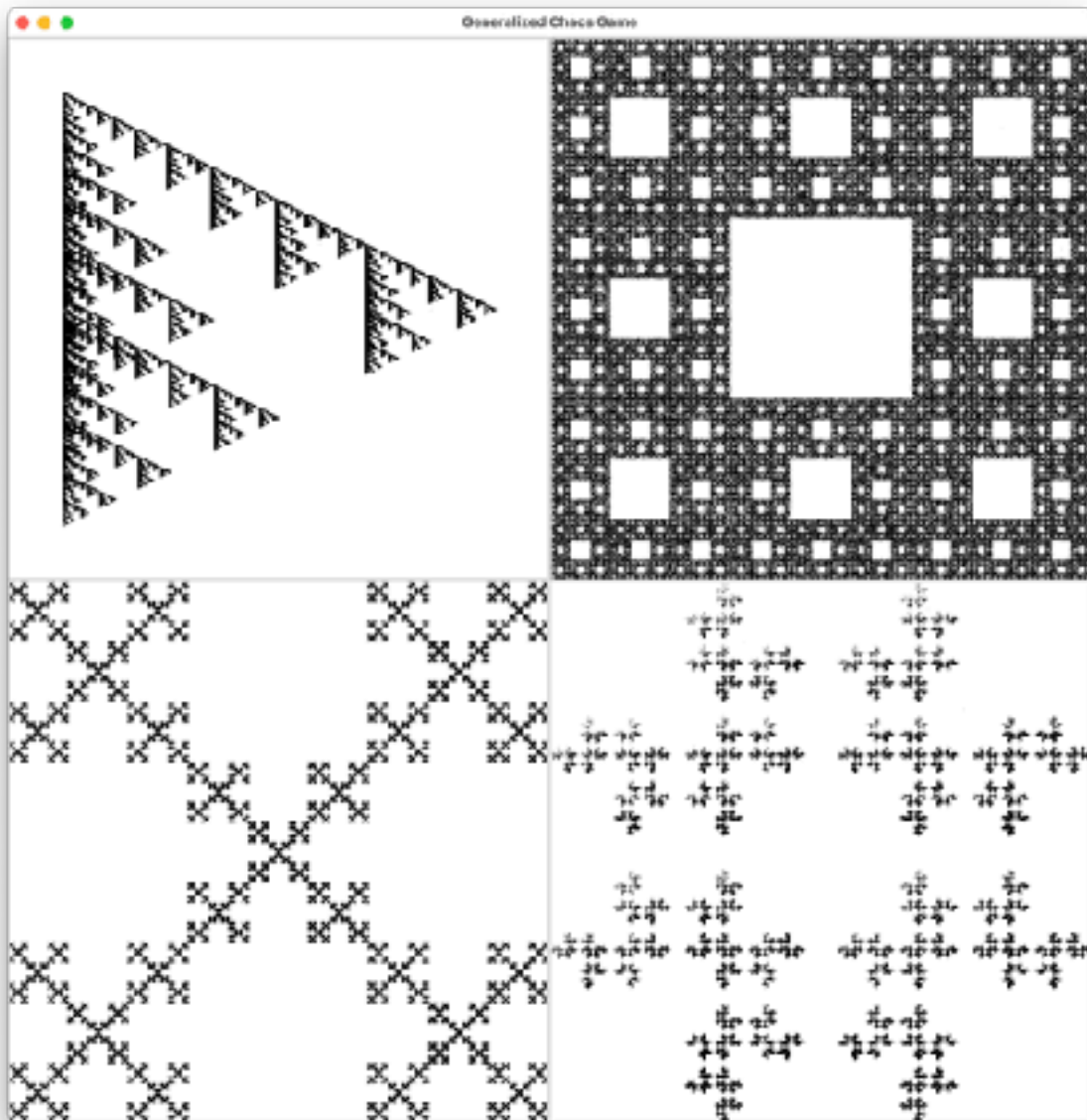
that iterates the evaluation of the starting `expression` until the value of the current expression is undefined so that its evaluation produces the result `null`. This method should return the count of how many steps of iterated evaluation were needed to reach the result `null`. For example, since the starting expression 1161175032222 evaluates to 16117503222, which evaluates to 611750322, which evaluates to 1175032, which evaluates to 17503 whose evaluation is then undefined, this method should return the answer 5 when starting from this `expression`.

One extra kink that you unfortunately have to consider in writing this method is that the iterated evaluation of some starting expressions will never terminate! For example, consider the **fixed point** expression 47431474312; inside the microcosm of Sultan's daughter, this expression is a [quine](#) that evaluates to itself, as the reader can readily verify. Your method must somehow detect such situations where the iterated evaluation eventually loops back to itself through one or more steps, and return -1 as the artificial "stuck in an infinite loop, will never terminate, kthx bye" answer for all such expressions.

Lab 77: Generalized Chaos Game

For testing: [ChaosGameMain.java](#)

With sufficient discipline to constrain its noisy excesses, shapes created by jumping randomly across the pixel image can end up far more regular than someone would initially assume. [Chaos game](#) is a classic general mechanism to generate all sorts of nifty fractal shapes just by randomly jumping around as if you were a headless chicken stuck in a *House of Pain* music video. Each jump is done towards a point chosen randomly from the given set of fixed points. The initial placement of these fixed points then determines the overall shape of the fractal and the Swiss cheese holes that this jumping around magically somehow always avoids, no matter how far you let it run. Starting from some arbitrary point on the image, choose one of these fixed points randomly. Move the current point halfway towards your chosen fixed point, and increment the visitation counter for the pixel that the current point lands on. Keep on keeping on until you have visited enough points to have exposed the image enough so that it stands out clearly from its white background.



In this lab, you will implement a slightly generalized version of the chaos game that our automated tester then uses to render some classic fractal shapes seen in the previous screenshot. In this generalization, all fixed points are not equal, but some of them are more equal than the others to that but each point can be randomly chosen with a different **frequency**. Furthermore, each point can have a separate **attraction weight** that determines how close the current point moves towards that chosen point in each jump.

In your labs project, create a new class named `ChaosGame`, and there the method

```
public static BufferedImage playChaosGame(
    long rounds, int n, int[] x, int[] y,
    int[] freq, double[] weight, boolean canRepeat
)
```

The first parameter `rounds` determines the number of jumps that will be made in total to generate the image, and the parameter `n` gives the width and the height of the resulting image in pixels. The parameter arrays `x` and `y` contain the x - and y -coordinates of the fixed points to jump towards.

The array `freq` contains the intended frequencies that determine how often each point will get randomly chosen in the long run. For example, if there are four fixed points for us to randomly jump towards, and the `freq` array equals $\{2, 3, 1, 4\}$, the first point will be chosen as the target roughly 20% of the time, the second point will be chosen roughly 30% of the time, and so on. This proportion is only statistical but not exact, since each individual jump must make the random choice between the target points **independently** of the past choices. As explained in the introduction to the earlier problem about Lemire dynamic frequency sampling, to choose the random fixed point with respect these given frequencies, you should generate a random integer d from $0, \dots, m - 1$, where m is the sum of frequencies of the fixed points, and then find the first fixed point for which the sum of frequencies up to and including that point is *strictly greater* than d . (Since these images will use only a handful of fixed points, you can do this with a single for-loop without any sophisticated optimizations by first once suitably preprocessing these frequencies.)

The array `weight` contains the attractive weights of each fixed point so that when jumping towards from the current point (x, y) towards the chosen target point (x', y') , this jump will land on the point $((1 - w) \times x + w \times x', (1 - w) \times y + w \times y')$. For example, if $w = 1/2$ as in the ordinary chaos game, the jump would land exactly halfway between the current point and the chosen target point.

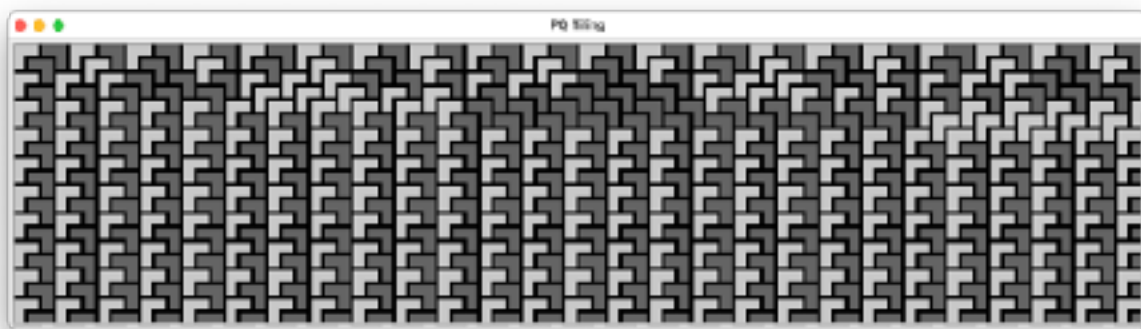
The parameter `canRepeat` determines whether the target point from the previous jump can be chosen again as the target point for the current jump. If this parameter is set to `false`, the choice of the current fixed point to jump towards must be rerun if it would lead to the previous target point.

The colours of the result image should be as in the above screenshot, with the fractals drawn in greyscale against white background. To compute the darkness of each pixel, you can consider every pixel that was visited eight or more times to be fully black, and every pixel whose count is zero to be fully white. The greyscale colours for the pixels whose counts fall between those two extremes can then be linearly interpolated between the full black and white colours.

Lab 78: Mind Your P's And Q's

For testing: [PQTilingMain.java](#)

Of all recreational math books that this author has consumed during the past quarter century or so, “[Mathematical Puzzles](#)” by [Peter Winkler](#) is the absolute unit and *numero uno*, earning a blurb to that effect from no less of a luminary than [Donald Knuth](#) himself! The problems and their elegant solutions presented in this book cut through all kinds of seemingly unbreakable Gordian knots reminiscent of the way a chainsaw cuts through butter, merely by applying simple flashes of insight that seem obvious in retrospect once they have been shown. (This is, of course, how you recognize genius ideas in all walks of life; they seem self-evident to be hardly even worth mentioning once pointed out, even though they sure as hell weren’t that self-evident five minutes before that!)



This lab has you implement a curious **tiling** of one quadrant of the infinite two-dimensional integer plane, using two types of **bent trominos** composed of three unit squares. In computer graphics, the y -coordinate conventionally increases downwards, opposite the usual convention in mathematics. We will therefore also present this problem vertically mirrored from the way that it was originally presented in the good professor Winkler’s problem book. The origin point of the infinite plane quadrant is located at the top left corner of the image, and the horizontal and vertical coordinates increase going right and down.

The two types of bent trominos available for filling the infinite plane quadrant can be referred to as P’s and Q’s. In the above screenshot, the P-tiles are rendered in light grey, whereas the Q-tiles are rendered in dark grey. These two types of tiles are the only types that you are allowed to use here. If the two other possible rotations of these tiles, the L-tiles and J-tiles, were also available in unlimited amounts, there would be an infinite number of ways to tile the plane quadrant without overlaps or holes. However, when restricted to use only these two particular types of P- and Q-tiles without rotations, care is needed in placing these tiles to avoid painting ourselves in a metaphorical corner. Under this restriction to use only P- and Q-tiles, there exists exactly one way to succeed in this task. Furthermore, this tiling turns out to have some very unexpected and counterintuitive properties that made it worthy of inclusion in the good professor Winkler’s book.

To get started, we quickly realize that the first row must necessarily be filled with alternating P- and Q-tiles. Placing two consecutive P-tiles or two consecutive Q-tiles anywhere on the first row would leave a unit gap between the legs of these tiles in the row immediately below. Such a gap would be

impossible to fill with P- or Q-tiles, since we are not allowed to rotate these tiles to make their legs point upwards to fill in the unit gap. The contents of the first row are therefore forced by the iron hand of the rules of this game.

After filling in the first infinite row, each gap of length two in the second row must be similarly filled with either P- and Q-tiles. In general, when filling the row n , the choice of tile used to fill each gap turns out to be forced by the exact same reasoning as in the first row, in that you may not leave a unit gap anywhere in the row $n + 1$ to be filled in next.

If we were tiling a half plane that was infinite to both north and south, every column could consist of a straight alternation of P-tiles and Q-tiles. However, for the quarter plane, even though the first row is as simple and periodic as it can be, the top edge of the quadrant breaks this symmetry and causes the entire forced tiling to become **aperiodic**, despite the fact that the top row is perfectly periodic! This is a good illustration of the phenomenon of how sneaking a small pebble of asymmetry under the stack of mattresses that the princess sleeps on will cause uncontrollable chaos to emerge from the initial perfect order. However, as can be seen in both screenshots, each vertical column will eventually become periodic, so this aperiodic tiling is not quite as exciting as it could potentially have been in the spirit of, say, [the Rule 110 of elementary cellular automata](#).

In your labs project, create a new class named `PQTiling`, and in there the method

```
public static BufferedImage render(int rows, int columns, int tile)
```

that creates and renders a view of this infinite tiling for the first `rows` and `columns`. The parameter `tile` gives the pixel width and height of each unit square on the plane. You should draw a black outline of width 30% of the `tile` width around each tromino, and fill the P-tiles with light grey and the Q-tiles with dark grey. The easiest way to achieve this would be to use the [Path2D.Double](#) shapes from the `java.awt.geom` package. Build each such shape object piecemeal from consecutive straight line segments, and then use `fill` and `draw` to render the internals and the outline of the completed closed shape at the desired position in the result image.



You might want to first define two `private` subroutines to render a P-shape and a Q-shape at the given location, to simplify the logic of your actual rendering. When you then compute the contents of the image with two nested loops, you should use a two-dimensional `boolean` array to keep track of the positions that have already been filled by some tile. You should also make this two-dimensional array to be slightly larger in both dimensions than needed, to avoid accidentally stepping out of the bounds of this array at the right and bottom edges of the image.

Lab 79: Die Hard With A Variance

JUnit: [BaysDurhamShuffleTest.java](#)

To ensure portability of results of any and all integer computations in all computing devices that have enough moxie to run the Java Virtual Machine, the class [java.util.Random](#) provided in the standard library to generate all kinds of pseudorandom quantities has been standardized to use the classic [linear congruential generator](#) algorithm with hardwired fixed parameters. This allows our automated JUnit tests used in these labs to generate the pseudorandom test cases with Random and expect to receive consistent hardwired checksums for all methods, regardless of which hardware and operating system these results are computed on, and whether these results are computed from the model solutions from the private top shelf reserve of your 'umble, very 'umble narrator, or from the solutions put together by all of you bratties and devotchkas toiling diligently at your desks.

Linear congruential generators are sufficiently indistinguishable from true randomness for the purposes of games and other applications whose aim is giving the human user the illusion of larger reality on top of what is really nothing but a simple deterministic state machine. However, the classic 1968 article "[Random numbers fall mainly in the planes](#)" by [George Marsaglia](#), the late author of the famous [DIEHARD test suite](#) for measuring the quality of pseudorandom number generators, not only established that punny titles for scientific articles well predate the snark of modern Internet, but that the statistical randomness properties of linear congruential generators leave a lot to be desired for the needs of serious simulations that in this day and age may require even trillions of random samples. Any systematic bias in the pseudorandom generator that generates these samples will then translate to some systematic bias in that simulation.

For the needs of this course, the class Random also works as a wonderful example of the **template method design pattern**; all its methods for generating random quantities internally use the same protected method `next` as the source of random bits that they refine into more complex random quantities. The template method pattern allows anyone to create subclasses of Random that use some more advanced algorithm of higher **entropy** to generate the pseudorandom bits, since the remaining methods such as `nextDouble` or `nextGaussian` inherited from Random will then work right out of the box as they are.

This lab has you implement two ways to produce statistically superior streams of pseudorandom numbers. We use techniques most famously discussed in "[Seminumerical Algorithms](#)", the Volume 2 of the classic doorstop "[The Art of Computer Programming](#)" by the starry dedoochka [Donald Knuth](#) himself. First, we will extend Random to create a subclass whose template method `next` uses the [Lagged Fibonacci generator](#) to produce its random integers. Second, we shall create a general purpose **decorator** subclass that can be placed in front of any past, present or future Random number generator, to use the ingeniously simple **Bays-Durham shuffling** algorithm to scramble the numbers returned from the underlying generator in a manner that greatly improves the statistical properties of the sequence. To get started, create a class in your labs project with the signature

```
public class LaggedFibonacci extends Random
```

As explained on the Wikipedia page linked above, the Lagged Fibonacci generator is initialized with two integers k and j that satisfy $k > j$, and the **seed** array Y that contains exactly k integer values that form the initial state of the generator. (Even though Java variable naming conventions reserve words with initial capital letters for classes, we shall follow the names used in TAOCP to make things easier for readers accustomed to mathematical notation.) The caller is responsible for filling in the initial contents of the seed array Y however they wish, except for the small requirement that at least one of the values in this array must be odd. (If all values in Y are even, it is easy to see from the **additive** formulas used to update the values that the generator will only produce even numbers from that point onwards; not exactly the uniform distribution of values that we hope for!)

The above Wikipedia page also collects some possible values for k and j that make the lowest-order bits of the generated pseudorandom numbers behave especially chaotically. Since the buffer size k can be arbitrary large, the state space and the resulting **period** of the pseudorandom generator can become truly astronomical so that the remaining lifetime of our physical universe until its inevitable heat death would not suffice for this pseudorandom number generation to even get close to ever repeating itself. The class should have a constructor

```
public LaggedFibonacci(int[] Y, int k, int j)
```

that initializes the corresponding data fields with the constructor parameters given. However, since we index the arrays starting from zero instead of starting from one as done in TAOCP, you should decrement the quantities k and j by one before assigning their values to the corresponding data fields. Furthermore, to ensure that you properly understand the difference between an object and a reference pointing to that object, this constructor should create and store a **defensive copy** of the array Y , instead of merely storing a reference to the original array object that got passed as the first argument to the constructor.

After completing this constructor, you only need to override the protected template method

```
@Override protected int next(int bits)
```

To keep life simple, this method should completely ignore its `bits` parameter, and always return a 32-bit integer `result` in the following fashion. Initialize your local variable `int result` from the result of evaluating $Y[k] + Y[j]$, and update the value of $Y[k]$ to also equal this `result`. Decrement both indices k and j , and should either index become zero or less than zero, make the value of that index equal the last position of the array Y . Having done this, return the stored `result` to the caller from its safekeeping. And that's all she wrote; you can now use the first half of the combined JUnit tester [BaysDurhamShuffleTest.java](#) to verify that your generator works as required.

The template method design pattern meshes especially nicely with the decorator design pattern, since the former completely eliminates the most annoying part of the latter pattern in Java; overriding all those methods that only consult the underlying client with the same arguments, and then pass on the result returned from the client to its own caller. Create a new class

```
public class BaysDurhamShuffle extends Random
```

Just like every decorator class everywhere, this class should contain a `private` reference to its underlying `client`, given to this decorator as a constructor parameter. The Bays–Durham shuffling algorithm maintains an internal buffer `V` that has room for `k` integers, initialized with the first `k` results from calling `client.nextInt()`. In addition, this algorithm maintains a `private` integer variable `Y`, initialized with one more call to `client.nextInt()`. Write the constructor

```
public BaysDurhamShuffle(Random client, int k)
```

to perform these initializations. Next, override the template method

```
@Override protected int next(int bits)
```

to again completely ignore its `bits` parameter and always return a 32-bit integer result computed in the following fashion, slightly modified here from the “Algorithm B” given in TAOCP to simplify the integer arithmetic involved. Start by calculating the index `j` to the buffer `V` using the expression

```
int j = (Y & 0x7FFFFFFF) % V.length;
```

where the strange-looking **bitwise arithmetic** formula is used to chop off the highest **sign bit** from the value of `Y`, to ensure that the result is nonnegative and can therefore be used as an array index. Knuth’s original discussion of the Algorithm B uses **scaling** of `Y` to make it fall in the range of legal buffer indices from 0 to `k – 1`, which requires some care to get to work correctly and avoid overflows when implemented with only integer arithmetic.

(Since the signed integers represented in two’s complement encoding necessarily contain one more value in the negative direction than in the positive direction due to the number zero being the odd man out in the middle, the expression `Math.abs(Y)` to achieve the non-negativity of the produced random number will silently fail when `Y` equals `Integer.MIN_VALUE`, the smallest possible negative integer. This edge case can lead to weird bugs in some applications that manifest once in a blue moon and are well nigh impossible to debug, unless you understand what is going on with the two’s complement encoding of fixed-size signed integers.)

Having computed the index `j` from `Y` to access the buffer `V` with, you next assign `Y` to have a new value equal to `V[j]`, and then assign the element `V[j]` to equal the result of again calling the method `client.nextInt()`. Return the new value of `Y` from its safekeeping as the next random number produced by the method `next`. With that, the fat lady has again sung you the song of her people so that you get to run the JUnit acceptance tests to verify that your implementation of Bays–Durham shuffle works as required.

Laws of neither nature nor man restrict the Bays–Durham shuffle to be used only on pseudorandom numbers. When applied with a buffer width `k` equal to 3, the perfectly orderly sequence of natural numbers 0, 1, 2, 3, ... gets scrambled into 0, 4, 1, 6, 5, 2, 9, 8, 10, 7, 13, 14, ..., and so on. The value 3 is missing since the initial value of `Y` will never get returned. However, this algorithm can be easily modified to not drop any values, merely by initializing `Y` from some external random source.

Bonus Lab 100: The Buddhabrot Is Silent

For testing: [BuddhabrotMain.java](#)

The famous [Mandelbrot set](#) is defined as the set of those initial points z_0 on the complex plane for which the iteration $z_{n+1} = z_n^2 - z_0$ does **not** escape to infinity. Since the Mandelbrot set itself is a black hole that is infinitely hairy around its edges, the rendered images are usually made more interesting by colouring the points outside the Mandelbrot set in bright colours according to how long the escape from those points takes to cross a certain distance threshold from the origin.

This instructor can still fondly remember the halcyon days of [Fractint](#) running on his first PC computer that could spend a better part of an hour rendering a simple zoom near the high level view into this fractal. With the exponential speedup of computing available to Joe Everyman, today's off-the-shelf desktop computers can render deep and complex parts of this fractal in almost real time. The instructor's own example program [Mandelbrot.java](#) uses multiple Java execution threads to carve out, classify and colour the escaping points outside and around the Mandelbrot set, so that no execution time is wasted iterating the sequence of complex points from any internal point of this black hole, except for the fencepost pixels at the edges of this set.

[Buddhabrot](#) is a variation of the Mandelbrot set whose computation is significantly heavier than rendering the plain old Mandelbrot set. Buddhabrot uses the same iterative formula for each starting point z_0 on the complex plane to determine the sequence of points whose escape properties are then investigated. However, the sequence z_n from the given starting point z_0 is not used to colour the pixel on which the point z_0 itself falls on. Instead, if the sequence z_n escapes within a certain preset iteration threshold, the colour of **every** point that the sequence visits along the way there is brightened a little bit. Starting from a black canvas and iterating through millions of sequences from randomly chosen starting points in this manner, the canvas slowly brightens up to display a figure of a meditating Buddha figure to bring out the energetic *ahhs* and *ohhs* of the mystically inspired audience of bohemians and similar sophistos. To make the Buddha figure sit up properly, the rendered image is canonically rotated 90 degrees counterclockwise; this is achieved simplest by mirroring and interchanging the roles of the real and imaginary axes of the complex plane.

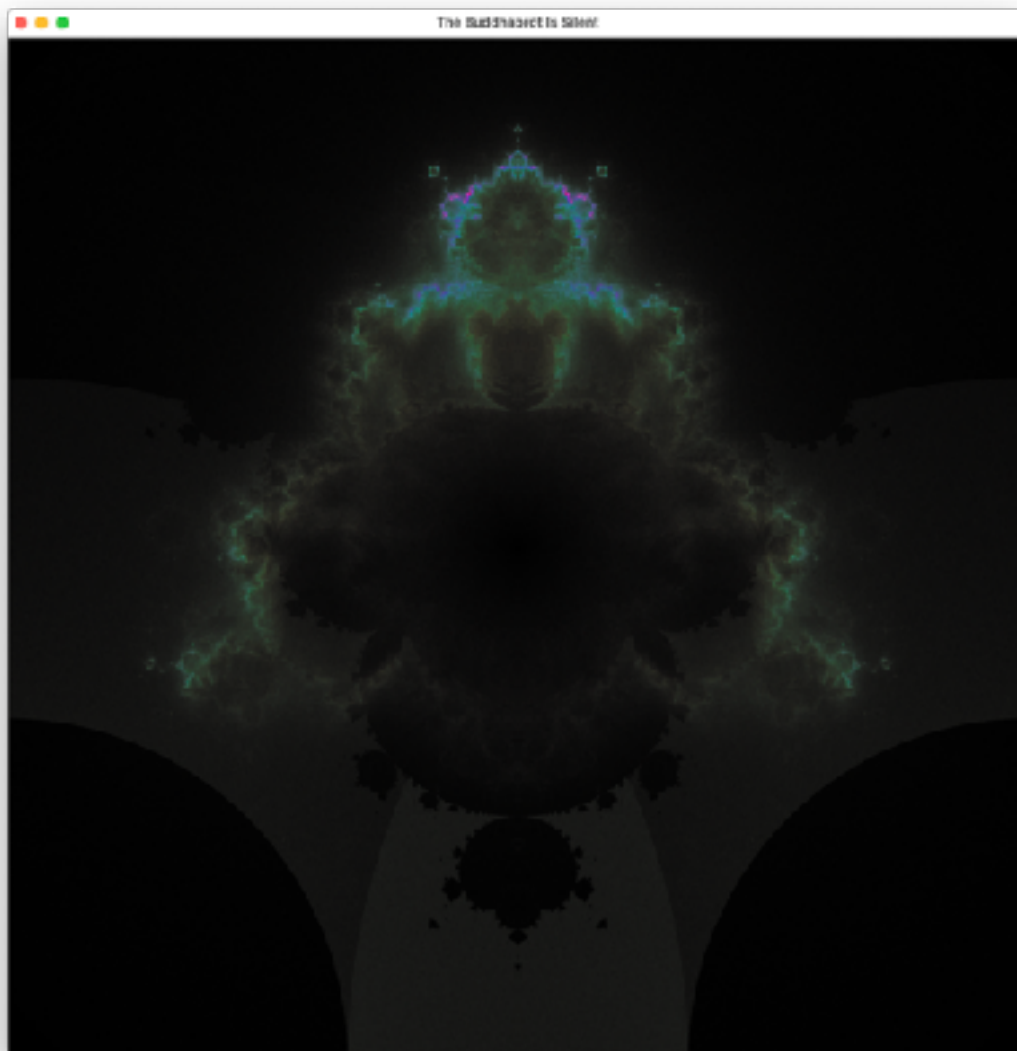
To make the fine structure of this fractal pop out, a simple **false colour technique** uses the length of the sequence z_n to determine which of the three colour channels is brightened for each pixel visited during that sequence. Instead of the classic red, green and blue colour channels, a false colour image can also be generated by using the hue, saturation and brightness of the HSB colour model to choose these colours. Using the modified **quartic** sequence formula $z_{n+1} = z_n^4 - z_0$ with the threshold limits set to 50 for the hue, 500 for the brightness, and 5000 for the saturation of the HSB colour of the pixel in the main class [BuddhabrotMain.java](#) along with the instructor's private model solution for a hundred million random point sequence SAMPLES produces the screenshot seen below.

In this last bonus lab to end this collection, the reader is invited to take seriously what gods made fun, by rendering your own variation of the Buddhabrot fractal using some artistic false colour scheme and sequence update rule freely chosen by you. Start by ensuring that you have the class `Complex` written in the earlier Newton-Raphson fractal lab to perform the necessary complex

plane iterations. In your project folder, create a new class named `Buddhabrot`, and in there the static method

```
public static BufferedImage renderBuddhabrot(  
    int size, long samples, int[] thresholds  
)
```

to render and return some kind of Buddhabrot image for the given parameters. This method should maintain a three-dimensional array of integer counters of dimensions of `[3][size][size]`, that is, one two-dimensional counter array for each of the three colour channels, to keep count of how many points have visited each pixel on the two-dimensional grid during the iteration of the complex point sequences. Remember to increment the visit count for each pixel only for those sequences that actually escape within the threshold of steps for that colour channel, as given in the parameter array `thresholds`. Once you have iterated a total of `samples` randomly initiated sequences, convert the visit count of each pixel for each colour component into a shade of that colour as a percentage of the visit count of that pixel relative to the visit count of the pixel visited the largest number of times. You might want to stretch these proportions with some nonlinear transformation to bring out the fine details in the strands.



Because of the heavy computational demands of rendering the Buddhabrot, many algorithmic optimizations will be their weight in gold in speeding up the computation. Before starting to code their solution, the reader might want to check out the series “[The Buddhabrot](#)” by David Aramant that discusses many of these techniques. Most importantly, since we are interested only in points z_0 from which the sequence visits a large number of points before escaping, it would be useful to sample the random starting points mostly from these areas of the complex plane. If the sequence formula is symmetric, the resulting Mandelbrot and Buddhabrot fractals will also be symmetric, producing the same result with a smaller number of random starting points. The article “[Rendering the Buddhabrot at 4K](#)” by Benedikt Bitterli picks up this ball and carries it even further past the goal post by animating the transitions between different slices into the same multidimensional complex.

To ensure that your code works correctly, you should initially test your computation with the ordinary Mandelbrot set sequence rule. Once you have crossed this motivational hurdle, you can plug in different update rules and colour schemes to explore the infinite possibilities of this wonderful family of fractals. Try to find some interesting variation that no human has cast their eyes upon before you, as if you were some modern day Magellan discovering a new terrain in the infinite space of possibilities. Modify the settings in the `main` method of the class [BuddhabrotMain.java](#) appropriately so that the final rendering of your artistic image has the `SIZE` of at least 1000-by-1000 pixels, and its contents are rendered with at least one hundred million `SAMPLES`. Make sure to include your rendered artwork in its glorious final form in the project folder that you submit to the instructor at the end of the course.

Bonus Lab 101: Remain, Remain, Remainder

JUnit: [RepeatingDecimalsTest.java](#)

As explained on the Wikipedia page "[Repeating Decimals](#)", decimal expansions of even seemingly simple rational numbers are infinitely long, except in those cases where the prime factors of the denominator are only 2 and 5, the prime factors of the familiar base 10 that we write out all decimal numbers. However, this infinite decimal expansion must eventually become **periodic** for every rational number; after some initial part, the infinite sequence of decimal digits will keep repeating the same finite block of decimal digits. For **irrational numbers**, such as π and the square root of two, the decimal expansion never becomes periodic.

For example, see the decimal expansion of $1/6$ equals $0.1666\dots$ with the initial part equal to 0.1 and the block of repeating decimals equal to 6. The initial part and the block of repeating decimals can each be arbitrarily long, since any such sequence of decimal digits could always be converted back to a rational number with an integer numerator and denominator. For example, the decimal expansion of $631/1665$ has the initial part 0.3, followed by the block of repeating digits 789.

In this bonus lab, you will write a Java method to extract the initial part and the block of repeating decimals from the given integer fraction a / b that satisfies $0 < a < b$. Since the block of repeating decimals of the decimal expansion may consist of up to $b - 1$ digits, this method cannot be implemented using the primitive integer and floating point arithmetic types of the Java language, since the Procrustean bed limited precision of these elementary data types will quickly run out even for some deceptively short and simple fractions. For example, the block of repeating decimals in the decimal expansion of $1/29$ consists of a block of 28 digits, far beyond the capabilities of these primitive data types. We must instead toil and moil through the drudgery of the **long division** algorithm to construct the result one digit at the time, even if doing so will give the reader elementary school flashbacks in the style of Norman Rockwell. Furthermore, we must store the result above the line using a `StringBuilder`, since the `StringBuilder` type will have no problem keeping track of the result of the long division even if it were to grow to millions of digits.

In your project folder, create a new class `RepeatingDecimals`, and there one method

```
public static String[] decimals(int a, int b)
```

Given the **numerator** and the **denominator** of the integer fraction a/b so that $0 < a < b$, this method must construct and return exactly two strings, given as one array. The first of these two strings will be the initial part of the decimal expansion of the fraction, and the second will be the repeating part. For example, given $a = 1$ and $b = 14$, the initial part of the result would be `"0.0"`, and the repeating part would be `"714285"`. The block of repeating decimals must be presented using the minimal period length, so the answer `"714285714285"`, though repeating just the same, would not be acceptable and sneak past the automated JUnit tester.

Your method should first declare a local variable `result` of the type `StringBuilder`, initialized to the empty string. Use this `result` variable to keep track of the digits that end up above the line

in the long division a/b , patiently appending these digits to the `result` one at the time as the long division process reveals these digits to you. Don't bother writing the decimal point character in the `result` quite yet, since we will put it there in the end once the `result` gets split into the initial and the repeating part of the decimal expansion of that fraction. Next, declare another local variable `int[ ] seen` whose length equals $10 \times b$ elements. You will use this array to keep track of the positions in the result whose digit computation each remainder was previously seen during the long division algorithm. Initially no remainders have been seen, so each element of this `seen` array starts as zero.

After these preliminaries, you should perform the cake walk of the familiar steps of the long division of a / b to generate the digits of the result above the line exactly one digit at the time, where a is your current remainder being divided by the fixed denominator parameter b . Should a become 0 at some point during the long division algorithm, you can safely conclude that the decimal expansion of the original fraction terminates at that position. The `result` accumulated so far becomes the initial part of the answer (remember to now put the decimal point in its correct place), and the block of repeating decimals of the terminating fraction is simply "0". For example, given the fraction $3/8$, the method should return the initial part "0.375" and the repeating part "0". (Of the two possible equivalent decimal expansions 0.375 and 0.374999... for this fraction, of course your method must return the terminating one.)

Your second chance to jump out of this carnival ride, needed in all those cases where the decimal expansion of the original fraction is nonterminating, arrives when your current remainder, stored in the variable a , has been previously spotted at some position after the first position during the long division. This is indicated by the value of the corresponding element `seen[ a ]` being nonzero. You may then stop and extract the initial part and the repeating part of the decimal expansion from the `result` that you have carefully constructed on the way leading up to this crescendo. The block of repeating decimals will start from the position `seen[ a ]`, everything before that position ending up as the initial part. (Again, you need to insert the decimal point in the correct position of the initial part.) The iron hand of the **pigeonhole principle** guarantees that some remainder must be the first one to repeat itself, which then allows you to terminate the construction of the decimal expansion of the original fraction.

In all remaining cases where your current remainder a has not yet shown its mug during the long division, update the element `seen[ a ]` to remember the current position of the digit being appended to the `result`. Then, append the single digit that is the result of the integer division a/b to the `result` (whenever $a < b$, this digit will trivially be zero), and update the new value of the remainder a for the next round to equal $10 * (a \% b)$. This simple approach will then generate the quotient above the line exactly one digit at the time.

Bonus Lab 102: Varshamov–Tenengolts Code

JUnit: [VarshamovTenengoltsTest.java](#)

“Programming isn’t so much about conveying to machine instructions for how to do a thing. It’s more about building clear mental models of the thing. That’s the hard part. Writing the code and running it can be a tool to shape your thinking.”

— François Chollet

This [recent gem from the Code Golf Stack Exchange](#) is not only an interesting problem in its own right, but the clever encoding scheme to solve a seemingly impossible problem feels almost magical in how nimbly it steps through the minefield of possibilities and manages to land on the correct square every time. Along with its illustrated techniques of bitwise arithmetic and building up sequences, this problem not only makes for a wonderful exercise for this course, but also illustrates the principles behind similar phenomena of efficient **coding** of messages for transmission over unreliable communication channels.

In this problem, [Alice](#) has a **message**, a **bit string** that she wishes to communicate to Bob across a channel controlled by their mutual **adversary** Mallory who would prefer this communication not to succeed. To throw a wrench in the works, Mallory is limited to dropping exactly one individual bit of her choosing from the communicated message, leaving no indication of the position where that bit was dropped, nor the original value of that bit. For example, if the message sent by Alice is 011010, Mallory must turn this into any one of the messages 11010, 01010, 01110, 01100 and 01101, depending on which bit she tactically chooses to drop.

The problem of **deletion-resistant encoding** is to come up with an agreed **encoding scheme** for Alice and Bob that allows Alice to add extra information bits to her message sufficient to guarantee that regardless of which bit of the encoded message Mallory chooses to drop, Bob can always reconstruct the original message from the surviving bits that got through! The reader can take a minute to think about this problem, and various potential ways to implement a Mallory-proof encoding that Alice and Bob can share. To thwart certain Byzantine schemes agreed between Alice and Bob, Mallory gets to see the entire message before she has to commit to the position she drops the bit from. You should also note that it makes no difference which bit inside a **run** of equal bits gets dropped, since all bits inside the same run are indistinguishable anyway. For example, Mallory can turn the single-run message 11111 only into 1111, the choice of which one of the five equal consecutive bits she drops making no difference to anything.

Explained in the paper “[Tutorial on algebraic deletion correction codes](#)” and elsewhere, the **systematic Varshamov–Tenengolts encoding** is a simplified variation of an encoding scheme conjectured to be optimal for this problem in the sense of adding the smallest possible number of bits to the original message to still render Mallory powerless to ever prevent Bob from reconstructing Alice’s original message. Before we look at this scheme and its implementation, we should note something about this implementation for the purposes that is of high educational value for this course. Unlike our usual convention to count the positions inside the sequence starting from zero, the encoding and decoding algorithms look much prettier when expressed with **one-based counting** of sequence positions, as if we were mathematicians instead of computer scientists. The **binary heap**

priority queue data structure is a more famous example of the same phenomenon. (Somebody really needs to gather up all these, and distill out the thing that those problems have fundamentally in common.) Since the Java position indices always start from zero, you need to adjust your code around this fact to behave as if the position counting had been one-based.

Another issue of educational value in this problem is the need for the genuine **modulus** operator, available in the Java standard library as the utility function `Math.floorMod`. Despite the common confusion about this issue, the modulus operator is not the same thing as the **remainder** operator, denoted by the percent symbol in Java, once the operands are negative. For example, the expression `-2%5` gives `-2`, whereas the expression `Math.floorMod(-2,5)` gives `3`. The modulus is never negative regardless of the signs of its operands, since the modular values repeat **cyclically** around zero in a **clock face** fashion, instead of being **reflected** around it. (Sometimes people say “modulo” when they really mean “remainder” because they want to sound all fancy and edumacated. The reader who is now aware of the difference will from now on surely resist such urges.)

In your project folder, create a new class named `VarshamovTenengolts`, in which you will write the following two methods. The first method expands Alice’s message for sending over the channel, and the second decodes the message received by Bob that is known to miss a bit somewhere.

```
public static String encode(String original)
public static String decode(String received)
```

The first step of encoding the original message is to fill a new `StringBuilder` instance with its zeros and ones, except that the (one-based) positions that are powers of two should be left unfilled for a moment. Marking such unfilled positions with asterisks as in the original Code Golf presentation of this problem, an example original message “01001” turns into the expanded message “**0*100*1”, the one-based positions 1, 2, 4 and 8 having been filled with asterisks. (Note the potential edge case bug lurking in that the modified message must not have a redundant asterisk at the end, but must stop at the last bit of the original message.)

Next, add up the (again one-based) positions where this expanded message contains a 1. For this expanded message “**0*100*1”, these positions would be 5 and 9, so the sum of these positions adds up to $5 + 9 = 14$. Take the negation of this sum modulo the length of the modified message plus one to compute the **checksum** for the expanded message. For example, $-14 \bmod (9 + 1)$ gives the checksum of 6 for this expanded message “**0*100*1”.

Next, use the bits of this checksum to fill in the asterisks of the modified message. You must perform this from left to right so that the least significant bit of the checksum lands on the first asterisk, followed by the more significant bits. Since the checksum 6 of this expanded message is 110_2 in binary, the expanded message becomes “010110001”. The expanded message that consists of zeros and ones only is now hot off the oven to be sent over the communication channel. Mallory to remove any one of its bits now, and we won’t care which one she does.

The decoding process that allows Bob to reconstruct the original message begins by computing the **Hamming weight** and the checksum of the received message. The Hamming weight is simply the

sum of all the ones in the message. The checksum is computed the exact same way as in the encoding process, except that the sum of positions that contain a 1 is computed modulo the length of the original message plus one, that is, `received.length()+2`.

For example, if Mallory turns the previous message "010110001" into "01011001" by dropping either one of its last zeros, the weight of the `received` message is 4. The sum of the positions that contain a one bit is $2 + 4 + 5 + 8 = 19$, from which $-19 \bmod 10$ gives the checksum of 1.

To detect the missing bit and its position, Bob has to simply compare the weight and checksum. If the checksum is less than or equal to the weight, this reveals that Mallory must have dropped a bit whose value is zero. This dropped zero bit can only be at a position where the number of ones to its right is given by the checksum. For example, since $1 < 4$, Mallory must have removed one of the zeros in the run of zeros before the last 1 in the `received` message.

On the other hand, had the checksum of the `received` message been strictly larger than its weight, this would reveal that Mallory dropped some bit whose value was one. As the math done by people well above this humble author's pay grade reveals, this dropped one bit can only have been in a position that has precisely $(checksum - weight - 1)$ zeros to its left. For example, should Mallory instead decided to turn the expanded message "010110001" into "01010001" by dropping either digit from the second run of ones, the weight of the `received` message is 3, and its new checksum equals $-(2+4+8) \bmod 10 = -14 \bmod 10 = 6$. Since $6 > 3$, Bob knows that Mallory dropped a one bit from the run that has exactly $6 - 3 - 1 = 2$ zero bits to its left.

Either way, Bob can now restore the missing bit to its rightful position in Alice's expanded message before Mallory got to stick her dirty hands on it. From this expanded message, Bob can leisurely extract the `original` message by skipping those one-based positions that are powers of two, and gathering the rest of the bits for his final answer to Regis.

(Don't ask me how any of this math works; really, it's all just magic. I, for one, welcome our new bitwise overlords. Like to remind them that as a trusted education personality, I can be helpful in rounding up others to toil in their underground silicon caves.)

Bonus Lab 103: In The End Was The Word

For testing: [ImageAsTextMain.java](#)

This bonus lab was inspired by a recent tweet that demonstrated the nifty image effect shown in the screenshot below, achieved by using nothing but standard CSS without any JavaScript. As all kinds of hidden things around us are slowly becoming Turing-complete and eventually AI-complete (“first gradually, then suddenly”) until we will eventually loop back to the beginning and have thousand demons to the left and ten thousand to the right of every individual human being, it might still be educational to do those same things the way that your ancestors back in the island of Java had to do them, uphill both ways rain or shine without complaining. More pragmatically, this lab showcases the [FontMetrics](#) utility class used by the Swing rendering engine to render text as graphics on the surface of a component.



In your labs folder, create a new class named `ImageAsText`. Inside that class, define the following static method to convert the given source image into a result image of equal size that contains only characters taken from the given text in a cyclic fashion. The colour that is used to render each character individually is the average of the pixel colours of the original image that reside inside the bounding box of that character. For simplicity, you can use linear averaging of red, green and blue, but artistic students can search for more accurate colour averaging formulas.

```
public static BufferedImage render(  
    BufferedImage source, Font font, String text, int skip  
)
```

To make these calculations correct even for fonts that are not monospace, the [FontMetrics](#) utility class provides the methods to perform these measurements on the fly. Once the `FontMetrics` object corresponding to the given font has been constructed (in a somewhat upside-down fashion, this is done with the method `getFontMetrics` defined in the `JComponent` class), its various methods can be used determine the pixel dimensions of text strings and individual characters ren-

dered in that font. The documentation of [FontMetrics](#) class explains what these methods are and how you should be using them. Make sure that your text rendering algorithm works even when the character glyphs in the font have greatly varying widths, such as with the beloved Papyrus font used in the left image of the above screenshot.

Your `render` method should render the characters of `text` individually to the image in inside the loop, iterating through the characters cyclically whenever it reaches the end. Start with the baseline point coordinates of $x = 0$ and y equal to the **ascent** of the font being used, that is, the distance from the baseline to the top of the tallest character of that font. (Also remember that in the coordinate system used in computer graphics, the y -coordinate increases downward.) Use the methods of `FontMetrics` to find out the dimensions of the box that the character would fill when rendered starting from that baseline point. Average the pixel colours inside the box to determine the colour of the character that you will render inside this box with the `drawString` method of the `Graphics` class. (Or the `Graphics2D` class from our present millennium, should you feel that rendering stuff with anti-aliasing turned on generally makes them look less jaggy and more appealing.)

After rendering the current character with the averaged colour, increment the baseline x -coordinate by the **advance** of that character. (Unless you are using a **monospace** font, you can't assume that all letters have the same width.) Should this advance move take the x -coordinate past the right edge of the image, bring the x -coordinate back to zero, and increment the y -coordinate by the amount of `skip` that was given as the last parameter to the method. Keep rendering the text characters in a cyclic fashion until your y -coordinate becomes greater than the height of the image plus `skip` to ensure that partial lines at the bottom edge also get rendered even if the descenders of the characters there reach below the image, after which the result can be returned to the caller fresh out of the oven.

Since each character is rendered individually, the algorithm cannot adjust for [kerning](#) or use [ligatures](#) that combine adjacent **graphemes** into a single **glyph** such as `fi` or `ſt` seen in more refined fonts more pleasing to the eye. Therefore, the reader could also implement a higher-resolution version of the previous scheme so that the text is first rendered into a blank result image one entire line at the time to allow the rendering engine to perform its magic, after which a loop through all pixels of the result image adjusts the colour of each pixel into the colour of that pixel in the original source image, adjusted by the brightness of that same pixel in the rendered text image. This operation ought to be implemented in its full *generalissimos* as a separate method to combine two images so that the second image acts as a translucent **stencil** that determines what parts of the first image show through in the result.

In[202]:

```
mosaic[img_] := Module[{pieces},
  pieces = ImagePartition[img, 20];
  ImageAssemble[Map[First[closest[ImageData[#], flags]] &, pieces, {2}]]
];
GraphicsGrid[{{mandrill, mosaic[mandrill]}}
```

Out[202]:



Another interesting version of this problem is to convert a grayscale image into a grid of characters so that all these characters are rendered using the same colour, but each character is chosen so that the shape of its glyph matches the underlying part of the image the optimal available way among all the available characters. This problem then becomes a rudimentary special case of the more general [photographic mosaic](#) problem.

The photographic mosaic problem would make for an exciting advanced exercise in Java programming, but the low level nature of the Java language would make this a far more thorny course project instead of mere single lab. At this point, the author can't resist the temptation of showcasing his old Wolfram solution for the photographic mosaic problem, essentially a one-liner, in the screenshot above. The list of `flags` was previously built from Mathematica's knowledge base to contain the flags of the nations of the Earth, but the same algorithm would work just as well for any set of image tiles. The `mandrill` is one of the free-to-use classic test images that, along with its more famous colleagues such as [Lena](#), are often used to demonstrate the behaviour of various image processing algorithms.

As they say in this industry, if you are not deeply ashamed when you look at the code that you wrote five years ago, you are not improving fast enough to stay abreast in the Red Queen's race that you have chosen to enter merely by reading this far. In that spirit, any Wolfram experts could surely further improve and optimize this solution. The first obvious simplification is to use a `With`-block instead of `Module`, to eliminate one line, and after that, inline the definition of `pieces` into the expression where it is used, to [turn the entire algorithm essentially into a tweet](#).

"Python, Java, R, Julia, Nim, Swift, Ruby, C#, ... I will never forget the look on their faces, all eight of them. Their faces dropped. All their clarity and power was drained right from their virtual machines. These languages had a reputation of breaking problems, but they knew that instant that they made a fatal mistake. Now youse can't leave."

— After *A Bronx Tale* (1993 film)

Bonus Lab 104: Beklemishev's Worms

JUnit: [BeklemishevWormTest.java](#)

♪ *Disk bound and down, loaded up and loopin'*
A-we gonna do what they say can't be done
We've got a long worm to build, and a short code to get there
I'm pointer bound, just watch ol' JVM run ♪

♪ *Keep your ops hard on the metal*
Son, never mind them breaks
Let it unroll out 'cause we got a worm to make
Hydras splitting in diagonals, computational arcana
And we'll bring it back no matter what it takes! ♪

This bonus lab is suitable only for students who have already implemented the Rope data structure from the lab 64, “Learning The Ropes I: Composition”, since the text strings in this problem will quickly grow to be so long that representing them as ordinary `String` objects of Java would soon run out of memory. (We don't need the comparison and hashing operations here since we are not storing these ropes into dynamic sets, so completing the second part of the ropes in lab 65 is not necessary for solving this fun little bonus lab.)

The Rope data structure shines especially bright next to the boring old `String` objects in situations where long text strings get produced by duplicating the same string a large number of times. Since the problem that this author first learned about in the post “[Beklemishev's Worms](#)” in the [Logic Matters blog](#) happens to do that very thing in spades, we can put our ropes to good use in this bonus lab to solve this offbeat problem. As explained in the linked post, Beklemishev's worms are sequences of digit strings that start from some given initial worm w_0 . Each worm is some string of digits, for example “2031”. Each step in the sequence applies the following deceptively simple rule to the current worm w_{m-1} to compute the next worm w_m in the sequence. The last digit of the current worm w_{m-1} is called its **head** that determines which one of the two possible ways is used to compute the next worm w_m in the sequence.

If the head of the current worm is 0, the next worm w_m is otherwise the same as the current worm w_{m-1} , except with its zero head chopped off. For example, the current worm “12340” turns into the next worm “1234”, regardless of the value of m .

If the head of the current worm is nonzero, find the last position k in the worm w_m that contains a digit that is strictly smaller than the head. For the example starting worm “2031” using zero-based indexing as usual, this gives $k=1$ for the position 1 that contains the digit zero. (For worms such as “432” whose body does not contain any digits smaller than their head, use $k = -1$.) Decrement the head of the current worm to create the **base** for the new worm; for this example worm w_0 , the base would be “2030”. Then, construct the new worm w_m by appending m identical copies of the base after position k . In this example, given that $m=1$, appending one copy of the part “30” of the base to the base itself produces the new worm “203030”.

Continuing the sequence the same way, the next worm w_2 is just "20303". Constructing w_3 from this worm, the new base is "20302" followed by three copies of the part "2" after the position 3 that contains the last digit that is less than the head 2, giving us "20302222". And so on.

As you can see, the operations of substring slicing and concatenation are both put in heavy use in this problem. However, since the simple ropes that we created in the lab 64 were not quite industrial strength in that they would automatically **rebalance** themselves once their shape gets too lopsided, we have to first implement a more efficient technique for repeated concatenation of the given rope with itself some large number of times, perhaps thousands or even millions of times.

In your labs project, create a new class named `BeklemishevWorm`. Before we implement the method to construct the next worm from the current one, write the above utility method

```
public static Rope repeat(Rope rope, int n)
```

that returns a new `Rope` that consists of the given `rope` concatenated with itself n times. When n is in the millions, it would be rather inefficient in both time and space to achieve this by concatenating `rope` to the current accumulation $n - 1$ times, thus creating $n - 1$ separate `Rope` objects in the heap. We will rather use the idea of binary power and realize that repeating the given `rope` n times is equivalent to first repeating that rope $n/2$ times (rounding down), and concatenating the result of that operation once with itself. When n is odd, you also need to concatenate that result once with the original `rope`. The base cases of this recursion are $n = 0$ when the method returns the empty rope, and $n = 1$ when the method just returns the original `rope` as is.

For example, to repeat "foo" four times, first recursively create a rope to represent "foofoo" of repeating it twice, and concatenate that rope with itself to get the desired "foofoofoofoo". This way, the number of new `Rope` objects created in this process remains **logarithmic** with respect to n , instead of ending up being **linear**. Repeating that original rope a hundred million times in such **repeated halving** fashion would create only 27 additional `Rope` objects, instead of a hundred million (minus one) vomited out by the blunt approach that keeps concatenating the original rope with the so far accumulated result one step at the time. Your Java Virtual Machine will eternally thank you for being this considerate.

Even better, you could even define an entire subclass `RepeatRope` to express such repetition as a single object whose methods `length` and `charAt` would compute that information from the corresponding methods of the child rope that is being repeated. All roads lead to Rome.

Having written and tested that method, we can finally grab this hydra by its heads and write the method that does all the heavy lifting in this lab:

```
public static final Rope wormStep(Rope worm, int m)
```

This method receives the current worm along with the index m of the next worm, and creates and returns the next worm in the sequence. With that one out of the way, we can write the method

```
public static int iterateWorm(String body, int rounds)
```

that iterates the sequence of worms from the body of the initial worm given as a `String`, for the given number of rounds. To keep testing simple, the method should return only the length of the final worm of the sequence. Once your method gets all the test cases right, I am willing to believe that you constructed all the worms correctly along the way.

As budding computer scientists, we should always appreciate some simple rule creating utter unpredictable chaos for us to rein in with code. This particular problem also turns out to be extremely weird from a logical perspective in that it has been proven that for any body of initial worm, this process will eventually terminate in that the worm becomes empty, which will happen once its body will contain only zeros. However, before reaching the goal, the length of the worm may reach some pretty dizzying heights during this iteration: the number of steps could easily be larger than a tower of exponentials reaching all the way to the Sun and beyond! The same will happen even if worms are generalized to lists of natural numbers, instead of a string of digits in some fixed base such as ten. However, this claim cannot be proven from the first order [Peano axioms of arithmetic](#) for natural numbers and their properties, for the reason that this claim turns out to be equivalent to the claim that the Peano axioms themselves are **consistent**, which by [Gödel's second incompleteness theorem](#) is unprovable from these axioms.

Generalizing worms into lists of natural numbers also allows us to define a function `Worm` so that `Worm(n)` equals the number of rounds required to reach the empty worm when starting from the initial single-digit worm `[ n ]`. The `Worm` function grows so unimaginably fast as n increases that it is impossible to even begin to describe its growth using only the terminology that would be relevant at this stage of your computer science studies. Interested readers can consult the [Googology Wiki](#) for more information about [Worm](#) and [many other similar otherworldly](#) (yet well-defined and finite for any finite n) fast-growing functions.

(This same Wiki can also hand little kids [the equivalents of MOABs and Tsar Bombas](#) in their contests of who can come up with the biggest number. Besides, wacky numbers such as “[gongulus](#)”, “[kungulus](#)” and “[meameamealokkapoowa oompa](#)” are also fun to say, even if the other kids in the playground would likely react to such numbers with predictable skepticism.)

Bonus Lab 105: Regex Fractals

For testing: [RegexFractalMain.java](#)

[Regular expressions](#) are a simple yet highly effective mechanism for pattern matching inside text strings. Regexes make many everyday text processing and analysis operations far easier compared to implementing these operations from scratch as loops and conditions.

In the Chang School computer science curriculum, regular expressions are covered in the course *CCPS 393 Introduction to Unix and C*, but their sheer universality makes them worth looking into also in our course. Once you have learned the idea and syntax that makes regular expressions tick and how they define the pattern that the regex engine searches inside the given string, this knowledge immediately translates across all languages and computing systems that you will ever encounter during your career. If we place our bets about the far future in accordance with [Lindy's law](#), regular expressions will be in heavy use long after today's freshman computer science students will one day receive their retirement gold watches from humanity's new robotic AI overlords.

Java offers a powerful regular expression matching engine in the package [java.util.regex](#), where the class [Pattern](#) defines a mechanism to compile regular expressions given as ordinary `String` objects into efficient state machines. The class [Matcher](#) then allows iteration through the positions of the text that contain a match for that regular expression. Just to keep things simple, the Java `String` objects these days come with a method to find out whether that string matches the regular expression somewhere as a simple yes/no answer, enough for most of the common use cases of regular expressions. However, if the same regular expression is repeatedly matched across a large number of strings, compiling it first into a `Pattern` eliminates the redundant effort to convert the regular expression into the same state machine every time.

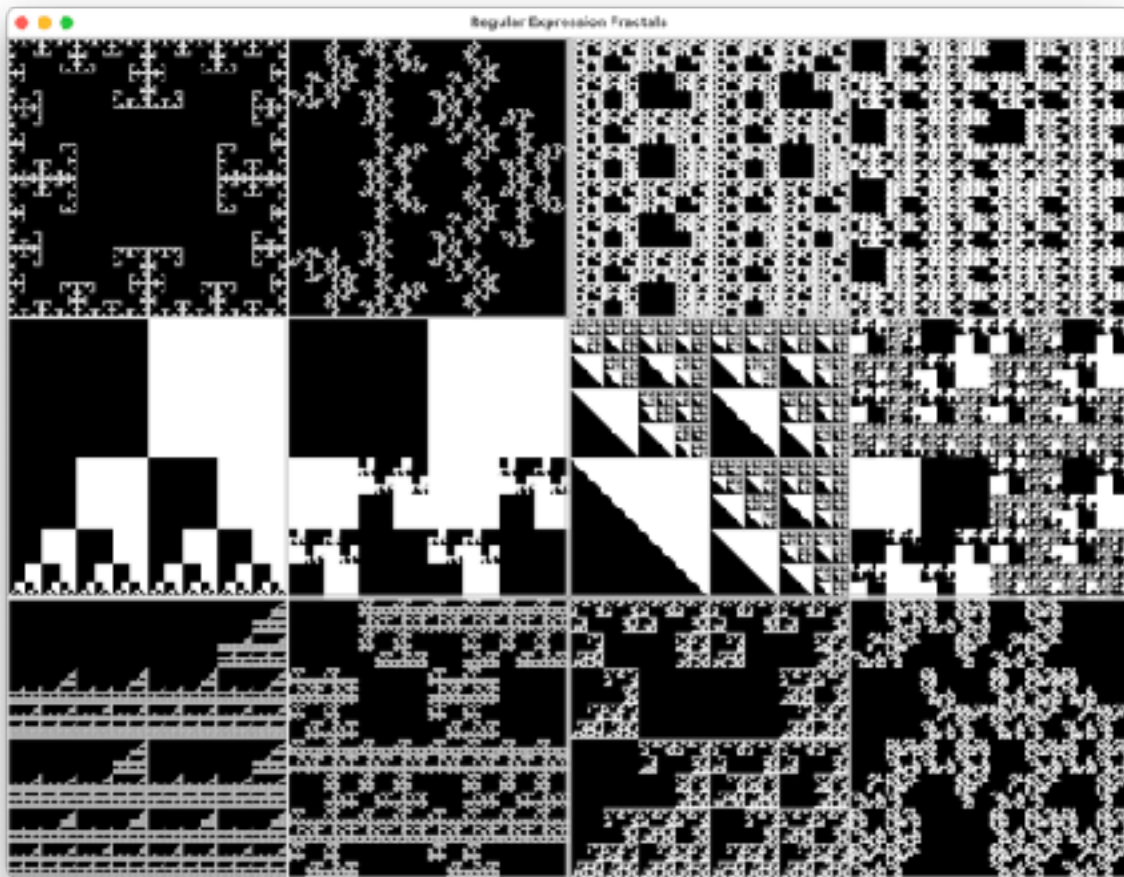
This lab uses regular expressions not to perform boring old text analysis and matching, but to render trippy fractals over a square grid of pixels, based on [a problem posted in r/dailyprogrammer](#). Converting regular expressions into fractal images sounds pretty far out lateral thinking, but actually turns out to be a surprisingly simple recursive process.

We normally use x - and y -coordinates to identify the position of each pixel inside a raster image. However, it is sometimes more fun to iterate through the pixels in a more whimsical order. If the image is a square whose side length 2^k is some power of two, that image can be recursively divided into quadrants. Let us therefore number these quadrants from 1 to 4 so that, say, the northeast quadrant is the number 1, the rest of the numbers given in counterclockwise order from there. Each quadrant is then divided into four sub-quadrants, which are then divided into four sub-sub-quadrants, and so on, until the area of a sub^{k-1} -quadrant covers exactly one pixel and no further subdivision is possible. In this scheme, the coordinates of each pixel are a string of k digits, each digit from 1 to 4. For example, if $k = 3$, the pixel 423 would be the southeast quadrant's northwest sub-quadrant's southwest sub-sub-quadrant. No pixel will escape our gaze.

In your lab project folder, create a new class named `RegexFractal`, and there the method

```
public static BufferedImage render(
    String regex, final int n, boolean rotate
)
```

that creates and returns an n -by- n pixel image whose each pixel is coloured either black or white according to whether the given `regex` accepts the digit coordinate string of that pixel; white for rejection, black for acceptance. (As in the famous vase illusion, which one is the foreground and the background is mere convention.) The screenshot below given by the `RegexFractalMain` class showcases some 256-by-256 pixel images of some regular expressions that were tactically chosen to produce pretty results when rendered in this manner.



If the additional parameter `rotate` is set to `true`, the rule for locating the x - and y -coordinates of the pixel from its digit coordinate string is slightly modified so that the numbering of the quadrants is rotated one step clockwise at every level of subdivision. For example, in the first quadrants, the southeast sub-quadrant is the number one, the remaining sub-quadrants numbered in clockwise order. After four levels of subdivision, the numbering is back to original. For each of the six regular expressions hardcoded into the `RegexFractalMain` demonstration class, the result without the rotation is displayed alongside the result of using this rotation.

Bonus Lab 106: Automatic For The Peephole

For testing: [DualTest.java](#)

“An ox might worry about a flea of ordinary size—a small creature of the first order of smallness. But he would probably not trouble himself about a flea’s flea; being of the second order of smallness, it would be negligible. Even a gross of fleas’ fleas would not be of much account to the ox.”

— Silvanus P. Thompson, “Calculus Made Easy”

Popular programming languages that most computer programmers are familiar with treat variables as boxes that store scalar values. In the languages that they actually use, all expressions are **eagerly evaluated** so that, for example, the assignment `a=b+c`; can be performed at all only if variables `a`, `b` and `c` have already been defined and have well-defined scalar values. When this assignment statement is executed, the new value of the variable `a` is set to equal the result of fully evaluating the expression on the right hand side. At no point during its lifetime can any variable be without a value, filling precisely the box stored in the computer memory, no more and no less.

However, actual high level **homoiconic languages** such as Lisp, Prolog and Wolfram allow variables to be bound to arbitrary **symbolic expressions** of that language, scalar values such as 42 being just a trivial special case of such **symbolic expressions**. Even if the symbols `b` and `c` appear the first time when the execution comes that assignment statement, the new value of the symbol `a` is then the symbolic expression `a+b`, since that is as far as it simplifies without any further knowledge of the values of `b` and `c`. Such expressions can of course at some point be simplified with various mechanistic **pattern matching rules**, but the important thing is that this simplification can be delayed until the symbols on the expression have themselves been associated to some expressions that then allow some pattern matching rule match and modify that entire expression.

In Wolfram Mathematica, the rules of calculus and other branches of mathematics have been baked in the language as pattern matching and substitution rules. The language can solve any integral or differential equation that you would find in an undergraduate math textbook, even if some of the variables have been left in the symbolic form. Sure, NumPy can invert a matrix, but can it do so when an element of the matrix is, say, $x^2 - a$ or some other symbolic expression? Irrationals such as π and even non-numbers such as ∞ behave inside these pattern matching rules just the same as any other symbols, their special semantic content in our minds giving them no special snowflake status in these calculations. For example, Euler’s famous identity $e^{\pi i} + 1$ equals exactly 0, and the infinite sum of negative powers of two equals exactly one, not one floating point precision iota more or less. (Let alone an infinite loop to the bottom, as would happen in Java.)

Symbolic pattern matching subsumes regular imperative programming as a special case, making the language more expressive than the [Blub programmers](#) at lower levels of abstraction can begin to conceptualize. The limits of my language mean the limits of my world. Sure, languages such as Python allow extensions such as SymPy that allow some variables to be handled in such a symbolic fashion. The question still remains why bother going through such a hassle only to turn your low-level language into a third rate imitation of Wolfram that is fully symbolic out of the box.

Down here in the Blubiverse, we will have to earn our daily bread with the sweat of our brow using floating point approximations that the demiurge blessed us with as our only tool to calculate mathematical quantities. This author does not hide his general distaste of floating point numbers and prefers airy symbolic abstractions whose end results can always be turned back into numerical approximations after the fact. Admittedly, same as with most ideologies devised by mortals, the ideologies themselves aren't quite as bad as their mindless followers make them to be. This author also immediately recalls a book on computing that started with an example that proudly used that one stupid formula with the floating point approximation of the golden ratio ϕ to "efficiently" compute Fibonacci numbers. (Hey, it worked great for me when $n = 10$.)

The practical value of floating point calculations is the same as the bright and garish colouring of all those poisonous animals, to immediately signal to others that nothing worthwhile is going to happen after digesting them. There is seldom any point reading or listening to the problems and solutions of floating point floozies who proudly write `(int) Math.pow(n, 2)` when they could have just said $n*n$. Sorry, life is just way too short for that. The reliability of this signal as an excuse to immediately stop reading is even better than seeing the word "input" used as a verb. (*"The program inputs the input that the user inputted."* So, nothing really wrong with that sentence? Let me count the ways.) Or, basically anything about math or computer science written on Microsoft Word. You will miss nothing valuable in your life or career by automatically skipping all such documents.

[But floating point is simply hard](#), and full of hidden deadly pitfalls behind every nook and cranny. The dark art of [numerical analysis](#) builds safe routes around the limitations of the pretence of using floating point numbers as if they were actual numbers that a naive person might blithely assume to satisfy rudimentary mathematical identities such as $(a + b) + c = a + (b + c)$. Clever algorithms keep the problems and their numerical results **well-conditioned**, but as they say, a turd can be polished up to only certain brightness and no brighter. When dealing with floating point numbers, we should just accept that they are not actually numbers, but something that we all pretend walks and quacks like some cartoon of a number, and even behaves as such in the short run provided that you just don't poke them too hard. Eventually the intermediate errors will accumulate so that any program that models some interesting system in our spiky and discontinuous reality will diverge catastrophically from the expected Platonic perfection. We wish to postpone that moment until we have extracted the useful numerical answer that we are seeking from that problem to begin with.

The modern technique of [automatic differentiation](#) helps maintain numerical stability in calculating **differentials** of functions, an operation that the big boys in **machine learning** need all the time to update their neural network weights with **backpropagation** as they updating the network weights in proportion to the differential to their local errors with the training data. In this lab, we implement a data type of [dual numbers](#) for this purpose. Dual numbers structurally resemble the **complex numbers** that we implemented back in the lab 73, "Newton-Raphson Fractals". Analogous to a complex number, a dual number consists of two real-valued components, the first component being the real value of that number. However, whereas the second part of a complex number is multiplied by i , that weird imaginary number ("Is that i now here in this room with us?") whose square equals minus one (and thus cannot itself be a real number, but must be something else altogether), the second component of a dual number is multiplied by a special **nilpotent** number called ϵ , a weird little feller that is greater than zero but whose square yet somehow equals... (*drum roll*)... zero!

Since the square of any nonzero real number is nonzero, ϵ cannot be a real number, but something else entirely. Compared to the good old infinitesimals such as dx that you encountered in your calculus classes, ϵ must surely feel just as weird as those infinitesimal numbers felt compared to ordinary real numbers; how can some number be greater than zero but less than any positive real number? This is where we have to throw out our intuitions of numbers as infinite continuums arranged on a finger-quotes “line”, and think at the level of formal algebraic equations without trying to wrap our brain around what those numbers, finger-quotes, “mean” and in what sense they, like, you know, “exist”, *maaan*. The lunkhead on the left end of the Bell curve of the midwit meme happily divides some formula by dx , as if dx really were a numerical quantity that allows such manipulations... yet the genius on the right edge of that curve does the same exact thing, safe in his knowledge that the theory of [hyperreal numbers](#) is mathematically just as sound as that of the grownup realies numbers! The midwit in the middle gnashes his teeth and insists *but muh limits* to achieve the same end results, just with way more work to finally get there.

Arithmetic operations of dual numbers are defined with formulas that make the first component of a dual number behave exactly as real numbers behave, with no ϵ ever appearing in these formulas. Therefore any computation that can be performed using real numbers, or more accurately (heh) with their floating point approximations in our imperfect fallen world, can be computed using dual numbers to give the same result in exact same steps, plus a constant bookkeeping factor. However, the formulas for the second component have been tactically chosen so that the coefficient ϵ in the second component equals the numerical differential of the formula that produced the real-valued result in the first component! These formulas work since ϵ disappears when squared, the same way as the square of dx disappears when compared to dx , no matter which real number you multiply it with. Viewed at the scale of real numbers, the square of dx might as well be zero, just like the square of ϵ . We get the same large scale results anyway for the standard (real) part of the differential.

Since any numerical computation can be expressed using only functions seen in the list below, performing that computation using dual numbers not only gives out that same result, but also the **numerical differential** of that result. Yes, Virginia, nature does not make jumps, and therefore **all numerical computer programs have automatic differentials**. Ladies, try not to slide off your chairs when you realize the implications of this fact and your eyes blaze with happiness resembling Vince McMahon in the last panel of that popular meme that he appears in. Once you have the data type for dual numbers in your language, you can write your numerical programs just like you did before, just using dual number data type everywhere that you previously used the type `double`. When you initialize a variable whose type is a dual number, initialize the second component of that variable to 1 if you want to compute the differential with respect to that variable. Initialize the second component of all other variables to 0. (Computing partial differentials of multiple variables would get a bit more, uh, convoluted.)

As a reward for you for reading this far through a continuous landscape that for us computer scientists feels as alien and hostile as the craters of the Moon, we can define and implement a Java data type to approximate dual numbers. In your project folder, create a new class

```
public final strictfp class Dual implements Comparable<Dual>
```

This class should have two `private double` fields `x` and `px` for the two components of the dual number; public accessor methods `getX` and `getPx` for those fields; no mutators of any kind, since we want this class to be **immutable**; constructors that can build a dual number with either one or two `double` parameters; the `toString` method; and then the bevy of mathematical functions

```
public Dual add(Dual other)
public Dual subtract(Dual other)
public Dual multiply(Dual other)
public Dual divide(Dual other)
public Dual sin()
public Dual cos()
public Dual exp()
public Dual log()
public Dual pow(double k)
public Dual abs()
```

similar in spirit to the methods in the example class [Fraction](#) seen in the lectures. The formulas for the first components of the resulting dual numbers are the same as the corresponding formulas for real numbers. Formulas for the second components are listed in the section “[Automatic differentiation using dual numbers](#)” of the Wikipedia page “[Automatic Differentiation](#)”. We also define this class to be `strictfp`, surely the least frequently used of all Java keywords in practice, to guarantee that floating point computations inside this class are done according to the [IEEE-754 standard](#) for floating point arithmetic. This keyword guarantees equal results for executing that same program in different machines. (Floating point coprocessors of off-the-shelf computers adhere to this standard anyway.) But even under this standard, $(0.1 + 0.2) + 0.3$ still does not equal $0.1 + (0.2 + 0.3)$, so two programmers who implement otherwise identical logic will get different results depending on the order they compute their intermediate results in.

All programs that deal with numerical quantities need to choose different execution paths based on comparing those quantities. Therefore, implement the `compareTo` and `equals` methods to look at only the first components of the numbers being compared. The second part of each `Dual` number is essentially just auxiliary measurement data gathered during execution, and it does not affect the execution in any way. Having completed all the above methods, write the following method to experience the thrill of getting to do your numerical computations using dual numbers:

```
public static final Dual hornerRule(Dual[] coefficients, Dual x)
```

Evaluate the polynomial with the given `coefficients` using the [Horner's rule](#) for best speed and numerical stability. Dual numbers are not a magic get-out-of-jail-free card for using inefficient and numerically unstable algorithms! The `coefficients` of the polynomial are encoded the same way as in the earlier labs 1 to 3 in the `Polynomial` class, so that the coefficient of the power x^i is given in `coefficients[i]`. The JUnit test class will create a bunch of random polynomials P along with their differential polynomials P' , and evaluate P with your method. The result of this evaluation is a dual number whose second part should be within the floating point precision from the value of the similarly evaluated value of the differential polynomial P' .

Bonus Lab 107: Inventory Sequence

JUnit: [InventorySequenceTest.java](#)

After all the highfalutin philosophical ramblings made in the previous bonus labs, here is a short and simple bonus lab problem inspired by yet another Numberphile video, this time “[A Number Sequence With Everything](#)” again featuring the mellow old grandpa Neil Sloane of the [On-Line Encyclopedia of Integer Sequences](#) fame. The reader should first watch this video to get an idea of what this self-describing **inventory sequence** is all about. The challenge of this lab is to generate the elements of this sequence efficiently in both time and space, so that your method can raze through billions of elements of this sequence like a hot chainsaw through butter in a mere couple of seconds.

The inventory sequence is generated in a loop that repeatedly counts up the natural numbers starting from zero. For the current number in your count, append the count of how many times that current number has been previously seen in this sequence, including its appearances in the current ascension, before moving on to the next number. However, every time that the count of how many times the current number has previously appeared in the sequence is zero, you must reset your natural numbers counter back to zero.

Consecutive ascensions colour-coded for your convenience, the inventory sequence starts with

0, 1, 1, 0, 2, 2, 2, 0, 3, 2, 4, 1, 1, 0, 4, 4, 4, 1, 4, 0, 5, 5, 4, 1, 6, 2, 1, 0, ...

whose second and fourth ascensions both illustrate how each ascension can aid itself along. For example, in the fourth ascension, since element 1 has at that point been already seen twice, appending that two makes the next value 2 to have been seen four times at this point in the sequence. However, even as these Sisyphean ascensions keep getting longer, they are all destined to eventually reach some previously unseen number, where they then fall back to zero to begin the next ascension.

In your lab project folder, create a new class named `InventorySequence`, and there one method

```
public static int[] inventorySequence(int start, int end)
```

that returns the elements of the inventory sequence from position `start` (inclusive) up to the position `end` (exclusive), where the position counting in the sequence starts from zero. The length of the integer array `result` returned by this method must therefore be exactly `end-start`, so you can create this `result` array as the first thing inside the method. Your method should still start generating elements from the first position to produce the expected elements correctly, but only return those from position `start` up to the position `end`.

To generate the next element of this sequence efficiently, your method should keep track of how many times each value has been previously seen in this sequence. A simple integer array of counters is the best technique to use here, seeing that the keys being counted are consecutive nonnegative integers. This counter array can start out small, since you can always expand it on the fly using the `Arrays.copyOf` utility method whenever large enough previously unseen elements make their

first appearances. Once your position count inside this sequence reaches the `start` position, start inserting your generated elements into the `result` array, so that once you reach the `end` position, your method can return the `result` array filled up with the most recent elements.

Bonus Lab 108: Rendering Piecewise Béziergons

For testing: [BeziergonMain.java](#)

Many curves that snake through the two-dimensional plane are easiest to define as **parametric curves** so that two functions $x(t)$ and $y(t)$ give the location of the point walking through the curve at time t . The time parameter t starts running from 0 and goes up to eleven, or whatever time the functions $x(t)$ and $y(t)$ come to the end point of that curve. The earlier lab 9 about rendering parametric Lissajous curves was a good illustration of this general technique.

The given continuous parametric curve can always be rendered piecewise by choosing a suitable discrete step size h , and calculating the coordinates $(x(0), y(0))$, $(x(h), y(h))$, $(x(2h), y(2h))$, ..., of these discrete points that the curve traipses through along its continuous journey. Provided that the step size h is small enough, these points can then be connected with just linear line segments or some higher-order piecewise curves. For both ease and simplicity of calculation, the parameter given to the parametric functions x and y is often scaled so that $h = 1$, so that these functions x and y only need to be evaluated at integer points.

The approximate rendering of the continuous curve is best done piecewise instead of trying to fit some giant polynomial of order $n - 1$ perfectly to these n sample points. The numerical instability of this approach along with the well-known [Runge's phenomenon](#) makes any such attempts, well, a *moot point* to begin with. In this bonus lab, we implement a method to render arbitrary parametric curves as [composite Bézier curves](#), made up piecewise with consecutive **cubic Bézier curves** that connect each two consecutive sample points $(x(t), y(t))$ and $(x(t + 1), y(t + 1))$ for integer values of t .

An individual cubic Bézier curve connects its starting point P_0 to its endpoint P_3 , and its shape along the way between these two points is determined by two additional control points P_1 and P_2 . When a long curve is rendered piecewise using cubic Bézier curves, we want to choose these control points P_1 and P_2 for each piecewise curve in a way to guarantee each consecutive piece to connect smoothly to the next piece, with no visual equivalents of stepping on a Lego piece at these connections. Defined in the standard library package [java.awt.geom](#), objects of type [CubicCurve2D.Double](#) represent shapes that are cubic Bézier curves, the coordinates of their control points expressed as primitive double values. The methods in this lab will also use the class [Point2D.Double](#) to represent individual points on the plane.

In your lab project folder, create a new class named `Beziergon`, where you will write two methods to perform the necessary calculations. The parametric curves that your methods will render are given as two function objects x and y of type [DoubleUnaryOperator](#), defined in the Java standard library package [java.util.function](#). The first method to write in your class is

```
public static List<Point2D.Double> computeCubicBezierPoints(  
    DoubleUnaryOperator x,  
    DoubleUnaryOperator y,  
    int n  
)
```

that gets as its arguments the functions x and y that define the parametric curve, and the number n of sample points that this function is supposed to go through. The functions x and y have been defined so that the parametric curve goes from its starting point $t = 0$ to its endpoint $t = n$. The piecewise curve that the next method will render will interpolate the points $(x(0), y(0))$, $(x(1), y(1))$, ..., $(x(n), y(n))$ on that curve, at the integer values of the parameter t .

This method should return some kind of `List<Point2D.Double>` collection object (most likely you will return an `ArrayList`) that contains a total of $3n + 1$ points. The zeroth point in this list is $(x(0), y(0))$, the third point is $(x(1), y(1))$, the sixth point is $(x(2), y(2))$, and so on. Between each pair of these exactly interpolated points, this method should add in the result list the two control points of the cubic Bézier curve that will be rendered between these two exactly interpolated endpoints.

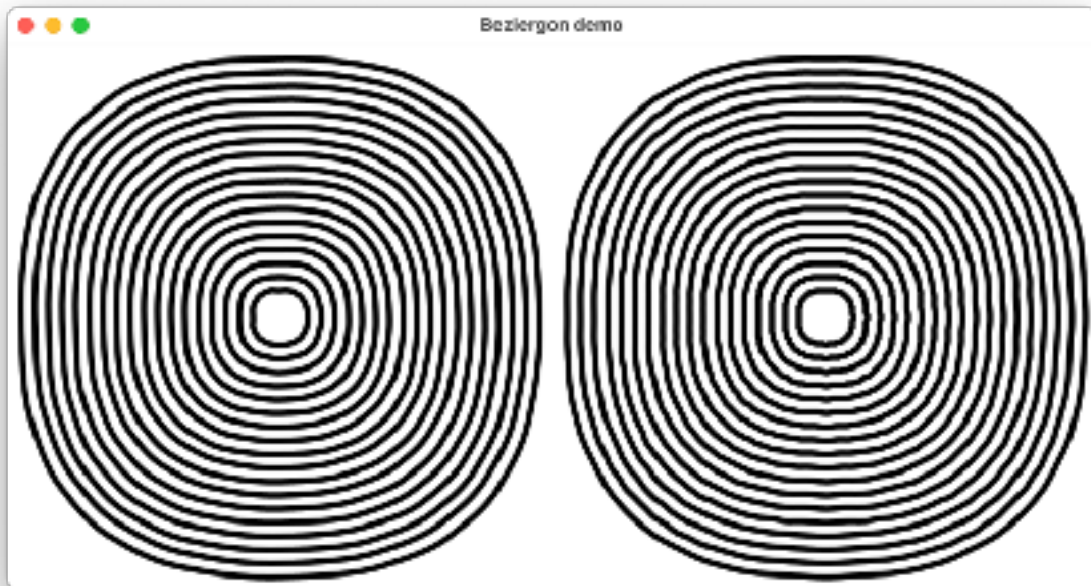
The second control point P_2 between each pair of the two exactly interpolated points $(x(t), y(t))$ and $(x(t + 1), y(t + 1))$ is given by the formula $(x(t + 2/3.0), y(t + 2/3.0))$. This second control point will therefore also lie exactly on the parametric curve. However, to ensure that all our piecewise cubic Bézier curves connect together smoothly, the first control point most emphatically is **not** given by the seemingly obvious formula $(x(t + 1/3.0), y(t + 1/3.0))$. Even though this would make both control points lie exactly on the curve, the piecewise curves would no longer connect together smoothly. (Since the cubic Bézier curve does not in general interpolate its two control points anyway, forcing this first control point to also lie exactly on the curve would be, heh, *pointless*.)

Instead, the first control point P_1 must be computed with a formula that guarantees the property denoted by the symbol C^1 , **velocity continuity** between the consecutive curve segments. The first control point after the point $(x(t), y(t))$ must lie in the mirror position relative to that $(x(t), y(t))$ when viewed from the second control point of the previous curve segment. In vector arithmetic, this continuity condition can be simply stated as $P_1 = P_0 + (P_0 - P_{-1}) = 2P_0 - P_{-1}$, where P_{-1} is the second control point P_2 of the previous curve segment. For the first curve segment that has no predecessor, you can use $(x(-1/3.0), y(-1/3.0))$ for that second control point. This will also guarantee that if your parametric curve is **closed** so that it ends exactly where it started, the curve segment that closes the loop will connect back smoothly to the first curve segment that started this whole mess.

Having completed the previous method, you can now write the method

```
public static Path2D.Double createCubicBezierCurve(
    List<Point2D.Double> points
)
```

Given the list of `points` computed from the previous method, this method should construct and return a `Path2D.Double` piecewise curve to the caller. This method should start by initializing a `result` object of type `Path2D.Double`, and use it to `moveTo` the first point in the list of `points`. After that, the method should `curveTo` to the next exactly interpolated point in the list of `points` three steps after the current point, using the two points listed between these two points in the list of `points` as the control points of the piecewise cubic Bézier curve.



The main method of the [BeziergonMain.java](#) class uses your two methods to interpolate a series of nested [superellipse](#) shapes, a family of geometric shapes beloved by this instructor. The expected result of using this main method when executing with your methods is shown in the above screenshot. The superellipse shapes on the left are rendered using twenty sample points per curve, whereas the shapes on the right are rendered using one hundred sample points per curve.

The obvious question why the shapes rendered on the left seem smoother to the eye despite having a smaller sample resolution of interpolated points is explained by the mechanism that was used to choose the two control points. Since the second control point P_2 of every curve segment was chosen to lie exactly on the parametric curve being rendered, and this parametric curve is convex, the first control point P_1 of the next curve segment will necessarily fall outside the curve, causing the wobbliness when the Bézier curve steers towards the second control point P_2 , and then corrects by steering back towards the endpoint P_3 of that curve segment to hit it exactly. Adding more sample points will not really help here, basically because circles, superellipses and other transcendental curves just aren't polynomials. (Eventually the curve segments would become smaller than one pixel anyway, turning the result into porridge of floating point mush.)

Such wobbliness is the necessary cost of doing business for each curve segment to connect to the next segment smoothly without discontinuities. Besides, this effect will inadvertently give these curves a pleasant and relaxed hand-drawn feel, in style of the famous [xkcd](#) webcomic. To eliminate both wobbliness and the curve segment discontinuities and still allow the rendered curve to interpolate every given sample point on the curve perfectly, more advanced curves such as [B-splines](#) or [NURBS](#) must be used as these piecewise curve segments.

Another issue illustrated with the superellipse curves is that choosing the sample points on regular intervals does not give the best possible results for curves where the speed of the curve, as computed from the differentials $x'(t)$ and $y'(t)$, varies greatly throughout the curve. Such curves are better sampled with an **adaptive subdivision** process that gives more precision in the tight parts of the curve, and uses a lower sampling resolution in the smooth sailing parts.

Bonus Lab 109: Soft Interruptions

For testing: [VeraMolnarMain.java](#)

In our age of miracles where neural networks judging the intermediate patterns created by the stable diffusion algorithm, these networks trained with a humongous corpus of both historical artworks and works of natural language, can generate artworks equal to and soon surpassing the talents of human masters, it is good to remember the pioneers of this field in a more simple and innocent time. Remember that these men and women had to create their artworks using computers and programming languages that, if the average coder of today accustomed to his compilers, IDE's and other tools were made to sit down to work with them, would probably run away crying for mommy.

This lab has you recreate two artworks of the late [Vera Molnár](#), one of the early pioneers of [generative art](#), an endeavour where the human artist first defines the framework of rules for the mindless machine to then follow. Aided by carefully controlled sprinkle of randomness that, same way as how we use fire, is used as a good servant instead of a bad master, the result will hopefully contain some sort of semantic content and ideas that were not present in the original code, thus raising all kinds of philosophical and other questions about the boundary between the artist and her tools.

Create a class named `VeraMolnar` where you will write two methods that create and return artworks to serve as imitations and tributes for [the life and career of Vera Molnár](#), since every budding computational artist, including yours truly, will end up unknowingly copying these forms anyway. The first method in this lab should generate a parameterized version of "[100 Trapèzes](#)", where the caller of the method gets to decide the `width` and the `height` of the image, and the number of `rows` and `columns` in the arrangement of its whimsical pieces. This method also receives the random number generator `rng` that it should make all its randomized decisions with. Your method should create the exact same result image every time that it is given the same random number generator. Of course, giving this method a different random number generator should then produce a different outcome, so that you can't cheat simply by preparing one nice image to return each time.

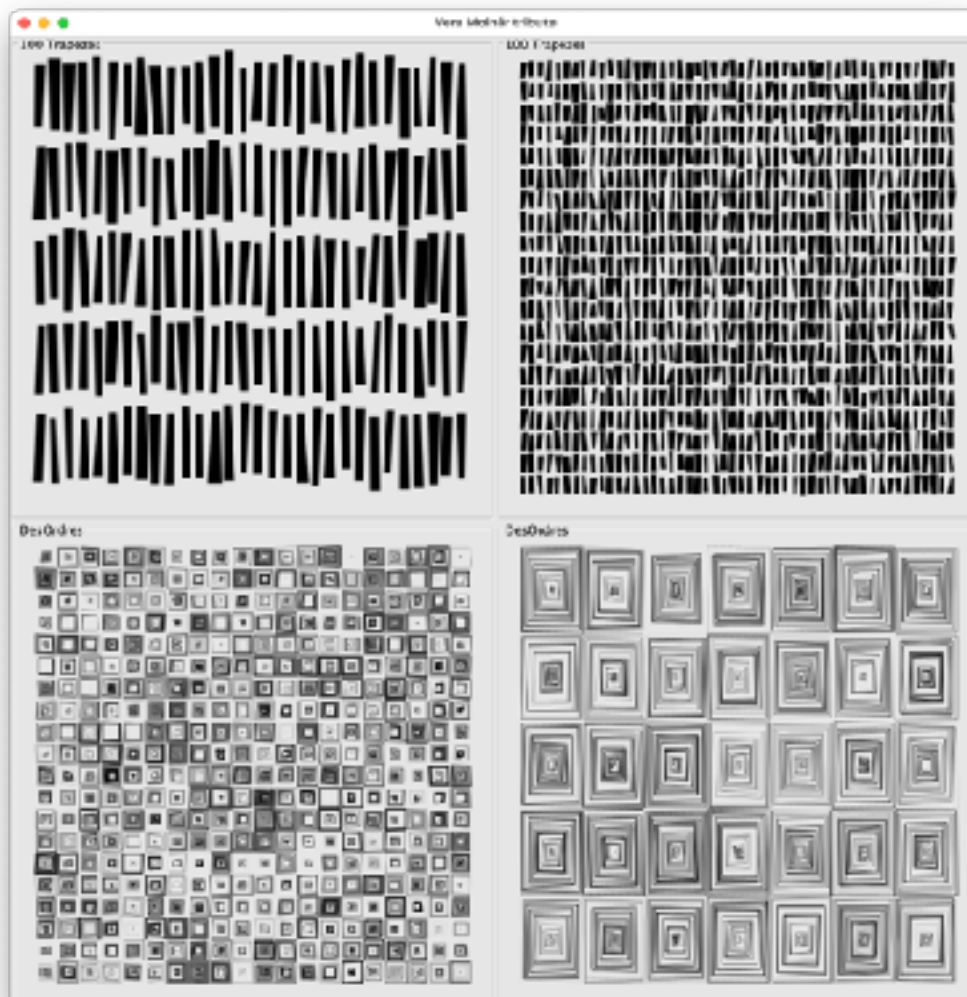
```
public static Image centTrapezes(  
    Random rng, int width, int height, int rows, int cols  
)
```

The example image at the top left of the screenshot below uses five rows and thirty columns, but your method should be prepared to handle any number of rows and columns. Your artwork should leave reasonable empty margins around the pieces (about five percent of the width and height seems good) in the `BufferedImage` that you create and return. Loop through the rows and columns and calculate the position of the four corners of each piece first assuming the piece to be a simple axis-aligned rectangle akin to the black keys of a piano, then randomly perturb each corner by a suitable amount. Create a [Path2D.Double](#) object to define the outline of the piece through these perturbed corners, and convert this path into an `Area` to be filled in the `Graphics2D` object connected to your result image. Of course, you should turn on **anti-aliasing**, instead of having your creation like some 8-bit eighties garbage. (That's also an aesthetic, yes, but not today, Satan.)

For aesthetic reasons, your method should also ensure that your pieces will never touch each other. How you guarantee this while allowing your pieces to lie close to each other is up to you. Come on, Vera managed to do that back in the year 1976 using only techniques available to her at the time, so surely you should be able to follow this feat with today's science fiction coding tools.

The second method in this lab should produce an imitation of "[\(Des\)Ordres](#)", actually an entire series of variations on a theme built on a shared idea and spirit, from which we imitate one piece that consists of nested rectangles drawn in grayscale. The parameters `width`, `height`, `rows` and `cols` play the exact same roles as they did in the previous method. The new parameter `n` determines how many layers of rectangles are drawn inside each rectangular onion, and the parameter `buzz` determines the extent that the corners of these rectangles are randomly perturbed. Use this parameter so that with the values of 0.3 for the bottom left and the 0.7 for the bottom right image, your artwork resembles those displayed in the screenshot below. Or, don't. WWVD?

```
public static Image desOrdres(
    Random rng, int width, int height,
    int rows, int cols, int n, double buzz
)
```



When rendering each layer inside the rectangular onion, flip a coin to decide whether that layer will be rendered at all. If the coin chooses to render that layer, choose a random grayscale value between 0 and 0.5, along with a random width for the `BasicStroke` to render that layer.

Bonus Lab 110: Twisty Little Passages, Reversible

For testing: [HuntAndKillMain.java](#)

[Jamis Buck](#), one of the forces behind the popular [Ruby on Rails](#) server side web application framework, has collected his writings on [maze generation](#) into an excellent little book "[Mazes for Programmers](#)" that inspired both this and the two following bonus labs.

Maze generation takes place in some **space** made up of **rooms**. The shape of the space that contains all these rooms determines which **neighbouring** rooms each room is **adjacent** to. Initially, each room is a **cell** that has been sealed off from all of its neighbours with walls. However, instead of being doomed to drink virtual amontillado, you can use your trusty pickaxe to dig a doorway between two adjacent rooms through the wall that separates them to make those rooms directly connected to each other.

Usually this space of rooms is a rectangular grid of integer coordinates so that each room (x, y) is adjacent to its four neighbours at the coordinates $(x + 1, y)$, $(x, y + 1)$, $(x - 1, y)$ and $(x, y - 1)$ to the four compass directions. However, most of these maze generation algorithms work for spaces and rooms that can be of any shape, requiring only that it is somehow possible to determine which rooms are adjacent to the given room, but needing no further information or assumptions about the spatial relationships that connect these rooms.

Mathematically, these rooms define a **graph** structure whose **nodes** are these rooms. Two nodes are connected by an **undirected edge** if and only if the corresponding rooms are adjacent to each other. Since a **maze** that allows exactly one path from any room to any other room is by definition some **spanning tree** of that graph, any graph algorithm that generates a spanning tree, either indirectly as a side effect of its execution (backtracking depth-first search, breadth-first search, Dijkstra's algorithm, A*), or directly (Kruskal's algorithm, Prim's algorithm, Borůvka's algorithm), can be used to construct a maze as a trivial side effect, especially if the decisions made by these algorithms during their execution are made randomly, for example by giving the edges of the graph random **weights**.

These classic graph algorithms will make more sense in a later course of graph theory and combinatorial algorithms, so we will not cover such algorithms here. Instead, in this first lab we will implement the fast and simple **hunt-and-kill algorithm** for maze generation, to be executed and rendered on the rectangular grid of rooms. Same as for other classic maze generation algorithms, Jamis Buck presents the hunt-and-kill algorithm in his blog post "[Maze Generation: Hunt-and-Kill algorithm](#)" that the reader might wish to check out before continuing with the lab.

The hunt-and-kill algorithm must somehow keep track of the rooms that it has already visited during the current maze generation. Initially, no rooms have been visited. The algorithm itself alternates between two stages "hunt" and "kill" that give this algorithm its name. For the initial trivial hunt that starts this algorithm, you can choose any room as this starting point, as it makes no difference which room you start the first killing stage. During each killing stage, look at all the neighbours of your current room. If at least one of these neighbours is still unvisited, choose one of these unvisited neighbours randomly. Dig a doorway from your current room to that previously unvisited neighbour, and make that neighbour your new current room where you continue the killing stage.

Eventually, your traversal through these rooms through your new doorways will lead you to a room whose all neighbours have already been visited during your algorithm, to end this killing stage. The hunting stage must then find some previously unvisited room, makes no difference which one, as long as that room has some previously visited room as a neighbour. This last condition is extremely important to ensure that the maze remains **connected**, that is, there exists exactly one path from any room to any other room in the maze, instead of the space being partitioned into separate **connected components**. Dig a doorway from your chosen previously unvisited room to any one of its neighbours that has already been visited, and start a new killing stage from the newly chosen room. Repeat these hunting and killing stages, until every room in the space has been visited and has become connected to the rest of the maze.

In your labs folder, create a new class named `HuntAndKill`. Before implementing the actual hunt-and-kill algorithm, write the following utility method to create a `BufferedImage` object depicting a rectangular space where every room is an island:

```
public static BufferedImage createGrid(  
    int gridWidth, int gridHeight, int cellWidth, int cellHeight,  
    float PEN  
)
```

where the parameters `gridWidth` and `gridHeight` determine how many columns and rows the grid of rooms contains, and the parameters `cellWidth` and `cellHeight` give the pixel size of each room when rendered. For example, if `gridWidth` equals 20 and `cellWidth` equals 50, the returned pixel image would be 1000 pixels wide. The parameter `PEN` gives the width of the pen used to draw the walls in pixels.

Having written this utility method, write the actual maze generation method

```
public static BufferedImage generateMaze(  
    Random rng, int gridWidth, int gridHeight,  
    int cellWidth, int cellHeight  
)
```

that receives the same parameters as the previous method, and the random number generator object `rng` that your method must use to make all its internal decisions during its execution. Giving your method the same `rng` object should have it produce the same maze every time, and giving it a different `rng` object should result in a different maze within the constraints of this algorithm.

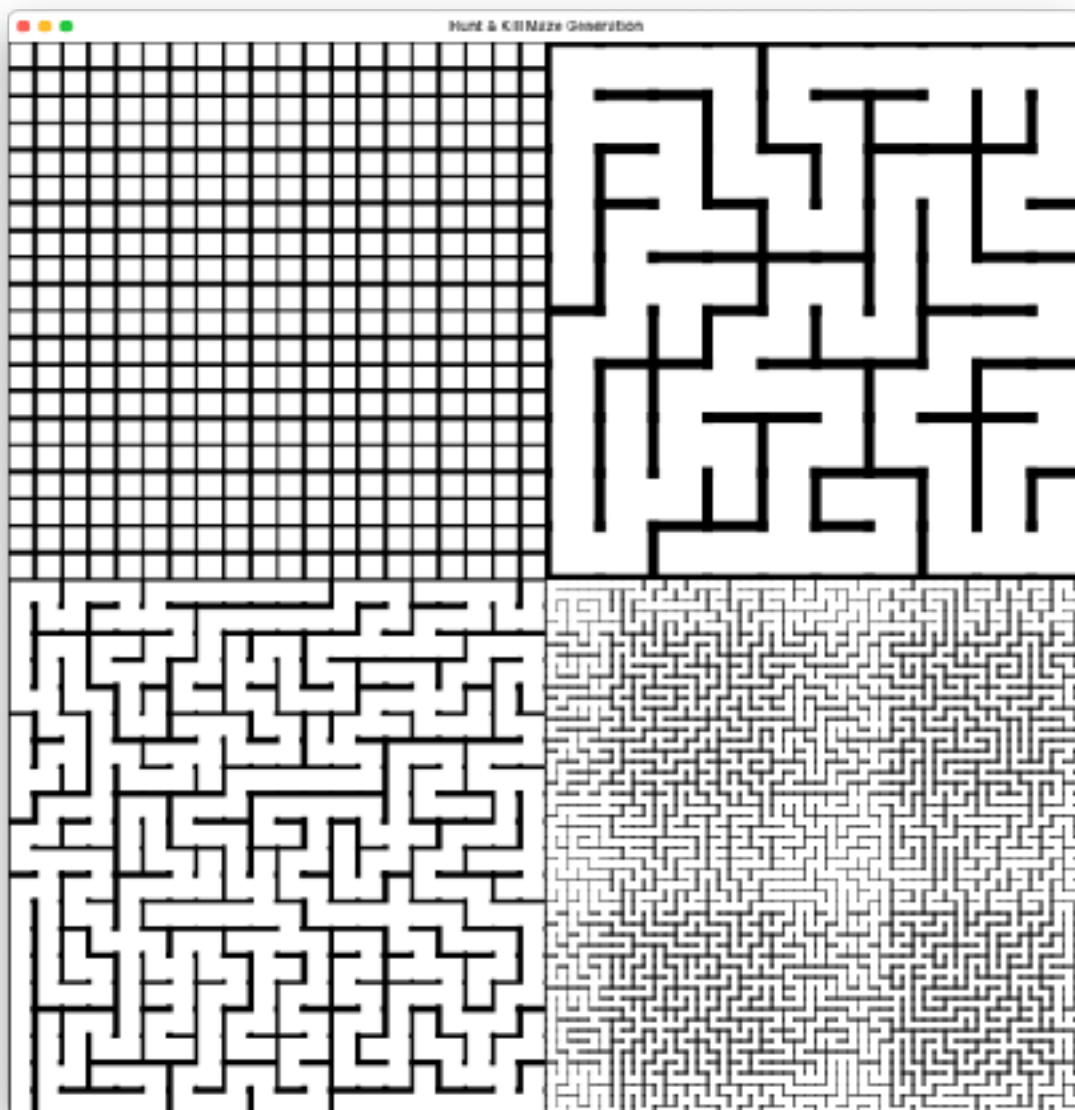
You get to decide for yourself how you want to keep track of the rooms that have already been visited. The simplest way would surely be to use a `boolean[ ][ ]` array for this purpose, but you can just as well use a `HashSet<Pair<Integer, Integer>>` collection for this purpose, once you bring in the example class [Pair<T, U>](#) from the lecture examples to your labs project to play the part that **tuples** would do for you in Python.

When implementing the hunt-and-kill algorithm, make sure to ensure its algorithmic **fairness**. During the killing stage, you should choose an unvisited neighbour randomly from all such neighbours of the current room. During the hunting stage, you should similarly choose the previously visited neighbour of your chosen unvisited room randomly among all its previously visited neighbours.

You might also want to separate the logic of digging the doorway between two rooms into a separate utility method. In the instructor's private model solution, this method has the signature

```
public static void connect(  
    Graphics2D g2, int x1, int y1, int x2, int y2,  
    int cellWidth, int cellHeight, float pen  
)
```

where the parameter `pen` is the width of the pen used to render the walls that separate the rooms, and `(x1, y1)` and `(x2, y2)` are the coordinates of the rooms to connect. Of course, many other ways to organize your logic would also be equally valid.



The `main` method in the class [`HuntAndKillMain.java`](#) provided by the instructor uses your methods to render the four images shown in the above screenshot. The image on top left can be used to test your `createGrid` utility method, whereas the other three images show the result returned from the `generateMaze` method for different grid sizes and random number generators. Of course your method doesn't have to make its random decisions the same way as the instructor's private model solution. However, your implementation must guarantee that the result is indeed a maze that has the same overall **texture** as the mazes returned by the hunt-and-kill algorithm tend to have, with lots of twists and turns and small dead ends all over the place.

As a final terminological point of note here, despite the common erroneous usage, the terms “[maze](#)” and “[labyrinth](#)” are not actually synonyms. A labyrinth is a **unicursal** special case of a maze that contains no branches anywhere, so that the rooms of a labyrinth form a single contiguous chain from beginning to end. A labyrinth therefore doesn't present a puzzle for its solver, but rather some kind of curated experience where the journey itself is more important than the destination. (The orchestra that needs the shortest time to finish playing a symphony is not necessarily the best possible orchestra.)

Bonus Lab 111: Dig-Through-Raster, Naked

For testing: [EllersAlgorithmMain.java](#)

This bonus lab continues on the theme of mazes from the previous bonus lab, but makes you implement a completely different algorithm to generate the random mazes. **Eller's algorithm** is restricted to operate on rectangular grids of rooms, since it doesn't work for arbitrarily shaped spaces. To compensate for this restriction, this offbeat algorithm has curious properties not shared by the other, more “mainstream” maze generation algorithms. For example, Eller's algorithm can lazily generate a rectangular maze that is infinite in one direction, since the generation of each row of rooms requires information only about the previous row of rooms, plus some additional bookkeeping data to identify groups of rooms that cannot be connected without creating a loop. This property makes Eller's algorithm suitable for some old school side-scrolling shoot-'em-up where the infinite playing field is generated one column at the time before that column slides into view.

Eller's algorithm is also explained wonderfully in Jamis Buck's blog post "[Maze Generation: Eller's Algorithm](#)" that the reader might again first want to check out before embarking on this lab. Furthermore, to let this lab have some educational historical value, we will parameterize this algorithm a bit as a tribute to [Frieder Nake](#), another historical pioneer of generative art, to imitate the style and spirit of the work "[Walk-through-Raster, Series 2](#)" and its variations.

In your labs folder, create a new class named `EllersAlgorithm`. This class contains only one method that is used to generate the maze, with the required signature

```
public static BufferedImage generateMaze(  
    Random rng, int gridWidth, int gridHeight,  
    int cellWidth, int cellHeight,  
    IntBinaryOperator digProb, IntBinaryOperator mergeProb  
)
```

The parameters `rng`, `gridWidth`, `gridHeight`, `cellWidth` and `cellHeight` play the same roles as they did in our earlier summer stock production of the hunt-and-kill algorithm in the previous bonus lab 110. The two new parameters `digProb` and `mergeProb` are function objects of type [java.util.function.IntBinaryOperator](#), representing two functions used to determine the probability that certain operations, explained later in this specification, are performed during the method execution, these probabilities depending on the row and column coordinates of the current room in the rectangular grid that is being processed. These functions return these probabilities as integers ranging from 0 to 100, inclusive, to represent these probabilities as everyday percentages. (To flip a weighted coin that says “yes” with the given integer percentage probability `p`, you can use the expression `rng.nextInt(100) < p`.)

As explained by Jamis Buck in his blog post linked above, Eller's algorithm generates the maze one row at the time, starting from the top row and marching from there towards the bottom row that will be processed in the end with a special rule that wraps up all the loose hanging threads and turns the final structure into a proper maze. Every room is again initially sealed off from its neigh-

bours with walls, some of which the algorithm will dig through during the execution to carve the random pathways through this space. As the algorithm processes the rows of rooms from top to bottom, the rooms in the current row will be processed from left to right. Each room will receive an **ID number** that identifies the earliest **ancestor** room visited by this algorithm from which there exists a path to the current room. The ID number of each room will either be inherited from its top or its left neighbour, or in case that room starts an entirely new path down the maze, gets initialized from a deli counter style ticket machine that produces a new ID number every time it is used. (Of course, in this algorithm this ticket machine is implemented as a simple local `int` counter variable.)

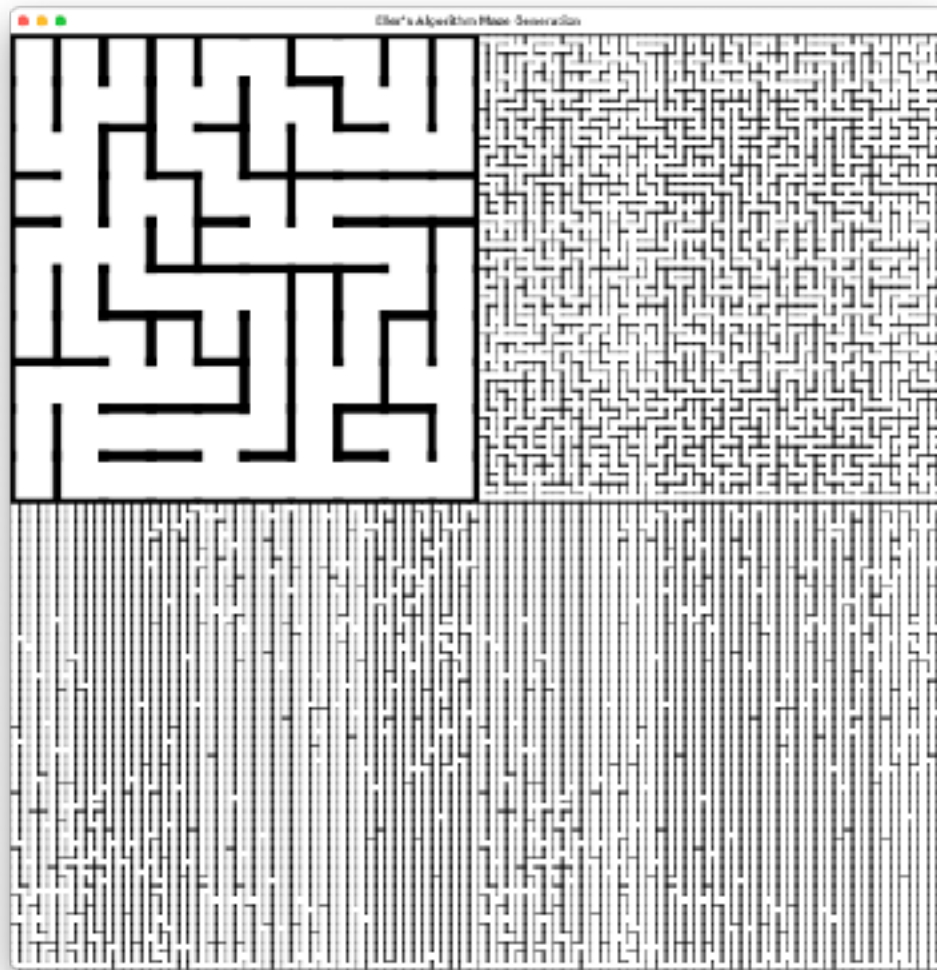
One absolutely inviolable rule about these ID numbers is necessary to understand right away: *if two adjacent rooms in the same row have been given the same ID number, this algorithm can never ever connect those two rooms by digging a doorway between them.* Never, ever, ever! Since somewhere in some earlier row those two rooms have a common ancestor from which there exists a separate path to both rooms, violating this rule by creating this horizontal connection would therefore necessarily create some loop, and thus invalidate the entire maze.

To start this process, each room in the first row is given a different ID number, being the ancestor of the pathway that emanates from that room down the maze. The inner loop to process each row iterates through this row from left to right. For each room in this row, use the function `digProb` to decide whether you are going to dig a doorway from that room to the room immediately below in the next row. If you choose to dig that doorway, the room below the current room is given the same ID as the room currently being processed. Otherwise the room below the current room is immediately given the next available ID number from the ticket machine.

The ID numbers of an entire block of consecutive rooms on the same row may be equal, in which case you are not allowed to dig a horizontal doorway between any of those rooms. However, when two horizontal neighbours have different ID numbers a and b , use the function `mergeProb` to decide whether you will dig a new doorway to connect those neighbours. If you choose to dig that doorway, the ID numbers of all rooms in the next consecutive block of rooms sharing their ID number b must then be updated to equal a , to let the algorithm know that all those rooms now have a new earlier common ancestor.

One more rule is required to guarantee that the resulting maze will be connected. For any consecutive block of rooms in the same row that share the same ID number, you have to dig down from at least one of these rooms, as otherwise the rooms in that block would not be reachable from any rooms that are not reachable from the common ancestor of that block identified by that ID number. Therefore, when you come to the end of such a consecutive block and have decided not to extend that block by digging a horizontal doorway into the next block of consecutive rooms with its own shared ID number, you have to dig down from either the current room (simple) or one randomly chosen room in the current block (more fair), unless you have already done such digging in at least one of those rooms.

Since we are not generating an infinite maze, the last row of the rectangular grid exists and must be handled in a special way to ensure that the resulting structure becomes a legitimate maze. Loop through the last row from left to right, but this time you have no choice about how you will dig the



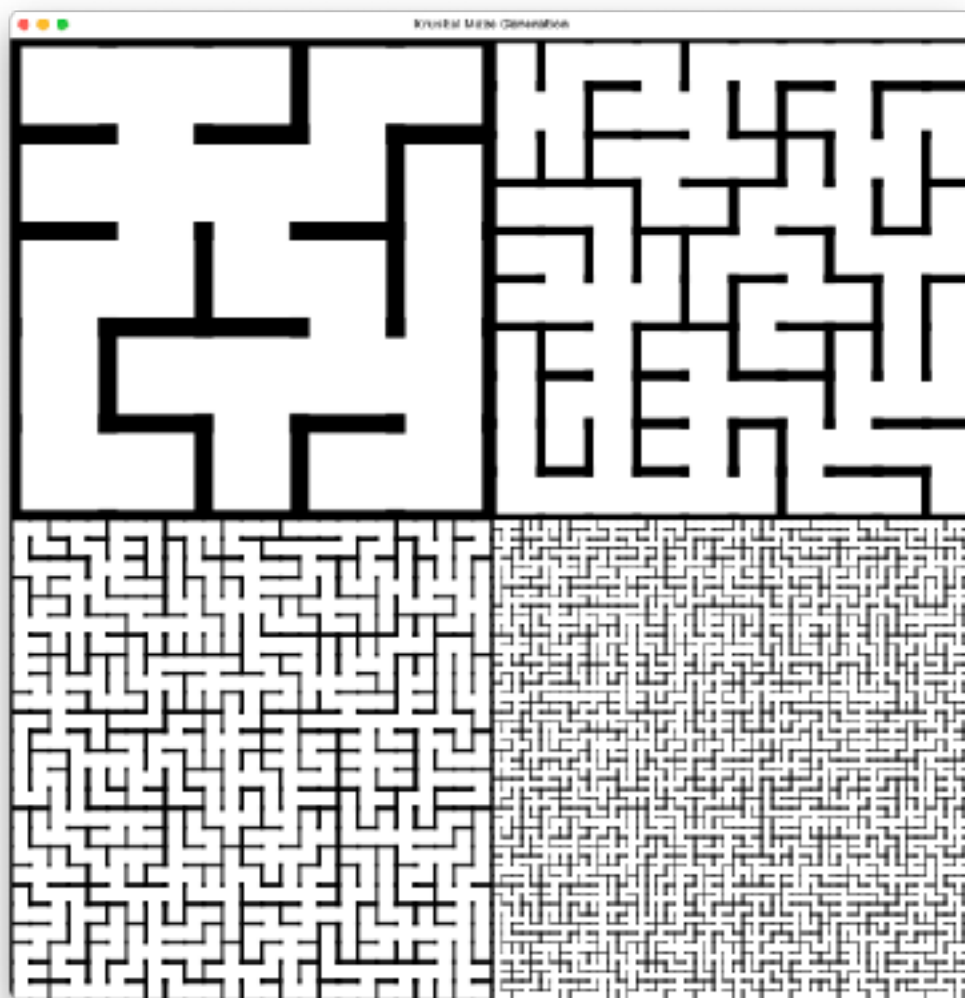
horizontal doorways. This time, you **must** dig a hole between every pair of consecutive rooms whose ID numbers are different. And that's all the fat lady sung; the maze is now complete.

The `main` method in the class [EllersAlgorithmMain.java](#) provided by the instructor uses your method to generate four mazes shown in the earlier screenshot. The two mazes on the top row use uniform probabilities for digging and merging decisions throughout the grid, whereas in the two mazes on the bottom row, these probabilities depend on the row and column coordinates of the current room. This produces a result whose texture is visibly different around the diagonal, similar to Frieder Nake's famous work linked to earlier. To fully imitate this work, you could also randomly remove some of the walls for an artistic effect, forcing the audience to imagine these missing walls to see the maze structure hiding underneath. Whatever this symbolizes and reveals about the tragedy of the human condition and our optimistic tendency to imagine comforting underlying support structures that maintain law and order where there actually aren't any, well, maybe some psychology or art professor could explain it all better than I can.

Bonus Lab 112: Kruskal Mazes

For testing: [KruskalMazeMain.java](#)

When the rooms and their adjacency relationships are considered a graph whose nodes are these rooms so that adjacent rooms are connected by edges, a maze is a spanning tree of that graph. The texture of the maze depends a lot on the graph algorithm used to construct this spanning tree. **Backtracking depth-first recursion** tends to create mazes consisting of long hallways that rarely branch. At the other extreme, giving each edge a random **weight** and then constructing the **minimum spanning tree** whose sum of weights is the smallest possible tends to produce mazes of very high branching and lots of short dead ends of length one, as illustrated in the screenshot below.



Along with [Prim's algorithm](#), itself a straightforward variation of the famous [Dijkstra's shortest path algorithm](#) with literally just one line changed to use the operation `min` instead of `+` to update frontier node values, [Kruskal's algorithm](#) is the other major minimum spanning tree algorithm that works for any graph, not just for simple **grid graphs** for which we are constructing mazes. The

reader might once again check out Jamis Buck's excellent presentation in his blog post "[Maze Generation: Kruskal's Algorithm](#)", especially the animations of this algorithm.

Kruskal's algorithm can be made vastly more efficient with aid of the [disjoint set union data structure](#). We already implemented this data structure in disguise in the earlier lab "48: The Curse of the Clumpino" under a different made-up term "clump" to denote a disjoint set of values, the data structure allowing such sets to be efficiently merged together. This makes this particular bonus lab especially suitable for students who have already solved that lab. However, Kruskal's algorithm doesn't require that data structure to get working correctly, if you don't mind using linear-time inner loops to do the work that the methods `sameClump`, `meld` and `clumpSize` in the `Clumps` data structure would do for you in constant time. Even without the aid of this wonderful little data structure, any off-the-shelf desktop computer from this century should be able to generate the mazes requested in the `main` method of the [KruskalMazeMain.java](#) class within a reasonable waiting time.

Kruskal's algorithm can be explained in terms of mazes with the realization that when you dig a doorway between two rooms a and b , you are not only creating a pathway between just those rooms to connect them together, but you are connecting every room that is already connected to room a to every room that is connected to room b . The disjoint set union data structure, implemented earlier as `Clumps`, can be used to efficiently keep track of reachability between rooms. Initially, every room is a prison cell that is not connected to any other room, corresponding to the initial state of the `Clumps` data structure where every element is its own singleton clump. During the execution of the algorithm, two rooms are in the same clump if and only if there exists a pathway that connects them. From this invariant we realize that we can never dig a doorway between two rooms in the same clump, as doing so would create a loop in the graph and invalidate the entire maze.

For a general graph, Kruskal's minimum spanning tree algorithm works by sorting the edges in ascending order by their weights (you can use for this purpose any sorting algorithm that you happen to have lying around), and creates a disjoint set union data structure where each node is initially its own singleton set. The algorithm then processes these edges in ascending order of weights. For each edge (a, b) , use the disjoint set data structure to determine whether the nodes a and b are already connected, that is, whether they are already members of the same set. If this is not the case, add the edge (a, b) to the minimum spanning tree, and update the disjoint set data structure by melding the nodes a and b , thus simultaneously melding everything connected to a to everything connected to b in one swoop. When all nodes belong to the same set, every node is somehow reachable from every other node, and the minimum spanning tree is complete.

In your project folder, create a new class `KruskalMaze`, and there the method

```
public static BufferedImage generateMaze(  
    Random rng, int gridWidth, int gridHeight,  
    int cellWidth, int cellHeight  
)
```

whose parameters play the exact same roles as they did back in the `generateMaze` method in the class `HuntAndKill` to create a random maze with the hunt-and-kill algorithm.

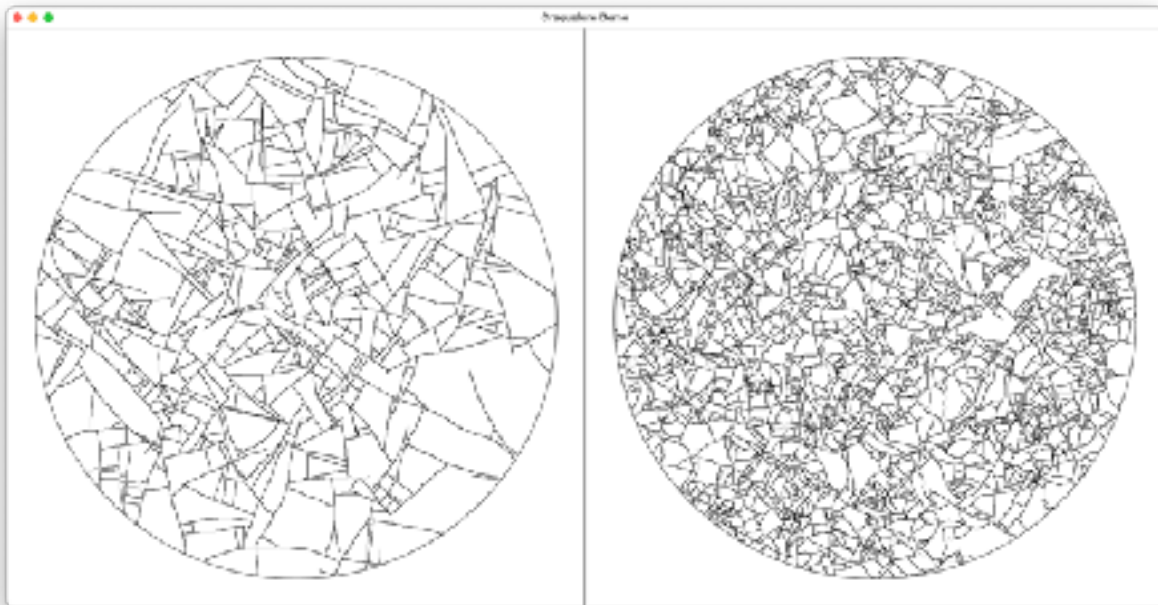
Unless you want to implement a generic version of the disjoint set data structure that can contain arbitrary types of elements, nodes of this graph need to be somehow encoded to nonnegative integers so that they can be stored inside the `Clumps` data structure. Since we are dealing with a grid graph that consists of exactly `gridHeight` rows with exactly `gridWidth` columns per row, this encoding can map the two-dimensional room coordinates (x, y) to a scalar integer e given by the formula $e = y * \text{gridWidth} + x$. To convert back to the original x - and y -coordinates from this scalar integer encoding e , use the formulas $x = e \% \text{gridWidth}$ and $y = e / \text{gridWidth}$. (Zero-based counting once again demonstrates its tremendous superiority over our everyday one-based counting by making these formulas nice and compact.)

When the grid coordinates of rooms encoded are into integers in this manner, you can build a disjoint set data structure to contain the consecutive integers from 0 to `gridWidth*gridHeight-1` as singleton sets. Each edge (e_1, e_2) can be expressed as a `Pair<Integer, Integer>` object from the [Pair](#) utility class discussed in the class about Java generic types. Since our task is to create a random maze, there is no point first making up random weights for these edges and then going through the task of sorting them; just build an `ArrayList<Pair<Integer, Integer>>` collection and fill it with all the edges of the grid graph in two nested for-loops, and use the utility method `Collections.shuffle` with the given `rng` to shuffle this list of edges in random order in linear time. After these preparations, Kruskal's algorithm itself will be a simple foreach-loop through the shuffled edges, consulting the `Clumps` data structure for each edge to determine if the two rooms connected by that edge are already connected via the previously dug doorways, and updating the global connection information for each new doorway that it digs.

Bonus Lab 113: Beautiful For Having Been Broken

For testing: [CraquelureMain.java](#)

This bonus lab was inspired by the page “[Fractures](#)” by Anders Hoff, part of [his excellent website on generative art algorithms](#), and by “[Branches: Nature's Patterns](#)”, a book by Philip Ball. The task in this lab is to simulate random patterns of [craquelure](#) for their artistic effect, as in the “[kintsukuroi](#)” style of Japanese art. Natural cracking can be simulated computationally using a **particle system** where random **particles** travel through the image leaving cracks in their wake. These particles will occasionally spawn new particles to make these cracks branch. Most importantly, a particle that hits an existing crack ceases to exist, causing the crack that it leaves behind to also end there.



In your project folder, create a new class `Craquelure`, and there the method

```
public static BufferedImage createImage(  
    Random rng, int w, int n, double buzz, double spawn  
)
```

that generates a new craquelure image that is `w` pixels wide and `w` pixels tall. Measured from the center of this image, the pattern should appear inside a circle whose radius is 45% of the width of the image, to leave a relaxed 10% margin around the edge.

To represent the individual particles, your method should define a local class `Particle` so that each particle has a position and a direction of movement. The direction of movement is best described as the angle `theta`, expressed in radians. Each particle should also have a positive integer ID that distinguishes it from other particles, assigned to that particle at construction. To keep track of collisions efficiently, we will trade some memory for time by defining an array

```
int[][] visited = new int[w][w];
```

to keep track of which particle (identified by its ID) has so far claimed a particular pixel in the image. Initially all elements of this array are zeroes, representing a fully intact surface before these particles start working on it.

Your method should start by creating *n* initial particles placed on random locations and directions inside the circle. As you create these particles, add them inside some kind of a queue. The main loop of this method should keep running as long as this queue is not empty. Pop the next particle out of this queue, and have this particle move a little bit to its current direction, claiming the next pixel that it moves into for itself. After making this move, the direction of this particle is slightly changed randomly, the magnitude of this change depending on the parameter *buzz*. However, if that pixel had already been visited by some other particle with a different ID, or if the current particle hits the circle boundary around the area where we form these cracks, the current particle should cease to exist in that it does not get pushed back to the back of the queue.

In case the reader has already blissfully forgotten the high school trigonometry needed in these calculations, the formulas that move the particle from its current position given by the two floating point variables *x* and *y* one unit into the direction of the angle *theta* are

```
x += Math.cos(theta);  
y += Math.sin(theta);
```

and a good formula to update the direction of the particle after this move would be

```
theta += rng.nextGaussian() * buzz;
```

The parameter *spawn* defines the probability that the current particle spawns a new particle after its move. A good way to make this decision is to test whether `rng.nextDouble() < spawn`. Make sure to move the new particle far enough from its parent so that it doesn't die immediately by falling in its parent's crack.

With anti-aliasing turned on and using a reasonable pen width to render the paths of these particles, your method should return results that resemble the following screenshot when tested out with the main method of the [CraquelureMain.java](#) class provided by the instructor. After getting the basic algorithm to work, the reader can try out variations such as having different particles move in different speeds, to better simulate the effect of glass cracking. With a splash of colour added to the rendering, the results might hopefully reach the level of “[Substrate](#)” by Jared Tarbell, evoking visuals of futuristic cityscapes, organic fields that “interrogate” the colonization of nature around us, and any other schmaltzy platitudes that artists so like to include in their vision statements. If you have previously implemented Lab 57 from this problem collection, you can use the **flood fill** algorithm to fill in the areas between cracks, choosing your colours randomly from some artistic colour palette.